Where Nehari's Theorem Survives

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Preface

This book organizes a body of research notes around one question: *where does Nehari's theorem survive in infinitely many variables, and why?* In one variable every bounded Hankel form has a bounded symbol. On $\mathbb{T}^\infty$ this fails, and the multiplicative Hilbert matrix sat for years exactly on the fault line. The chapters record what is now known on either side of it.

The book assumes the standard theory: Hardy spaces of Dirichlet series and the Bohr lift, Helson's inequality, the Cole–Gamelin estimate, weighted Bergman spaces of Dirichlet series in the sense of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt, and Perfekt's recent Fejér–Riesz inequality. Known results are quoted, not reproved, and attributed where we know their source. For the remaining results we have not completed a literature search, so "new" here means only that we did not find them; it is not a priority claim. Failed approaches appear only as brief remarks, where they explain why a result has the shape it has.

How the proofs were checked. The results and most proofs were developed with the assistance of large language models and checked line by line in separate model runs. Results that rest on a computation say so in their statement or proof; rigorous computations use interval or exact rational arithmetic, and uncertified computations are marked as numerical. A *probabilistic certificate* is a computation whose conclusion holds except on an event of explicitly bounded probability, under stated assumptions on the random-number generator and on floating-point arithmetic. It is not a proof. None of these is formal verification, and none replaces reading by a specialist: *no proof in this book has been independently verified.* The proofs most in need of such a reading are the envelope argument of section 8.1, the Schatten characterization Theorem 5.7, and the Hellinger–Bessel chain of chapters 1 and 2 on which most of the book depends.

The parts are arranged as an argument. Part I proves the stability theorem for point evaluations and the Hellinger–Bessel inequality, and applies them to Carleson measures. Part II uses them, with sign averages, to prove Nehari-type theorems for structured Helson matrices. Part III shows where Nehari's theorem fails, in Schatten classes, and why the known constructions stop where they do. Part IV returns to the borderline object, the multiplicative Hilbert matrix, and to the constants and coefficient inequalities that remain open there.

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Overview

The problem

A Helson matrix $M(\alpha) = (\alpha(mn))_{m,n\ge1}$ is a Hankel form in infinitely many variables: under the Bohr lift it becomes $(F,G) \mapsto \int FG\,\overline{g}\,\mathrm{d}m$ on $H^2(\mathbb{T}^\infty)$, for any g whose analytic coefficients are α. On the circle Nehari's theorem says that every bounded Hankel form has such a g in L^∞, of the same norm. On $\mathbb{T}^\infty$ this fails: the best constant on $\mathbb{T}^d$ grows at least exponentially in d [44], and there are bounded Helson matrices with no bounded symbol at all. The one-variable proof goes through inner–outer factorization, $H^1 = H^2 \cdot H^2$, and that is exactly what is lost.

This book asks which Helson matrices still have bounded symbols, and why. Duality reduces the question to a Carleson embedding: a positive matrix $M(\alpha)$ with $\alpha(n) = \int z^{\kappa(n)}\,\mathrm{d}\mu$ has a bounded symbol as soon as $\int |F|\,\mathrm{d}\mu \lesssim \|F\|_{H^1}$, whereas boundedness of the matrix is the same estimate in H^2. So the question becomes: when does an H^2 Carleson measure on $\mathbb{D}^\infty$ embed H^1? In one variable the answer is always, by factorization; here we need a replacement.

The starting point is Perfekt's theorem [45]: the multiplicative Hilbert matrix $(1/(\sqrt{mn}\log mn))$, which is the moment matrix of Lebesgue measure on the curve $\gamma(s) = (p^{-s})_p$, has a bounded symbol. His proof localizes point evaluations with Möbius maps and Hoeffding's inequality, and works for H^1.

Separation: the Hellinger–Bessel inequality

Part I replaces factorization by a one-line inequality and a Gram estimate. For a point $r \in \mathbb{D}^\infty \cap \ell^2$ with reproducing kernel K_r, every $F \in H^p$ and every $p > 0$ satisfy

$$|F(r)|^{p/2} \le \int |F|^{p/2} |K_r|\,\mathrm{d}m$$

(Theorem 1.7). Equivalently, a function that is efficient at r must have $|F|^p$ close, in Hellinger distance, to the harmonic measure $\omega_r = |K_r|^2 dm/\|K_r\|^2$. Since $|F|^{p/2}$ lies in L^2, the evaluations at many points can then be added up by Bessel's inequality for the vectors $|K_r|$. This gives the *Hellinger–Bessel inequality* (Theorem 2.1): for every positive measure μ and every $p > 0$,

$$\int |F|^p\,\mathrm{d}\mu \le \|\mathcal{G}_\mu\|\,\|F\|_p^p, \qquad \mathcal{G}(r,s) = \|K_r\|\|K_s\|\,H(\omega_r,\omega_s).$$

The condition does not see p, and it holds in the Hardy, Bergman and Gaussian geometries alike. Everything quantitative then rests on computing Hellinger affinities. In each coordinate they are hypergeometric, and across coordinates they multiply,

so $H(\omega_r, \omega_s) \le e^{-E(r,s)/4}$ with $E = \sum_p \rho_p^2$ the sum of squared pseudo-hyperbolic distances. Along the prime curve E is, up to a bounded error, the hyperbolic distance in the half-plane (Theorem 1.6).

The consequences in Part I:

- the Fejér–Riesz inequality $\int_0^1 |f(1/2+\sigma)|^p \, \mathrm{d}\sigma \le (5+2\sqrt{2})\|f\|^p_{\mathscr{H}^p}$ for every $p > 0$ (Theorem 2.5), and, keeping the phases of one prime, the constant $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ on $(0, 2)$, which is exactly the limit of the method there (Theorem 2.8);
- Carleson estimates for measures along curves in all three geometries and on prime sets of any density, with constant below 21 for the prime curve (Theorem 2.10);
- a complete description of Carleson measures on families separated in energy: for these, H^p Carleson for one p is the same as for all p, and the same as the one-point condition (Proposition 2.13 and Corollary 2.12);
- on vertical segments of the critical line, the local embedding constant is $O(\varepsilon^{-1/2})$ at distance ε for every p (Theorem 3.4); for $p = 2$ the constant C_0 of $\int_0^1 |f(1/2+it)|^2 \, \mathrm{d}t \le C_0 \|f\|^2_{\mathscr{H}^2}$ satisfies $2\pi < C_0 < 2\pi + 10^{-3}$ and is attained (Proposition 3.7).

Two structural facts organize the chapter. The extremals of point evaluation in infinitely many variables are exactly $cK_r^{2/p}$ (Proposition 1.11), and in every geometry the harmonic-measure overlap is squeezed between the reproducing-kernel overlap and its square root, $|\langle k_r, k_s\rangle| \le H(\omega_r, \omega_s) \le |\langle k_r, k_s\rangle|^{1/2}$ (Proposition 1.3), with equality on the right for Gauss. The method has limits. The Gram condition is not necessary: angular averages in one coordinate are Carleson for every p while $\mathcal{G}_\mu$ is unbounded (chapter 2). And near the critical line its constant is exactly $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.24$, against π for the corresponding H^2 estimate.

Symmetry: sign averages

The second mechanism is independent. For $g \in H^p(\mathbb{T})$ and $|x| < 1$,

$$\frac{|g(x)|^p + |g(-x)|^p}{2} \le \frac{\|g\|_p^p}{1 - |x|^4},$$

sharply, and this tensorizes (Proposition 4.7). Averaging signs in a set of coordinates therefore costs $\prod(1 - |x_p|^4)^{-1}$, which is finite for $x \in \ell^4$, whereas a single evaluation needs $x \in \ell^2$. Part II combines the two mechanisms. Keep the primes of a set R of density θ on the curve and average the signs of the rest. The resulting family of positive Helson matrices interpolates between the multiplicative Hilbert matrix (R all primes) and its full sign average (R empty). For every member, boundedness, the H^1 Carleson condition and the existence of a bounded symbol are equivalent to $\nu((1/2, 1/2+t]) = O(t^\theta)$, with a constant depending only on R (Theorem 4.2). On the curve, the Nehari constant is below 43; for the full sign average it is at most $\zeta(2)$.

The same family has a complete compactness theory. There is a Hartman-type theorem: compactness is the little-o condition, and the symbol then lies in $C(\mathbb{T}^\infty) + N$ (Theorem 4.10). There is an essential Nehari theorem: the essential

norm is comparable to the essential size of the symbol, and equals $\pi w(1/2)$ for measures with a density (Theorem 4.12). Finally, for the curve matrix compressed to the integers generated by R, the whole distribution of the spectrum is read off from the dyadic profile $a = (e^{j\theta}\nu(I_j))_j$: for every $0 < p < \infty$ and $0 < q \le \infty$, it lies in $S_{p,q}$ if and only if $a \in \ell^{p,q}$ (Theorem 5.13). For R all primes this is the curve matrix itself. In particular, positive curve matrices occur in every Schatten class and all have bounded symbols. Off the curve the spectrum collapses: root-exponential decay for compressions to sparse prime sets, and $1/(n \log n)$ for the full sign average. The constructions and compactness arguments carry over to Bergman and Fock spaces, where Kulikov's contractive embedding, and its Gaussian limit, replace Carleman's inequality (chapter 6).

Failure: the threshold 4

Part III turns to Schatten-class forms with no bounded symbol. Helson's theorem rules them out in S_2. Brevig and Perfekt found them in S_p for $p > 5.74$, and Bampouras was the first to show that they exist for every $p > 4$ [4]. We found the same bound independently, with the same constant-plus-product family; chapter 7 gives both proofs. His proof integrates over scales and is shorter; ours fixes the scale by large deviations in the Gaussian model and is Legendre dual to his.

The point of Part III is the threshold itself. *These methods stop at* 4*; whether the true threshold is* 4 *is open.* For the constant-plus-product family, no test function and no choice of coefficients beats S_4, for any n (Theorem 8.1). For every polynomial on every torus $\|\varphi\|_{L^4} \le \|H_\varphi\|_{S_4}$, with equality in the Gaussian model, so scale integration can never go below 4 (Corollary 7.12). Affine symbols, even quadratic symbols of rank at most two, and zero-free symbols are stuck in the same way. Every family of bounded degree in a bounded number of variables is excluded on an interval above 2 (Theorem 8.10). For the Gaussian method everything comes down to one concrete question: is $|\int f\overline{\varphi}\,\mathrm{d}\mu| \le \|f\|_{L^1(\mu)}\|\varphi\|_{L^4(\mu)}$ for Gaussian polynomials (Problem 6)? A counterexample would go below 4; a proof would show that Gaussian blocks cannot, but would not decide p_*, which is 4 exactly when forms in S_4 have bounded symbols (Proposition 7.8). Bampouras's Paley–Wiener threshold, also 4, appears to have a different source: square-root cancellation in L^1 combined with parabolic scaling, not a Gaussian moment (section 8.5).

The borderline: is the constant π?

Part IV returns to the multiplicative Hilbert matrix. In one variable four inequalities share the constant π: Hilbert's, Fejér and Riesz's, Hardy's, and the bound for $\sum a_k/(k+1)$ on H^1, all because every H^1 function is a product of two H^2 functions (Proposition 9.1). On $\mathbb{T}^\infty$ the first survives with the same constant and factorization fails, and chapters 9 to 11 measure what that costs. The norm C of the functional $\sum a_n/(\sqrt{n}\log n)$ on $\mathscr{H}_0^1$ is finite by Perfekt's theorem, satisfies $\pi \le C \le 5.515$ (Proposition 9.14), and no counterexample to $C = \pi$ uses only primes below 10^{216}; there even the sum with absolute values on the coefficients is below π. Near-extremal vectors for the Hilbert matrix vary slowly under multiplication, so no fixed finite-prime factor can raise an extremal square above π (Theorem 9.3

and Corollary 9.4); the problem is equivalent to a Hankel completion on the ordered rationals (Theorem 9.5); and the six per cent excess over the H^2 benchmark seen in two to four primes is an Ortega-Cerdà–Seip effect of linear symbols, which tends to $\pi/(2\sqrt{2})$ for two large primes and says nothing about π (Proposition 9.10).

Chapter 10 turns to the relatives of ℓ. The embedding of Part I settles a conjecture of Bondarenko, Brevig, Saksman and Seip, who had treated $\beta \neq 1/p$ for $p \leq 2$, and more: for every $0 < p < \infty$, the fractional primitive $\sum a_n/(\sqrt{n}(\log n)^\beta)$ is bounded on $\mathscr{H}^p$ if and only if $\beta > 1/p$, or $\beta = 1/p$ and $p \leq 1$, exactly as on the disc; on a prime set of density θ the threshold is θ/p (Corollary 10.10 and Theorem 10.9). The companion question with absolute values on the coefficients is proved on every piece of $\mathscr{H}^1$ that a degree constraint visible to the Euler product cuts out (Theorem 10.2). On a prime set of density θ the absolute sum is bounded for every logarithmic exponent $\beta > \theta$, and the signed sum at $\beta = \theta$ (Corollary 10.7), so only the endpoint is open.

Is $C = \pi$? Very probably not. For quadratic combinations of Euler kernels at many scales near the critical line, the H^1 norm has an exact Gaussian limit (Theorem 11.3), so C is at least a constant of a log-correlated Gaussian field, and we conjecture equality (Conjecture 1). A probabilistic certificate puts that constant at 3.1450 or more, with failure probability at most $2 \cdot 10^{-30}$ (remark 11.10); a deterministic proof is open. The same Gaussian witnesses show that this excess would survive removing every continuous symbol (Corollary 11.7), and if ℓ does not attain its norm, then C equals its distance to the functionals of finitely many coefficients (Corollary 11.14).

Reading guide

Chapters 1 and 2 are the foundation. Part II depends on them throughout; chapter 3 is not needed later except for Theorem 3.1, used in chapter 10. Part III is independent of Parts I and II. In Part IV, chapter 9 uses chapter 2; chapter 10 needs chapter 2, Theorem 3.1 and Helson's inequality; chapter 11 uses chapter 9 and the essential norms of Part II. Each chapter ends with what remains open, and chapter B collects the open problems.

Part 1

Replacing factorization: point evaluations

CHAPTER 1

Harmonic measure, energy and stability

How much of the torus does a point evaluation use, and when can two point evaluations share it? Section 1.2 answers this for one point: the mass of an extremal function is exactly the harmonic measure of the point. Section 1.3 answers it for two points: harmonic measures overlap by an amount exponentially small in an energy, with an exact formula in one variable and a quantitative form of Kakutani's dichotomy. Section 1.5 identifies that energy, along the Dirichlet half-plane, with hyperbolic distance. Section 1.6 proves the converse: a function that is nearly extremal at a point distributes its mass like that point's harmonic measure, in every H^p. Chapter 2 turns this into the Hellinger–Bessel inequality.

Three geometries run in parallel: the Hardy space of the torus, Bergman spaces of the polydisc, and the Fock space. The Gaussian case is the flat model, where every overlap is an exact exponential. Statements are made on finitely many coordinates, with constants independent of their number, and pass to $\mathbb{T}^\infty$ by approximation.

What is new, to our knowledge, is this point of view: one change-of-variables rule for three geometries reduces every statement about one or two point evaluations to a computation with harmonic measures. In this form the stability theorem, its two-point and many-point consequences, and the two-sided identification of energy with hyperbolic distance are new; the real-axis case of the latter, a lower bound for the energy along the prime curve, is in Perfekt's paper [45]. The basic inequality $|F(r)|^{p/2} \le \int |F|^{p/2} |K_r| \, \mathrm{d}\lambda$ is elementary: it is the mean-value argument behind the Cole–Gamelin estimate, and because $|K_r|$ is a product over coordinates it tensorizes, so it holds in infinitely many variables with no loss. What is new is reading it as a lower bound for a Hellinger affinity, which turns questions about many point evaluations into Hilbert-space geometry (chapter 2).

1.1. Three geometries

Each geometry consists of a product probability measure λ, the closure X^p of the analytic polynomials in $L^p(\lambda)$, a reproducing kernel K_r of X^2, and a coordinatewise family of maps T_r with $T_r(0) = r$.

	λ per coordinate	$K_r(z)$ per coordinate	$T_r(w)$ per coordinate
Hardy	arc length dm on $\mathbb{T}$	$(1-\overline{r}z)^{-1}$	$(r+w)/(1+\overline{r}w)$
Bergman	$m_\beta = (\beta-1)(1-\lvert z\rvert^2)^{\beta-2}\,\mathrm{d}A$	$(1-\overline{r}z)^{-\beta}$	$(r+w)/(1+\overline{r}w)$
Gauss	$\gamma = \pi^{-1}e^{-\lvert z\rvert^2}\,\mathrm{d}A$ on $\mathbb{C}$	$e^{\overline{r}z}$	$r+w$

Here $\beta > 1$, and dA is area measure on $\mathbb{D}$, normalized to have mass one in the Bergman row. Write $k_r = K_r/\|K_r\|_2$ for the normalized kernel. Then

$$\|K_r\|_2^2 = \prod_p (1-|r_p|^2)^{-1}, \quad \prod_p (1-|r_p|^2)^{-\beta}, \quad \exp\Big(\sum_p |r_p|^2\Big)$$

in the three cases. All three have the same change-of-variables rule.

LEMMA 1.1. *In each geometry, for every r,*

$$\lambda \circ T_r = \|K_r\|_2^2 \, |K_r \circ T_r|^{-2} \, \lambda,$$

and $K_r \circ T_r$ is holomorphic and zero-free on a neighbourhood of the closed domain.

PROOF. In one coordinate:

- *Hardy.* $K_r(T_r w) = (1+\overline{r}w)/(1-|r|^2)$, and the Jacobian of T_r on $\mathbb{T}$ is $(1-|r|^2)/|1+\overline{r}w|^2$.
- *Bergman.* The density of $m_\beta \circ T_r$ is $(1-|r|^2)^\beta |1+\overline{r}w|^{-2\beta}$, and $K_r(T_r w) = \big((1+\overline{r}w)/(1-|r|^2)\big)^\beta$.
- *Gauss.* $\gamma(r+dw) = e^{-|r|^2 - 2\Re(\overline{r}w)}\gamma(dw)$, and $K_r(r+w) = e^{|r|^2+\overline{r}w}$.

In each case the identity is checked by multiplying out. Products over coordinates preserve it. □

1.2. One point

A *representative* of evaluation at r is a function ψ with $\int F\overline{\psi}\,\mathrm{d}\lambda = F(r)$ for every analytic polynomial F. Every representative has $\|\psi\|_\infty$ at least the norm of evaluation on X^1. Two representatives sit at opposite extremes.

- *The positive representative* $|k_r|^2 = |K_r|^2/\|K_r\|_2^2$. In the Hardy case this is the Poisson kernel P_r. It has L^1 norm one, but it is unbounded, with $\sup P_r = \prod_p (1+|r_p|)/(1-|r_p|)$, which is infinite unless $\sum|r_p| < \infty$.
- *The flat representative* $\Phi_r = \|K_r\|_2^2 \, K_r/\overline{K_r}$, of constant modulus $\|K_r\|_2^2$, introduced in the Hardy case by Perfekt [45]. That it represents evaluation is the case $D = \Omega$ of Lemma 1.5 below.

In the Hardy case the norm of evaluation is due to Cole and Gamelin [21], who identified $\mathbb{D}^\infty \cap \ell^2$ as the set of points where evaluation on $H^1(\mathbb{T}^\infty)$ is bounded.

PROPOSITION 1.2. *In each geometry, evaluation at r has norm exactly $\|K_r\|_2^2$ on X^1. The flat representative is extremal:*

$$\min\{\|\psi\|_\infty : \psi \text{ represents evaluation at } r\} = \|K_r\|_2^2 = \|\Phi_r\|_\infty.$$

The function $F_r = K_r^2/\|K_r\|_2^2$ attains the norm, with $F_r(r) = \|K_r\|_2^2$, $\|F_r\|_1 = 1$, and

$$|F_r|\,\mathrm{d}\lambda = \omega_r := |k_r|^2\,\mathrm{d}\lambda = (T_r)_*\lambda.$$

PROOF. The flat representative gives $|F(r)| \le \|K_r\|_2^2 \|F\|_1$. For F_r we have $\|F_r\|_1 = \|K_r\|_2^2/\|K_r\|_2^2 = 1$ and $F_r(r) = \|K_r\|_2^4/\|K_r\|_2^2$. So the norm is $\|K_r\|_2^2$, and no representative has smaller sup norm. The identity $|k_r|^2\lambda = (T_r)_*\lambda$ is Lemma 1.1, read backwards. □

We call ω_r the *harmonic measure* of r. In the Hardy case it is the usual product of Poisson measures, and in the Gaussian case it is the standard Gaussian translated to r. Proposition 1.2 says that the mass of the extremal function at r is distributed exactly like ω_r. So the question of where near-extremal functions live becomes a question about harmonic measures.

1.3. Two points

Let $\rho(a,b) = |a-b|/|1-\overline{a}b|$ be the pseudo-hyperbolic distance on $\mathbb{D}$, and define the *energy* between two points:

$$E(r,s) = \sum_p \rho(r_p, s_p)^2 \quad \text{(Hardy, Bergman)}, \qquad E(r,s) = \sum_p |r_p - s_p|^2 \quad \text{(Gauss)}.$$

Two numbers measure how much two points share. In X^2 it is the overlap $|\langle k_r, k_s\rangle|$ of the normalized kernels. In X^1 it is the Hellinger affinity

$$H(r,s) = \int \sqrt{\mathrm{d}\omega_r \, \mathrm{d}\omega_s} = \int |k_r||k_s| \, \mathrm{d}\lambda$$

of the harmonic measures.

PROPOSITION 1.3. *The two overlaps compare as follows.*

(1) Exact X^2 overlaps.

$$|\langle k_r, k_s\rangle|^2 = \prod_p (1-\rho_p^2), \quad \prod_p (1-\rho_p^2)^\beta, \quad e^{-E(r,s)}$$

in the three cases, where $\rho_p = \rho(r_p, s_p)$.

(2) Comparison. *In every geometry, $H(r,s) \geq |\langle k_r, k_s\rangle|$.*

(3) Exact X^1 overlaps. *In the Gauss case $H(r,s) = e^{-E(r,s)/4}$. In the Hardy and Bergman cases $H(r,s) = \prod_p h_\beta(\rho_p)$, with $\beta = 1$ for Hardy and*

$$h_\beta(\rho) = (1-\rho^2)^{\beta/2} {}_2F_1\Big(\frac{\beta}{2}, \frac{\beta}{2}; \beta; \rho^2\Big);$$

for $\beta = 1$ we write $h = h_1 = (2/\pi)(1-\rho^2)^{1/2}\mathbf{K}(\rho)$, with $\mathbf{K}$ the complete elliptic integral of the first kind.

(4) Two-sided comparison. *In every geometry,*

$$|\langle k_r, k_s\rangle| \leq H(r,s) \leq |\langle k_r, k_s\rangle|^{1/2},$$

with equality on the right in the Gauss case. Consequently $H(r,s) \leq e^{-E(r,s)/4}$ in the Hardy and Gauss cases, and $H(r,s) \leq e^{-\beta E(r,s)/4}$ in the Bergman case; in the Hardy case $h(\rho) \leq (1-\rho^2)^{1/4} \leq 1 - \rho^2/4$.

PROOF. (1) In one coordinate, $|\langle k_a, k_b\rangle|^2 = (1-|a|^2)(1-|b|^2)/|1-\overline{a}b|^2 = 1 - \rho(a,b)^2$, and the Bergman kernel is the βth power of the Hardy kernel. In the Gauss case, $|\langle k_r, k_s\rangle|^2 = e^{2\Re\langle s,r\rangle - |r|^2 - |s|^2} = e^{-|r-s|^2}$.

(2) $\int |k_r||k_s| \, \mathrm{d}\lambda \geq |\int k_r \overline{k_s} \, \mathrm{d}\lambda|$.

(3) In the Gauss case ω_r is the complex Gaussian centered at r with identity covariance, and the Hellinger affinity of two such measures is $e^{-|r-s|^2/4}$. In the other two cases, Möbius invariance of the pseudo-hyperbolic distance and of the

construction $\omega_a = (T_a)_* m$ reduce one coordinate to $a = 0$, $b = c$ with $|c| = \rho$, and $h_\beta(\rho) = (1-\rho^2)^{\beta/2}\|(1-\bar{c}z)^{-\beta/2}\|^2_{A^2_\beta}$, which is the stated hypergeometric expression.

(4) The left inequality is (2). On the right both sides are products over coordinates. In the Gauss case $H = e^{-E/4} = |\langle k_r, k_s\rangle|^{1/2}$. In the other two cases, with $a = \beta/2$ and $x = \rho^2$, the claim is ${}_2F_1(a, a; 2a; x) \le (1-x)^{-a/2}$, which holds coefficient by coefficient: both series start at 1, and the ratios of consecutive coefficients are $(a+k)^2/((2a+k)(k+1))$ and $(a/2+k)/(k+1)$, where $(a+k)^2 \le (2a+k)(a/2+k)$ reduces to $0 \le ak/2$. Finally $(1-x)^{a/2} \le e^{-ax/2}$, and $(1-x)^{1/4} \le 1 - x/4$ by Bernoulli's inequality. □

So in every geometry the overlap of harmonic measures decays exponentially in the energy, and it is never smaller than the X^2 overlap. The Gauss case is the exact model. In the Hardy and Bergman cases the upper rates are $1/4$ and $\beta/4$, respectively; both are optimal at small energy, and near $\rho = 1$ the affinities are far smaller. For constants near the boundary the relevant Hardy bound is $h \le (1-\rho^2)^{1/4}$, from part (4).

Corollary 1.4 (Kakutani's dichotomy, quantitatively). *In the Hardy geometry on infinitely many coordinates, ω_r and ω_s are mutually singular if and only if $E(r,s) = \infty$. In particular ω_r is absolutely continuous with respect to Haar measure exactly when $r \in \ell^2$, and any two points of $\mathbb{D}^\infty \cap \ell^2$ have equivalent harmonic measures, with*

$$2\big(1 - H(r,s)\big) \le \|\omega_r - \omega_s\| \le 2\sqrt{1 - H(r,s)^2}.$$

Proof. By Kakutani's theorem [33], two product measures with equivalent factors are equivalent or mutually singular according as the product of the factor affinities is positive or zero. By Proposition 1.3 that product is positive exactly when $\sum \rho_p^2 < \infty$. Take $s = 0$ for the second statement. The total variation bounds are the standard Hellinger inequalities. □

This explains two things. First, it explains the domain $\mathbb{D}^\infty \cap \ell^2$ of Cole and Gamelin: it is where harmonic measure lives on the torus at all. Second, it shows that between two points of $\mathbb{D}^\infty \cap \ell^2$ mutual singularity is only ever approximate. Any argument that separates point evaluations must therefore be quantitative in the energy. The rest of the chapter makes it so.

1.4. Localization

The flat representative spreads its mass over the whole space. The key observation, due to Perfekt [45] in the Hardy case, is that it can be cut down to the image under T_r of any rotation-invariant set, at the price of the measure of that set. Lemma 1.1 makes the proof the same in all three geometries.

Lemma 1.5 (Localization). *Let D have positive measure and be invariant under the common rotations $w \mapsto e^{i\vartheta} w$. Then*

$$\psi_{r,D} = \frac{\|K_r\|_2^2}{\lambda(D)} \frac{K_r}{\overline{K_r}} \mathbf{1}_{T_r(D)}$$

represents evaluation at r, and $|\psi_{r,D}| = \|K_r\|_2^2/\lambda(D)$ on its support $T_r(D)$.

PROOF. Substitute $z = T_r w$ and use Lemma 1.1. Since $\overline{K_r}/K_r \cdot |K_r|^{-2} = K_r^{-2}$,

$$\int F\,\overline{\psi_{r,D}}\,\mathrm{d}\lambda = \frac{\|K_r\|_2^4}{\lambda(D)}\int_D G\,\mathrm{d}\lambda, \qquad G = \frac{F\circ T_r}{(K_r\circ T_r)^2}.$$

The function G is holomorphic near the closed domain (in the Gauss case it is entire, with growth at most $e^{c|w|}$). Since D and λ are invariant under common rotations, averaging $G(e^{i\vartheta}w)$ over ϑ shows $\int_D G\,\mathrm{d}\lambda = G(0)\lambda(D) = F(r)\lambda(D)/\|K_r\|_2^4$. □

The cost of localization is the factor $1/\lambda(D)$. As long as D has measure at least $1/2$, the localized representative is at most twice the flat one, and it lives on $T_r(D)$. That set carries at least half of the harmonic measure $\omega_r = (T_r)_*\lambda$.

1.5. The hyperbolic plane on the prime curve

Lift the right half-plane $\{\Re s > 1/2\}$ to the torus by $s \mapsto \gamma(s) = (p^{-s})_p$. Write d for the hyperbolic distance in the half-plane $\{\Re w > 0\}$, $w = s - 1/2$, with metric $|\mathrm{d}w|/\Re w$.

THEOREM 1.6. *Uniformly for $0 < \Re w, \Re w' \le 1$ and $|\Im w - \Im w'| \le 1$,*

$$E\big(\gamma(\tfrac12 + w), \gamma(\tfrac12 + w')\big) = 2\log\cosh\frac{d(w,w')}{2} + O(1).$$

PROOF. Put $a_p = p^{-1/2-w}$, $b_p = p^{-1/2-w'}$. Then $\rho_p^2 = |a_p - b_p|^2/|1 - \overline{b_p}a_p|^2$, and $|a_p - b_p|^2(1 + |a_p b_p|)^{-2} \le \rho_p^2 \le |a_p - b_p|^2(1 - |a_p b_p|)^{-2}$. The difference contributes $O(\sum_p p^{-2}) = O(1)$. Expanding,

$$\sum_p |a_p - b_p|^2 = \mathcal{P}(1 + 2\Re w) + \mathcal{P}(1 + 2\Re w') - 2\Re\,\mathcal{P}(1 + w + \overline{w'}),$$

where $\mathcal{P}(1+z) = \sum_p p^{-1-z} = \log\zeta(1+z) + O(1) = \log(1/z) + O(1)$ uniformly for $0 < \Re z \le 2$ and $|\Im z| \le 1$, since ζ has only the simple pole at 1 there. The sum is therefore $\log\big(|w + \overline{w'}|^2/(4\Re w \Re w')\big) + O(1)$. For the right half-plane, $\cosh^2(d/2) = |w + \overline{w'}|^2/(4\Re w \Re w')$. □

So near the critical line, the geometry that governs harmonic measures on $\mathbb{T}^\infty$ is the hyperbolic plane, and $H(\omega, \omega') \lesssim \cosh(d/2)^{-1/2} \approx e^{-d/4}$. Hyperbolic area grows like e^d, while the number of points within distance d along a curve going to the boundary grows only like d. That difference explains why curves carry H^p-Carleson measures for every p (chapter 2), while vertical segments, which are horocycles with $e^{d/2}$ growth, fail at $p = 1$ (chapter 3).

1.6. Stability of point evaluation

For $F \in X^p$ define the efficiency at r, $\mathfrak{e}_r^{(p)}(F) = |F(r)|^p/(\|K_r\|_2^2\|F\|_p^p) \in [0,1]$.

THEOREM 1.7 (Stability). *In each geometry, for $0 < p < \infty$, nonzero $F \in X^p$ and every point r,*

$$H\Big(\frac{|F|^p\lambda}{\|F\|_p^p}, \omega_r\Big) \ge \mathfrak{e}_r^{(p)}(F)^{1/2}, \qquad \textit{equivalently} \qquad \int |F|^{p/2}|K_r|\,\mathrm{d}\lambda \ge |F(r)|^{p/2},$$

PROOF. By Lemma 1.1, $K_r \circ T_r$ is holomorphic and zero-free near the closed domain, so $G = (F \circ T_r)\left(\|K_r\|_2^2/(K_r \circ T_r)^2\right)^{1/p}$ is holomorphic there, and $|G|^{p/2} = |F \circ T_r|^{p/2}\|K_r\|_2 |K_r \circ T_r|^{-1}$. Substituting $z = T_r w$ and using the cocycle $\lambda \circ T_r = \|K_r\|_2^2 |K_r \circ T_r|^{-2}\lambda$,

$$\int |F|^{p/2}|K_r|\,\mathrm{d}\lambda = \int |F \circ T_r|^{p/2}\,\frac{\|K_r\|_2^2}{|K_r \circ T_r|}\,\mathrm{d}\lambda = \|K_r\|_2 \int |G|^{p/2}\,\mathrm{d}\lambda.$$

Since $\log|G|$ is plurisubharmonic, $|G|^{p/2}$ is subharmonic on complex lines through 0, and λ is invariant under common rotations. So $\int |G|^{p/2}\,\mathrm{d}\lambda \geq |G(0)|^{p/2} = |F(r)|^{p/2}\|K_r\|_2^{-1}$, using $K_r(r) = \|K_r\|_2^2$. Hence $\int |F|^{p/2}|K_r|\,\mathrm{d}\lambda \geq |F(r)|^{p/2}$. The affinity form follows on dividing by $\|K_r\|_2 \|F\|_p^{p/2}$. □

Taking $H \leq 1$ recovers the point estimate $|F(r)|^p \leq \|K_r\|_2^2 \|F\|_p^p$. The theorem is that estimate with a defect term. For Banach-valued F the same proof applies with $\log\|F\|_X$ in place of $\log|F|$. In one variable it is a stability form of the factorization $F = Ih^{2/p}$. In several variables F need not factor, but the inequality is linear in $|F|^{p/2}$ and the weight $|K_r|$ is a product, so it tensorizes; the square root $|F|^{p/2}$ plays the role of the H^2 vector h.

COROLLARY 1.8 (Two-point necessity). *In the Hardy and Gauss geometries, if $\mathfrak{e}_r^{(p)}(F), \mathfrak{e}_s^{(p)}(F) \geq 1-\eta$ with $\eta < 1/2$, then $E(r,s) \leq 4\log\big(1/(1-2\eta)\big) = 8\eta + O(\eta^2)$. In the Bergman geometry the same holds with E replaced by βE.*

PROOF. The Hellinger angle $\arccos H$ is a metric (the angle between square-root densities in L^2). With $\mu = |F|^p\lambda/\|F\|_p^p$, $\arccos H(\omega_r, \omega_s) \leq 2\arccos\sqrt{1-\eta}$, so $H(\omega_r, \omega_s) \geq 1 - 2\eta$. Combine with $H \leq e^{-E/4}$ (Proposition 1.3). □

The bound is sharp up to a factor 2. The function $K_t^{2/p}$ at the coordinatewise hyperbolic midpoint t of r and s has efficiency $1 - E/4 + O(E^2)$ at both points. Along the prime curve, a Dirichlet series is $(1-\eta)$-efficient only on a set of scales of logarithmic diameter at most $7 + 4\log(1/(1-2\eta))$. So separation in energy is not only sufficient for adding up evaluations (chapter 2) but necessary.

For more than two points the same Hilbert-space picture does better than iterating the triangle inequality.

COROLLARY 1.9 (Many points). *In each geometry let $F \in X^p$ be nonzero, let $r_1, \ldots, r_N$ be points, $\mathfrak{e}_i = \mathfrak{e}_{r_i}^{(p)}(F)$ and $H_{ij} = H(\omega_{r_i}, \omega_{r_j})$. Then for all $a_1, \ldots, a_N \geq 0$,*

$$\Big(\sum_i a_i\sqrt{\mathfrak{e}_i}\Big)^2 \leq \sum_{i,j} a_i a_j H_{ij}.$$

In particular, if every $\mathfrak{e}_i \geq 1-\eta$, the average of H_{ij} over the pairs $i < j$ is at least $1 - N\eta/(N-1)$.

PROOF. The square roots v of $|F|^p\lambda/\|F\|_p^p$ and w_i of ω_{r_i} are unit vectors in $L^2(\lambda)$ with $\langle v, w_i\rangle \geq \sqrt{\mathfrak{e}_i}$ by Theorem 1.7. So $\sum_i a_i\sqrt{\mathfrak{e}_i} \leq \langle v, \sum_i a_i w_i\rangle \leq \|\sum_i a_i w_i\|_2$. For the second statement take equal weights. □

Iterating the two-point estimate gives only $1-2\eta$ for each pair; for $N>2$ the average bound is better, and combined with $H\le e^{-E/4}$ it forces some pair to have energy at most $-4\log(1-N\eta/(N-1))$.

Equality and near-extremizers

COROLLARY 1.10. *In each geometry, let $0<p<\infty$ and $F\in X^p$ with $F(r)\neq 0$.*

(1) $\mathfrak{e}_r^{(p)}(F)=1$ if and only if $F=cK_r^{2/p}$ for a constant c.

(2) With total variation $\|\cdot\|_{\mathrm{TV}}$ the total mass of the difference,

$$\Big\|\frac{|F|^p\lambda}{\|F\|_p^p}-\omega_r\Big\|_{\mathrm{TV}}\le 2\sqrt{1-\mathfrak{e}_r^{(p)}(F)}.$$

PROOF. (1) Equality forces equality in $\int|G|^{p/2}\,\mathrm{d}\lambda\ge|G(0)|^{p/2}$ in the proof above. Disintegrating λ over complex lines through 0, the one-variable slices $g_w(\zeta)=G(\zeta w)$ satisfy $\int_{\mathbb{T}}|g_w|^{p/2}\,\mathrm{d}m=|g_w(0)|^{p/2}\neq 0$ for almost every w. A zero of g_w in $\mathbb{D}$ would make the inequality strict, so g_w is zero-free and $g_w^{p/2}$ is analytic with $\int|g_w^{p/2}|=|\int g_w^{p/2}|$; hence $g_w^{p/2}$ has constant argument on $\mathbb{T}$ and is constant. So G is constant, which is $F=cK_r^{2/p}$. (2) For probability measures, $\|\mu-\nu\|_{\mathrm{TV}}\le 2\sqrt{1-H(\mu,\nu)^2}$, and $H\ge\lambda^{1/2}$ by Theorem 1.7. □

So an $(1-\eta)$-efficient function at r distributes its mass like ω_r up to $2\sqrt{\eta}$ in total variation. The converse fails. In the Hardy geometry, let I be a Möbius map in one coordinate with $|I(r)|=(1-\eta)^{1/p}$ and put $F=IK_r^{2/p}$. Then $|F|^p\,\mathrm{d}m/\|F\|_p^p=\omega_r$ exactly, while $\mathfrak{e}_r^{(p)}(F)=1-\eta$. Modulus information cannot distinguish near-extremizers from inner perturbations of the extremizer; this is the phase blindness that limits the method in chapter 9.

The proofs above use T_r in every coordinate, which is harmless when r has finitely many nonzero coordinates. The equality statement holds on the whole domain of bounded point evaluations, in each geometry.

PROPOSITION 1.11 (Equality in infinitely many variables). *In each geometry, let $0<p<\infty$, let $r\in\ell^2$, with $r\in\mathbb{D}^\infty$ in the Hardy and Bergman cases, and let $F\in X^p$ be nonzero. Then $\mathfrak{e}_r^{(p)}(F)=1$ if and only if $F=cK_r^{2/p}$ for a constant c.*

PROOF. For s with finitely many nonzero coordinates put

$$V_sF=(F\circ T_s)\Big(\frac{\|K_s\|_2}{K_s\circ T_s}\Big)^{2/p},$$

with the branch positive at 0. By Lemma 1.1, V_s is an isometry of X^p onto itself, with $V_sF(0)=F(s)\|K_s\|_2^{-2/p}$. Let $r^{(N)}$ keep the first N coordinates of r. On a polynomial in the first N variables, $V_{r^{(M)}}P=V_{r^{(N)}}P\cdot B_{N,M}$ for $M>N$, where

$$B_{N,M}=\prod_{N<j\le M}k_{-r_j}(z_j)^{2/p},$$

because the formulas in the proof of Lemma 1.1 give $\|K_s\|_2/(K_s\circ T_s)=k_{-s}$ in each geometry. The logarithms of the factors are independent, with mean $O(|r_j|^2)$ and L^2 norm $O(|r_j|)$, so $\log B_{N,M}\to 0$ in L^2 uniformly in M, because $\sum_{j>N}|r_j|^2\to 0$. Also

$|B_{N,M}|^p$ is the squared modulus of a unit vector of X^2 close to 1, hence uniformly integrable; so $B_{N,M} \to 1$ in L^p uniformly in M. By density, $V_{r^{(N)}}$ converges strongly to an isometry V_r of X^p onto itself, with inverse obtained in the same way, and $V_r F(0) = F(r)\|K_r\|_2^{-2/p}$.

So $g = V_r F$ satisfies $|g(0)|^p = \|g\|_p^p$. Setting the coordinates beyond M to zero does not increase the X^p quasi-norm, by subharmonicity in those coordinates and rotation invariance of λ, and fixes $g(0)$; so each truncation g_M attains equality as well. In finitely many variables a nonconstant h with $h(0) \neq 0$ has a local power $h^{p/2}$ near 0, and Parseval on a small torus gives $\int |h(\rho z)|^p \, \mathrm{d}m > |h(0)|^p$. The means of $|h|^p$ over tori increase in each radius, so $\int |h|^p \, \mathrm{d}\lambda > |h(0)|^p$ in each geometry, which contradicts equality. Hence every g_M is the constant $g(0)$, and so is g. Then $F = V_r^{-1} g(0)$, a multiple of $K_r^{2/p}$. □

REMARK 1.12 (Density of evaluations). Point evaluations at the curve points $\gamma(\sigma)$, $\sigma > 1$, already span a weak-$*$ dense subspace of $(H^1(\mathbb{T}^\infty))^*$, by uniqueness of Dirichlet coefficients. So superpositions of point evaluations, the functionals controlled in chapter 2, are weak-$*$ dense in the dual.

Open

QUESTION 1. Is the factor 2 in Corollary 1.8 necessary? What is the sharp many-point version of Corollary 1.9?

CHAPTER 2

The Hellinger–Bessel inequality

When can point evaluations be added up in H^p, for all p at once? The answer has a single source. By the stability theorem of chapter 1, the square root of $|F|^p$ is at least as aligned with the square root of the harmonic measure ω_r as the efficiency of F at r requires. Bessel's inequality then bounds the sum over many points by the Gram norm of the square roots $\sqrt{\omega_r}$. The results below are consequences of this one inequality; they hold for every $p > 0$, and the constants are explicit and small. To our knowledge the inequality itself, its sharpness on prime curves (Corollary 2.11) and the analysis of what it loses (Proposition 2.16) are new.

Throughout, the setting is one of the three geometries of section 1.1, and X^p is the analytic closure in $L^p(\lambda)$. The modulus of the reproducing kernel satisfies $|K_r|^2 = \|K_r\|_2^2 \, \mathrm{d}\omega_r/d\lambda$.

2.1. The inequality

Why this replaces factorization. In one variable a Carleson estimate in H^p is proved by writing $F = Ih^{2/p}$ with I inner and $h \in H^2$, so that $|h|^2 = |F|^p$, and applying the H^2 estimate to h. On $\mathbb{T}^\infty$ the analytic H^2 function h does not exist. The stability theorem uses instead the vector $|F|^{p/2} \in L^2(\lambda)$, which always exists and is not analytic. Analyticity enters once, in the pointwise inequality $|F(r)|^{p/2} \le \langle |F|^{p/2}, |K_r| \rangle$ of Theorem 1.7; everything after it is Hilbert-space geometry. If F is to be efficient at many points, the one vector $|F|^{p/2}$ must be correlated with many vectors $|K_r|$, and Bessel's inequality forbids this unless the $|K_r|$ are nearly orthogonal.

The price is the passage from K_r to $|K_r|$. The moduli keep positivity and lose the cancellation between kernels: $\langle |K_r|, |K_s| \rangle \ge |\langle K_r, K_s \rangle|$, and by Proposition 1.3 it can be as large as the square root of the kernel overlap. This is why the Gram condition below is not necessary (Proposition 2.16), and why near the critical line it gives the constant $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ (Theorem 2.8) where the H^2 estimate gives π.

THEOREM 2.1 (Hellinger–Bessel). *Let μ be a positive Borel measure on the set of admissible points, with the ℓ^2 topology. Let $\mathcal{G}_\mu$ be the integral operator on $L^2(\mu)$ with kernel*

$$\mathcal{G}(r,s) = \langle |K_r|, |K_s| \rangle = \|K_r\|_2 \|K_s\|_2 \, H(\omega_r, \omega_s).$$

Then for every $0 < p < \infty$ and $F \in X^p$,

$$\int |F(r)|^p \, \mathrm{d}\mu(r) \le \|\mathcal{G}_\mu\| \, \|F\|_p^p.$$

Moreover, for every positive weight w on $\operatorname{supp}\mu$,

$$\|\mathcal{G}_\mu\| \le \operatorname*{ess\,sup}_{r\in\operatorname{supp}\mu} \frac{1}{w(r)} \int \mathcal{G}(r,s)\, w(s)\, \mathrm{d}\mu(s).$$

With $w(r) = \|K_r\|_2$ the right side becomes $\sup_r \int H(\omega_r,\omega_s)\|K_s\|_2^2\,\mathrm{d}\mu(s)$.

PROOF. Put $u = |F|^{p/2}$, so $\|u\|_2^2 = \|F\|_p^p$. By the stability theorem (Theorem 1.7), in the form $\int |F|^{p/2}|K_r|\,\mathrm{d}\lambda \ge |F(r)|^{p/2}$, we have $|F(r)|^p \le \langle u, |K_r|\rangle^2$. The map $r \mapsto |K_r| \in L^2(\lambda)$ is continuous, because $\big\||K_r|/\|K_r\|_2 - |K_s|/\|K_s\|_2\big\|_2^2 = 2 - 2H(\omega_r,\omega_s)$ and $H \ge |\langle k_r, k_s\rangle|$. Let $V : L^2(\mu) \to L^2(\lambda)$, $Vc = \int c(r)|K_r|\,\mathrm{d}\mu(r)$. Then $V^*u(r) = \langle u, |K_r|\rangle$ and $V^*V = \mathcal{G}_\mu$. Hence

$$\int |F(r)|^p\,\mathrm{d}\mu \le \|V^*u\|^2 \le \|V\|^2\|u\|^2 = \|\mathcal{G}_\mu\|\,\|F\|_p^p.$$

The weighted Schur test needs the row bound only μ-almost everywhere, since the kernel is non-negative. □

In X^2 the corresponding statement is Bessel's inequality for the kernels themselves, with $\langle K_r, K_s\rangle$ in place of $\mathcal{G}(r,s)$. Since $H(\omega_r,\omega_s) \ge |\langle k_r, k_s\rangle|$ (Proposition 1.3), the X^p condition is the stronger one. It is a single, p-independent condition that is sufficient for every p simultaneously.

COROLLARY 2.2 (Signed and complex measures). *If μ is a complex measure with $S = \operatorname{ess\,sup}_{|\mu|} \int H(\omega_r,\omega_s)\|K_s\|_2^2\, d|\mu|(s) < \infty$, then $h \mapsto \int h\,\mathrm{d}\mu$ is bounded on X^1 with norm at most S, and the Helson matrix with moments $\alpha(n) = \int r^{\kappa(n)}\,\mathrm{d}\mu(r)$ has a bounded symbol of norm at most S.*

PROOF. $|\int h\,\mathrm{d}\mu| \le \int |h|\, d|\mu|$. Apply Theorem 2.1 with $p = 1$, then Hahn–Banach. □

2.2. The Gram kernel along the curve

Let $\gamma(s) = (p^{-s})_p$ and $r(\sigma) = \gamma(1/2 + \sigma)$.

LEMMA 2.3. *The one-coordinate Hardy affinity satisfies $h(\rho) \le (1-\rho^2)^{1/4}$.*

PROOF. This is the case $\beta = 1$ of part (4) of Proposition 1.3. □

LEMMA 2.4 (Gram kernel). *In the Hardy geometry,*

$$\mathcal{G}(r(\sigma), r(\tau)) \le \zeta(1+2\sigma)^{1/4}\zeta(1+2\tau)^{1/4}\zeta(1+\sigma+\tau)^{1/2}.$$

PROOF. In one coordinate with $a = p^{-1/2-\sigma}$, $b = p^{-1/2-\tau}$, $1 - \rho(a,b)^2 = (1-a^2)(1-b^2)/(1-ab)^2$. So by Lemma 2.3 the factor $(1-a^2)^{-1/2}(1-b^2)^{-1/2}h(\rho)$ is at most $(1-a^2)^{-1/4}(1-b^2)^{-1/4}(1-ab)^{-1/2}$. Take the product over p. □

Near the boundary the kernel is homogeneous of degree -1. With $\zeta(1+x) \approx 1/x$,

$$\mathcal{G} \approx \frac{(\sigma\tau)^{-1/4}}{\sqrt{2}\,(\sigma+\tau)^{1/2}},$$

and in the variable $t = \log(\sigma/\tau)$, after the unitary $f \mapsto \sqrt{\sigma} f$, it becomes the convolution kernel $1/(2\sqrt{\cosh(t/2)})$. Its integral,

$$\int_{\mathbb{R}} \frac{\mathrm{d}t}{2\sqrt{\cosh(t/2)}} = B\big(\tfrac{1}{4}, \tfrac{1}{2}\big) = \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \approx 5.244,$$

is the best constant the inequality can give near the boundary. The same chain with $|\langle K_r, K_s\rangle|$ in place of $\mathcal{G}$, which is the H^2 statement, gives the Carleman kernel with norm exactly π. The factor $5.244/\pi \approx 1.67$ is the price of discarding the phases of F (section 9.4).

2.3. Fejér–Riesz inequalities in every $\mathscr{H}^p$

Perfekt [45] proved the case $p = 1$ of the next theorem, with an unspecified constant (of order 10^5 if traced through the proof), by localizing the flat representative of chapter 1 and controlling the overlaps through separation and concentration estimates. The Hellinger–Bessel inequality replaces that machinery by a single Bessel inequality and gives every p at once. Since $p < 1$ is included and $H^p(\mathbb{T}^\infty)$ admits no factorization, interpolation from $p = 1$ and $p = 2$ is not available; for $p \notin 2\mathbb{N}$ these inequalities seem not to have been recorded before.

THEOREM 2.5. *For every $0 < p < \infty$ and $f \in \mathscr{H}^p$,*

$$\int_0^1 |f(1/2 + \sigma)|^p \, \mathrm{d}\sigma \le (5 + 2\sqrt{2}) \, \|f\|_{\mathscr{H}^p}^p.$$

PROOF. Apply Theorem 2.1 to Lebesgue measure on $(0, 1)$, pushed forward by r. It suffices to bound the operator norm of the kernel $\overline{\mathcal{G}}(\sigma, \tau) = (1 + 1/\big(2\sigma\big))^{1/4}(1 + 1/\big(2\tau\big))^{1/4}(1+1/\big(\sigma+\tau\big))^{1/2}$, which dominates $\mathcal{G}$ by Lemma 2.4 and $\zeta(1+x) \le 1+1/x$. Split $(0, 1) = I_0 \cup I_1$ with $I_0 = (0, 1/10)$.

- *On I_0^2:* $\overline{\mathcal{G}} \le 6/5 \cdot 1/\sqrt{2}(\sigma\tau)^{-1/4}(\sigma + \tau)^{-1/2}$. The latter kernel is unitarily a convolution by $1/(2\sqrt{\cosh(t/2)}) \le (\sqrt{2}/2)e^{-|t|/4}$, of norm at most $4\sqrt{2}$. So the block has norm $< 24\sqrt{2}/5 < 7$.
- *Off the diagonal block:* with $q(s) = 1 + (2s)^{-1/2}$ we have $\overline{\mathcal{G}}(s, t)^2 \le q(s)q(t)(1 + 1/\big(s + t\big))$. For $s \in I_0$, $t \in I_1$ use $(s + t)^{-1} \le t^{-1}$. The Hilbert–Schmidt norm squared is at most $\int_{I_0} q \cdot \int_{I_1} q(1 + t^{-1}) = (1/10 + 1/\sqrt{5})(9/10 + \log 10 + \sqrt{20} - 1/\sqrt{5}) < 4$.
- *On I_1^2:* use $(s + t)^{-1} \le (2\sqrt{st})^{-1}$. The Hilbert–Schmidt norm squared is at most $A_1^2 + (1/2)C_1^2 < 8$, with $A_1 = \int_{I_1} q < 1.87$ and $C_1 = \int_{I_1} q(s)s^{-1/2} \, \mathrm{d}s < 3$.

The norm of the block operator is at most the top eigenvalue of the symmetric matrix with rows $(7, 2)$ and $(2, 3)$, which is $5 + 2\sqrt{2}$. Positivity of the kernels gives $\|\mathcal{G}\| \le \|\overline{\mathcal{G}}\|$. □

For $p = 1$ this is Perfekt's inequality with an explicit constant, and for the multiplicative Hilbert matrix it gives a bounded symbol with an explicit norm.

COROLLARY 2.6 (Perfekt's inequality [45]). *For every $f \in \mathscr{H}^1$,*

$$\Big|a_1 + \sum_{n\ge 2} \frac{a_n}{\sqrt{n}\log n}\Big| \le \Big(5 + 2\sqrt{2} + \sum_{n\ge 2} \frac{1}{n^{3/2}\log n}\Big)\|f\|_{\mathscr{H}^1} < 9\,\|f\|_{\mathscr{H}^1}.$$

In particular the norm C of this functional on $\mathscr{H}^1_0$ satisfies $\pi \le C < 9$, the lower bound coming from the multiplicative Hilbert matrix on $\mathscr{H}^2$ [16], and by the Hahn–Banach theorem the multiplicative Hilbert matrix has a bounded symbol of norm less than 9. Proposition 9.14 lowers 9 to 5.515.

PROOF. The left side is $|\int_0^1 f(1/2+\sigma)\,\mathrm{d}\sigma + \int_1^\infty (f(1/2+\sigma) - a_1)\,\mathrm{d}\sigma|$. Use Theorem 2.5 with $p = 1$ on the first integral and $|a_n| \le \|f\|_1$ on the second. The series is about 1.113. □

One exact prime, and the local constant of the method

The Gram bound loses the phases of all coordinates. In one coordinate they can be kept, by factorizing slices.

THEOREM 2.7. *Write $z = (z_2, w)$ and $a_\sigma = 2^{-1/2-\sigma}$, and let*

$$\mathcal{K}(\sigma,\tau) = \zeta(1+2\sigma)^{1/4}\zeta(1+2\tau)^{1/4}\zeta(1+\sigma+\tau)^{1/2}\Big(\frac{(1-a_\sigma^2)(1-a_\tau^2)}{(1-a_\sigma a_\tau)^2}\Big)^{1/4}.$$

For $0 < \delta \le \infty$, $0 < p < \infty$ and $F \in H^p(\mathbb{T}^\infty)$, $\int_0^\delta |F(r(\sigma))|^p\,\mathrm{d}\sigma \le \|\mathcal{K}_\delta\|\,\|F\|_p^p$, where $\mathcal{K}_\delta$ is the integral operator with kernel $\mathcal{K}$ on $L^2(0,\delta)$.

PROOF. For a polynomial F, factor each slice $F(\cdot, w) = I_w G_w^{2/p}$ with I_w inner and $G_w \in H^2(\mathbb{T})$ outer, and put $U(w) = G_w$, so $U \in L^2(\mathbb{T}^{\{p\ge 3\}}; H^2(\mathbb{T}))$ with $\|U\|^2 = \|F\|_p^p$. Stability in the variables w (Theorem 1.7) and $|F(a,w)|^{p/2} \le |G_w(a)| = \|K_a\|\,|\langle G_w, k_a\rangle|$ give $|F(r(\sigma))|^{p/2} \le \|K_{r(\sigma)}\|\langle U, v_\sigma\rangle$ with the unit vector $v_\sigma(w) = \theta_\sigma(w)P_{r'(\sigma)}(w)^{1/2}k_{a_\sigma}$, θ_σ a phase. Then $|\langle v_\sigma, v_\tau\rangle| \le |\langle k_{a_\sigma}, k_{a_\tau}\rangle|\,H(\omega_{r'(\sigma)}, \omega_{r'(\tau)})$: the prime 2 contributes $(1-\rho_2^2)^{1/2}$ instead of the affinity $h(\rho_2) \le (1-\rho_2^2)^{1/4}$. With Lemma 2.3 for $p \ge 3$ this gives $\|K_{r(\sigma)}\|\|K_{r(\tau)}\||\langle v_\sigma, v_\tau\rangle| \le \mathcal{K}(\sigma,\tau)$, and the Bessel argument of Theorem 2.1 concludes. □

The factorization is not analytic in the remaining variables, so the argument cannot be repeated in a second coordinate.

THEOREM 2.8 (The local constant). *$\|\mathcal{K}_2\| = \Gamma(1/4)^2/\sqrt{2\pi}$. Hence, for every $0 < p < \infty$ and $f \in \mathscr{H}^p$,*

$$\int_0^2 |f(1/2+\sigma)|^p\,\mathrm{d}\sigma \le \frac{\Gamma(1/4)^2}{\sqrt{2\pi}}\,\|f\|_{\mathscr{H}^p}^p.$$

PROOF. *Lower bound.* As $\sigma, \tau \to 0$, $\mathcal{K}$ is asymptotic to

$$L_0(\sigma,\tau) = 2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2},$$

homogeneous of degree -1, with Mellin multiplier $2^{-1/2}B(1/4,1/4) = \Gamma(1/4)^2/\sqrt{2\pi}$ on the critical line; test with $\sigma^{-1/2+it}$ cut off to (e^{-2T}, e^{-T}).

Upper bound. Schur's test with $w(\tau) = \tau^{-1/2}e^{3\tau/10}$. For $\sigma \le 1/100$ it is analytic: $\zeta(1+x) \le x^{-1}(1 + 7x/12 + x^2/12)$ gives $\mathcal{K} \le L_0$ times explicit factors

$1 + O(\sigma + \tau)$, and the saving $\int_2^\infty L_0(\sigma,\tau)\tau^{-1/2}\,\mathrm{d}\tau \asymp \sigma^{1/4}$ beats the loss $O(\sigma^{1/4})$ from these factors, the margin being 0.0926 at $\sigma = 1/100$. On $[1/100, 2]$ the row sums were bounded in ball arithmetic (`arb`) on 268 cells of ratio at most 1.02, with smallest margin 0.177. □

This replaces $5 + 2\sqrt{2}$ on $(0,1)$ by the exact near-boundary constant of the method on $(0,2)$. Numerically $\|\mathcal{K}_\delta\|$ stays equal to the same value up to some $\delta_* \in (2,3)$ and then grows.

2.4. Measures on curves

A set R of primes has *density* $\theta \in [0,1]$ *with constant* B if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta \log\frac{1}{t}\Big| \le B \qquad (0 < t \le 2). \tag{2.1}$$

All primes have density 1. Unions of residue classes have rational densities. Sets with $\sum_{p\in R} 1/p < \infty$ have density 0. Let $r_R(\sigma) = (p^{-1/2-\sigma})_{p\in R}$.

LEMMA 2.9 (Energy and kernel growth). *Let R have density $\theta \in (0,1]$ with constant B, and $u, v \in (0,1]$. In each geometry,*

$$E\big(r_R(u), r_R(v)\big) \ge \theta|\log u - \log v| - \theta\log 4 - 4B.$$

The kernel norm $\|K_{r_R(\sigma)}\|_2^2$ is at most $e^{1+B}(2\sigma)^{-\theta}$, $e^{\beta(1+B)}(2\sigma)^{-\beta\theta}$ and $e^B(2\sigma)^{-\theta}$ in the Hardy, Bergman and Gauss cases. We write κ for the exponent: θ, $\beta\theta$ and θ respectively.

PROOF. Since $\rho(a,b) \ge |a-b|$ for $a, b \in [0,1)$, every energy is at least

$$\sum_{p\in R} (p^{-1/2-u} - p^{-1/2-v})^2.$$

Expanding gives three values of $\sum_{p\in R} p^{-1-t}$, with $t \in \{2u, 2v, u+v\} \subset (0,2]$. By (2.1) the sum is at least $\theta\log((u+v)^2/(4uv)) - 4B$, and $(u+v)^2/(4uv) \ge e^{|\log u - \log v|}/4$. For the kernels, use $-\log(1-x) \le x + x^2$ for $x \le 1/2$, $\sum_p p^{-2} < 1$, and (2.1) at $t = 2\sigma$. □

THEOREM 2.10 (Curves in three geometries). *Let R have density $\theta \in (0,1]$ with constant B, and let κ be the kernel exponent. There is $C = C(\theta, B, \text{geometry})$ such that, for every $0 < p < \infty$ and every F in the coordinates of R:*

(1) (block maximum) for every choice of $\tau_j \in [e^{-j-1}, e^{-j}]$,

$$\sum_{j\ge 0} e^{-j\kappa}|F(r_R(\tau_j))|^p \le C\|F\|_p^p;$$

(2) (measures) for every positive ν on $(0,1]$ with $[\nu]_\kappa = \sup_t \nu((0,t])/t^\kappa < \infty$,

$$\int |F(r_R(\sigma))|^p \,\mathrm{d}\nu(\sigma) \le C[\nu]_\kappa \|F\|_p^p.$$

For all primes in the Hardy geometry, (2) holds with $C < 21$.

PROOF. In each geometry $H(\omega_r, \omega_s) \le e^{-cE(r,s)}$, with $c = 1/4$ (Hardy and Gauss, Proposition 1.3) or $c = \beta/4$ (Bergman, Proposition 1.3).

(1) Apply Theorem 2.1 to $\mu = \sum_j e^{-j\kappa}\delta_{r_R(\tau_j)}$ with the weight $\|K\|$. Since $\|K_{r_R(\tau_k)}\|_2^2 \le C_K e^{(k+1)\kappa}$, the row sums are at most
$$C_K e^{\kappa} \sum_k \min\bigl(1, e^{-c(\theta(|j-k|-1)-\theta\log 4 - 4B)}\bigr),$$
which is bounded uniformly in j by Lemma 2.9.

(2) Apply Theorem 2.1 to ν itself with the weight $\|K\|$, and split $(0,1]$ into the blocks $I_k = (e^{-k-1}, e^{-k}]$. Then $\int_{I_k}\|K\|^2\,\mathrm{d}\nu \le C_K e^{(k+1)\kappa}[\nu]_\kappa e^{-k\kappa}$, while for $\sigma \in I_j$ and $\tau \in I_k$ the affinity is at most $\min(1, e^{-c(\theta(|j-k|-1)-C_2)})$. Sum over k.

For all primes in the Hardy geometry, Perfekt's estimate $\sum_p (p^{-1/2-u} - p^{-1/2-v})^2 > |\log u - \log v| - 7$ and $\zeta(1+2\sigma) \le 3/(2\sigma)$ give the explicit bound $(3/2)e\bigl(17 + 2e^{-1/4}/(1-e^{-1/4})\bigr) < 99$ in both parts. For (2) the Gram kernel does better. By Lemma 2.4, the Schur row at σ with the weight $\|K_\tau\| = \zeta(1+2\tau)^{1/2}$ is at most $\int g_\sigma\,\mathrm{d}\nu$ with $g_\sigma(\tau) = \zeta(1+2\sigma)^{-1/4}\zeta(1+2\tau)^{3/4}\zeta(1+\sigma+\tau)^{1/2}$, which decreases in τ. Since $\nu((0,t]) \le [\nu]t$, the layer-cake formula gives $\int g_\sigma\,\mathrm{d}\nu \le [\nu]\int_0^1 g_\sigma\,\mathrm{d}\tau$, and $1/x \le \zeta(1+x) \le 3/x$ for $0 < x \le 2$ gives
$$\int_0^1 g_\sigma\,\mathrm{d}\tau \le 3^{5/4}\int_0^\infty (2\sigma)^{1/4}(2\tau)^{-3/4}(\sigma+\tau)^{-1/2}\,\mathrm{d}\tau = 3^{5/4}\frac{\Gamma(1/4)^2}{\sqrt{2\pi}} < 20.71.$$

□

In the block maximum the points can be chosen to maximize $|F|$ in each block, which is what atomic measures need (chapter 4). Averaging over the position in the block recovers $\int_0^1 |F(r_R(\sigma))|^p \sigma^{\kappa-1}\,\mathrm{d}\sigma \lesssim \|F\|_p^p$. For prime sets this is the Fejér–Riesz inequality with weight $\sigma^{\theta-1}$; for the Bergman geometry with all primes it is $\int_0^1 |f(1/2+\sigma)|^p \sigma^{\beta-1}\,\mathrm{d}\sigma \lesssim \|f\|^p_{\mathscr{A}^p_\beta}$.

On the curves themselves the condition of Theorem 2.10 is also necessary, so there the Hellinger–Schur bound loses nothing.

COROLLARY 2.11 (Carleson measures on curves). *Let R have density $\theta \in (0,1]$ with constant B, let κ be the kernel exponent, and let ν be a positive measure on $(0,1]$. If $\int |F(r_R(\sigma))|^p\,\mathrm{d}\nu(\sigma) \le M\|F\|_p^p$ for some $0 < p < \infty$, then $[\nu]_\kappa \le CM$. Hence the following are equivalent: the embedding holds for one p; it holds for every p; $[\nu]_\kappa < \infty$; and the Gram operator $\mathcal{G}_\nu$ is bounded.*

PROOF. Test with $F_t = K_{r_R(t)}^{2/p}\|K_{r_R(t)}\|_2^{-2/p}$, of norm one. For $\sigma \le t$ the kernel estimates behind Lemma 2.9 give $|F_t(r_R(\sigma))|^p = |K_{r_R(t)}(r_R(\sigma))|^2/\|K_{r_R(t)}\|_2^2 \ge ct^{-\kappa}$, so $\nu((0,t]) \le CMt^\kappa$. Part (2) of the proof of Theorem 2.10 bounds $\|\mathcal{G}_\nu\|$ by $C[\nu]_\kappa$, and Theorem 2.1 closes the circle. □

2.5. Further consequences

Four notions of separation appear in this chapter, and they are not interchangeable. Ordered by the way the argument uses them:

(1) *Energy.* Boundedly many near neighbours, and $\sum_{\text{far } k} e^{-E(r^j,r^k)/4} < \infty$ uniformly in j.
(2) *Summable rows.* $\sup_j \sum_k H(\omega_{r^j}, \omega_{r^k}) < \infty$.
(3) *Bounded affinity matrix.* $(H(\omega_{r^j}, \omega_{r^k}))$ is bounded on ℓ^2; this is all that Theorem 2.1 needs.
(4) *Geometric separation.* Pseudo-hyperbolic distances bounded below.

Here (1) implies (2) by Proposition 1.3, and (2) implies (3) by Schur's test. The converse of the second step fails: in the tree family of Proposition 2.13 every row diverges while the matrix is bounded. And (4) controls only nearest neighbours, not how many points lie at each distance: the separated grids of Proposition 2.16 carry Carleson measures whose Gram operators are unbounded, because the long tails of the affinities accumulate. The prime curve works because overlap decays exponentially in the hyperbolic distance (Theorem 1.6) while the number of points within distance d along the curve grows only linearly in d, so the rows of the affinity matrix are geometric series.

COROLLARY 2.12 (Energy-separated families). *Let (r^j) be points with*

$$S = \sup_j \sum_k H(\omega_{r^j}, \omega_{r^k}) < \infty,$$

and let $c_j \geq 0$. The following are equivalent:

(1) $\sum c_j |F(r^j)|^p \lesssim \|F\|_p^p$ for one, equivalently every, $p > 0$;
(2) $\sum c_j |F(r^j)|^2 \lesssim \|F\|_2^2$;
(3) $c_j \lesssim \|K_{r^j}\|_2^{-2}$.

PROOF. The one-point condition and Theorem 2.1 with the weight $\|K\|$ give every p. Testing (1) with $K_{r^j}^{2/p}/\|K_{r^j}\|_2^{2/p}$ gives the one-point condition (3), and so does testing (2) with K_{r^j}. The summability of the affinities holds, for instance, when there are boundedly many near neighbours and $\sum_{\text{far } k} e^{-E_{jk}/4} < \infty$. □

Summability is used only to bound the affinity matrix on ℓ^2, and that bound is all the argument needs.

PROPOSITION 2.13 (Bounded affinity matrices). *(1) Let (r^j) be points whose affinity matrix $\mathbf{H} = (H(\omega_{r^j}, \omega_{r^k}))_{j,k}$ is bounded on ℓ^2, and let $c_j \geq 0$. Then $\sum c_j |F(r^j)|^p \lesssim \|F\|_p^p$ for one $p > 0$ if and only if $c_j \lesssim \|K_{r^j}\|_2^{-2}$, and then the Gram operator is bounded and the embedding holds for every $p > 0$.*

(2) In the Hardy and Gauss geometries there are energy-separated families with $\mathbf{H}$ bounded on ℓ^2 but $\sum_k H(\omega_{r^j}, \omega_{r^k}) = \infty$ for every j.

PROOF. (1) Testing with $K_{r^j}^{2/p}/\|K_{r^j}\|_2^{2/p}$, as in Corollary 2.11, gives $c_j \|K_{r^j}\|_2^2 \leq C_p$. For the measure $\mu = \sum_j c_j \delta_{r^j}$, the map $f \mapsto (\sqrt{c_j} f(r^j))_j$ identifies $L^2(\mu)$ with ℓ^2 and turns $\mathcal{G}_\mu$ into $D\mathbf{H}D$, with D diagonal and $D_{jj} = (c_j \|K_{r^j}\|_2^2)^{1/2}$. So $\|\mathcal{G}_\mu\| \leq C_p \|\mathbf{H}\|$, and Theorem 2.1 gives every p.

(2) Let T be the tree in which every vertex has degree 3, with adjacency operator A, and let $c = 3/5$. The weighted Schur test with weight $2^{-|v|/2}$, where $|v|$ is the distance to a fixed root, gives $\|A\| \leq 2\sqrt{2}$. For $v \neq w$ exactly one neighbour of

v is closer to w, so $h_w(v) = c^{d(v,w)}$ satisfies $((1+2c^2)I - cA)h_w = (1-c^2)e_w$, and $h_w \in \ell^2(T)$ since $2c^2 < 1$. As $(1+2c^2)I - cA \geq (1-\sqrt{2}c)^2 I$, the matrix $(c^{d(v,w)})$ equals $(1-c^2)((1+2c^2)I - cA)^{-1}$ and is bounded, while its row sums $1 + 3c\sum_{n\geq 0}(2c)^n$ diverge.

Index the coordinates by the edges of T. To the vertex v assign the point r^v with coordinate t on each edge of the path from the root to v and 0 elsewhere. In the Hardy geometry choose t with $h(t) = c$; it exists because h is continuous, $h(0) = 1$, and $h(t) \leq (1-t^2)^{1/4}$ (Lemma 2.3). In the Gauss geometry choose t with $e^{-t^2/4} = c$. Shared edges contribute affinity 1 and the $d(v,w)$ edges of the symmetric difference contribute c each, so $H(\omega_{r^v}, \omega_{r^w}) = c^{d(v,w)}$ by Proposition 1.3, and $E(r^v, r^w) = t^2 d(v,w)$. □

With $c_v = \|K_{r^v}\|_2^{-2}$ the Gram operator of the tree family is $\mathbf{H}$ itself, so it is bounded and every embedding holds. The Hellinger–Schur rows with the weight $\|K\|$ are the divergent row sums of $\mathbf{H}$, so a different weight is needed to show boundedness.

COROLLARY 2.14 (Independent general frequencies). *Let $b_j > 0$ be linearly independent over $\mathbb{Q}$, let $0 < q_j < 1$ with $\sum_j [-\log(1-q_j^2) - q_j^2] < \infty$, put $r_j(\sigma) = q_j e^{-b_j\sigma}$, and suppose that $A(\sigma) = \sum_j q_j^2 e^{-2b_j\sigma}$ satisfies $|A(\sigma) - \theta\log(1/\sigma)| \leq M$ on $(0,1]$. Then for every $p > 0$,*

$$\int_0^1 |F(r(\sigma))|^p \, \sigma^{\theta-1} \, \mathrm{d}\sigma \leq C\|F\|_p^p.$$

PROOF. The proof of Theorem 2.10 uses only the analogue of (2.1). For Hardy spaces of general Dirichlet series $\sum a_n e^{-\lambda_n s}$ whose frequencies are generated by the b_j, this is the corresponding Fejér–Riesz inequality, by the identification of Defant and Schoolmann [22] with $H^p(\mathbb{T}^\infty)$. □

REMARK 2.15. Two further consequences need no new argument.

- *Vector-valued H^p.* For X-valued F, the function $\|F\|_X^{p/2}$ is plurisubharmonic. So stability, and with it every result above, holds for $H^p(\mathbb{T}^\infty; X)$ and every Banach space X.
- *Translation-average Bergman spaces.* Averaging Theorem 2.5 over horizontal translations gives $\int_0^1 |f(1/2+t)|^p \mu([0,t])\,\mathrm{d}t \leq (5+2\sqrt{2})\|f\|_{\mathscr{A}^p_\mu}^p$ for the spaces of Gómez-Cabello, Lefèvre and Queffélec.

2.6. What the Gram bound sees

The inequality replaces each evaluation by the positive function $|K_r|$, and so forgets the phases of the kernels. The cleanest way to see what this costs is to compare, in one coordinate, the exact Carleson constant $C(\mu) = \sup \int |F|^p \,\mathrm{d}\mu / \|F\|_p^p$ with $\|\mathcal{G}_\mu\|$ for measures that interpolate between a point and a circle.

PROPOSITION 2.16. *In the Hardy geometry, in one coordinate and for $0 < r < 1$, let μ_k be the average of the point masses at the k points ωr, $\omega^k = 1$, and let m_r be normalized arc length on $\{|z| = r\}$. For every $0 < p < \infty$:*

(1) *$C(\mu_k) = 1/(1-r^{2k})$ and $C(\mu_k) \le \|\mathcal{G}_{\mu_k}\| \le kC(\mu_k)$. For fixed k the ratio tends to 1 as $r \to 1$ and as $r \to 0$.*
(2) *$C(m_r) = 1$, while $\|\mathcal{G}_{m_r}\| \ge c\log^2\big(1/(1-r)\big)$.*
(3) *For $\mu = (6/\pi^2)\sum_j j^{-2} m_{1-e^{-j}}$, $C(\mu) \le 1$ while $\mathcal{G}_\mu$ is unbounded.*
(4) *Separation does not help. Every probability measure ν on $\{|z| = r\}$ has $\|\mathcal{G}_\nu\| \ge c\log^2\big(1/(1-r)\big)$. For every $\delta < 1$ there are radii $r_j \to 1$ and grids ν_j, each the normalized counting measure on n_j equally spaced points of $\{|z| = r_j\}$, such that any two points of the union are at pseudo-hyperbolic distance at least δ, and $\mu = \sum_j j^{-3/2}\nu_j$ satisfies $C(\mu) < \infty$ for every p while $\mathcal{G}_\mu$ is unbounded.*

Proof. (1) The constant is Lemma 4.6, sharp. The Gram matrix is circulant with positive entries $(1-r^2)^{-1}h(\rho(r,\omega r))/k$, so its norm is the row sum $(1-r^2)^{-1}k^{-1}\sum_\omega h(\rho(r,\omega r))$. This lies between $(1-r^2)^{-1}/k$ and $(1-r^2)^{-1}$, and $k(1-r^2) \ge 1-r^{2k} \ge 1-r^2$. As $r \to 1$, $\rho(r,\omega r) \to 1$ for $\omega \ne 1$, so $\sum_\omega h(\rho(r,\omega r)) \to 1$ and the ratio tends to $\lim(1-r^{2k})/(k(1-r^2)) = 1$. As $r \to 0$, both sides tend to 1. (2) Subharmonicity gives $\int |F|^p \,\mathrm{d}m_r \le \|F\|_p^p$, with equality for constants. With V as in the proof of Theorem 2.1, $\|\mathcal{G}\| = \|V^*\|^2 \ge \int \langle 1, |K_z|\rangle^2 \,\mathrm{d}m_r(z)$, and $\langle 1, |K_z|\rangle = \int_{\mathbb{T}} |1-\overline{z}\zeta|^{-1}\,\mathrm{d}m(\zeta) \asymp \log(1/(1-r))$. (3) Subharmonicity on each circle gives $C(\mu) \le 1$. The lower bound $\|\mathcal{G}_\mu\| \ge \int \langle 1, |K_z|\rangle^2 \,\mathrm{d}\mu(z)$ from (2) is linear in μ, and here it is $\asymp \sum_j j^{-2}\cdot j^2 = \infty$. (4) The bound in (2) used only that $\langle 1, |K_z|\rangle$ depends on $|z|$ alone, so it holds for every probability measure on the circle. Let $\lambda, \eta > 0$, put $a_j = e^{-\lambda j}$, $r_j = 1 - a_j$ and $n_j = \lceil \eta/a_j \rceil$, and keep $j \ge j_0$ with $\eta/a_{j_0} \ge 2$. Since $|F|^p$ is subharmonic, averaging the Poisson kernel over the grid gives $\int |F|^p \,\mathrm{d}\nu_j \le (1+r_j^{n_j})(1-r_j^{n_j})^{-1}\|F\|_p^p$, and $r_j^{n_j} \le e^{-\eta}$; the weights $j^{-3/2}$ are summable. For points on different circles, $\rho(z,w) \ge \rho(|z|,|w|) \ge (a_j - a_k)/(a_j + a_k) \ge \tanh(\lambda/2)$ when $j < k$. On one circle, two distinct points differ by an angle θ with $|\sin(\theta/2)| \ge 2/n_j \ge a_j/\eta$, and $1 - r_j^2 \le 2a_j$, so

$$\rho^2 = \frac{4r_j^2 \sin^2(\theta/2)}{(1-r_j^2)^2 + 4r_j^2\sin^2(\theta/2)} \ge \frac{r_j^2/\eta^2}{1 + r_j^2/\eta^2}.$$

Both lower bounds exceed δ once λ is large and η small. Finally $\|\mathcal{G}_{\nu_j}\| \gtrsim \lambda^2 j^2$, so the compression of $\mathcal{G}_\mu$ to the jth grid has norm $\gtrsim \lambda^2 j^{1/2}$. □

So the Gram bound loses even when each kernel's support, of angular width about $1-r$, contains at most one point of the measure. The loss is in the tails: the exact formula of Proposition 1.3 gives $h(\rho) \sim (2/\pi)(1-\rho^2)^{1/2}\log\big(4(1-\rho^2)^{-1/2}\big)$ as $\rho \to 1$, so on a grid the affinity to the mth neighbour is about $m^{-1}\log m$, and the rows of the (circulant) affinity matrix sum to about $\log^2(1/(1-r))$. Thus the affinity matrix is unbounded on ℓ^2, which is exactly the hypothesis of Proposition 2.13 that fails. The loss comes from directions in which the kernels rotate: on a circle, $K_{\omega r}$ and K_r differ by a phase, and the cancellation that makes $\int |F|^p \,\mathrm{d}m_r \le \|F\|_p^p$ trivial is invisible to $|K|$. Along the prime curve the kernels at different heights have no relative rotation, every coordinate being real and positive, and Theorem 2.10 is

efficient there. Rotational averages are handled instead by the symmetry mechanism of section 4.4.1, which uses analyticity in the rotated variable.

Open

QUESTION 2. On prime curves, and for families whose affinity matrix is bounded on ℓ^2, the Hellinger–Schur condition is necessary (Corollary 2.11 and Proposition 2.13); for merely separated families it is not (part (4) of Proposition 2.16). Is there a geometric condition on the family, between the two, under which it is necessary? What exactly does the passage from K_r to $|K_r|$ lose?

QUESTION 3. What is the sharp constant in Theorem 2.5? Gram methods give exactly $\Gamma(1/4)^2/\sqrt{2\pi}$ near the boundary (Theorem 2.8), and exactness in finitely many primes does not lower this. For $p = 2$ and $a_1 = 0$ the truth is at most π, the norm of the multiplicative Hilbert matrix; for general f the constant term changes the kernel to $\zeta(1 + \sigma + \tau)$, and no such bound is known.

CHAPTER 3

Carleson measures near the critical line

Question. Which measures in $\mathbb{D}^\infty \cap \ell^2$, or in the half-plane $\mathbb{C}_{1/2}$, are Carleson measures for H^p, and when does the answer depend on p?

The Hellinger–Bessel inequality (chapter 2) gives a sufficient condition that does not see p, and it settles discrete families separated in energy completely (Proposition 2.13). This chapter tests how far it reaches. It handles twisted points, and hence complex measures on the half-plane, at no extra cost; and on vertical lines it gives the upper rate $\varepsilon^{-1/2}$ for every p. Its role for the rest of the book is Theorem 3.1, which handles twisted points and so complex measures on the half-plane (used in chapter 10); the rest of the chapter tests the Gram condition on vertical lines, where factorization is lost in a different way. Apart from the embedding theorems quoted in Proposition 3.3, the results of the chapter are, to our knowledge, new.

3.1. Twists and the half-plane

For a Dirichlet series f and a completely multiplicative $\chi : \mathbb{N} \to \mathbb{T}$ write $f_\chi = \sum a_n \chi(n) n^{-s}$. Vertical translation is the case $\chi(n) = n^{-iy}$. The point is that twisting the evaluation points costs nothing: kernel norms depend only on moduli, and energies only increase, since $|a_p(r,s)| \geq \big||r_p| - |s_p|\big|$.

THEOREM 3.1 (Twisted block maximum). *Let R have density $\theta \in (0,1]$ with constant B. For every $0 < p < \infty$ and every $f \in \mathscr{H}^p$ supported on the R-integers,*

$$\sum_{j \geq 0} e^{-j\theta} \max_{\tau \in \overline{I_j}} \sup_\chi |f_\chi(1/2+\tau)|^p \leq C \|f\|_{\mathscr{H}^p}^p,$$

$$\int_0^1 \sup_\chi |f_\chi(1/2+\sigma)|^p \sigma^{\theta-1} \, \mathrm{d}\sigma \leq C \|f\|_{\mathscr{H}^p}^p,$$

with C as in Theorem 2.10.

PROOF. For a polynomial the supremum over χ is a maximum over a finite torus. Choose maximizing (τ_j, χ_j) and the points $r^j = (\chi_j(p) p^{-1/2-\tau_j})_p$. The proof of Theorem 2.10 (1) uses the points only through $\|K_{r^j}\|_2$ and lower bounds on the energies by moduli. The second estimate follows from the first, since $\int_{I_j} \sigma^{\theta-1} \, \mathrm{d}\sigma = e^{-j\theta}(1 - e^{-\theta})/\theta \leq e^{-j\theta}$. □

COROLLARY 3.2 (Complex measures on the half-plane). *Let μ be a complex measure on $\mathbb{C}_{1/2}$ with*

$$[\mu] = \sup_{0<t\le 1} \frac{|\mu|(\{1/2 < \Re s \le 1/2+t\})}{t} < \infty.$$

Then $\int |f|^p \, d|\mu| \le (C[\mu] + \zeta(3)|\mu|(\mathbb{C}_{1/2}))\|f\|^p_{\mathscr{H}^p}$ for every $0 < p < \infty$, and the Helson matrix with moments $\alpha(n) = \int n^{-s}\,\mathrm{d}\mu(s)$ has a bounded symbol.

PROOF. On the strip $\Re s - 1/2 \in I_j$, $|f(s)| \le \max_\tau \sup_\chi |f_\chi(1/2+\tau)|$. Apply Theorem 3.1; for $\Re s > 3/2$ use $|f(s)|^p \le \zeta(2\Re s)\|f\|_p^p$. With $p = 1$, Hahn–Banach gives the symbol. □

The strip condition is sufficient but not necessary.

PROPOSITION 3.3 (Necessity on the half-plane). *(1) An $\mathscr{H}^p$ Carleson measure satisfies the box condition $|\mu|(\{\Re s - 1/2 \le \delta,\ |\Im s - y| \le \delta\}) \lesssim \delta$. For $2 \le p < \infty$ and μ supported in a bounded box above the line, the box condition is also sufficient: this is the local embedding theorem of Chen, Fang, Guo, Hou, Shao and Zhou [19], which for $p = 2$ goes back to the Montgomery–Vaughan inequality, combined with Carleson's theorem and the box test of Olsen and Saksman [42].*

(2) The strip condition is not necessary for $\mathscr{H}^2$: with λ_ε Lebesgue measure on $\{1/2 + \varepsilon + it : 0 \le t \le 1\}$ and $\varepsilon_k = e^{-k^2}$, the measure $\sum_k 2^{-k}\lambda_{\varepsilon_k}$ is $\mathscr{H}^2$-Carleson.

(3) There is an explicit $\mathscr{H}^2$- but not $\mathscr{H}^1$-Carleson measure of the form $\sum_k 2^{-k}\lambda_{\varepsilon_k}$.

PROOF. (1) Test with $\zeta(s + 1/2 + \delta - iy)^{2/p}$ truncated to finitely many primes, using $|\zeta(1+w)| \ge 1/|w| - 3$, and let the truncation grow. (2) The local embedding for $\mathscr{H}^2$ and contractivity of horizontal translation make each λ_ε uniformly Carleson. (3) The norms $N_1(\varepsilon) = \sup \int |f|\,\mathrm{d}\lambda_\varepsilon / \|f\|_{\mathscr{H}^1}$ are finite but unbounded, for otherwise $\int_0^1 |f(1/2+it)|\,\mathrm{d}t \lesssim \|f\|_{\mathscr{H}^1}$, which Harper's theorem [25] rules out. Choose ε_k with $N_1(\varepsilon_k) \ge 8^k$. □

Perfekt and Pushnitski [46] observed that the $\mathscr{H}^1$ and $\mathscr{H}^2$ Carleson classes differ, and asked whether they coincide for measures on real points. The measure in part (3) is an explicit witness, but it lives on complex points. For real points the question is Problem 2, and by chapter 4 and Proposition 2.13 every family we can handle has the same Carleson class for all p.

3.2. Vertical lines

The most prominent open problem about Carleson measures for Dirichlet series concerns a single segment of the critical line.

PROBLEM 1 (The embedding problem; Saksman and Seip [49, Problem 2.1]). *For which $0 < p < \infty$ is there a constant C_p with $\int_0^1 |f(1/2+it)|^p\,\mathrm{d}t \le C_p\|f\|^p_{\mathscr{H}^p}$ for every Dirichlet polynomial f?*

The answer is yes for $p = 2$ and hence for $p \in 2\mathbb{N}$, by applying the case $p = 2$ to $f^{p/2}$. It is no for every $0 < p < 2$: Harper's low-moment estimates [25] show that for $f = \sum_{n \le x} n^{-s}$ the pth power of the $\mathscr{H}^p$ norm is smaller than $x^{p/2}$ by a power of $\log\log x$, while $\int_0^1 |f(1/2+it)|^p \, \mathrm{d}t \asymp x^{p/2}$. Chen, Fang, Guo, Hou, Shao and Zhou [19] have announced an affirmative answer for every $p > 2$, by reducing the dual problem to a log-correlated Gaussian field and a critical branching random walk; if their proof stands, the answer is exactly $p \ge 2$. We study the problem at a fixed distance from the line. For $0 < \varepsilon \le 1/2$ let

$$N_p(\varepsilon) = \sup_{f \ne 0} \frac{\int_0^1 |f(1/2+\varepsilon+it)|^p \, \mathrm{d}t}{\|f\|_{\mathscr{H}^p}^p}.$$

For $p \in 2\mathbb{N}$ these are bounded, and for $p = 1$ they are unbounded (Proposition 3.3). Pointwise evaluation gives only $N_p(\varepsilon) \le \zeta(1+2\varepsilon) \approx (2\varepsilon)^{-1}$. The following theorem gains a factor $\varepsilon^{1/2}$, for every p; it is our contribution to Problem 1.

THEOREM 3.4. *There is an absolute C with $N_p(\varepsilon) \le C\varepsilon^{-1/2}$ for all $0 < p < \infty$ and $0 < \varepsilon \le 1/2$.*

PROOF. Let $r_t = \gamma(1/2+\varepsilon+it)$. Then $\|K_{r_t}\|_2^2 = \zeta(1+2\varepsilon) \le \varepsilon^{-1}$ for all t. By Theorem 1.6 and Proposition 1.3, with $\cosh^2(d/2) = 1 + (t-u)^2/(4\varepsilon^2)$ for two points at the same height,

$$H(\omega_{r_t}, \omega_{r_u}) \le C_0 \big(1 + (t-u)^2/(4\varepsilon^2)\big)^{-1/4} \qquad (|t-u| \le 1).$$

Apply Theorem 2.1 to $\zeta(1+2\varepsilon)^{-1} \, \mathrm{d}t$ on $[0,1]$; the Schur bound is $C_0 \sup_t \int_0^1 (1 + (t-u)^2/(4\varepsilon^2))^{-1/4} \, \mathrm{d}u \le C_0(4\varepsilon + 4\sqrt{2\varepsilon})$, and multiplying back by $\zeta(1+2\varepsilon)$ gives $C\varepsilon^{-1/2}$. □

The exponent $1/2$ is what one point evaluation per unit hyperbolic length gives: the hyperbolic length of $[0,1]$ at height ε is $1/\varepsilon$, and the affinity decays like $e^{-d/4}$, so the Gram operator sees $\varepsilon^{-1} \cdot \varepsilon^{1/2}$. What the true rate is for $p < 2$, or even whether it is a power, is open; Harper's work on low moments suggests $(\log 1/\varepsilon)^{1/4}$ at $p = 1$.

Translating vertically, the theorem answers the remaining question about the strip condition: it is not necessary, for any p.

COROLLARY 3.5. *The measure $\mathrm{d}\mu = \varepsilon^{-1/4} e^{-|t|} \, \mathrm{d}\varepsilon \, \mathrm{d}t$ on $\{1/2+\varepsilon+it : 0 < \varepsilon < 1/4, \ t \in \mathbb{R}\}$ is an $\mathscr{H}^p$ Carleson measure for every $0 < p < \infty$, but $\mu(\{1/2 < \Re s \le 1/2+\delta\}) = (8/3)\delta^{3/4}$, so it violates the strip condition.*

PROOF. Vertical translation preserves the $\mathscr{H}^p$ norm, so Theorem 3.4 bounds $\int_k^{k+1} |f(1/2+\varepsilon+it)|^p \, \mathrm{d}t$ by $C\varepsilon^{-1/2}\|f\|_{\mathscr{H}^p}^p$ for every integer k. Summing against $e^{-|t|}$ and integrating $\varepsilon^{-3/4}$ over $(0, 1/4)$ gives the embedding. □

The measure of Corollary 3.5 also satisfies the Hellinger–Schur condition of Theorem 2.1, though with a weight other than $\|K\|$. Let $r = \gamma(1/2+a+iy)$ and $s = \gamma(1/2+\varepsilon+it)$, with $\gamma(z) = (q^{-z})_q$. Since $|r_q s_q| \le 1/q$, we have $\rho(r_q, s_q)^2 \ge (1-2/q)|r_q - s_q|^2$, and $\Re \sum_q q^{-z} = \log|\zeta(z)| + O(1)$ for $\Re z > 1$. Hence

$$E(r,s) \ge \log\zeta(1+2a) + \log\zeta(1+2\varepsilon) - 2\log|\zeta(1+a+\varepsilon+i(t-y))| - C.$$

With $H \le e^{-E/4}$ (Proposition 1.3), $\zeta(1+x) \le 1 + 1/x$ for $x > 0$, and $|\zeta(1+w)| \le C/|w|$ for $|w| \le 3$ and $\le C\log(2+|w|)$ otherwise, the Gram kernel satisfies $\mathcal{G}(r,s) \le C(a\varepsilon)^{-1/4}k(t-y)$, where $k(\tau) = |\tau|^{-1/2}$ for $|\tau| \le 2$ and $k(\tau) = \log^{1/2}(2+|\tau|)$ otherwise. Schur's test with the weight $\varepsilon^{-1/4}e^{|t|/2}$ bounds every row by

$$C\int_0^{1/4} \varepsilon^{-3/4}\,\mathrm{d}\varepsilon \cdot \sup_y e^{-|y|/2}\int_{\mathbb{R}} e^{-|t|/2}k(t-y)\,\mathrm{d}t < \infty.$$

So the Gram condition alone already proves Corollary 3.5.

For $p = 2$ the limit of $N_2(\varepsilon)$ is the constant of the local embedding on the line itself. The author's thesis [51, Section 4.4] suggested the value 2; in fact the constant lies just above 2π, and it can be located quite precisely.

PROPOSITION 3.6 (The constant at $p = 2$). *Let C_0 be the least constant with $\int_0^1 |P(1/2+it)|^2\,\mathrm{d}t \le C_0\|P\|^2_{\mathscr{H}^2}$ for all Dirichlet polynomials P. Then $N_2(\varepsilon) \le C_0$ for every ε, $N_2(\varepsilon) \to C_0$ as $\varepsilon \to 0$, and*

$$2\pi \le C_0 < 2\pi + 1/8.$$

PROOF. Horizontal translation is a contraction on $\mathscr{H}^2$, so $N_2(\varepsilon) \le C_0$; conversely, for a fixed polynomial P the integral over $1/2+\varepsilon+[0,1]i$ tends to the one over the line, so $\liminf N_2(\varepsilon) \ge C_0$.

For the lower bound, write $\zeta(1+w) = 1/w + \psi(1+w)$ with ψ bounded on $\{|w| \le 2\}$, and test $N_2(\varepsilon)$ with $f(s) = \zeta(1/2+\eta+s-i/2)$, for which $\|f\|^2_{\mathscr{H}^2} = \zeta(1+2\eta)$. With $\delta = \eta + \varepsilon$,

$$\int_0^1 |f(1/2+\varepsilon+it)|^2\,\mathrm{d}t = \int_{-1/2}^{1/2} \frac{\mathrm{d}v}{\delta^2+v^2} + O\big(\log(1/\delta)\big) = \frac{\pi}{\delta} + O\big(\log(1/\delta)\big),$$

and $\zeta(1+2\eta) = 1/(2\eta) + O(1)$. Taking $\eta = \sqrt{\varepsilon}$, the quotient tends to 2π as $\varepsilon \to 0$.

For the upper bound, let $T_\varepsilon a(t) = \sum_n a_n n^{-1/2-\varepsilon-it}$ on $[0,1]$, so that $N_2(\varepsilon) = \|T_\varepsilon\|^2$. Then $T_\varepsilon T_\varepsilon^*$ has kernel $\zeta(1+2\varepsilon+i(t-u)) = k(t-u)+\psi(1+2\varepsilon+i(t-u))$, where $k(x) = 1/(2\varepsilon+ix) = \int_0^\infty e^{-(2\varepsilon+ix)y}\,\mathrm{d}y$ and $\psi(s) = \zeta(s) - 1/(s-1)$. Convolution with k on $L^2(\mathbb{R})$ is the Fourier multiplier $2\pi e^{2\varepsilon\xi}\mathbf{1}_{\xi<0}$, which lies below $2\pi\mathbf{1}_{\xi<0}$. Compressing to $[0,1]$, $T_\varepsilon T_\varepsilon^* \le S + \Psi_\varepsilon$, where S is the compression of $2\pi\mathbf{1}_{\xi<0}$ and Ψ_ε has kernel $\psi(1+2\varepsilon+i(t-u))$. In terms of the finite Hilbert transform $\mathcal{H}f(t) = \mathrm{PV}\int_0^1 f(u)(t-u)^{-1}\,\mathrm{d}u$, $S = \pi I - i\mathcal{H}$.

The operator S can be diagonalized. The substitution $t = e^x/(1+e^x)$ gives a unitary map $Uf(x) = f(t)(\mathrm{d}t/\,\mathrm{d}x)^{1/2}$ from $L^2(0,1)$ onto $L^2(\mathbb{R})$. Since $t-u = \sinh((x-y)/2)/(2\cosh(x/2)\cosh(y/2))$, it carries $\mathcal{H}$ to convolution with $\mathrm{PV}\,1/(2\sinh(x/2))$, the Fourier multiplier $-i\pi\tanh(\pi\xi)$. So S becomes multiplication by $\pi(1-\tanh(\pi\xi))$, and $0 \le S \le 2\pi$. The constant function goes to $1/(2\cosh(x/2))$, whose Fourier transform is $\pi\,\mathrm{sech}(\pi\xi)$. Substituting $s = \pi(1-\tanh(\pi\xi))$,

$$\langle \phi(S)1, 1\rangle = \frac{1}{2\pi}\int_0^{2\pi} \phi(s)\,\mathrm{d}s$$

for bounded Borel ϕ: the spectral measure of 1 is uniform on $[0, 2\pi]$.

Let $J = 1 \otimes 1$ and $c > 0$. For $\lambda > 2\pi$ the rank-one operator $c(\lambda - S)^{-1/2}J(\lambda - S)^{-1/2}$ has norm $c\langle(\lambda-S)^{-1}1,1\rangle = (c/2\pi)\log(\lambda/(\lambda-2\pi))$. When this is less than

1, $\lambda - S - cJ > 0$. Hence

$$S + cJ \le \lambda_c = \frac{2\pi}{1 - e^{-2\pi/c}}.$$

Take $c = \psi(1) = \gamma$, Euler's constant. Then $\lambda_\gamma < 2\pi + 2 \cdot 10^{-4}$ and $T_\varepsilon T_\varepsilon^* \le \lambda_\gamma + R_\varepsilon$, where $R_\varepsilon = \Psi_\varepsilon - \gamma J$ has kernel $\psi(1 + 2\varepsilon + i(t-u)) - \psi(1)$. By the Euler–Maclaurin formula,

$$\psi(s) = \frac{1}{2} + \frac{s}{12} - \frac{s(s+1)}{2} \int_1^\infty \widetilde{B}_2(x) x^{-s-2}\,\mathrm{d}x,$$

where $\widetilde{B}_2$ is the periodic Bernoulli function, with $|\widetilde{B}_2| \le 1/6$. Differentiating, and using $\int_1^\infty x^{-3}\,\mathrm{d}x = 1/2$ and $\int_1^\infty x^{-3}\log x\,\mathrm{d}x = 1/4$, for $s = 1 + iv$ with $|v| \le 1$

$$|\psi'(s)| \le \frac{1}{12} + \frac{\sqrt{13}}{24} + \frac{\sqrt{10}}{48} < \frac{3}{10},$$

and the same bound holds up to $O(\varepsilon)$ for $1 \le \Re s \le 1 + 2\varepsilon$. So the kernel of R_ε is at most $(3/10 + O(\varepsilon))|t-u| + O(\varepsilon)$. Its Hilbert–Schmidt norm is at most $(3/10)(\int_0^1 \int_0^1 (t-u)^2\,\mathrm{d}t\,\mathrm{d}u)^{1/2} + O(\varepsilon) = 3/(10\sqrt{6}) + O(\varepsilon) < 0.1225 + O(\varepsilon)$. Hence $N_2(\varepsilon) < 2\pi + 0.0002 + 0.1225 + O(\varepsilon)$, and letting $\varepsilon \to 0$ gives the upper bound. □

The upper bound improves the constant $5\pi/2$ that follows from Brevig's weighted inequality $\pi^{-1}\int_{\mathbb{R}} |f(1/2+it)|^2 (1+t^2)^{-1}\,\mathrm{d}t \le 2\|f\|_{\mathscr{H}^2}^2$.

The proof shows more. The edge 2π is the top of the spectrum of S, and the constant function sits at the edge with a logarithmically divergent resolvent. So any positive multiple of J pushes an eigenvalue above 2π, by the exponentially small amount $2\pi/(e^{2\pi/c} - 1)$. The true remainder Ψ is not a multiple of J, so this heuristic does not decide whether $C_0 > 2\pi$. A computation does.

PROPOSITION 3.7 (Above the edge). *Let $A\colon \mathscr{H}^2 \to L^2(0,1)$ be the bounded extension of $P \mapsto P(1/2 + it)$.*

(1) $AA^* = S + \Psi$, where Ψ has the continuous kernel $\psi(1 + i(t-u))$.
(2) If $C_0 > 2\pi$, then C_0 is an isolated eigenvalue of AA^* of finite multiplicity, and there is $f \in \mathscr{H}^2$ with $\|Af\|^2 = C_0\|f\|_{\mathscr{H}^2}^2$.
(3) $2\pi + 2.8 \cdot 10^{-7} < C_0 < 2\pi + 10^{-3}$.

PROOF. (1) With D_ε the diagonal operator $n^{-\varepsilon}$, $T_\varepsilon = AD_\varepsilon$ and $D_\varepsilon^2 \to I$ strongly, so $T_\varepsilon T_\varepsilon^* \to AA^*$ strongly. In the proof of Proposition 3.6, the compressed multipliers converge strongly to S, and $\Psi_\varepsilon \to \Psi$ uniformly.

(2) Since Ψ is compact and $S \le 2\pi$, the essential spectrum of AA^* lies in $[0, 2\pi]$. Hence the spectrum above 2π consists of isolated eigenvalues of finite multiplicity, and $C_0 = \|AA^*\|$ is one of them. If $AA^*g = C_0 g$ with $g \ne 0$, then $f = A^*g$ is nonzero and $\|Af\|^2 = \langle AA^*g, AA^*g\rangle = C_0\|f\|^2$.

(3) For the lower bound, take $g = \sum_{n<192} a_n P_n(2t-1)$ with rational a_n, real for even n and imaginary for odd n, found by an eigenvalue computation. By (1), $\langle AA^*g, g\rangle = \pi\|g\|^2 + \langle -iHg, g\rangle + \langle \Psi g, g\rangle$. The Hilbert form is exact: $\int_0^1 P_n(2t-1)\,(HP_m(2\cdot -1))(t)\,\mathrm{d}t$ equals $2/((n-m)(n+m+1))$ when $n+m$ is odd, and 0 otherwise. On $|w| \le 1$, $\psi(1+w)$ is replaced by the degree-96 Taylor polynomial of an Euler–Maclaurin approximant with cutoff 64. The error is below

$2 \cdot 10^{-18}$ by the remainder formula and Cauchy's estimate, and the moments of g are exact. Exact rational arithmetic then gives $\langle AA^* g, g\rangle > (2\pi + 2.8 \cdot 10^{-7})\|g\|^2$.

For the upper bound, repeat the proof of Proposition 3.6 with the rank-two model $c + ia(t-u)$ in place of γJ, where c and a are rationals close to γ and $-\gamma_1$ and $\psi(1+w) = \gamma - \gamma_1 w + \cdots$. Its positive part is $\mu\, q \otimes q$ with q in the span of 1 and $t - 1/2$. Under U, $t - 1/2$ goes to a multiple of ξ times the image of 1, so $\langle (\lambda - S)^{-1} q, q\rangle$ is an explicit integral over $[0, 2\pi]$ with logarithmic and dilogarithmic terms. At $\lambda = 2\pi + 1/340000$, $\mu\langle (\lambda - S)^{-1} q, q\rangle < 0.999$, in exact arithmetic. The remainder kernel has Hilbert–Schmidt norm below $9.7 \cdot 10^{-4}$, by exact integration of the Taylor polynomial above. □

Open

(1) The exact value of C_0. By Proposition 3.7 it exceeds 2π by between $2.8 \cdot 10^{-7}$ and 10^{-3}.
(2) The rate of $N_p(\varepsilon)$ for $p < 2$; is it $(\log 1/\varepsilon)^{1/4}$ at $p = 1$?
(3) Do the H^1 and H^2 Carleson classes coincide for measures on $(-1, 1)^\infty$ (Perfekt–Pushnitski)? Every family treated here and in chapter 4 says yes.
(4) A version of Proposition 2.13 for continuous families with many near neighbours, such as uncountably many sign patterns at one scale, beyond the product structure of Proposition 4.7.

Part 2

Where Nehari's theorem survives: structured Helson matrices

CHAPTER 4

Positive Helson matrices on the half-line

In one variable a non-negative Hankel matrix is a moment matrix. Widom characterized the bounded ones by a Carleson condition on the moment measure [54], and Nehari's theorem then supplies a bounded symbol for free. In infinitely many variables Nehari's theorem fails, but the known counterexamples, built from tensor powers of small symbols such as $z_1 + z_2$, have indefinite Hankel forms [44]. This chapter asks whether positivity rescues the theorem.

Two mechanisms produce bounded symbols for positive Helson matrices, and they behave very differently.

- *Separation.* On the curve $\gamma(s) = (2^{-s}, 3^{-s}, 5^{-s}, \dots)$, points at different scales have nearly mutually singular harmonic measures. The Hellinger–Bessel inequality of chapter 2, which replaces Perfekt's localization argument [45], turns this into an H^1 Carleson embedding. It requires the moment measure to satisfy $\nu((1/2, 1/2 + t]) \lesssim t$, exactly the condition for boundedness.
- *Symmetry.* Averaging a point over independent signs in each coordinate kills every pair (m, n) except those with mn a square, in H^1 as well as in H^2. This needs no condition at all near $s = 1/2$.

The main result of the chapter, Theorem 4.2, runs both mechanisms at once. Keep a set R of primes on the curve and average the signs of the others. If R has density θ among the primes, the condition becomes $\nu((1/2, 1/2 + t]) \lesssim t^\theta$. Within the whole family, boundedness, the H^1 embedding and the existence of a bounded symbol are equivalent, with a constant depending only on R. The curve ($\theta = 1$) and the full sign average ($\theta = 0$) are the two endpoints. The same interpolation describes the compact members (Theorem 4.10), which is a Hartman-type theorem whose condition moves from $o(t)$ at one endpoint to no condition at the other. On the curve the essential norm is comparable to the essential size of the symbol (Theorem 4.12); for measures with a density it equals $\pi w(1/2)$, and the essential spectrum is $[0, \pi w(1/2)]$ (Corollary 4.14). To our knowledge the interpolating family, the main theorem with its uniform constant, the sign-average mechanism and the Hartman-type theorem are new.

Section 4.1 sets up the dictionary between symbols, functionals and Carleson measures. Section 4.2 defines the family and states the results. Sections 4.3 and 4.4 prove the two mechanisms separately; the second, averaging over roots of unity, is developed there for every order $k \geq 2$ and every p, with exact spectra and decimation. Then section 4.5 combines them. Section 4.6 treats compactness,

section 4.7 collects consequences, and section 4.8 discusses what remains of the general question.

4.1. Positive Helson matrices and Carleson measures

Throughout, $M(\alpha) = (\alpha(mn))_{m,n\ge 1}$ is a Helson matrix, and $z^{\kappa(n)}$ is the Bohr lift of n^{-s}. For a Dirichlet polynomial $h(s) = \sum_k h_k k^{-s}$ with Bohr lift H, put

$$\Lambda_\alpha(h) = \sum_k h_k\, \alpha(k).$$

For finitely supported $a, b \in \ell^2$ with Dirichlet polynomials f, g, the form of $M(\alpha)$ is $\sum_{m,n} \alpha(mn) a_m b_n = \Lambda_\alpha(fg)$.

DEFINITION. A *bounded symbol* for $M(\alpha)$ is a function $g \in L^\infty(\mathbb{T}^\infty)$ with $\int_{\mathbb{T}^\infty} z^{\kappa(n)} g \,\mathrm{d}m = \alpha(n)$ for every $n \ge 1$.

For real α this agrees, after complex conjugation, with the convention of [46], where a symbol satisfies $\langle A, z^{\kappa(n)}\rangle = \alpha(n)$. The bilinear form is more convenient here.

LEMMA 4.1. *$M(\alpha)$ has a bounded symbol if and only if Λ_α extends to a bounded functional on $\mathscr{H}^1$. In that case*

$$\|M(\alpha)\| \le \inf\{\|g\|_\infty : g \text{ a bounded symbol}\} = \|\Lambda_\alpha\|_{(\mathscr{H}^1)^*}.$$

PROOF. If g is a bounded symbol, then $\Lambda_\alpha(h) = \int H g \,\mathrm{d}m$ on polynomials, so $|\Lambda_\alpha(h)| \le \|g\|_\infty \|h\|_{\mathscr{H}^1}$. Conversely, a bounded functional on $\mathscr{H}^1$, which the Bohr lift identifies with a closed subspace of $L^1(\mathbb{T}^\infty)$, extends by the Hahn–Banach theorem to L^1 with the same norm, and is represented by some $g \in L^\infty$ whose moments are $\Lambda_\alpha(n^{-s}) = \alpha(n)$. Finally, $\|fg\|_{\mathscr{H}^1} \le \|f\|_{\mathscr{H}^2}\|g\|_{\mathscr{H}^2}$ gives $\|M(\alpha)\| \le \|\Lambda_\alpha\|$. □

Positivity

Perfekt and Pushnitski [46] show that, under a mild growth condition, $M(\alpha) \ge 0$ exactly when α is a moment sequence $\alpha(n) = \int t^{\kappa(n)} \,\mathrm{d}\mu(t)$ of a positive measure μ on $\mathbb{R}^\infty$. Then $M(\alpha) = \mathcal{N}^*\mathcal{N}$ with $\mathcal{N} : \ell^2 \to L^2(\mu)$, $(a_n) \mapsto \sum_n a_n t^{\kappa(n)}$, so $M(\alpha)$ is bounded exactly when μ is a Carleson measure for $H^2(\mathbb{T}^\infty)$:

$$\int |F|^2 \,\mathrm{d}\mu \le C\|F\|_{H^2}^2.$$

On the other side, if μ is a Carleson measure for $H^1(\mathbb{T}^\infty)$, then $|\Lambda_\alpha(h)| = |\int H \,\mathrm{d}\mu| \le C\|h\|_{\mathscr{H}^1}$, and Lemma 4.1 gives a bounded symbol. So for non-negative Helson matrices, Nehari's theorem follows from the inclusion

$$\{H^2\text{-Carleson measures}\} \subseteq \{H^1\text{-Carleson measures}\} \tag{4.1}$$

for the moment measures in question. The reverse inclusion always holds: apply an H^1 embedding to F^2, using $\|F^2\|_{H^1} = \|F\|_{H^2}^2$. In one variable (4.1) is trivial, because every H^1 function is a product of two H^2 functions with $\|F\|_1 = \|G\|_2\|H\|_2$. On $\mathbb{T}^\infty$ that factorization fails, and (4.1) is false for general measures: arc length on a vertical segment of the critical line is a counterexample, by Harper's work [25,

45], and chapter 3 gives one in the interior of $\mathbb{D}^\infty$. Whether this can happen for measures on real points is the following problem.

PROBLEM 2 (Positive Nehari; Perfekt and Pushnitski [46, Section 4]). *Does every bounded non-negative Helson matrix have a bounded symbol? Equivalently, does (4.1) hold for every positive measure on $(-1,1)^\infty$ whose moments define a bounded Helson matrix?*

4.2. Partially averaged curve measures

The real half-line $s > 1/2$ lifts to the curve

$$\gamma(s) = (p^{-s})_p \in (0,1)^\infty \cap \ell^2,$$

and $F(\gamma(s)) = f(s)$ for every Dirichlet series f with Bohr lift F. Split the primes into two sets: R, the primes kept on the curve, and Q, the primes whose signs are averaged. Let $\varepsilon = (\varepsilon_p)_{p\in Q}$ be independent uniformly distributed signs, and write $\varepsilon\gamma(s)$ for the point with coordinates $\varepsilon_p p^{-s}$ for $p \in Q$ and p^{-s} for $p \in R$.

For a finite positive Borel measure ν on $(1/2,\infty)$, let $\mu_{\nu,R}$ be the image of $\nu \otimes \mathbb{P}_\varepsilon$ under $(s,\varepsilon) \mapsto \varepsilon\gamma(s)$. Its moments are

$$\alpha_{\nu,R}(n) = \int \mathbb{E}_\varepsilon\big[\varepsilon^{\kappa(n)}\big] n^{-s}\,\mathrm{d}\nu(s) = \mathbf{1}\big[n_Q \text{ is a square}\big] \int n^{-s}\,\mathrm{d}\nu(s),$$

where n_Q is the part of n composed of primes in Q. So $M(\alpha_{\nu,R}) \geq 0$, and $\int |F|\,\mathrm{d}\mu_{\nu,R} = \int \mathbb{E}_\varepsilon |F(\varepsilon\gamma(s))|\,\mathrm{d}\nu(s)$.

The size of R is measured by its density among the primes. Say that R has *density $\theta \in [0,1]$ with constant B* if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta \log\frac{1}{t}\Big| \leq B, \qquad 0 < t \leq 2. \tag{4.2}$$

All primes have density 1, since $\sum_p p^{-1-t} = \log(1/t) + O(1)$. Finite sets, and more generally sets with $\sum_{p\in R} 1/p < \infty$, have density 0. The primes in a union of a residue classes modulo q coprime to q have density $a/\varphi(q)$, by the prime number theorem in arithmetic progressions, and the odd-indexed primes have density $1/2$. Put

$$[\nu]_\theta = \sup_{0<t\leq 1} \frac{\nu((1/2, 1/2+t])}{t^\theta},$$

which for $\theta = 0$ is just $\nu((1/2,3/2])$.

THEOREM 4.2. *Let R have density $\theta \in [0,1]$ with constant B, and let ν be a finite positive Borel measure on $(1/2,\infty)$. The following are equivalent.*

(1) $M(\alpha_{\nu,R})$ is bounded on ℓ^2.
(2) $[\nu]_\theta < \infty$.
(3) $\mu_{\nu,R}$ is a Carleson measure for $H^1(\mathbb{T}^\infty)$.
(4) $M(\alpha_{\nu,R})$ has a bounded symbol.

Moreover there is $C = C(\theta, B)$ such that

$$\|M(\alpha_{\nu,R})\| \leq \inf\|g\|_\infty \leq C\,\|M(\alpha_{\nu,R})\|,$$

the infimum being over all bounded symbols. If $\theta = 0$, all four conditions hold for every finite ν.

The two endpoints are the motivating cases.

COROLLARY 4.3 (The curve). *For R the set of all primes, $\alpha_\nu(n) = \int n^{-s}\,\mathrm{d}\nu(s)$, and $[\nu] = [\nu]_1$: the matrix $M(\alpha_\nu)$ is bounded if and only if $[\nu] < \infty$, if and only if $\int |f|\,\mathrm{d}\nu \lesssim \|f\|_{\mathscr{H}^1}$, if and only if it has a bounded symbol. One may take $\inf\|g\|_\infty \le (2 \cdot 20.71 + \zeta(3))\|M(\alpha_\nu)\| < 43\,\|M(\alpha_\nu)\|$.*

The equivalence of the first two conditions is due to Perfekt and Pushnitski [46, Theorem 5.1]. The multiplicative Hilbert matrix [16] comes from Lebesgue measure, which is not finite but differs from a finite measure by a Hilbert–Schmidt term (Corollary 4.19). The constant does not depend on ν; by contrast, the best constant in Nehari's theorem on $\mathbb{T}^d$ is at least $(\pi^2/8)^{d/4}$ [44].

COROLLARY 4.4 (Full sign averages). *For $R = \emptyset$ and every finite positive ν, the Helson matrix with $\alpha(n) = \mathbf{1}_\square(n) \int n^{-s}\,\mathrm{d}\nu(s)$, where $\mathbf{1}_\square$ is the indicator of the squares, is bounded and has a bounded symbol with*

$$\inf\|g\|_\infty \le \zeta(2)\,\nu((1/2,\infty)) \le \zeta(2)\,\|M(\alpha)\|.$$

So the full sign averages satisfy Nehari's theorem with constant at most $\zeta(2) \approx 1.645$, uniformly in ν. That is far better than the constant for the curve. Only the single orbits of Proposition 4.8, where the constant is exactly 1, do better.

4.3. Mechanism I: separation

The difficulty on the curve is a logarithm. Point evaluation at $\gamma(1/2+\sigma)$ has norm $\zeta(1+2\sigma) \asymp 1/\sigma$ on $\mathscr{H}^1$, by the Cole–Gamelin estimate [21]. If $[\nu] < \infty$, then ν puts mass $O(e^{-j})$ on the scale $1/2 + (e^{-j-1}, e^{-j}]$, so each scale contributes $O(\|f\|_{\mathscr{H}^1})$ to $\int |f|\,\mathrm{d}\nu$, and there are $\log(1/\delta)$ scales above height δ. One function cannot be near-extremal at many scales at once: the harmonic measures at different scales have exponentially small Hellinger affinity (chapter 1), and the Hellinger–Bessel inequality turns this into a Carleson estimate (chapter 2). For the Hardy geometry the kernel exponent equals the density, $\kappa = \theta$, and Theorem 2.10 with $p = 1$ reads as follows.

LEMMA 4.5 (Block-maximal evaluation on prime sets). *Let R have density $\theta \in (0,1]$ with constant B. There is $C_1 = C_1(\theta, B)$ such that every $f \in \mathscr{H}^1$ whose coefficients are supported on integers composed of primes in R satisfies*

$$\sum_{j\ge 0} e^{-j\theta} \max_{e^{-j-1} \le t \le e^{-j}} |f(1/2+t)| \le C_1 \|f\|_{\mathscr{H}^1},$$

and $\int_{(1/2,3/2]} |f|\,\mathrm{d}\nu \le C_1 [\nu]_\theta \|f\|_{\mathscr{H}^1}$ for every positive ν. For R the set of all primes one may take $C_1 = 99$ in the first bound and $C_1 < 20.71$ in the second (Theorem 2.10).

PROOF. Both are Theorem 2.10 for $p = 1$, applied to the Bohr lift; the maximum is attained since f is continuous on each closed block, and Fatou's lemma passes from polynomials to $\mathscr{H}^1$. □

For all primes, Perfekt's localization argument [45, Section 2], with Hoeffding's inequality [29] controlling the overlaps, gives the lemma with $C_1 = 3eL$, $L = 10^5$. The Gram route of chapter 2 needs no disjointness of the localized sets, which is why its constant is so much smaller.

Taking R to be all primes and ν Lebesgue measure on $(1/2, 3/2]$ recovers Perfekt's inequality $\int_0^1 |f(1/2+\sigma)|\,\mathrm{d}\sigma \lesssim \|f\|_{\mathscr{H}^1}$. The block maximum is the strengthening a general measure needs: ν may be atomic, and an average says nothing about any particular point.

4.4. Mechanism II: symmetry

Averaging a point over roots of unity in some of its coordinates is a second way of bounding evaluations, independent of separation. It works for every p, and it reaches points outside ℓ^2.

4.4.1. Averages over roots of unity

Between a single evaluation and the full circle average there is a ladder: average over the kth roots of unity.

LEMMA 4.6. *For $k \geq 2$, $0 < p < \infty$, $x \in \mathbb{D}$ and $g \in H^p(\mathbb{T})$,*

$$\frac{1}{k} \sum_{\omega^k = 1} |g(\omega x)|^p \leq \frac{\|g\|_{H^p}^p}{1 - |x|^{2k}},$$

with equality for $g = (1 - \overline{x}^k z^k)^{-2/p}$. For $k = 2$ this is $(|g(x)|^p + |g(-x)|^p)/2 \leq \|g\|_p^p/(1 - |x|^4)$.

PROOF. Write $g = Ih^{2/p}$ with I inner and h outer, so $|g(\omega x)|^p \leq |h(\omega x)|^2$ and $\|h\|_2^2 = \|g\|_p^p$. Split h by residues of the exponent, $h(z) = \sum_{r<k} z^r h_r(z^k)$, so that $\|h\|_2^2 = \sum_r \|h_r\|_2^2$. Averaging over the roots of unity removes the cross terms:

$$\frac{1}{k} \sum_{\omega^k = 1} |h(\omega x)|^2 = \sum_{r<k} |x|^{2r} |h_r(x^k)|^2 \leq \sum_{r<k} \frac{\|h_r\|_2^2}{1 - |x|^{2k}} = \frac{\|h\|_2^2}{1 - |x|^{2k}},$$

by the point estimate at x^k. Equality holds when only h_0 is present and is the reproducing kernel at x^k, and $I = 1$. □

As $k \to \infty$ the constant tends to 1, which is the trivial bound for the full circle average. For a single point ($k = 1$) the constant is $1/(1 - |x|^2)$, the Cole–Gamelin bound.

PROPOSITION 4.7 (Root-of-unity averages in every H^p). *Let $0 < p < \infty$, let Q be a set of coordinates with orders $k_q \geq 2$ and points $x_q \in \mathbb{D}$, and fix the remaining coordinates at w. With independent ω_q uniform on the k_qth roots of unity,*

$$\mathbb{E}_\omega |F((\omega_q x_q)_{q \in Q}, w)|^p \leq \prod_{q \in Q} \frac{1}{1 - |x_q|^{2k_q}} \int_{\mathbb{T}^Q} |F(\zeta, w)|^p \,\mathrm{d}m(\zeta).$$

For $x = \gamma(s)$ with $\Re s \geq 1/2$ and all $k_q \geq k$, the constant is at most $\zeta(k)$; for signs it is $\zeta(4\Re s) \leq \zeta(2)$.

PROOF. Induct on the coordinates, applying Lemma 4.6 in one and Fubini in the rest; $\prod_q (1-q^{-2k_q\Re s})^{-1} \le \prod_q (1-q^{-k})^{-1}$. □

So averaging over kth roots in each coordinate replaces the domain ℓ^2 of point evaluations by ℓ^{2k}: the averaged evaluation is bounded on every H^p at points where no single evaluation is, and the higher the order, the more of $\mathbb{D}^\infty$ it reaches. The analytic average is decimation in disguise: $\mathbb{E}_\omega F(\omega x) = (D_{k,Q}F)(x^k)$ in the notation of section 4.4.2; the proposition controls instead the average of $|F|^p$, which decimation does not.

A single orbit can be computed exactly. For $k \ge 2$ and $s \ge 1/2$ let $M_{k,s}(m,n) = \mathbf{1}[mn \text{ is a } k\text{th power}](mn)^{-s}$.

PROPOSITION 4.8 (Root-of-unity orbits). *For $0 < p < \infty$,*

$$\|M_{k,s}\|_{S_p}^p = \zeta(2ks)^p \prod_\ell \big(1 + (k-1)\ell^{-ksp}\big),$$

the product over primes, so $M_{k,s} \in S_p$ exactly when $ksp > 1$. Moreover $\|M_{k,s}\| = \zeta(2ks)$, and $M_{k,s}$ has a bounded symbol of norm exactly $\zeta(2ks)$. For $k \ge 3$ the matrix is indefinite, and every finite positive mixture $\int M_{k,s}\,\mathrm{d}\nu(s)$ over $s \ge 1/2$ is trace class.

PROOF. The matrix is the tensor product over primes ℓ of the matrices $(\mathbf{1}[k \mid a+b]x^{a+b})_{a,b\ge 0}$ with $x = \ell^{-s}$. Split the exponents by their residues mod k. Residue 0 against 0 gives the rank-one block $(x^{k(\alpha+\beta)})$ with singular value $1/(1-x^{2k})$; residue r against $k-r$, $1 \le r < k$, gives $x^k w \otimes w$ with $w = (x^{k\alpha})_\alpha$. So the singular values are $1/(1-x^{2k})$ once and $x^k/(1-x^{2k})$ with multiplicity $k-1$, and for $k \ge 3$ the pairs $r \ne k-r$ give eigenvalues of both signs. Tensoring gives the S_p norm and the operator norm. For the symbol, the functional is $F \mapsto \mathbb{E}_\omega F(\omega x) = (D_k F)(x^k)$, and decimation is contractive on H^1, being an average of rotations followed by the measure-preserving map $z \mapsto z^k$, so its norm is at most $\prod_\ell (1-\ell^{-2ks})^{-1} = \zeta(2ks)$. For $k \ge 3$ and $s \ge 1/2$, the trace norm is at most $\zeta(3)\prod_\ell (1 + (k-1)\ell^{-3/2})$. □

So an individual orbit has Nehari constant exactly one, even when it is indefinite, and at $s = 1/2$ it lies in S_p exactly for $p > 2/k$.

4.4.2. Decimation

For $k \ge 2$ and a set Q of coordinates let $D_{k,Q}$ replace each exponent ν_q, $q \in Q$, by $k\nu_q$ in the coefficient index: $\widehat{D_{k,Q}F}(\nu) = \widehat{F}(\tilde{\nu})$. For Q all primes, $D_k \sum a_n n^{-s} = \sum a_{n^k} n^{-s}$.

PROPOSITION 4.9. *$\|D_{k,Q}\|_{H^p \to H^p} = 1$ for $1 \le p \le \infty$. For $0 < p < 1$ and $|Q| = d$, $\|D_{k,Q}\| = k^{d(1/p-1)}$; so $D_{k,Q}$ is unbounded for infinite Q. If $\alpha(n^k) = \beta(n)$ and $\alpha = 0$ off kth powers, the functionals Λ_α and Λ_β on H^1 have equal norms, hence equal optimal symbol norms.*

PROOF. $(D_{k,Q}F) \circ \pi_k$ is the average of F over kth roots of unity in the Q-coordinates, where π_k raises those coordinates to the kth power and preserves Haar measure. For $p \ge 1$ this is a conditional expectation. For $p < 1$ the upper bound is p-subadditivity; the lower bound uses outer functions of modulus 1 on an arc

whose k rotations are disjoint and δ elsewhere, tensored d times. For the functionals, $\Lambda_\alpha = \Lambda_\beta \circ D_k$ and $D_k J_k = I$ with the isometry $J_k G = G \circ \pi_k$. □

Decimation identifies analytic sign averages exactly,

$$\mathbb{E}_\varepsilon F((\varepsilon_q q^{-s})_Q, (p^{-s})_R) = (D_{2,Q}F)((q^{-2s})_Q, (p^{-s})_R),$$

but it bounds $|\mathbb{E}_\varepsilon F|$, not $\mathbb{E}_\varepsilon |F|$ (already $F = z_q$ has $D_2 F = 0$), so it does not give Proposition 4.7.

For the functional, and hence for the symbol, this already suffices. Since $\Lambda_{\alpha_{\nu,R}}$ is linear, $\Lambda_{\alpha_{\nu,R}}(h) = \int (D_{2,Q}H)((q^{-2s})_Q, (p^{-s})_R)\,\mathrm{d}\nu(s)$, and $D_{2,Q}$ is contractive on H^1. The bent curve $s \mapsto ((q^{-2s})_Q, (p^{-s})_R)$ lies in ℓ^2, its kernel norms exceed those of γ_R by at most the factor $\prod_{q\in Q}(1-q^{-4s})^{-1} \le \zeta(2)$, and its Hellinger affinities are at most those of γ_R, being products with factors at most 1. So the Schur bound in the proof of Theorem 2.10 bounds $\Lambda_{\alpha_{\nu,R}}$ on $\mathscr{H}^1$ directly: averaging the signs moves the Q-coordinates from $\gamma(s)$ to $\gamma(2s)$, and only the singularity of the coordinates in R is left. The two-point lemma is needed only for the stronger Carleson statement, which bounds $\mathbb{E}_\varepsilon |F|$ rather than $|\mathbb{E}_\varepsilon F|$.

4.4.3. Signs

The second mechanism is the case $k_q = 2$ of the root-of-unity averages of section 4.4.1. For sign averages Lemma 4.6 reads $(|g(x)| + |g(-x)|)/2 \le \|g\|_{H^1}/(1-|x|^4)$, with equality for $g = (1-\overline{x}^2 z^2)^{-2}$, and Proposition 4.7 gives, for an analytic polynomial F, a set Q of coordinates with points $x_p \in \mathbb{D}$, and the remaining coordinates fixed at w,

$$\mathbb{E}_\varepsilon \big|F\big((\varepsilon_p x_p)_{p\in Q}, w\big)\big| \le \prod_{p\in Q} \frac{1}{1-|x_p|^4} \int_{\mathbb{T}^Q} \big|F(\zeta, w)\big|\,\mathrm{d}m(\zeta). \tag{4.3}$$

For $x = \gamma(s)$ the constant is at most $\zeta(4s) \le \zeta(2)$. Two features matter here. The constant does not blow up as $s \to 1/2$, and $\sum |x_p|^4 < \infty$ is weaker than $x \in \ell^2$: the sign-averaged evaluation is bounded on H^1 at points where no single point evaluation is.

4.5. Proof of the main theorem

The two mechanisms fit together because they act on different coordinates. Point evaluation at $\gamma(1/2+\sigma)$ costs $\zeta(1+2\sigma) \asymp 1/\sigma$ on $\mathscr{H}^1$, and the cost is a product over primes. Averaging the signs of the primes in Q removes their share of this singularity at the bounded price $\prod_{q\in Q}(1-q^{-4\Re s})^{-1} \le \zeta(2)$, since sign averages need only ℓ^4. What survives is the share of R, which is $\zeta_R(1+2\sigma) \asymp \sigma^{-\theta}$. Separation along the curve then handles exactly this residual singularity, and the condition $\nu((1/2, 1/2+t]) = O(t^\theta)$ matches it. So θ measures how much of the evaluation singularity survives averaging, and the two endpoints are visible before any estimate: at $\theta = 1$ nothing is averaged, and at $\theta = 0$ nothing survives.

(1) ⇒ (2). Assume $\theta > 0$; for $\theta = 0$ there is nothing to prove. Test with $a_n = n^{-1/2-\delta}$, $0 < \delta \le 1$, on the integers composed of primes in R. For such m, n

the Q-part of mn is 1, so $\alpha_{\nu,R}(mn) = \int (mn)^{-s}\,\mathrm{d}\nu(s)$. By Tonelli's theorem, with $\zeta_R(w) = \prod_{p\in R}(1-p^{-w})^{-1}$,

$$\sum_{m,n} \alpha_{\nu,R}(mn)a_m a_n = \int \zeta_R(1/2+\delta+s)^2\,\mathrm{d}\nu(s) \geq \zeta_R(1+2\delta)^2\,\nu((1/2,1/2+\delta]).$$

The left side is at most $\|M(\alpha_{\nu,R})\|\zeta_R(1+2\delta)$, and by (4.2) $\zeta_R(1+2\delta) \geq e^{-B}(2\delta)^{-\theta}$. Hence $[\nu]_\theta \leq 2e^B\|M(\alpha_{\nu,R})\|$. We also record $\nu((1/2,\infty)) = \alpha_{\nu,R}(1) \leq \|M(\alpha_{\nu,R})\|$.

(2) $\Rightarrow$ (3). Let F be an analytic polynomial. For fixed s, (4.3) with the R-coordinates fixed at $\gamma_R(s)$ gives

$$\mathbb{E}_\varepsilon|F(\varepsilon\gamma(s))| \leq \zeta(2)\int_{\mathbb{T}^Q}|F_\zeta(\gamma_R(s))|\,\mathrm{d}m(\zeta),$$

where $F_\zeta = F(\zeta,\cdot)$ is a polynomial in the R-coordinates. It is the Bohr lift of a Dirichlet polynomial f_ζ supported on integers composed of primes in R, with $f_\zeta(s) = F_\zeta(\gamma_R(s))$. Integrate in ν and use Tonelli:

$$\int |F|\,\mathrm{d}\mu_{\nu,R} \leq \zeta(2)\int_{\mathbb{T}^Q}\Big(\int |f_\zeta|\,\mathrm{d}\nu\Big)\,\mathrm{d}m(\zeta).$$

If $\theta > 0$, split $(1/2,3/2]$ into the blocks $1/2+I_j$ with $I_j = (e^{-j-1},e^{-j}]$. Since $\nu(1/2+I_j) \leq [\nu]_\theta e^{-j\theta}$, Lemma 4.5 gives $\int_{(1/2,3/2]}|f_\zeta|\,\mathrm{d}\nu \leq C_1[\nu]_\theta\|F_\zeta\|_{L^1(\mathbb{T}^R)}$. If $\theta = 0$, the point estimate gives $|f_\zeta(s)| \leq \zeta_R(2\Re s)\|F_\zeta\|_1 \leq e^{1+B}\|F_\zeta\|_1$ for every $s > 1/2$. On $(3/2,\infty)$, $|f_\zeta| \leq \zeta(3)\|F_\zeta\|_1$. Since $\int_{\mathbb{T}^Q}\|F_\zeta\|_{L^1(\mathbb{T}^R)}\,\mathrm{d}m(\zeta) = \|F\|_{L^1}$, we obtain

$$\int |F|\,\mathrm{d}\mu_{\nu,R} \leq \zeta(2)\big(C_1[\nu]_\theta + \zeta(3)\,\nu((1/2,\infty))\big)\|F\|_{H^1} \tag{4.4}$$

for $\theta > 0$, and $\int|F|\,\mathrm{d}\mu_{\nu,R} \leq \zeta(2)e^{1+B}\nu((1/2,\infty))\|F\|_{H^1}$ for $\theta = 0$. Fatou's lemma extends both to $H^1(\mathbb{T}^\infty)$.

(3) $\Rightarrow$ (4). For polynomials, $\Lambda_{\alpha_{\nu,R}}(h) = \int H\,\mathrm{d}\mu_{\nu,R}$. Apply Lemma 4.1.

(4) $\Rightarrow$ (1). Lemma 4.1.

The constant. Combine Lemma 4.1, (4.4) and the bounds from the first step. Corollary 4.3 follows with $\zeta(2)$ omitted, since $Q = \emptyset$, and with $C_1 < 20.71$, $[\nu] \leq 2\|M\|$. Corollary 4.4 follows from (4.3) directly, since then $R = \emptyset$ and $\int|F|\,\mathrm{d}\mu_{\nu,\emptyset} \leq \int \zeta(4s)\,\mathrm{d}\nu(s)\,\|F\|_{H^1}$. □

4.6. Compactness

In one variable, Hartman's theorem [26] says that a Hankel operator is compact exactly when its symbol can be taken in $H^\infty + C$, and Sarason's theorem [50] says that this space is closed. Let

$$N = \Big\{g \in L^\infty(\mathbb{T}^\infty) : \int z^\kappa g\,\mathrm{d}m = 0 \text{ for every multi-index } \kappa \geq 0\Big\},$$

so that the bounded symbols of a given $M(\alpha)$ form a coset $g+N$. Call a functional on $\mathscr{H}^1$ a *polynomial functional* if it has the form $h \mapsto \sum_{k\leq K} h_k\beta(k)$.

THEOREM 4.10. *Let R have density $\theta \in [0,1]$ with constant B, and let ν be a finite positive Borel measure on $(1/2,\infty)$ with $[\nu]_\theta < \infty$. The following are equivalent.*

(1) $M(\alpha_{\nu,R})$ is compact.

(2) $\nu((1/2,1/2+t]) = o(t^\theta)$ *as* $t \to 0^+$.
(3) $\Lambda_{\alpha_{\nu,R}}$ *lies in the* $(\mathscr{H}^1)^*$*-closure of the polynomial functionals.*
(4) Some, equivalently every, bounded symbol lies in $C(\mathbb{T}^\infty) + N$*, which is closed (Proposition 4.11).*

In particular, if $\theta = 0$*, then* $M(\alpha_{\nu,R})$ *is compact for every finite* ν.

The condition interpolates in the same way as in Theorem 4.2. At the curve endpoint it is $o(t)$; at the sign-average endpoint it is vacuous, because $\nu((1/2,1/2+t]) \to 0$ for every finite measure. The full sign averages of Corollary 4.4 are therefore all compact, while the multiplicative Hilbert matrix, on the curve, is not.

Proof. (1) $\Rightarrow$ (2). Assume $\theta > 0$. The unit vectors k_δ proportional to $(n^{-1/2-\delta})$ on the R-integers tend to 0 weakly as $\delta \to 0^+$, because $\zeta_R(1+2\delta) \to \infty$. A compact operator maps them to a norm-null family, and the computation in the proof of (1) $\Rightarrow$ (2) above gives $\nu((1/2,1/2+\delta]) \le e^B (2\delta)^\theta \langle M(\alpha_{\nu,R})k_\delta, k_\delta\rangle$.

(2) $\Rightarrow$ (3). Fix $0 < \varepsilon \le 1$ and write $\nu = \nu_\varepsilon + \nu^\varepsilon$, with ν_ε the restriction to $(1/2,1/2+\varepsilon]$. Then $[\nu_\varepsilon]_\theta \le \eta(\varepsilon) := \sup_{0<u\le\varepsilon} \nu((1/2,1/2+u])/u^\theta$, which tends to 0 by (2), and the estimates in the proof of Theorem 4.2 give $\|\Lambda_{\alpha_{\nu_\varepsilon,R}}\| \le \zeta(2) C_1 \eta(\varepsilon)$, or $\le \zeta(2) e^{1+B} \nu((1/2,1/2+\varepsilon])$ when $\theta = 0$. The measure ν^ε lives on $[1/2+\varepsilon,\infty)$, so $0 \le \alpha_{\nu^\varepsilon,R}(n) \le \nu((1/2,\infty))\, n^{-1/2-\varepsilon}$. By Cauchy–Schwarz and Helson's inequality $\sum_k |h_k|^2/d(k) \le \|h\|^2_{\mathscr{H}^1}$ [27],

$$\Big|\Lambda_{\alpha_{\nu^\varepsilon,R}}(h) - \sum_{k\le K} h_k \alpha_{\nu^\varepsilon,R}(k)\Big| \le \Big(\sum_{k>K} d(k)\alpha_{\nu^\varepsilon,R}(k)^2\Big)^{1/2} \|h\|_{\mathscr{H}^1},$$

which tends to 0 as $K \to \infty$ because $\sum_k d(k) k^{-1-2\varepsilon} = \zeta(1+2\varepsilon)^2$.

(3) $\Rightarrow$ (1). A polynomial functional with coefficients $\beta(k)$, $k \le K$, gives the Helson matrix $(\beta(mn))$, which vanishes unless $m, n \le K$ and so has finite rank. By Lemma 4.1, $\|M(\alpha) - M(\beta)\| \le \|\Lambda_\alpha - \Lambda_\beta\|$.

(3) $\Leftrightarrow$ (4). Let $q : L^\infty \to L^\infty/N \cong (\mathscr{H}^1)^*$ be the quotient map, and let V be the space of symbols $\sum_{k\le K} \beta(k)\overline{z}^{\kappa(k)}$ of polynomial functionals. A trigonometric monomial z^μ lies in V if $\mu \le 0$ and in N otherwise, so $V + N$ contains all trigonometric polynomials and $\overline{q(V)} = \overline{q(C(\mathbb{T}^\infty)+N)}$. For a subspace $W \supseteq N$ one has $q^{-1}(\overline{q(W)}) = \overline{W}$: if $\|q(x - w_k)\| \to 0$ with $w_k \in W$, choose $n_k \in N$ with $\|x - w_k - n_k\|_\infty \to 0$. Take $W = C(\mathbb{T}^\infty) + N$; the statement holds for one element of the coset $g + N$ exactly when it holds for all. □

Proposition 4.11. $C(\mathbb{T}^\infty) + N$ *is closed in* $L^\infty(\mathbb{T}^\infty)$*. Hence the symbols of the compact members of the family can be chosen in* $C(\mathbb{T}^\infty) + N$ *itself.*

Proof. For $k \ge 1$ let E_k be the conditional expectation onto the first k coordinates and F_k convolution on $\mathbb{T}^k$ with the product of Fejér kernels of order k, and put $T_k = F_k E_k$. Then $\|T_k\|_{L^\infty \to L^\infty} \le 1$, T_k maps L^∞ into trigonometric polynomials, and $T_k c \to c$ uniformly for $c \in C(\mathbb{T}^\infty)$, by density of finite-coordinate polynomials. Moreover $T_k N \subseteq N$: E_k is self-adjoint and sends z^κ either to itself or to 0, and F_k multiplies Fourier coefficients. So for $c \in C$ and $n \in N$, $\operatorname{dist}(c, C \cap N) \le \|c - T_k n\|_\infty \le \|c - T_k c\|_\infty + \|c - n\|_\infty$, and letting $k \to \infty$ gives $\operatorname{dist}(c, C\cap N) = \operatorname{dist}(c, N)$. Thus $C/(C \cap N) \to L^\infty/N$ is isometric, its image $(C+N)/N$ is complete, hence

closed, and $C + N$ is its preimage. This is Rudin's argument for $H^\infty + C$, which needs only the maps T_k, not an algebra structure. □

Essential norms

Compactness is the qualitative statement; the quantitative one compares the essential norm $\|M\|_{\mathrm{e}}$ with the essential size of the symbol,

$$\mathcal{E}_{\mathrm{sym}}(M) = \operatorname{dist}_{L^\infty}\big(g, C(\mathbb{T}^\infty) + N\big),$$

which does not depend on the bounded symbol g chosen. A continuous symbol gives a compact matrix, being a uniform limit of trigonometric polynomials, and N gives the zero matrix, so $\|M\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M)$ always. On the curve the converse holds up to an absolute constant.

THEOREM 4.12 (Essential Nehari on the curve). *Let R be all primes, let ν be finite and positive with $[\nu] < \infty$, and put $\ell_\nu = \limsup_{t\to 0^+} \nu((1/2, 1/2+t])/t$. Then*

$$\frac{1}{2}\ell_\nu \le \|M(\alpha_\nu)\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M(\alpha_\nu)) \le \frac{\Gamma(1/4)^2}{\sqrt{2\pi}}\,\ell_\nu \le \frac{2\Gamma(1/4)^2}{\sqrt{2\pi}}\,\|M(\alpha_\nu)\|_{\mathrm{e}}.$$

PROOF. The lower bound is the test in (1) ⇒ (2) of Theorem 4.10, with $\delta\zeta(1+2\delta) \to 1/2$ and the weak nullity of k_δ. For the upper bound, split $\nu = \nu_\varepsilon + \nu^\varepsilon$ as in (2) ⇒ (3), and put $A_\varepsilon = \sup_{t\le\varepsilon} \nu((1/2, 1/2+t])/t$. The functional of ν^ε is a norm limit of polynomial functionals. For ν_ε apply Theorem 2.1 with $p = 1$ and the weight $\|K_\tau\|$: by Lemma 2.4 the row at σ is at most $\int g_\sigma \,\mathrm{d}\nu_\varepsilon$ with $g_\sigma(\tau) = \zeta(1+2\sigma)^{-1/4}\zeta(1+2\tau)^{3/4}\zeta(1+\sigma+\tau)^{1/2}$, which decreases in τ, so the layer-cake formula bounds it by $A_\varepsilon \int_0^\varepsilon g_\sigma \,\mathrm{d}\tau$. Since $x^{-1} \le \zeta(1+x) \le (1+x)/x$, the integral is at most $(1+2\varepsilon)^{5/4}\Gamma(1/4)^2/\sqrt{2\pi}$, as in the proof of Theorem 2.10. Through the quotient map q, $\mathcal{E}_{\mathrm{sym}}$ is the distance of $q(g)$ to the closure of the polynomial functionals, which is at most this bound. Let $\varepsilon \to 0$. □

The same proof works on every prime set.

COROLLARY 4.13. *Let R have density $\theta \in (0,1]$ with constant B, let $[\nu]_\theta < \infty$, and put $\ell_{\nu,\theta} = \limsup_{t\to 0^+} \nu((1/2, 1/2+t])/t^\theta$. Then*

$$e^{-B}2^{-\theta}\ell_{\nu,\theta} \le \|M(\alpha_{\nu,R})\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M(\alpha_{\nu,R})) \le \zeta(2)C_1\ell_{\nu,\theta},$$

with $C_1 = C_1(\theta, B)$ from Lemma 4.5.

PROOF. For the lower bound, test with the normalized kernels k_δ of $\gamma_R(1/2+\delta)$ in the coordinates of R, which are weakly null since $\zeta_R(1+2\delta) \to \infty$: $\langle Mk_\delta, k_\delta\rangle \ge \zeta_R(1+2\delta)\nu((1/2, 1/2+\delta])$ and $\zeta_R(1+2\delta) \ge e^{-B}(2\delta)^{-\theta}$. For the upper bound, split ν as in the proof above and use (4.4) for ν_ε, which has no mass on $(3/2, \infty)$. □

COROLLARY 4.14. *If $d\nu = w\,\mathrm{d}s$ near $1/2$, with w right-continuous at $1/2$, then $M(\alpha_\nu)$ has essential spectrum $[0, \pi w(1/2)]$; in particular $\|M(\alpha_\nu)\|_{\mathrm{e}} = \pi w(1/2)$.*

PROOF. Let $\nu_0 = w(1/2)\mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. The positive and negative parts of $\nu - \nu_0$ have $o(t)$ mass near $1/2$, so their matrices are compact by Theorem 4.10. Extending ν_0 to $(1/2, \infty)$ changes the matrix, after deleting the first row and column, by a

Hilbert–Schmidt term, and leaves $w(1/2)$ times the multiplicative Hilbert matrix, whose essential spectrum is $[0,\pi]$ [47]. □

The same decomposition compares symbols. Let M be the multiplicative Hilbert matrix.

COROLLARY 4.15. *Under the hypotheses of Corollary 4.14,* $\mathcal{E}_{\mathrm{sym}}(M(\alpha_\nu)) = w(1/2)\,\mathcal{E}_{\mathrm{sym}}(M)$.

PROOF. In the proof of Corollary 4.14, $M(\alpha_\nu) - w(1/2)M$ is a sum of compact curve matrices. By Theorem 4.10 each has a symbol in $C(\mathbb{T}^\infty)+N$, which is closed (Proposition 4.11), so the symbols of $M(\alpha_\nu)$ and $w(1/2)M$ have the same distance to $C(\mathbb{T}^\infty)+N$. □

By Corollary 11.7, the essential size of the symbol is at least $w(1/2)C_{\mathrm{Gauss}}$, while the essential norm is $\pi w(1/2)$. So an excess $C_{\mathrm{Gauss}} > \pi$ would give a strict essential Nehari gap for every curve measure with a density continuous and positive at $1/2$. For the multiplicative Hilbert matrix $\mathscr{M}$ itself, the same splitting of the functional $\int_0^\infty (f(1/2+\sigma)-a_1)\,\mathrm{d}\sigma$ gives $\pi \le \mathcal{E}_{\mathrm{sym}}(\mathscr{M}) \le \Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$: the essential Nehari gap lies in $[1, 1.67]$, and its upper end is exactly the price of discarding phases in section 2.2.

So for a curve measure with a density the essential norm is read off from a single number, and the symbol cannot be made essentially smaller than $\pi w(1/2)$ by any choice in the coset. For the multiplicative Hilbert matrix this gives $\mathcal{E}_{\mathrm{sym}} \ge \pi$: *no* bounded symbol is within distance less than π of $C(\mathbb{T}^\infty)+N$, a statement about the whole coset that the sharp-constant problem (chapter 11) only asks about at the point 0.

4.7. Consequences

Let R have density θ, Q its complement, orders $k_q \ge 2$ on Q, and ν a finite positive measure on $(1/2,\infty)$. Let $\mu_{\nu,R}$ be the law of $\big((\omega_q q^{-s})_{q\in Q}, (p^{-s})_{p\in R}\big)$ under $d\omega \times d\nu(s)$. Its moments are $\alpha(n) = \mathbf{1}[k_q \mid v_q(n)\ \forall q\in Q]\int n^{-s}\,\mathrm{d}\nu(s) \ge 0$. For signs, $k_q = 2$, these are the measures of Theorem 4.2, and then the Carleson measures of the family are the same for every p.

COROLLARY 4.16 (One Carleson class for all p). *For every $0<p<\infty$, $\mu_{\nu,R}$ is a Carleson measure for $H^p(\mathbb{T}^\infty)$ if and only if $[\nu]_\theta = \sup_t \nu((1/2,1/2+t])/t^\theta < \infty$, and then, with $k=\min_q k_q$,*

$$\int |F|^p\,\mathrm{d}\mu_{\nu,R} \le \zeta(k)\big(C[\nu]_\theta + \zeta(3)\nu((1/2,\infty))\big)\|F\|_p^p.$$

In particular the Helson matrix $(\alpha(mn))$ has a bounded symbol of norm at most the constant for $p=1$.

PROOF. Sufficiency: Proposition 4.7 in the Q-coordinates, then Theorem 2.10 (2) on each slice $F(\zeta,\cdot) \in H^p(\mathbb{T}^R)$, then Fubini. Necessity: functions of the R-coordinates only do not see the averaging, and testing with $K^{2/p}$ for kernels at $\gamma_R(1/2+\delta)$ reduces to the H^2 test with K, which gives $[\nu]_\theta \le 2e^B C_2$, C_2 the H^2 Carleson constant, as in section 4.5. The symbol comes from Hahn–Banach. □

Compactness needs a Montel lemma in infinitely many variables.

LEMMA 4.17. *Let $0 < p < \infty$. Every bounded sequence in $H^p(\mathbb{T}^\infty)$ has a subsequence that converges uniformly on every compact subset of $\mathbb{D}^\infty \cap \ell^2$, compact in the ℓ^2 topology.*

PROOF. Point evaluation satisfies $|F(r)| \le \|F\|_p \prod_j (1-|r_j|^2)^{-1/p}$, which is at most $\|F\|_p \exp(M^2/(p(1-\rho^2)))$ when $\|r\|_2 \le M$ and $\|r\|_\infty \le \rho < 1$. A compact set K has $\sup_K \|r\|_2 < \infty$ and $\sup_K \|r\|_\infty < 1$, and the same bounds hold, with room to spare, on a fixed ℓ^2-neighbourhood of K and of its truncations $r^{(D)} = (r_1, \dots, r_D, 0, \dots)$. So the sequence is uniformly bounded there, and Cauchy's estimate along lines bounds its derivatives uniformly. Since $\sup_K \|r - r^{(D)}\|_2 \to 0$, we get $|F_n(r) - F_n(r^{(D)})| \to 0$ uniformly in n and $r \in K$. The coefficients of a bounded sequence are bounded, so a diagonal subsequence converges coefficientwise; on each $\mathbb{D}^D$ the finite-dimensional Montel theorem then gives uniform convergence on compact sets. Combining the two statements, the subsequence is uniformly Cauchy on K. □

PROPOSITION 4.18 (Compact embeddings). *In the setting of Corollary 4.16, suppose $[\nu]_\theta < \infty$.*

(1) If the embedding $H^p \to L^p(\mu_{\nu,R})$ is compact for some $0 < p < \infty$, then $\nu((1/2, 1/2+t]) = o(t^\theta)$.
(2) If $\nu((1/2, 1/2+t]) = o(t^\theta)$, the embedding is compact for every $0 < p < \infty$.

PROOF. (1) Let K_δ be the kernel at $\gamma_R(1/2+\delta)$ and $f_\delta = K_\delta^{2/p}/\|K_\delta\|_2^{2/p}$, of norm one. At every point of the support, $f_\delta \to 0$ as $\delta \to 0$, because $K_\delta(z)$ stays bounded there while $\|K_\delta\|_2 \to \infty$. If the embedding is compact, every sequence $\delta_n \to 0$ has a subsequence along which f_{δ_n} converges in $L^p(\mu)$, necessarily to 0, the pointwise limit. But $\int |f_\delta|^p d\mu \ge \zeta_R(1+2\delta)\nu((1/2, 1/2+\delta]) \gtrsim \delta^{-\theta}\nu((1/2, 1/2+\delta])$, as in section 4.5. (2) Split ν at $1/2+\delta$. By Corollary 4.16, the part near $1/2$ has embedding norm at most a constant times $\sup_{t \le \delta} \nu((1/2, 1/2+t])t^{-\theta} + \nu((1/2, 1/2+\delta])$, which tends to 0 with δ. The far part is the image of a compact subset K_δ of $\mathbb{D}^\infty \cap \ell^2$: its points have coordinates of modulus at most $q^{-1/2-\delta}$, whose squares have uniformly summable tails, and the root phases range over a compact product. Given a bounded sequence in H^p, Lemma 4.17 provides a subsequence converging uniformly on K_δ, hence in L^p of the finite far measure. Since the near parts are uniformly small, the images form a Cauchy sequence; for $p < 1$ use the p-subadditivity of $\int |\cdot|^p$. □

For signs the Helson matrix is positive and the rest of this chapter develops its full theory. For some $k_q \ge 3$ it is not: the measure lives on complex points, and already the one-variable Hankel matrix of $\mathbf{1}[3 \mid n]r^n$ has the indefinite block $\left(\begin{smallmatrix} 0 & r^3 \\ r^3 & 0 \end{smallmatrix}\right)$. Boundedness of the matrix and the H^2 Carleson condition then need not coincide, and the corollary gives the bounded symbol directly from the H^1 embedding.

So the inclusion (4.1) holds throughout the family, and Problem 2 has a positive answer for every measure of the form $\mu_{\nu,R}$.

The best-known special case is the multiplicative Hilbert matrix, with entries $1/(\sqrt{mn}\log mn)$ for $m, n \ge 2$.

Problem 3 (Saksman and Seip [49, Problem 3.2]). *Does the multiplicative Hilbert matrix have a bounded symbol?*

Perfekt [45] answered Problem 3 in the affirmative. The family of Theorem 4.2 contains the matrix, up to a harmless correction, and so gives a second proof.

Corollary 4.19 (Perfekt [45, Corollary 1.2]). *The multiplicative Hilbert matrix has a bounded symbol: there is $g \in L^\infty(\mathbb{T}^\infty)$ with $\int g\,\mathrm{d}m = 1$ and $\int z^{\kappa(n)} g\,\mathrm{d}m = 1/(\sqrt{n}\log n)$ for $n \geq 2$.*

Proof. Take R to be all primes and $\nu = \mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. Then $[\nu] = 1$, $\alpha_\nu(1) = 1$, and

$$\alpha_\nu(n) = \frac{1}{\sqrt{n}\log n} - c_n, \qquad c_n = \frac{n^{-3/2}}{\log n} \qquad (n \geq 2).$$

Corollary 4.3 gives a bounded symbol for α_ν. The functional $h \mapsto \sum_{n\geq 2} h_n c_n$ is bounded on $\mathscr{H}^1$ by Helson's inequality, since $\sum_n d(n) n^{-3} < \infty$, so it too has a bounded symbol by Lemma 4.1. The sum of the two symbols works. □

Perfekt's Tauberian argument also gives convergence of the coefficient series in its natural order, which this proof does not.

The same subtraction applied to the odd-indexed primes ($\theta = 1/2$) gives a bounded symbol for the matrix $\big(\mathbf{1}[(mn)_Q \in \square]/(\sqrt{mn}\log mn)\big)_{m,n\geq 2}$, the multiplicative Hilbert matrix with the signs of the even-indexed primes averaged out.

4.8. The positive-Nehari question

Problem 2 therefore has a positive answer for every measure $\mu_{\nu,R}$, and for discrete measures on energy-separated families with weights bounded by the inverse kernel norm (chapter 3). The separation mechanism controls the coordinates kept on a curve. The symmetry mechanism controls the coordinates whose signs are averaged, and costs only the factor $\zeta(2)$. Rotations $z \mapsto \varepsilon z$ of $\mathbb{T}^\infty$ preserve every H^p norm, so each statement also holds for a fixed rotated copy of the family. Since a positive measure dominates each of its parts, finite sums of such measures are covered as well.

Explicit compressions to $\mathcal{S}_R$ at every borderline are listed in section 5.2, and Proposition 5.10 compares them with the averaged matrices.

The first case not covered is a countable superposition. A counterexample would need mass that is neither separated in energy nor symmetric under sign changes in any fixed set of coordinates, for instance many rotated copies of the curve with individually bounded constants $[\nu_\varepsilon]$ whose sum diverges.

Question 4. Let $\mu = \sum_\varepsilon (\varepsilon\gamma)_*\nu_\varepsilon$ be a countable superposition of measures on rotated copies of γ whose moments define a bounded Helson matrix. Is μ a Carleson measure for $H^1(\mathbb{T}^\infty)$?

CHAPTER 5

Spectral, Schatten and rank structure

Question. What do positive Helson matrices look like spectrally, and how far can the Nehari constant grow among matrices of bounded rank?

The answer to the first question is a dictionary. For a curve matrix the dyadic profile $a_j = e^{j\theta}\nu(I_j)$ of the measure determines everything: boundedness is $a \in \ell^\infty$, compactness is $a \in c_0$ (chapter 4), and membership in S_p is $a \in \ell^p$ for every $p > 0$, and likewise for every Lorentz class (Theorems 5.7 and 5.13). For the compression of the multiplicative Hilbert matrix to a prime set of density $\theta < 1$ the spectrum decays root-exponentially; for the full sign average $s_n \sim 1/(2n \log n)$. The second question has a short answer at the level of S_2, where Helson's inequality is exactly a weak-factorization identity, and a gap between the exponents 0.174 and 1/2.

Four different spectral regimes appear below, and it helps to keep them apart.

- *Scale invariance.* When the companion kernel is asymptotically homogeneous at the critical line, the Mellin transform diagonalizes the model and the essential spectrum is an interval (Theorem 5.4).
- *Periodicity in logarithmic scale.* An oscillating prime density makes the model periodic, and Floquet theory gives bands (Proposition 5.5).
- *A weaker singularity.* The Hilbert matrix of a prime set of density $\theta < 1$ has a kernel with singularity $t^{-\theta}$, which is integrable against $t\,\mathrm{d}t$; Taylor approximation then gives root-exponential decay (Theorem 5.16).
- *Many blocks.* The full sign average splits into one block for each squarefree q, each of rank one up to a trace-class error, and the asymptotics count the leading eigenvalues of the blocks (Theorem 5.17 and Proposition 5.18).

Restriction to a sparse prime set and sign averaging both simplify the matrix, but in different ways. Restriction removes primes and so weakens the singularity of a single kernel: few singular values survive. Averaging keeps every prime but cuts the matrix into infinitely many blocks, and then the number of blocks, not the smoothness of one kernel, sets the decay, which is only polynomial.

Chavan, Sahu and Sinha [18] study Helson matrices induced by measures and give Schatten criteria in terms of the entries $\alpha(mn)$; the criteria below are in terms of the measure. To our knowledge the following are new: the dictionary between the profile and the Schatten–Lorentz classes for every $p > 0$, with its pointwise corollaries and the separated-atom theorem; the exact asymptotics of the full sign average; the critical spectrum for general prime sets, including the band structure for oscillating densities. The S_2 endpoint in the last section, where Helson's inequality gives bounded symbols exactly, is where chapter 7 starts.

5.1. The companion kernel

Throughout, R is a set of primes, $\mathcal{S}_R$ the multiplicative semigroup it generates, and $\zeta_R(w) = \prod_{p\in R}(1-p^{-w})^{-1}$. In this chapter a curve measure is a finite positive measure ν on $(0,\infty)$, shifted so that $u=0$ is the critical line:

$$\alpha_\nu(n) = \int_0^\infty n^{-1/2-u}\,\mathrm{d}\nu(u), \qquad M_{\nu,R} = \big(\alpha_\nu(mn)\big)_{m,n\in\mathcal{S}_R}.$$

For R all primes we write M_ν. With this shift, $M_{\nu,R}$ is the compression of the matrix $M(\alpha_{\nu,R})$ of chapter 4 to $\ell^2(\mathcal{S}_R)$: for $m,n\in\mathcal{S}_R$ the Q-part of mn is 1. Density of R is defined as in (2.1). Put $I_j = (e^{-j-1}, e^{-j}]$ for $j\ge 0$.

LEMMA 5.1. *Let $\mathcal{N}:\ell^2(\mathcal{S}_R)\to L^2(\nu)$, $\mathcal{N}e_n(u) = n^{-1/2-u}$, on finitely supported vectors. Then $M_{\nu,R} = \mathcal{N}^*\mathcal{N}$ as forms, and whenever either side is bounded, $M_{\nu,R}$ and the integral operator on $L^2(\nu)$ with kernel $\zeta_R(1+u+v)$ have the same nonzero spectrum with multiplicities.*

PROOF. $\langle \mathcal{N}e_n, \mathcal{N}e_m\rangle_{L^2(\nu)} = \int (mn)^{-1/2-u}\,\mathrm{d}\nu(u) = \alpha_\nu(mn)$, and $\mathcal{N}\mathcal{N}^*$ has kernel $\sum_{n\in\mathcal{S}_R} n^{-1-u-v} = \zeta_R(1+u+v)$. The nonzero spectra of $\mathcal{N}^*\mathcal{N}$ and $\mathcal{N}\mathcal{N}^*$ coincide. □

The kernel $\zeta_R(1+u+v)$ is a Hankel kernel in the additive variable $u+v$, singular like $(u+v)^{-\theta}$ at the origin. Every result in the first half of this chapter is a statement about this one kernel restricted to ν.

PROPOSITION 5.2 (Traces). *For every finite positive ν and integer $k\ge 1$,*

$$\operatorname{Tr} M_{\nu,R} = \int \zeta_R(1+2u)\,\mathrm{d}\nu(u),$$

$$\operatorname{Tr} M_{\nu,R}^k = \int \prod_{i=1}^{k} \zeta_R(1+u_i+u_{i+1})\,\mathrm{d}\nu(u_1)\cdots d\nu(u_k),$$

with $u_{k+1}=u_1$, both sides possibly infinite. In particular $\|M_{\nu,R}\|_{S_2}^2 = \iint \zeta_R(1+u+v)^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v)$.

PROOF. Restrict ν to $[\varepsilon,\infty)$. Then $\mathcal{N}$ is Hilbert–Schmidt, the formula is the trace of the kth power of the companion, and the compressions to $n\le N$ increase to it. Monotone convergence as $N\to\infty$ and $\varepsilon\to 0$ gives the general case. □

COROLLARY 5.3. *Let R have density $\theta\in[0,1)$. The multiplicative Hilbert matrix restricted to $\mathcal{S}_R\setminus\{1\}$ is positive and trace class, with*

$$\operatorname{Tr} = \frac{1}{2}\int_0^\infty \big(\zeta_R(1+t)-1\big)\,\mathrm{d}t.$$

More generally, $\big(\Gamma(\kappa)(mn)^{-1/2}(\log mn)^{-\kappa}\big)$ on $\mathcal{S}_R\setminus\{1\}$ is trace class for $\kappa>\theta$, bounded and not compact for $\kappa=\theta>0$, and unbounded for $\kappa<\theta$.

PROOF. The matrix is $M_{\nu,R}$ for Lebesgue measure ν on $(0,\infty)$ with the index 1 removed, and Proposition 5.2 with $\zeta_R(1+2u)-1$ in place of $\zeta_R(1+2u)$ gives the trace, which is finite because $\zeta_R(1+t)\lesssim t^{-\theta}$ near 0 and $\zeta_R(1+t)-1\le 2^{1-t}(\zeta(2)-1)$ for $t\ge 1$. The second family is $\nu = u^{\kappa-1}\,\mathrm{d}u$. Its block profile is $a_j \asymp e^{-j(\kappa-\theta)}$. The

part of ν on $(1,\infty)$ has infinite mass, but once the index 1 is removed its companion kernel $\zeta_R(1+u+v)-1$ decays exponentially, so it contributes a trace-class term; for the part on $(0,1]$, Theorem 5.7 below, with Theorems 4.2 and 4.10, gives the three cases. □

At the critical exponent the matrix is bounded but not compact. For the multiplicative Hilbert matrix the essential spectrum is $[0,\pi]$ [47], and the Mellin structure of the companion kernel suggests a universal answer.

PROBLEM 4 (A universal spectral constant). *Let the companion kernel of a positive Helson matrix grow like $c\,x^{-\kappa}$ at the boundary. Is its critical essential spectrum always $[0, c\,\Gamma(\kappa/2)^2/\Gamma(\kappa)]$?*

The answer is yes when the kernel and the weight vary regularly at the boundary.

THEOREM 5.4 (Critical essential spectrum). *Let $\kappa>0$, let $h\ge 0$ be continuous on $(0,\infty)$ with $x^{\kappa}h(x)\to c$ as $x\to 0$, and let $\nu=g(u)\,\mathrm{d}u$ be a positive measure on $(0,\infty)$ with $u^{1-\kappa}g(u)\to b$ as $u\to 0$. Suppose that $\iint_{\max(u,v)\ge\delta} h(u+v)^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v)<\infty$ for every $\delta>0$. Then the integral operator T on $L^2(\nu)$ with kernel $h(u+v)$ is bounded, and*

$$\sigma_{\mathrm{ess}}(T)=\Big[0,\; bc\,\frac{\Gamma(\kappa/2)^2}{\Gamma(\kappa)}\Big].$$

PROOF. Multiplication by $g^{1/2}$ is unitary from $L^2(\nu)$ onto $L^2(0,\infty)$ and carries T to the kernel $K(u,v)=g(u)^{1/2}h(u+v)g(v)^{1/2}$. Compare it with $K_0(u,v)=(uv)^{(\kappa-1)/2}(u+v)^{-\kappa}$, which is homogeneous of degree -1. The Mellin transform $f\mapsto\int_0^\infty f(u)u^{-1/2-i\xi}\,\mathrm{d}u$ is a multiple of a unitary map onto $L^2(\mathbb{R})$ and carries K_0 to multiplication by

$$m(\xi)=\int_0^\infty t^{\kappa/2-1+i\xi}(1+t)^{-\kappa}\,\mathrm{d}t=\frac{|\Gamma(\kappa/2+i\xi)|^2}{\Gamma(\kappa)},$$

a continuous even function, decreasing in $|\xi|$, with maximum $m_0=\Gamma(\kappa/2)^2/\Gamma(\kappa)$ at $\xi=0$ and limit 0 at infinity. So K_0 is bounded with $\sigma(K_0)=\sigma_{\mathrm{ess}}(K_0)=[0,m_0]$. Let P_δ be multiplication by the indicator of $(0,\delta)$. The functions $f_n(u)=n^{-1/2}u^{-1/2+i\xi}\mathbf{1}_{(e^{-2n},e^{-n})}(u)$ are, up to $o(1)$ in norm, approximate eigenvectors of K_0 with eigenvalue $m(\xi)$; they tend weakly to zero and are eventually supported in $(0,\delta)$. Hence $\sigma_{\mathrm{ess}}(P_\delta K_0 P_\delta)=[0,m_0]$ for every $\delta>0$.

Now $K=P_\delta K P_\delta+R_\delta$, where R_δ has kernel supported in $\{\max(u,v)\ge\delta\}$ and is Hilbert–Schmidt by hypothesis. On $(0,\delta)^2$ the hypotheses give $|K-bcK_0|\le\varepsilon(\delta)K_0$ pointwise, with $\varepsilon(\delta)\to 0$. A kernel dominated by a positive kernel has at most its norm, so $\|P_\delta(K-bcK_0)P_\delta\|\le\varepsilon(\delta)m_0$. In particular T is bounded. Now put $D=K-bcP_1K_0P_1$. For $\delta\le 1$, the operator $D-P_\delta DP_\delta$ is Hilbert–Schmidt: its K part by hypothesis, and its model part because K_0 is square integrable on $(0,1)^2\setminus(0,\delta)^2$. Since $\|P_\delta DP_\delta\|\le\varepsilon(\delta)m_0\to 0$, D is a norm limit of compact operators, hence compact: the operator is the boundary model $bcP_1K_0P_1$ modulo compacts, and $\sigma_{\mathrm{ess}}(T)=\sigma_{\mathrm{ess}}(bcP_1K_0P_1)=[0,bcm_0]$. □

For a prime set R with

$$c_R = \lim_{x\to 0} x^\theta \zeta_R(1+x) \quad \text{existing},$$

for instance when $\sum_{p\in R} p^{-1-t} = \theta\log(1/t) + C + o(1)$, take $\nu = u^{\theta-1}\,\mathrm{d}u$ and $h(x) = \zeta_R(1+x) - 1$, which decays exponentially. By Lemma 5.1 the matrix $\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)$ on $\mathcal{S}_R \setminus \{1\}$ and this operator have the same nonzero spectrum, so

$$\sigma_{\mathrm{ess}}\Big(\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)_{m,n\in\mathcal{S}_R\setminus\{1\}}\Big) = \Big[0,\, c_R\frac{\Gamma(\theta/2)^2}{\Gamma(\theta)}\Big].$$

For R all primes, $c_R = 1$, $\theta = 1$, and this is the interval $[0,\pi]$ of the multiplicative Hilbert matrix found by Perfekt and Pushnitski [47]; the proof above is the standard Mellin argument for kernels homogeneous of degree -1. The theorem applies verbatim to any positive family whose companion kernel and weight vary regularly at the boundary, and so answers Problem 4 in that setting.

Ma, Lu and Zu [38, Theorem 3.3] prove a sharper statement under stronger hypotheses. For kernels $w(x)b(x+y)w(y)$ whose normalized profiles $t^\kappa b(t)$ and $t^{(1-\kappa)/2}w(t)$ are C^2, have limits at 0 and at ∞, and satisfy $|(t\partial_t)^2 f(t)| \lesssim (1+|\log t|)^{-2}$, they find the same essential spectrum, with a contribution from each end, and by a Mourre estimate they exclude singular continuous spectrum and show that each compact window inside the open bands contains only finitely many eigenvalues. For the Bergman forms with weights d_α on all primes this gives essential and absolutely continuous spectrum $[0,\Gamma(\alpha/2)^2/\Gamma(\alpha)]$, the case $\kappa = \alpha$ of the theorem above, together with finitely many simple eigenvalues above it, which occur exactly when α exceeds a threshold $\alpha_* \in (1,2)$ [38, Theorem 1.1]. Theorem 5.4 asks only for limits, which is what lets it reach prime sets, where ζ_R need not be smooth in the required sense; it says nothing about the finer spectral type. If only the bounded remainder (2.1) is known, $\psi(L) = \log\big(x^\theta\zeta_R(1+x)\big)$, $L = \log(1/x)$, may oscillate without converging, and then the answer changes.

PROPOSITION 5.5 (Oscillating densities). *Let R have density $\theta \in (0,1]$, and suppose that $\psi(L) - \psi_P(L) \to 0$ as $L\to\infty$ for a continuous P-periodic function ψ_P. Let T_P be the operator on $L^2(\mathbb{R})$ with kernel*

$$\Big(2\cosh\frac{s-t}{2}\Big)^{-\theta} e^{\psi_P(\Lambda(s,t))}, \qquad \Lambda(s,t) = -\log(e^{-s}+e^{-t}).$$

Then the essential spectrum of the critical matrix

$$\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)_{m,n\in\mathcal{S}_R\setminus\{1\}}$$

is $\sigma(T_P)$, the closure of the union over $k\in[0,2\pi)$ of the spectra of the compact fibre operators on $L^2(0,P)$ with kernels $\sum_{n\in\mathbb{Z}} K_P(s,t+nP)e^{ikn}$, where K_P is the kernel of T_P.

PROOF. In the variable $u = e^{-s}$ the companion operator of Lemma 5.1, with $\nu = u^{\theta-1}\,\mathrm{d}u$, becomes the operator on $L^2(\mathrm{d}s)$ with kernel $(2\cosh((s-t)/2))^{-\theta}e^{\tilde\psi(\Lambda(s,t))}$, where $x^\theta(\zeta_R(1+x)-1) = e^{\tilde\psi(\log(1/x))}$ and $\tilde\psi - \psi \to 0$. Since $\Lambda(s+P,t+P) = \Lambda(s,t)+P$, the operator T_P commutes with translation by P, and the Floquet–Bloch

decomposition gives its spectrum as stated; the fibres are Hilbert–Schmidt because the kernel decays like $e^{-\theta|s-t|/2}$. As in the proof of Theorem 5.4, the part of the companion operator with $\min(s,t) \le s_0$ is Hilbert–Schmidt, and on $\{s,t > s_0\}$ its kernel differs from that of T_P by at most $\varepsilon(s_0)$ times the kernel of T_P, with $\varepsilon(s_0) \to 0$. So the essential spectrum is that of the compression of T_P to a half-line. The pieces of T_P coupling $(-\infty, s_0)$ with (s_0, ∞) are Hilbert–Schmidt, by the exponential decay, so the compressions to the two half-lines together have essential spectrum $\sigma(T_P)$; and Weyl sequences for T_P, truncated Bloch waves translated by multiples of P, can be placed in (s_0, ∞). Hence the compression to (s_0, ∞) has essential spectrum $\sigma(T_P)$ as well. □

When ψ_P is constant, the fibres are diagonalized by the Mellin transform, the bands of T_P fill $[0, c_R\Gamma(\theta/2)^2/\Gamma(\theta)]$, and Theorem 5.4 is recovered. Genuine oscillations occur for prime sets. If $R_P = \{p : \{\log\log p/P\} < \theta\}$, then by the prime number theorem ψ is asymptotically the P-periodic smoothing of the indicator of $\{y : \{y/P\} < \theta\}$ by the kernel e^{-e^z}, whose kth Fourier coefficient carries the factor $\Gamma(2\pi ik/P)$, and R_P has density θ with bounded remainder. Numerically, for $\theta = 1/2$ and $P = 8$, where ψ oscillates between ± 0.53, the two largest bands of T_P are $[3.137, 8.236]$ and $[0.359, 0.790]$, stable to four digits under refinement of the discretization; for $P = 12$ they are $[7.43, 10.80]$ and $[1.03, 1.46]$. So for densities with bounded remainder the critical essential spectrum is very probably not an interval in general, which would answer Problem 4 in the negative. The reduction to T_P is proved.

REMARK 5.6 (A certified gap). For $\theta = 1/2$ and $P = 8$ one gap can be certified. Let $c_R = \lim_{L\to\infty}(\log\zeta_R(1 + e^{-L}) - L/2 - \psi_0(L))$, with ψ_0 the periodic profile above, normalized to mean zero. Truncating the Floquet fibres to 128×128 matrices on a mesh of 129 Bloch parameters, enclosing their entries in ball arithmetic, and bounding the truncation and mesh errors gives

$$\sigma_{\mathrm{ess}} \cap \Big(\frac{189}{100}e^{c_R}, \frac{211}{100}e^{c_R}\Big) = \varnothing$$

for the critical matrix of R_8. Since the essential spectrum contains 0 and points above this interval, it is not an interval. The certificate is published with the book's numerics (chapter A). It does not locate the band edges or exclude eigenvalues in the gap.

5.2. Schatten classes along the curve

Put

$$a_j(\nu) = e^{j\theta}\nu(I_j), \qquad j \ge 0.$$

Chapter 4 shows that $M_{\nu,R}$ is bounded exactly when a is bounded, and compact exactly when $a_j \to 0$. The trace formula shows that $M_{\nu,R}$ is trace class exactly when $\int_0^1 u^{-\theta}\,\mathrm{d}\nu < \infty$, that is, when $a \in \ell^1$. The intermediate classes interpolate.

The profile is the natural quantity. On the block I_j one evaluation along the curve costs $\zeta_R(1 + 2u) \asymp e^{j\theta}$, so a_j is the mass of the block times the size of an evaluation there; it is, up to constants, the trace of the piece M_j of the matrix that comes from I_j. Two facts turn this into a spectral classification. Within one

block, the kernel $k(u)$ depends analytically on u across a region of fixed hyperbolic size, so M_j is of rank one up to geometrically small corrections (5.3): one block contributes one principal singular value, of size about a_j. Across blocks, normalized kernels at distant scales are nearly orthogonal (Lemma 2.4), so these principal values interact only weakly. The singular values of $M_{\nu,R}$ are therefore, up to constants and geometric errors, the numbers a_j. Nothing in the profile records how the mass is arranged inside a block, which is why it determines the distribution of the spectrum (Theorem 5.13) but not individual singular values (Proposition 5.11).

THEOREM 5.7 (Schatten classes of curve matrices). *Let R have density $\theta \in (0,1]$ with constant B, and let $1 \le p < \infty$. There are $c, C > 0$ depending only on θ, B and p such that, for every finite positive ν on $(0,\infty)$,*

$$c \sum_{j\ge 0} a_j(\nu)^p \le \|M_{\nu,R}\|_{S_p}^p \le C\Big(\sum_{j\ge 0} a_j(\nu)^p + \nu((1,\infty))^p\Big).$$

In particular $M_{\nu,R} \in S_p$ if and only if $a(\nu) \in \ell^p$. The same holds for every $0 < p < \infty$, and for every Lorentz class $S_{p,q}$, by Theorem 5.13 below.

PROOF. The part of ν on $(1,\infty)$ gives a positive matrix whose trace is at most $\zeta(3)\,\nu((1,\infty))$. So assume that ν lives on $(0,1]$.

Upper bound. Let $\nu_j = \nu|_{I_j}$ and, for those j with $a_j > 0$, let $B_j = M_{\nu_j,R}/a_j$. For a finitely supported complex sequence c put $Tc = \sum_j c_j B_j$, so that $M_{\nu,R} = Ta$. Since $\zeta_R(1+2u) \le e^{1+B}(2u)^{-\theta}$,

$$\|B_j\|_{S_1} = \operatorname{Tr} B_j \le e^{1+B} 2^{-\theta} e^{(j+1)\theta} \nu(I_j)/a_j \le e^{2+B},$$

so $\|Tc\|_{S_1} \le e^{2+B}\|c\|_{\ell^1}$. For real c and finitely supported x, positivity gives $|\langle Tc\,x, x\rangle| \le \|c\|_\infty \langle M_{\nu',R}\,x, x\rangle$ with $\nu' = \sum_j \nu_j/a_j$. Every block of ν' has $a_j(\nu') = 1$, so $\nu'((0,t]) \le (e^\theta/(1-e^{-\theta}))\,t^\theta$ and Theorem 2.10 with $p = 2$ bounds $\|M_{\nu',R}\|$ by a constant. Splitting complex c into real and imaginary parts, $\|Tc\| \le C\|c\|_{\ell^\infty}$. Complex interpolation, with $[\ell^1, \ell^\infty]_{1-1/p} = \ell^p$ and $[S_1, B(H)]_{1-1/p} = S_p$, gives $\|Tc\|_{S_p} \le C\|c\|_{\ell^p}$. Apply this to $c = a$.

Lower bound. For $0 < \delta \le 1$ let $k_\delta(n) = n^{-1/2-\delta}\zeta_R(1+2\delta)^{-1/2}$, a unit vector in $\ell^2(\mathcal{S}_R)$. For $u \le \delta$ one has $\zeta_R(1+\delta+u) \ge \zeta_R(1+2\delta)$, so

$$\langle M_{\nu,R}k_\delta, k_\delta\rangle = \frac{\int \zeta_R(1+\delta+u)^2\,\mathrm{d}\nu(u)}{\zeta_R(1+2\delta)} \ge \zeta_R(1+2\delta)\,\nu((0,\delta]) \ge e^{-B}(2\delta)^{-\theta}\nu((0,\delta]).$$

With $\delta_j = e^{-j}$ this is at least $e^{-B}2^{-\theta}a_j$. The density bounds, used as in Lemma 2.9, give

$$\langle k_{\delta_j}, k_{\delta_k}\rangle = \frac{\zeta_R(1+\delta_j+\delta_k)}{\sqrt{\zeta_R(1+2\delta_j)\zeta_R(1+2\delta_k)}} \le e^{2B+1}\Big(\frac{2\sqrt{\delta_j\delta_k}}{\delta_j+\delta_k}\Big)^\theta \le 2e^{2B+1}e^{-\theta|j-k|/2}.$$

By Schur's test the Gram matrix of the whole sequence $(k_{\delta_j})_{j\ge 0}$ is bounded, with norm at most $C_V = 2e^{2B+1}(1+e^{-\theta/2})/(1-e^{-\theta/2})$, so the synthesis map $V : e_j \mapsto k_{\delta_j}$ has $\|V\|^2 \le C_V$. For $p \ge 1$ the diagonal of an S_p operator lies in ℓ^p with norm at

most its S_p norm, hence

$$\sum_j \langle M_{\nu,R} k_{\delta_j}, k_{\delta_j}\rangle^p \le \|V^* M_{\nu,R} V\|_{S_p}^p \le C_V^p \|M_{\nu,R}\|_{S_p}^p.$$

This lower bound needs only that the kernels form a Bessel sequence; the separated-atom theorem below needs a Gram matrix bounded from below, and therefore a stronger separation. □

COROLLARY 5.8. *With $F(t) = \nu((0,t])$ and $0 < \theta \le 1$:*

(1) $M_{\nu,R} \in S_1$ if and only if $\int_0^1 u^{-\theta}\,\mathrm{d}\nu(u) < \infty$;
(2) $M_{\nu,R} \in S_2$ if and only if $\int_0^1 F(t)^2 t^{-1-2\theta}\,\mathrm{d}t < \infty$;
(3) if $\theta = 0$, then $M_{\nu,R}$ is trace class for every finite ν, with $\operatorname{Tr} M_{\nu,R} \le e^{1+B}\nu((0,\infty))$.

PROOF. (1) and (2) restate $a \in \ell^1$ and $a \in \ell^2$; for (2), note $e^{j\theta}F(e^{-j}) = \sum_{k\ge j} e^{-(k-j)\theta} a_k$, a convolution of a with a summable kernel, so $(e^{j\theta}F(e^{-j})) \in \ell^2$ exactly when $a \in \ell^2$. (3) is the trace formula with $\zeta_R(1+2u) \le e^{1+B}$. □

Part (1) is the trace formula. For R all primes the boundedness criterion that the profile refines is due to Perfekt and Pushnitski [46, Theorem 5.1].

Part (3) sharpens the $\theta = 0$ case of Theorem 4.10, which gave only compactness.

REMARK 5.9 (Positive matrices in every Schatten class have symbols). Take $\nu = \sum_j j^{-1/p} e^{-j\theta}\delta_{e^{-j}}$. Then $M_{\nu,R}$ lies in S_q exactly for $q > p$, and it has a bounded symbol: Theorem 4.2 gives one for $M(\alpha_{\nu,R})$, and its conditional expectation onto the coordinates in R has moments $\alpha_\nu(n)\mathbf{1}_{\mathcal{S}_R}(n)$ and no larger norm. So positive curve matrices occupy every Schatten class, each with a bounded symbol. The counterexamples of chapter 7, which exist in S_p for every $p > 4$, are therefore necessarily not of this form. Inside the positive family the S_p condition is a pure size condition on ν; the S_p failure of Nehari's theorem is a phenomenon of cancellation.

The matrices $M_{\nu,R}$ of this chapter are compressions to $\mathcal{S}_R$. The sign-averaged matrices $M(\alpha_{\nu,R})$ of chapter 4, on all of $\ell^2(\mathbb{N})$, have the same Schatten classes for $p > 1$ but not at $p \le 1$.

PROPOSITION 5.10. *Let R have density $\theta \in [0,1]$, let ν be a finite positive measure on $(0,\infty)$, and put $W = M(\alpha_{\nu,R})$, with the shift of this chapter.*

(1) For $p > 1$, $\|M_{\nu,R}\|_{S_p} \le \|W\|_{S_p} \le \zeta(2)\zeta(p)^{1/p}\|M_{\nu,R}\|_{S_p}$.
(2) $\operatorname{Tr} W = \int \zeta(1+2u)\,\mathrm{d}\nu(u)$, so $W \in S_1$ exactly when $\int_0^1 u^{-1}\,\mathrm{d}\nu < \infty$, whatever R is.
(3) If $\theta < 1$ and $\nu((0,1]) > 0$, then $W \notin S_p$ for $p \le 1/3$.

PROOF. Let Q be the complement of R and write $m = m_R q a^2$ with q squarefree and $q, a \in \mathcal{S}_Q$. The entry $\alpha_{\nu,R}(mn)$ vanishes unless m and n have the same q, so $W = \bigoplus_q W_q$, and $W_q = \mathcal{N}_q^* \mathcal{N}_q$ with $\mathcal{N}_q e_{(m_R,a)}(u) = m_R^{-1/2-u}(qa^2)^{-1/2-u}$. Hence $\mathcal{N}_q\mathcal{N}_q^* = \sum_a w_{qa^2} K w_{qa^2}$, where K is the companion operator of Lemma 5.1 and w_n is multiplication by $n^{-1/2-u}$, of norm at most $n^{-1/2}$. For $p \ge 1$ the triangle inequality gives $\|W_q\|_{S_p} \le \zeta(2) q^{-1}\|K\|_{S_p}$, and summing pth powers over q gives (1)

for $p > 1$; the block $q = a = 1$ gives the lower bound. For (2), every n^2 has a square Q-part. For (3), testing $\mathcal{N}_q \mathcal{N}_q^*$ with $\mathbf{1}_{(0,1]}$ gives an eigenvalue at least $q^{-3}\nu((0,1])$ in each block, and $\sum_q q^{-1} = \prod_{p\in Q}(1 + 1/p) = \infty$ when $\theta < 1$. □

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The criteria of chapter 4 and Theorem 5.7 turn every borderline into an explicit matrix. Let R have density $\theta \in (0, 1]$.

(1) *Power weights.* $d\nu = u^{\kappa-1}\,\mathrm{d}u$ gives $\alpha(n) = \Gamma(\kappa)n^{-1/2}(\log n)^{-\kappa}$ and $a_j \asymp e^{-j(\kappa-\theta)}$: in every S_p for $\kappa > \theta$, bounded but not compact for $\kappa = \theta$, unbounded for $\kappa < \theta$ (Corollary 5.3). For R all primes and $\kappa = 1$ this is the multiplicative Hilbert matrix.

(2) *Logarithmic weights.* $d\nu = u^{\theta-1}(\log(e/u))^{-\alpha}\,\mathrm{d}u$ on $(0, 1]$ has $a_j \asymp (1 + j)^{-\alpha}$. It is bounded exactly for $\alpha \geq 0$, compact exactly for $\alpha > 0$, and in S_p, $p > 0$, exactly for $\alpha p > 1$. For R all primes, Miheisi and Pushnitski [39] found the exact eigenvalue asymptotics $\lambda_n \sim \kappa(\alpha)n^{-\alpha}$ for the closely related weights $1/(\sqrt{n}\log n(\log\log n)^{\alpha})$.

(3) *Iterated logarithms.* With an extra factor $(\log(e\log(e/u)))^{-\beta}$ at $\alpha = 1$, $a_j \asymp (1 + j)^{-1}\log^{-\beta}(e + j)$: compact and in every S_p, $p > 1$, and trace class exactly for $\beta > 1$.

(4) *Sparse atoms.* $\nu = \sum_j c_j\delta_{e^{-Lj}}$ has $a_{Lj} = e^{Lj\theta}c_j$ and $a = 0$ elsewhere. With $c_j = e^{-Lj\theta}j^{-1/q}$ the singular values are exactly of order $n^{-1/q}$ (Theorem 5.12 below), so $M \in S_p$ iff $p > q$, for every $p > 0$.

(5) *Same profile, different spectra.* One atom in a block gives a rank-one matrix; two atoms in the same block, with the same total mass, give rank two (Proposition 5.11). So the profile cannot see individual singular values, and Theorem 5.12 needs its separation hypothesis.

Each example has a bounded symbol whenever it is bounded, with the constant of Theorem 4.2.

5.3. What the block profile determines

The profile $a(\nu)$ determines the operator norm, compactness and the S_p norms for $p \geq 1$, up to constants. It does not determine finer spectral information.

PROPOSITION 5.11. *Let R have density $\theta \in (0, 1]$ with constant B.*

(1) $c\sup_j a_j(\nu) \leq \|M_{\nu,R}\| \leq C\sup_j a_j(\nu) + C\nu((1,\infty))$.

(2) If ν consists of N atoms at distinct points of $(0,\infty)$, then $\operatorname{rank} M_{\nu,R} = N$. *In particular measures with the same profile can have different ranks, and no statement about individual singular values depends on $a(\nu)$ alone.*

PROOF. (1) The lower bound is the test in the proof of Theorem 5.7; the upper bound is Theorem 2.10 with $[\nu]_\theta \lesssim \sup_j a_j$. (2) $M = \mathcal{N}^*\mathcal{N}$ with $\mathcal{N}$ mapping into $L^2(\nu) \cong \mathbb{C}^N$, and the vectors $(n^{-1/2-u_i})_{n\in\mathcal{S}_R}$ are linearly independent by uniqueness of Dirichlet coefficients, so $\mathcal{N}$ is onto. □

Individual singular values are controlled once each atom has its own scale.

THEOREM 5.12 (Separated atoms). *Let R have density $\theta \in (0,1]$ with constant B. There is $L = L(\theta, B)$ such that, if $\nu = \sum_i c_i \delta_{u_i}$ with $u_i \in (0,1]$ and $|\log u_i - \log u_j| \geq L$ for $i \neq j$, then*

$$\frac{1}{2} b_n^* \leq s_n(M_{\nu,R}) \leq \frac{3}{2} b_n^* \qquad (n \geq 1),$$

where $b_i = c_i \zeta_R(1+2u_i) \asymp c_i u_i^{-\theta}$ and b^* is its decreasing rearrangement. In particular $M_{\nu,R} \in S_p$ if and only if $b \in \ell^p$, for every $0 < p < \infty$.

PROOF. By Lemma 5.1 the nonzero spectrum of $M_{\nu,R}$ is that of the matrix $\big(\sqrt{c_i c_j}\,\zeta_R(1+u_i+u_j)\big)_{i,j} = D^{1/2}\widetilde{K}D^{1/2}$, with $D = \operatorname{diag}(b_i)$ and $\widetilde{K}$ the Gram matrix of the normalized kernels k_{u_i}. As in the proof of Theorem 5.7, $\widetilde{K}_{ij} \leq 2e^{2B+1}(2\sqrt{u_i u_j}/(u_i+u_j))^{\theta} \leq 2^{1+\theta} e^{2B+1} e^{-\theta|\log u_i - \log u_j|/2}$. At most two indices j have $|\log u_i - \log u_j| \in [mL, (m+1)L)$, so for L large Schur's test gives $\|\widetilde{K} - I\| \leq 1/2$. Hence $D/2 \leq D^{1/2}\widetilde{K}D^{1/2} \leq 3D/2$, and the min-max principle compares eigenvalues. □

So the profile cannot determine individual singular values, but it does determine their distribution up to dilations, and with it every Schatten and Lorentz class. For a positive sequence or operator x write $n_x(t) = \#\{n : x_n^* > t\}$, with x^* the decreasing rearrangement of the sequence or the singular values of the operator, and let $\ell^{p,q}$ and $S_{p,q}$ be the Lorentz sequence spaces and the corresponding Schatten–Lorentz classes, $0 < p < \infty$, $0 < q \leq \infty$; $S_{p,p} = S_p$, and $S_{p,\infty}$ is weak S_p.

THEOREM 5.13 (Distribution of the spectrum). *Let R have density $\theta \in (0,1]$ with constant B, and let ν be a finite positive measure on $(0,1]$. There are $C, \varepsilon > 0$ depending only on θ and B such that, for all $\tau > 0$ and $0 < r \leq 1$,*

$$n_a(\tau) \leq C\, n_M(\varepsilon\tau), \tag{5.1}$$

$$n_M(C\tau) \leq C_r\, \tau^{-r} \sum_{a_j > \tau} a_j^r, \tag{5.2}$$

where $M = M_{\nu,R}$, $a = a(\nu)$, and C_r depends also on r. Consequently, for every finite positive ν on $(0,\infty)$, every $0 < p < \infty$ and every $0 < q \leq \infty$,

$$\|M_{\nu,R}\|_{S_{p,q}} \asymp \|a(\nu)\|_{\ell^{p,q}} + \nu((1,\infty)),$$

with constants depending only on θ, B, p and q. In particular Theorem 5.7 holds for every $0 < p < \infty$.

The proof rests on two estimates. The first is that one block is almost of finite rank: with $\nu_j = \nu|_{I_j}$ and $M_j = M_{\nu_j,R}$, there is $\eta = \eta(\theta, B) < 1$ with

$$s_{N+1}(M_j) \leq C a_j \eta^{2N} \qquad (N \geq 0), \qquad \text{hence} \qquad \operatorname{Tr} M_j^r \leq C_r a_j^r. \tag{5.3}$$

Indeed, $M_j = \mathcal{N}_j^* \mathcal{N}_j$ by Lemma 5.1, and $\mathcal{N}_j^*$ is synthesis $f \mapsto \int k(u) f(u)\, \mathrm{d}\nu_j(u)$ by the $\ell^2(\mathcal{S}_R)$-valued function $k(u) = (n^{-1/2-u})_n$, which is analytic in $\Re u > 0$ with $\|k(u)\|^2 = \zeta_R(1 + 2\Re u)$. Write $u = t(c+z)$ with $t = e^{-j}$ and $c = (1+e^{-1})/2$; then I_j corresponds to $|z| \leq \delta = (1-e^{-1})/2$, and on $|z| = 1/2$ we have $\Re u \geq te^{-1}/2$, so $\|k\| \leq Ct^{-\theta/2}$ there. Truncating the Taylor series in z after N terms gives an

operator of rank at most N, and by Cauchy's estimate its Hilbert–Schmidt distance to $\mathcal{N}_j^*$ is at most $C(t^{-\theta}\nu(I_j))^{1/2}(2\delta)^N = C\sqrt{a_j}\,\eta^N$ with $\eta = 2\delta$.

The second estimate is the concavity inequality $\operatorname{Tr}(A+B)^r \le \operatorname{Tr}A^r + \operatorname{Tr}B^r$ for positive A, B and $0 < r \le 1$. It follows from $x^r = (\sin \pi r/\pi)\int_0^\infty t^{r-1}x(x+t)^{-1}\,\mathrm{d}t$ and $\operatorname{Tr}A(A+B+t)^{-1} \le \operatorname{Tr}A(A+t)^{-1}$.

Proof of Theorem 5.13. The idea is to count the blocks with large profile, which by the block estimate (5.3) carry essentially one singular value each, and to absorb all other blocks into an operator of small norm.

Upper estimate. Let $J = \{j : a_j > \tau\}$ and split $M = M' + M''$, where M' is the matrix of ν restricted to the blocks in J and M'' that of the rest; both are positive, since $\nu \mapsto M_{\nu,R}$ is linear and positivity-preserving. Every block of the second measure has profile at most τ, so part (1) of Proposition 5.11 gives $\|M''\| \le C\tau$. The first is the sum of the block matrices M_j, $j \in J$, so concavity and (5.3) give $\operatorname{Tr}M'^r \le \sum_{j\in J}\operatorname{Tr}M_j^r \le C_r\sum_{j\in J}a_j^r$. By Weyl's inequality and Chebyshev's inequality for the eigenvalues of M', $n_M((C+1)\tau) \le n_{M'}(\tau) \le \tau^{-r}\operatorname{Tr}M'^r$.

Lower estimate. Let F be any finite set of N blocks with $a_j > \tau$, and let $\mu = \sum_{j\in F}(\tau/a_j)\nu_j \le \nu$, so that $Q = M_{\mu,R} \le M$ and every nonzero entry of $a(\mu)$ equals τ. Then $\|Q\| \le C\tau$, $\operatorname{Tr}Q \ge cN\tau$ by Proposition 5.2 and $\zeta_R(1+2u) \ge cu^{-\theta}$, and $\operatorname{Tr}Q^r \le C_rN\tau^r$ for $r = 1/2$ by (5.3) and concavity. The eigenvalues of Q that are at most $\varepsilon\tau$ contribute at most $(\varepsilon\tau)^{1/2}\operatorname{Tr}Q^{1/2} \le C\varepsilon^{1/2}N\tau$ to the trace. With ε so small that this is at most $cN\tau/2$, at least $cN/(2C)$ eigenvalues of Q exceed $\varepsilon\tau$, and by the min-max principle so do as many eigenvalues of M. Since F was arbitrary, $n_a(\tau) \le (2C/c)\,n_M(\varepsilon\tau)$.

Lorentz norms. The quasi-norm $\|x\|_{\ell^{p,q}}^q$ is comparable to $\int_0^\infty \tau^{q-1}n_x(\tau)^{q/p}\,\mathrm{d}\tau$ for $q < \infty$, and $\|x\|_{\ell^{p,\infty}} = \sup_\tau \tau n_x(\tau)^{1/p}$, the same for singular values. The lower bound for M follows at once from (5.1). For the upper bound fix $r < \min(p,1)$ and split the sum in (5.2) into the bands $2^l\tau < a_j \le 2^{l+1}\tau$:

$$\tau^p n_M(C\tau) \le C_r\sum_{l\ge 0} 2^{-l(p-r)}(2^l\tau)^p n_a(2^l\tau).$$

Raise to the power q/p and integrate against $\mathrm{d}\tau/\tau$, which is invariant under the dilations $\tau \mapsto 2^l\tau$; use subadditivity of $x \mapsto x^{q/p}$ if $q \le p$ and Minkowski's inequality in $L^{q/p}$ if $q > p$. The geometric factor makes the series converge. For $q = \infty$ take suprema. Finally, the part of ν on $(1,\infty)$ gives a matrix in every $S_{p,q}$ with norm at most $C\nu((1,\infty))$: write $k(u) = e_1 + \psi(u)$ with $\|\psi(w)\|^2 = \zeta_R(1+2\Re w) - 1 \le C4^{-\Re w}$, apply the Taylor argument on the unit intervals $[m, m+1)$, $m \ge 1$, to ψ, and absorb the rank-one part e_1 by $s_{n+1}(X+Y) \le s_n(X)$ for $\operatorname{rank}Y \le 1$. The entry $\alpha_\nu(1) \ge \nu((1,\infty))$ gives the reverse bound. □

In the Banach range of exponents the upper half of the theorem also follows from Olsen's local embedding of $\mathscr{H}^2$ into a weighted Dirichlet space of the half-plane [43], together with Luecking's description of Schatten class Toeplitz operators there [37]. The argument above avoids interpolation, and so covers $p < 1$, $q < 1$ and weak S_1 directly.

Two pointwise consequences answer the question of what lies between the profile and individual singular values.

COROLLARY 5.14. *Let R and ν be as in Theorem 5.13.*

(1) If the profile is doubling, $a_n^* \le D a_{2n}^*$ for all n, then $s_n(M_{\nu,R}) \asymp a_n^*$ for all n, with constants depending only on θ, B and D.

(2) If ν is purely atomic with at most K atoms in each block, and b is the sequence of Theorem 5.12, then for all n

$$c\, b^*_{\lceil Cn \rceil} \le s_n(M_{\nu,R}) \le C\, b_n^*,$$

with c, C depending only on θ, B and K.

PROOF. (1) Iterating the doubling condition gives $a_m^* \le D(n/m)^A a_n^*$ for $m \le n$, with $A = \log_2 D$. Take $\tau = a_n^*$ and $r < \min(1, 1/A)$ in (5.2). Since $n_a(\tau) < n$, $\sum_{a_j > \tau} a_j^r \le \sum_{m<n} (D(n/m)^A \tau)^r \le C\tau^r n$, so $n_M(Ca_n^*) \le Cn$, that is $s_{Cn}(M) \le Ca_n^*$; doubling removes the index change. The lower bound follows in the same way from (5.1) with $\tau = a_n^*/2$.

(2) Let L be as in Theorem 5.12. Colour each atom by the residue of its block index modulo $L+1$ and its position, among at most K, inside the block. Atoms of the same color are L-separated, so each of the at most $K(L+1)$ colored parts M_c satisfies Theorem 5.12. Since $M = \sum_c M_c$ is a sum of positive operators, Weyl's inequality gives $n_M(K(L+1)t) \le \sum_c n_{M_c}(t) \le n_b(2t/3)$, the upper bound. Conversely $M \ge M_c$ for every c, and among the $K(L+1)n$ largest entries of b some color has at least n, so $s_n(M) \ge b^*_{K(L+1)n}/2$. □

The index change in (2) cannot be removed, even for $K = 1$: two atoms with the same normalized mass, at $e^{-L}(1 \pm \delta)$ on either side of a block boundary, give almost the same rank-one operator, so the second singular value of the pair is of order δ times the first. Placing such pairs at rapidly separated scales with masses 2^{-j^2} and $\delta_j \to 0$ gives one measure with $s_{2j}/a^*_{2j} \to 0$. Nor does the profile determine the decay scale: with the same blocks and normalized masses e^{-j}, one atom per block gives $s_n \asymp e^{-n}$, while j atoms spread in the jth occupied block give $s_n = e^{-\Theta(\sqrt{n})}$, the root-exponential rate permitted by (5.2). Both measures lie in the same Lorentz classes, as they must.

5.4. Eigenvalue decay

Below the curve the spectrum is far smaller than Theorem 5.7 requires.

LEMMA 5.15. *Let h be completely monotone on $(0,\infty)$ with $h(t) \le Ct^{-\theta}$ on $(0,1]$ for some $\theta < 1$ and $h(t) \le Ce^{-ct}$ on $[1,\infty)$. The integral operator with kernel $h(u+v)$ on $L^2(0,\infty)$ has $s_n \le C' e^{-c'\sqrt{n}}$.*

PROOF. Bernstein's representation gives $|h^{(m)}(t)| \le m!(2/t)^m h(t/2)$. Fix m. Discard the outputs $u < e^{-m}$ and $u > m$, whose Hilbert–Schmidt norms are $O(e^{-cm})$ because $\theta < 1$ and the tail is exponential. Cover $[e^{-m}, m]$ by $O(m)$ intervals $[a, (1+\delta)a]$ with $\delta < 1/2$, and on each replace $h(u+v)$ by its Taylor polynomial of degree $m-1$ in u about a. Each piece has rank m, and the remainders

have total Hilbert–Schmidt norm squared at most $C(2\delta)^{2m}\int_0^\infty th(t)^2\,\mathrm{d}t$. A rank-$Cm^2$ operator is thus within Ce^{-cm} in norm. □

THEOREM 5.16. *If R has density $\theta<1$, the multiplicative Hilbert matrix on $\mathcal{S}_R\setminus\{1\}$ has $s_n\le Ce^{-c\sqrt{n}}$.*

PROOF. By Lemma 5.1, with Lebesgue measure and the index 1 removed, the companion kernel is $h_R(u+v)$ with $h_R(t)=\zeta_R(1+t)-1$, completely monotone, $O(t^{-\theta})$ at 0 and $O(2^{-t})$ at infinity. Apply Lemma 5.15. □

At the other end of the family of chapter 4, where all signs are averaged, the decay is only polynomial, and its exact asymptotics can be computed.

THEOREM 5.17 (The full sign average). *Let A be the matrix with entries $\mathbf{1}_{\square}(mn)/(\sqrt{mn}\log mn)$ for $m,n\ge 2$. Then*

$$s_n(A)\sim\frac{1}{2n\log n}\qquad(n\to\infty).$$

In particular $A\in S_p$ exactly for $p>1$, and A lies in weak S_1 but not in S_1.

PROOF. Write $m=qa^2$, $n=qb^2$ with q squarefree. Then $A=\bigoplus_q A_q$ with $A_q(a,b)=(2qab\log(qab))^{-1}$, where for $q=1$ only $a,b\ge 2$ occur. For $q\ge 2$ let $L=\log q$, $u_t=(a^{-1-t})_{a\ge1}$, and let T be exponential with rate L. Then $A_q=(2q)^{-1}\int_0^\infty e^{-Lt}u_t\otimes u_t\,\mathrm{d}t=(2qL)^{-1}\mathbb{E}[u_T\otimes u_T]$. Subtract the rank-one mean, $C_q=(2qL)^{-1}v\otimes v$ with $v=\mathbb{E}u_T$, whose entries are $v(a)=a^{-1}L/(L+\log a)$. The remainder $R_q=A_q-C_q$ is a covariance, hence positive, and since $\|\partial_t u_t\|_2$ is bounded and $\mathbb{E}T^2=2/L^2$,

$$\operatorname{Tr}R_q=\frac{\mathbb{E}\|u_T-v\|_2^2}{2qL}\le\frac{\mathbb{E}\|u_T-u_0\|_2^2}{2qL}\le\frac{C}{q\log^3 q}.$$

The eigenvalue of C_q is $c_q=\|v\|_2^2/(2qL)=\zeta(2)/(2q\log q)\,(1+O(1/\log q))$, and the squarefree integers up to x number $x/\zeta(2)+o(x)$, so $C=\bigoplus_q C_q$ has $n_C(s)\sim 1/(2s\log(1/s))$ as $s\to0$.

Let R be the sum of the R_q and the block A_1. For $q\le Q=s^{-1/2}$ use $R_q\le A_q\le(2/q)A_2$ and Lemma 5.15: the companion kernel of A_2 is $e^{-(u+v)\log 2/2}\zeta(2+u+v)/4$, completely monotone and exponentially decaying, so $s_j(A_2)\le Ce^{-c\sqrt{j}}$ and $n_{R_q}(s)\le C\log^2(1/s)$; the block A_1, with kernel $(\zeta(2+u+v)-1)/2$, is the same. For $q>Q$ use $n_{R_q}(s)\le\operatorname{Tr}R_q/s$. Altogether

$$n_R(s)\le Cs^{-1/2}\log^2(1/s)+\frac{C}{s}\sum_{q>Q}\frac{1}{q\log^3 q}=o\Big(\frac{1}{s\log(1/s)}\Big).$$

Since $C\le A=C+R$, Weyl's inequality gives $n_C(s)\le n_A(s)\le n_C((1-\varepsilon)s)+n_R(\varepsilon s)$ for every $\varepsilon\in(0,1)$, so $n_A(s)\sim1/(2s\log(1/s))$, which inverts to the claim. □

The order $1/n$ of the singular values of a full sign average is decided by a single number, the mass of ν on the critical line itself.

Proposition 5.18 (Endpoint law for full sign averages). *Let ν be a finite positive measure on $[0,\infty)$ and $A_\nu(m,n)=\mathbf{1}_\square(mn)\int (mn)^{-1/2-u}\,\mathrm{d}\nu(u)$ for $m,n\ge 1$. Then A_ν is bounded,*

$$\lim_{n\to\infty} n\,s_n(A_\nu)=\nu(\{0\}),$$

and $A_\nu\in S_1$ if and only if $\int \zeta(1+2u)\,\mathrm{d}\nu(u)<\infty$. In particular $s_n(A_\nu)=o(1/n)$ when $\nu(\{0\})=0$.

Proof. As in the proof of Theorem 5.17, $A_\nu=\bigoplus_q A_q$ over squarefree q, with $A_q=q^{-1}\int q^{-2u}v_u\otimes v_u\,\mathrm{d}\nu(u)$ and $v_u=(a^{-1-2u})_{a\ge1}$. Put $L_q=\int q^{-2u}\,\mathrm{d}\nu$ and let $\bar v_q$ be the mean of v_u under $q^{-2u}\,\mathrm{d}\nu/L_q$. Then $A_q=C_q+R_q$ with $C_q=q^{-1}L_q\bar v_q\otimes\bar v_q$ and $R_q\ge0$ a covariance, and $\operatorname{Tr}R_q\le q^{-1}\int q^{-2u}\|v_u-v_0\|_2^2\,\mathrm{d}\nu$. Since $\|v_u-v_0\|_2^2\le C\min(u^2,1)$ and $\sum_q q^{-1-2u}\le 1+1/(2u)$, the sum of the R_q is trace class. The eigenvalue $c_q=q^{-1}L_q\|\bar v_q\|_2^2$ of C_q is at most $\zeta(2)\nu([0,\infty))/q$, which gives boundedness. As $q\to\infty$, $L_q\to b=\nu(\{0\})$, and if $b>0$ the weights concentrate at $u=0$, so $\bar v_q\to v_0$ and $qc_q\to\zeta(2)b$; if $b=0$, $qc_q\to0$. Since the squarefree integers have density $1/\zeta(2)$, $n_C(s)\sim b/s$, or $n_C(s)=o(1/s)$ if $b=0$. Trace class gives $n_R(\varepsilon s)=o(1/s)$, and $n_C(s)\le n_{A_\nu}(s)\le n_C((1-\varepsilon)s)+n_R(\varepsilon s)$ by Weyl's inequality. Finally $A_\nu\ge0$ and $\operatorname{Tr}A_\nu=\sum_m\int m^{-1-2u}\,\mathrm{d}\nu$. □

So along the family $\mu_{\nu,R}$ of chapter 4, with ν Lebesgue measure, the spectrum moves from $[0,\pi]$ (all coordinates on the curve) to $1/(2n\log n)$ (all signs averaged). The compression to $\mathcal{S}_R$ of a sparse set R decays root-exponentially (Theorem 5.16), but the averaged matrices with R and its complement both infinite are not computed.

5.5. Rank and Hilbert–Schmidt structure

For a submonoid $S\subseteq\mathbb{N}$ let $d_S(k)=\#\{(m,n)\in S^2:mn=k\}$. The space $W_2(S)$ is the completion of polynomials under

$$\|F\|_{W_2(S)}=\inf\Big\{\Big(\sum_j\|G_j\|_2^2\|H_j\|_2^2\Big)^{1/2}:F=\sum_j G_jH_j\Big\},$$

with (G_j) and (H_j) orthogonal families in $H^2(S)$; equivalently the quotient of $H^2(S)\otimes_2 H^2(S)$ under multiplication.

Theorem 5.19 (Helson [27], restated). *For $F=\sum_{k\in S}a_k\chi_k$,*

$$\|F\|_{W_2(S)}^2=\sum_{k\in S}\frac{|a_k|^2}{d_S(k)}.$$

Hence $\sum_k|a_k|^2/d_S(k)\le C^2\|F\|_1^2$ is equivalent, with the same constant, to the statement that every Hilbert–Schmidt S-Hankel form H_b has a bounded symbol of norm at most $C\|H_b\|_{S_2}$.

Proof. A tensor $U=(u_{mn})$ maps to F when $\sum_{mn=k}u_{mn}=a_k$ for each k. The fibres are disjoint, and Cauchy–Schwarz in each fibre gives $\sum_{mn=k}|u_{mn}|^2\ge|a_k|^2/d_S(k)$, with equality for the constant choice $u_{mn}=a_k/d_S(k)$. The dual norm is $\sum_k d_S(k)|b_k|^2=\|H_b\|_{S_2}^2$, and Hahn–Banach supplies the symbol. □

For $S = \mathbb{N}$ this is Helson's inequality [27] read as $H^1(\mathbb{T}^\infty) \hookrightarrow W_2$ contractively. Classical weak factorization is the W_1 endpoint, with cost $\sum_j \|G_j\|_2 \|H_j\|_2$, and it fails on $\mathbb{T}^\infty$ [44]; Nehari's theorem in S_q corresponds to embedding into $W_{q'}$. The divisor weight therefore supplies exactly the S_2 theorem and nothing more.

COROLLARY 5.20. *(1) If $K_S = \sup_{k\in S} d(k)/d_S(k) < \infty$, the intrinsic W_2 estimate holds with constant $\sqrt{K_S}$; for $S = \{1, 2^2, 2^3, \dots\}$, $K_S = 2$.*

(2) If S is freely generated by integers with $\mathbb{Z}$-independent exponent vectors, the estimate holds with constant 1.

Write $\mathcal{N}(H)$ for the least norm of a bounded symbol of a Helson matrix H, and $C(r) = \sup\{\mathcal{N}(H)/\|H\| : \operatorname{rank} H \le r\}$.

THEOREM 5.21 (Bounded rank). *$\mathcal{N}(H) \le \sqrt{r}\,\|H\|$ whenever $\operatorname{rank} H \le r$, independently of the number of variables. Moreover*

$$r^{\log_2(2/\sqrt{\pi})} \lesssim C(r) \le \sqrt{r}, \qquad \log_2(2/\sqrt{\pi}) = 0.1742\dots$$

PROOF. $\|H\|_{S_2} \le \sqrt{r}\|H\|$, and Theorem 5.19 with $S = \mathbb{N}$, $C = 1$. For the lower bound, let $P_d = z_1 + \cdots + z_d$ and $M_d = \|P_d\|_{L^1(\mathbb{T}^d)}$. The Helson matrix of the degree-one coefficients of P_d has rank 2 and norm $\sqrt{d}$, and every symbol φ satisfies $d = |\int \varphi \overline{P_d}| \le \|\varphi\|_\infty M_d$; the symbol $(d/M_d)P_d/|P_d|$ is admissible, by rotation and permutation symmetry, so $\mathcal{N} = d/M_d$ exactly. Tensoring k copies on disjoint variables, as in [44], gives rank 2^k, norm $d^{k/2}$ and $\mathcal{N} = (d/M_d)^k$; by conditional expectation, extra variables do not help. The central limit theorem gives $M_d/\sqrt{d} \to \sqrt{\pi}/2$. □

The tensor examples are the only lower bounds we have, and their exponent is $\log_2$ of the ratio $\|Z\|_2/\|Z\|_1$ for a complex Gaussian Z. Any improvement must use matrices that are not tensor products of rank-two blocks.

REMARK 5.22 (Maximal forms). Call $H_\varphi(F, G) = \int FG\overline{\varphi}\,\mathrm{d}m$, with $\varphi \in H^2 \cap L^\infty$, maximal if $\|H_\varphi\| = \|\varphi\|_\infty$ and minimal if $\|H_\varphi\| = \|\varphi\|_2$. The one-variable argument of [13] transfers to $\mathbb{T}^\infty$: if a maximal form attains its norm at F, G, equality in $|\int FG\overline{\varphi}| \le \|\varphi\|_\infty \|FG\|_1$ forces $|\varphi| = \|\varphi\|_\infty$ on $\{FG \ne 0\}$, which has full measure because a nonzero $H^1(\mathbb{T}^\infty)$ function vanishes only on a null set [1, Corollary 2]. So compact maximal Helson matrices are minimal.

Open

(1) Characterize the measures with $s_n(M_{\nu,R}) \asymp a_n^*$ without an index change; doubling profiles (Corollary 5.14) and separated atoms (Theorem 5.12) are two sufficient conditions.
(2) The true exponent of $C(r)$, between 0.174 and 1/2.
(3) The spectral scale for R and its complement both infinite.
(4) Certify the spectral gaps of Proposition 5.5 for R_P, and describe the critical essential spectrum when ψ is not asymptotically periodic.

CHAPTER 6

Bergman and Fock spaces of Dirichlet series

Question. What changes when Haar measure on $\mathbb{T}^\infty$ is replaced by a Bergman or Gaussian product measure?

Less than one might expect. Separation along curves is the Hellinger–Bessel inequality, which holds in all three geometries (chapter 2). Symmetry under sign changes needs a substitute for inner-outer factorization, and a contractive embedding into a space with heavier weight supplies it: Carleman's inequality for Hardy, Kulikov's theorem [35] for Bergman, and its $\beta \to \infty$ limit, proved below, for Gauss. Each embedding gives a Helson-type inequality, a two-point lemma with constant $1 + O(|x|^4)$, and the approximation step of a Hartman theorem. With these, the positive Nehari theorem and its compactness counterpart hold in every geometry, and the Bergman multiplicative Hilbert form has a bounded symbol.

New here, to our knowledge, are the Gauss limit of Kulikov's theorem and the Gauss Helson inequality, the two-point constants outside the Hardy case, the compactness criterion for every $\beta \geq 1$, the sign-averaged endpoint in the Bergman and Gauss geometries, and the bounded symbol for the Bergman multiplicative Hilbert form.

6.1. Spaces of Dirichlet series and their symbols

We keep the three geometries of section 1.1 and identify a Dirichlet series $f(s) = \sum a_n n^{-s}$ with its Bohr lift $F(z) = \sum a_n z^{\kappa(n)}$, so that $f(s) = F(\gamma(s))$ with $\gamma(s) = (p^{-s})_p$. Write $\mathscr{H}^p$, $\mathscr{A}^p_\beta$ and $\mathscr{F}^p$ for the closures of the Dirichlet polynomials in L^p of Haar measure, of m_β^∞ and of γ^∞. The monomials are orthogonal in each L^2 space, with

$$\|n^{-s}\|^2_{\mathscr{H}^2} = 1, \qquad \|n^{-s}\|^2_{\mathscr{A}^2_\beta} = \frac{1}{d_\beta(n)}, \qquad \|n^{-s}\|^2_{\mathscr{F}^2} = [n]! := \prod_p \kappa_p(n)!,$$

where $\zeta(s)^\beta = \sum d_\beta(n) n^{-s}$. The Bergman spaces are those of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [10]. For $\beta = 2$ they are the spaces of Bailleul and Lefèvre [2]. The Gauss spaces are the Fock-type spaces of Dirichlet series; along the curve γ, point evaluation on $\mathscr{F}^2$ has norm $\exp\bigl(\sum_p p^{-2\Re s}/2\bigr)$.

Let X be one of these families and α a sequence. The *weighted Helson form* with moments α is

$$H_\alpha(f, g) = \sum_k (fg)_k \, \alpha(k) = \langle fg, \varphi \rangle_{X^2}, \qquad \varphi = \sum_n \frac{\alpha(n)}{\|n^{-s}\|^2_{X^2}} \, n^{-s}.$$

For the Bergman spaces these are the forms of [10] and [38], with $\varrho(n) = d_\beta(n)\alpha(n)$. A *bounded symbol* is a function $g \in L^\infty(\lambda)$ with $\int z^{\kappa(n)} g \, d\lambda = \alpha(n)$ for every n. Its orthogonal projection onto X^2 is then φ.

LEMMA 6.1. *H_α has a bounded symbol if and only if $h \mapsto \sum_k h_k \alpha(k)$ is bounded on X^1, and then $\|H_\alpha\| \le \inf \|g\|_\infty$, which equals the norm of that functional.*

PROOF. As Lemma 4.1, using $\|FG\|_{L^1(\lambda)} \le \|F\|_{L^2(\lambda)} \|G\|_{L^2(\lambda)}$. □

6.2. Contractive embeddings

THEOREM 6.2 (Kulikov). *For every $\beta \ge 1$ and every analytic polynomial g in one variable, $\|g\|_{L^2(m_{2\beta})} \le \|g\|_{L^1(m_\beta)}$, where m_1 is arc length on $\mathbb{T}$.*

This is [35, Corollary 1.3] in the case $p = 1$, $q = 2$; our normalization of m_β agrees with Kulikov's. For $\beta = 1$ it is Carleman's inequality $\|g\|_{A^2} \le \|g\|_{H^1}$. For integer β it also follows from repeated applications of [10, Theorem 1].

Let $\gamma_a = (a/\pi) e^{-a|z|^2} \, dx \, dy$ on $\mathbb{C}$, so $\gamma = \gamma_1$.

PROPOSITION 6.3 (Gauss as the limit of Bergman). *For every analytic polynomial g in one variable, $\|g\|_{L^2(\gamma_2)} \le \|g\|_{L^1(\gamma_1)}$.*

PROOF. Apply Theorem 6.2 to $g_\beta(z) = g(\sqrt{\beta} z)$ and substitute $z = w/\sqrt{\beta}$. With dA normalized area,

$$\|g_\beta\|_{L^1(m_\beta)} = \frac{\beta - 1}{\beta} \int_{|w| < \sqrt{\beta}} |g(w)| \Big(1 - \frac{|w|^2}{\beta}\Big)^{\beta - 2} \frac{dx \, dy}{\pi} \longrightarrow \|g\|_{L^1(\gamma_1)},$$

and similarly $\|g_\beta\|^2_{L^2(m_{2\beta})} \to \|g\|^2_{L^2(\gamma_2)}$, since $(2\beta - 1)\beta^{-1}(1 - |w|^2/\beta)^{2\beta - 2} \to 2e^{-2|w|^2}$. Dominated convergence applies because $(1 - x/\beta)^{\beta - 2} \le e^{-x/2}$ for $\beta \ge 4$. □

The rescaling $z = w/\sqrt{\beta}$ is the general principle behind this: the Bergman geometry with weight β tends to the Gauss geometry as $\beta \to \infty$, coordinates measured in units of $\beta^{-1/2}$. Several statements in this chapter could be obtained for the Gauss geometry by passing to that limit, but the direct proofs are just as short.

Each embedding tensorizes. Applying it in one coordinate at a time and using Minkowski's integral inequality, as in [10], gives contractive embeddings on infinitely many coordinates. Reading off the L^2 norm in coefficients:

COROLLARY 6.4 (Helson inequalities). *For every $f = \sum a_n n^{-s}$:*

$$\sum_n \frac{|a_n|^2}{d(n)} \le \|f\|^2_{\mathscr{H}^1}, \qquad \sum_n \frac{|a_n|^2}{d_{2\beta}(n)} \le \|f\|^2_{\mathscr{A}^1_\beta}, \qquad \sum_n |a_n|^2 \frac{[n]!}{2^{\Omega(n)}} \le \|F\|^2_{L^1(\gamma^\infty)}.$$

The first is Helson's inequality [27], and the second was proved for $\beta = 2$ by Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [10]. The third uses that the monomials of $L^2(\gamma_2)$ have $\|z^k\|^2 = k!/2^k$.

Kulikov also records the full Hardy scale [35, Theorem 5.1]: for $0 < p \le 2$,

$$\sum_n \frac{|a_n|^2}{d_{2/p}(n)} \le \|f\|^2_{\mathscr{H}^p}.$$

His proof tensorizes the one-variable contraction $H^p \hookrightarrow A^2_{2/p}$. This is valid also for $p < 1$, because the Minkowski exponent in the induction is $2/p \geq 1$.

6.3. Sign averages

PROPOSITION 6.5 (Two-point lemmas). *Let $x \in \mathbb{C}$, with $|x| < 1$ in the first two cases, and let g be an analytic polynomial. Then $(|g(x)|+|g(-x)|)/2 \leq c(x)\|g\|_{L^1(\lambda)}$, with*

$$c(x) = \frac{1}{1-|x|^4}, \qquad c(x) = \Lambda_\beta(|x|^2)^{1/2}, \qquad c(x) = \cosh(2|x|^2)^{1/2}$$

in the Hardy, Bergman and Gauss cases, where $\Lambda_\beta(t) = \big((1-t)^{-2\beta}+(1+t)^{-2\beta}\big)/2 = 1 + \beta(2\beta+1)t^2 + O(t^4)$. The Hardy constant is sharp.

PROOF. The Hardy case is Lemma 4.6 with $k = 2$. In the other two, by Cauchy–Schwarz and the contractive embedding into a space Y with kernel K^Y,

$$\frac{1}{2}\big(|g(x)| + |g(-x)|\big) \leq \Big(\frac{1}{2}\big(|\langle g, K^Y_x\rangle|^2 + |\langle g, K^Y_{-x}\rangle|^2\big)\Big)^{1/2} \leq \big(\frac{1}{2}\|G\|\big)^{1/2}\|g\|_Y,$$

where G is the Gram matrix of $K^Y_{\pm x}$. Its entries are real and positive: $K^Y(x,x)$ on the diagonal and $K^Y(x,-x)$ off it, so $\|G\| = K^Y(x,x) + K^Y(x,-x)$. For $Y = A^2_{2\beta}$ the kernel is $(1-\overline{w}z)^{-2\beta}$, giving $2\Lambda_\beta(|x|^2)$. For $Y = L^2(\gamma_2)$ it is $e^{2z\overline{w}}$, giving $e^{2|x|^2} + e^{-2|x|^2}$. □

All three constants are $1 + O(|x|^4)$. That is what lets them tensorize over the primes, where $\sum_p |x_p|^4 \leq \sum_p p^{-2} < \infty$. A single point evaluation, by contrast, costs $\|K_x\|_2^2 = 1 + O(|x|^2)$. Comparing the Hardy cases is instructive: the route through Carleman's inequality gives $\Lambda_1(t)^{1/2} = 1 + 3t^2/2 + O(t^4)$ instead of the sharp $1 + t^2 + O(t^4)$. Passing through L^2 loses only in the coefficient of $|x|^4$. In the Bergman case the whole constant can be recovered from the L^1 norm near $x = 0$, and in both cases the obvious lower bound fails once $K_x + K_{-x}$ has a zero.

The same route works in every L^p. Kulikov's theorem holds in the full range [35, Corollary 1.3]: for $0 < p < q$ and $\beta > 1$, $\|g\|_{L^q(m_{q\beta/p})} \leq \|g\|_{L^p(m_\beta)}$, and the rescaling in the proof of Proposition 6.3 turns the case $q = 2$ into $\|g\|_{L^2(\gamma_{2/p})} \leq \|g\|_{L^p(\gamma_1)}$.

PROPOSITION 6.6 (Two-point lemmas in every L^p). *In the Bergman case with $\beta > 1$, and in the Gauss case, let $0 < p < \infty$, write $p = kq$ with $k = \lceil p/2 \rceil$ and $0 < q \leq 2$, and let $t = |x|^2$. For every analytic polynomial g,*

$$\frac{1}{2}\big(|g(x)|^p + |g(-x)|^p\big) \leq c_p(x)\|g\|^p_{L^p(\lambda)}, \qquad c_p(x) = \Lambda_{\beta/q}(t)^{q/2} \quad \text{or} \quad \cosh(2t/q)^{q/2},$$

and $c_p(x) = 1 + O(|x|^4)$.

PROOF. Apply the case q to g^k, since $|g^k|^q = |g|^p$. For $q \leq 2$, the power-mean inequality gives $\frac{1}{2}(|g(x)|^q + |g(-x)|^q) \leq (\frac{1}{2}(|g(x)|^2 + |g(-x)|^2))^{q/2}$, and, as in the proof of Proposition 6.5, the inner average is at most half the Gram norm of the kernels at $\pm x$ in $A^2_{2\beta/q}$, respectively $L^2(\gamma_{2/q})$, times the square of the norm there, which is at most $\|g\|^2_{L^q}$. The Gram norms are $2\Lambda_{\beta/q}(t)$ and $2\cosh(2t/q)$. □

COROLLARY 6.7. *In the Bergman geometry with $\beta > 1$ and in the Gauss geometry, the sign-averaged measure $\mu_{\nu,R}$ of Theorem 6.14 is a Carleson measure for X^p for some $p > 0$ if and only if it is for every $p > 0$, if and only if $[\nu]_\kappa < \infty$.*

PROOF. Sufficiency: Proposition 6.6 in each averaged coordinate, with Fubini, costs $\prod_q c_p(\gamma_q(s))$, which is bounded since $\sum_q |\gamma_q(s)|^4 \le \sum_q q^{-2}$; then part (2) of Theorem 2.10 on each slice, as in Corollary 4.16. Necessity: test with $K^{2/p}$ for the kernel K at $\gamma_R(1/2+\delta)$ in the coordinates of R, as in Corollary 4.16. □

PROPOSITION 6.8 (Sharp two-point constants). *In the Bergman and Gauss cases, let $c(x)$ be the least constant in Proposition 6.5, let K be the kernel of X^2 (that is, of A^2_β, or of $L^2(\gamma)$ in the Gauss case), and let $\mu = \int |z|^4 \, \mathrm{d}\lambda$, which is $2/(\beta(\beta+1))$ and 2 respectively.*

1. *$c(x) \ge L(x) := (K(x,x) + K(x,-x))/2$, that is, $\big((1-|x|^2)^{-\beta} + (1+|x|^2)^{-\beta}\big)/2$ in the Bergman case and $\cosh(|x|^2)$ in the Gauss case.*
2. *$c(x) = 1 + |x|^4/\mu + o(|x|^4)$ as $x \to 0$; that is, $1 + \beta(\beta+1)|x|^4/2 + o(|x|^4)$ and $1 + |x|^4/2 + o(|x|^4)$.*
3. *In the Bergman case, $c(x) = L(x)$ whenever $\beta \arctan|x| \le \pi/2$: for every x when $\beta \le 2$, and for $|x| \le \tan(\pi/(2\beta))$ when $\beta > 2$.*
4. *$c(x) > L(x)$ in the Bergman case when $|x| > \tan(\pi/(2\beta))$, which can happen only for $\beta > 2$, and in the Gauss case for every $x \ne 0$.*

PROOF. By rotation invariance we may take $x = r > 0$. Put $H = K_r + K_{-r}$, $A = K(r,r)$ and $B = K(r,-r)$, so that $H(\pm r) = A + B = 2L$ and $\|H\|_2^2 = 4L$.

(1) The function $g = H^2$ lies in the closure of the polynomials in L^1, and $\|g\|_1 = \|H\|_2^2 = 4L$, $g(\pm r) = 4L^2$. So its two-point quotient is L.

(2) The lower bound is (1). For the upper bound, let $g = \sum a_n z^n$ be a polynomial with $\|g\|_1 = 1$ and two-point quotient $N \ge 1$, rotated so that $a_0 \ge 0$; note $a_0 \le 1$. Let $w_n = \|z^n\|_Y^2$ in the space Y of the proof of Proposition 6.5, and $q = 1 - a_0^2$. The contractive embedding gives $\sum_{n\ge1} w_n |a_n|^2 \le q$, so by Cauchy–Schwarz

$$\sum_{n\ge k} |a_n| \rho^n \le \sqrt{q}\, \Phi_k(\rho), \qquad \Phi_k(\rho)^2 = \sum_{n\ge k} \rho^{2n}/w_n \le K^Y(\rho,\rho),$$

and $\Phi_k(r) = O(r^k)$ as $r \to 0$. Let $e = (g(r) + g(-r))/2$ and $o = (g(r) - g(-r))/2$. Then $e = a_0 + a_2 r^2 + O(\sqrt{q}\, r^4)$, $o = O(\sqrt{q}\, r)$, and

$$N \le \big(|e|^2 + |o|^2\big)^{1/2}.$$

Since $a_0 \le 1 - q/2$ and $|a_2| \le \sqrt{q/w_2}$, the condition $N \ge 1$ forces $\sqrt{q} = O(r^2)$. Hence $o = O(r^3)$ and $N \le a_0 + |a_2| r^2 + O(r^6)$.

It remains to bound $1 - a_0$ below. Let $h = g - a_0$. For complex h and $a_0 > 0$, $|a_0 + h| \ge a_0 + \Re h$, and $|a_0 + h| \ge a_0 + \Re h + (\Im h)^2/(2(a_0 + |h|))$. Since $\int \Re h \, \mathrm{d}\lambda = 0$, for any disc $D_\rho = \{|z| \le \rho\}$,

$$1 - a_0 = \int \big(|a_0 + h| - a_0 - \Re h\big) \, \mathrm{d}\lambda \ge \frac{1}{2(1+\eta)} \int_{D_\rho} (\Im h)^2 \, \mathrm{d}\lambda,$$

where $\eta = \sup_{D_\rho}|h| \le \sqrt{q}\,\Phi_1(\rho)$. On the circle of radius t the mean of $(\Im h)^2 = (|h|^2 - \Re h^2)/2$ is $\frac12\sum_{n\ge1}|a_n|^2t^{2n} \ge \frac12|a_2|^2t^4$. Hence $1 - a_0 \ge |a_2|^2\mu(\rho)/(4(1+\eta))$ with $\mu(\rho) = \int_{D_\rho}|z|^4\,\mathrm{d}\lambda$. Let ρ tend to the boundary slowly enough that $r^2\Phi_1(\rho)\to 0$. Then $\eta\to0$ and $\mu(\rho)\to\mu$, so

$$N \le 1 - \frac{\mu}{4}(1-o(1))|a_2|^2 + |a_2|r^2 + O(r^6) \le 1 + (1+o(1))\frac{r^4}{\mu}.$$

(3) By continuity in r for a fixed polynomial, we may assume $\beta\arctan r < \pi/2$. Here $H(z) = 2(1-r^2z^2)^{-\beta/2}\cosh(\beta\operatorname{artanh}(rz))$ and $|\Im\operatorname{artanh}(rz)| \le \arctan r$ on the closed disc, so H has no zeros near the closed disc. By reproduction, $\int h|H|^2\,\mathrm{d}\lambda = 2L(h(r)+h(-r))$ for h analytic near the closed disc. So the real measure $\sigma = |H|^2\lambda - 2L(\delta_r+\delta_{-r})$ has zero mass and zero analytic moments, and its logarithmic potential $U(w) = \int\log|w-z|\,\mathrm{d}\sigma(z)$ vanishes for $|w|\ge1$; it is continuous off $\pm r$, since the density of λ lies in L^p for some $p>1$, and tends to $+\infty$ at $\pm r$.

We show $U\ge0$. Angular integration gives, with $\rho_\beta(t) = (\beta-1)(1-t)^{\beta-2}$,

$$\int\frac{\overline{H(z)}}{w-z}\,\mathrm{d}\lambda(z) = \frac1w\int_0^{|w|^2}H(t/w)\rho_\beta(t)\,\mathrm{d}t = \sum_{\varepsilon=\pm1}\frac{1-(1-|w|^2)^{\beta-1}(1-\varepsilon r\overline{w})^{1-\beta}}{w-\varepsilon r},$$

and reproduction applied to $(H(z)-H(w))/(w-z)$ then gives

$$2\partial_w U(w) = -(1-|w|^2)^{\beta-1}H(w)\Big(\frac{(1-r\overline{w})^{1-\beta}}{w-r} + \frac{(1+r\overline{w})^{1-\beta}}{w+r}\Big).$$

At a critical point the bracket vanishes, so $|w+r|/|w-r| = (|1-r\overline{w}|/|1+r\overline{w}|)^{\beta-1}$; if $\Re w\ne0$ one side exceeds 1 and the other does not, so critical points lie on the imaginary axis. For $w=iy$, $0<y<1$, put $a=\arctan(ry)$ and $\gamma=\beta-1$. Then $H(iy) = 2(1+r^2y^2)^{-\beta/2}\cos(\beta a) > 0$ and

$$\partial_y U(iy) = \frac{2(1-y^2)^\gamma H(iy)}{(r^2+y^2)(1+r^2y^2)^{\gamma/2}}\big(r\sin(\gamma a) - y\cos(\gamma a)\big).$$

The ratio $\arctan(y/r)/\arctan(ry)$ decreases in y, because the ratio $(1+r^2y^2)/(r^2+y^2)$ of the derivatives does, so it exceeds its value $(\pi/2-\arctan r)/\arctan r > \gamma$ at $y=1$. Hence $\gamma a < \arctan(y/r)$, the bracket is negative, and $U(iy) > U(i) = 0$; by symmetry the same holds for $y<0$. If U were negative somewhere, it would have a negative minimum in the disc away from $\pm r$, which is a critical point; there are none off the imaginary axis, and $U>0$ on it. So $U\ge0$.

If u is smooth and subharmonic near the closed disc, then

$$\int u\,\mathrm{d}\sigma = \frac{1}{2\pi}\int U\Delta u\,\mathrm{d}A \ge 0.$$

For a polynomial g put $f = g/H^2$ and apply this to $u = (|f|^2+\varepsilon^2)^{1/2}$. Letting $\varepsilon\to0$,

$$\|g\|_1 = \int|f||H|^2\,\mathrm{d}\lambda \ge 2L\big(|f(r)|+|f(-r)|\big) = \frac{|g(r)|+|g(-r)|}{2L}.$$

With (1), $c(r) = L$.

(4) Let $a = ic$ be a zero of H with $c > 0$ inside the domain: in the Bergman case $z = i\tau/r$, $\tau = \tan(\pi/(2\beta))$, is a zero, inside when $r > \tau$; in the Gauss case $H = 2\cosh(rz)$ vanishes at $i\pi/(2r)$. Let $F_t = H^2 + itH/(z-a)$ for real t, which lies in the closure of the polynomials. Then $F_t(\pm r) = 4L^2 + 2itL/(\pm r - a)$, and both moduli have derivative $-2Lc/(r^2+c^2)$ at $t = 0$. Since $|F_t| - |F_0| = O(t|H/(z-a)|)$, the norm has derivative $\Re(iJ)$, $J = \int \overline{H}/(z-a)\,\mathrm{d}\lambda$. Write $\mathrm{d}\lambda = w(u)\,\mathrm{d}u\;\mathrm{d}\theta/(2\pi)$ with $u = |z|^2$. On a circle $|z| < c$ expand $1/(z-a)$ in powers of z/a, and on $|z| > c$ in powers of a/z; only the inside contributes, and since H is even with real coefficients,

$$\Re(iJ) = -\frac{1}{c}\int_0^{c^2} H(iu/c)\,w(u)\,\mathrm{d}u.$$

Over the whole range of u this integral is known. With $\omega = r/c$,

$$\frac{1}{2}\int H(iu/c)w(u)\,\mathrm{d}u = \Re\int (1-i\omega u)^{-\beta}w(u)\,\mathrm{d}u = \Re\frac{1}{1-i\omega} = \frac{1}{1+\omega^2},$$

by Euler's integral for ${}_2F_1(\beta,1;\beta;i\omega)$, and in the Gauss case by $\int_0^\infty e^{i\omega u-u}\,\mathrm{d}u = (1-i\omega)^{-1}$. The quotient $Q_t = N_t/\|F_t\|_1$ equals L at $t = 0$, so

$$Q_0' = -\frac{c}{2(r^2+c^2)} + \frac{1}{4c}\int_0^{c^2} H(iu/c)w(u)\,\mathrm{d}u = -\frac{1}{4c}\int_{u>c^2} H(iu/c)w(u)\,\mathrm{d}u.$$

In the Gauss case $H(iu/c) = 2\cos(\omega u)$ and $rc = \pi/2$, so $Q_0' = re^{-c^2}/(2(r^2+c^2)) > 0$. In the Bergman case $\frac{1}{2}H(iu/c) = \Re(1+i\omega u)^{-\beta}$. Substitute $u = \tan\phi/\omega$ and put $\tan\phi_1 = \omega$; then $1 - u = \sin(\phi_1-\phi)/(\sin\phi_1\cos\phi)$, the lower limit becomes $\phi = \pi/(2\beta)$ because $\omega c^2 = \tau$, and

$$Q_0' = \frac{\beta-1}{2c\,\omega\sin^{\beta-2}\phi_1}\int_0^\alpha \sin(\beta(\alpha-\psi))\sin^{\beta-2}\psi\,\mathrm{d}\psi = \frac{\sin^\beta\alpha}{2c\,\omega\sin^{\beta-2}\phi_1},$$

where $\alpha = \phi_1 - \pi/(2\beta)$. The integral equals $\sin^\beta\alpha/(\beta-1)$: both sides vanish to first order at $\alpha = 0$ and solve $y'' + \beta^2 y = \beta\sin^{\beta-2}\alpha$. Since $\tau < \omega < 1/\tau$, $0 < \alpha < \pi/2$, so $Q_0' > 0$. In both cases the quotient of F_t exceeds L for small $t > 0$. □

For $\beta = 3$, $Q_0' = c(3|x|^2-1)^3/(16|x|^2(3|x|^4+1))$ with $c^2 = 1/(3|x|^2)$.

REMARK 6.9 (A dual certificate). Near $x = 0$ there is a shorter proof of (3), which also covers the Hardy space. Here $K(z,w) = (1-z\overline{w})^{-\beta}$. Put $O = K_r - K_{-r}$ and $\varphi = O/H = \tanh(\beta\,\mathrm{artanh}(rz))$. The Taylor coefficients of artanh are nonnegative, and those of tanh have the moduli of those of tan, which are nonnegative; so on the disc

$$|\varphi| \le \tan(\beta\,\mathrm{artanh}\,r) \le 1/2.$$

In particular H has no zeros on the closed disc. For a polynomial g let $G = g/H$, which is analytic on a neighbourhood of the closed disc, and let e, o be as in the proof of (2). Since $\langle G, H\rangle = G(r) + G(-r)$ and $H(\pm r) = 2L$,

$$e = L\int g\,\frac{\overline{H}}{H}\,\mathrm{d}\lambda, \qquad o = L\big(G(r) - G(-r)\big).$$

Let $\rho = (2L/A)\Re\varphi$, a real function with $|\rho| \le L/A \le 1$. Then

$$\int G\overline{H}\rho\,\mathrm{d}\lambda = \frac{L}{A}\Big(\langle G\varphi, H\rangle + \langle G, \varphi H\rangle\Big) = \frac{L}{A}\big(1+\varphi(r)\big)\big(G(r)-G(-r)\big) = G(r)-G(-r),$$

because φ is odd, $\varphi H = O$ and $\varphi(r) = (A-B)/(A+B)$. So $o = L\int g(\overline{H}/H)\rho\,\mathrm{d}\lambda$. Now $(|g(r)|+|g(-r)|)/2 = \max_\theta |e\cos\theta - io\sin\theta|$, and

$$|e\cos\theta - io\sin\theta| = L\left|\int g\,\frac{\overline{H}}{H}(\cos\theta - i\rho\sin\theta)\,\mathrm{d}\lambda\right| \le L\|g\|_1,$$

since $|\cos\theta - i\rho\sin\theta| \le 1$. This needs $\tan(\beta\,\mathrm{artanh}|x|) \le 1/2$. With the Hardy measure and $\beta = 1$ it gives $|\rho| \le 2|x|/(1+|x|^2) \le 1$ for every x, a second proof of the Hardy constant.

In (3) and (4) together, the square bound is sharp exactly up to the first zero of $K_x + K_{-x}$: beyond it the squares H^2 stop being extremal because a zero of H can be moved in one of the two factors; in the Gauss case the gain $e^{-\pi^2/(4|x|^2)}$ is invisible to every order at $x = 0$, consistent with (2).

Remark 6.10 (Neumann ovals). For $\beta = 2$, where λ is normalized area measure, (3) holds on the whole disc and has a geometric meaning. With $\phi(z) = z/(1-x^2z^2)$, so that $2\phi' = H$, the inequality $U \ge 0$ says that the Neumann oval $\phi(\mathbb{D})$ is a quadrature domain for subharmonic functions, with two equal nodes at $\pm x/(1-x^4)$: $\pi^{-1}\int_{\phi(\mathbb{D})} u\,\mathrm{d}A \ge (L/2)(u(t)+u(-t))$. This classical fact (see Nishry–Wennman [41]) can also be proved by shrinking the oval to two tangent discs.

Proposition 6.11. *Let F be an analytic polynomial, let Q be a set of coordinates with points x_p, $p \in Q$, and fix the other coordinates at points w_p. With independent signs ε_p,*

$$\mathbb{E}_\varepsilon\big|F\big((\varepsilon_p x_p)_{p\in Q}, w\big)\big| \le \prod_{p\in Q} c(x_p) \int \big|F(\zeta, w)\big|\,\mathrm{d}\lambda^Q(\zeta).$$

For $x = \gamma_Q(s)$ with $\Re s \ge 1/2$ the constant is at most

$$C_{\mathrm{sign}} = \zeta(2), \qquad C_\beta = \prod_p \Lambda_\beta(1/p)^{1/2}, \qquad \prod_p \cosh(2/p)^{1/2} \le e^{\sum_p p^{-2}} < 1.58.$$

Proof. As Proposition 4.7, by induction on the coordinates of Q with Fubini. For the Gauss bound use $\cosh y \le e^{y^2/2}$. □

For comparison, the Carleman route at $\beta = 1$ gives $C_1 = \zeta(2)^{3/2}\zeta(4)^{-1/2} \approx 2.03$ instead of $\zeta(2)$, and $C_2 \approx 7.26$.

The constant in Proposition 6.11 is $\prod_{p\in Q} c(x_p)$ exactly, and on the Bergman side it can now be computed.

Corollary 6.12 (Sharp sign averages). *In the Bergman case with $1 < \beta \le \pi/(2\arctan(2^{-1/2}))$, the best constant in Proposition 6.11, uniformly over $\Re s \ge 1/2$,*

over Q and over F, is

$$C_\beta^{\mathrm{sharp}} = \prod_p L(p^{-1/2}) = \prod_p \frac{(1-p^{-1})^{-\beta} + (1+p^{-1})^{-\beta}}{2},$$

and it is attained in the closure of the polynomials. For $\beta = 2$, $C_2^{\mathrm{sharp}} = \zeta(2)^3/\zeta(4) = 5\pi^2/12 \approx 4.11$. In the Hardy case the constant $\zeta(2)$ is sharp in the same sense.

PROOF. For product polynomials $F = \prod_{p\in Q} g_p(z_p)$, with no other coordinates, both sides of Proposition 6.11 factor, so the best constant for a finite Q is $\prod_{p\in Q} c(x_p)$. Since $c \le \Lambda_\beta^{1/2}$, the factors are $1+O(p^{-2})$, and the product over all primes converges. The condition on β gives $\beta \arctan(p^{-1/2}) \le \pi/2$ for every prime, so $c(p^{-\sigma}) = L(p^{-\sigma})$ by Proposition 6.8(3), and L increases in $|x|$, so $\sigma = 1/2$ is the worst case. At $\beta = 2$, $L(p^{-1/2}) = (1-p^{-4})/(1-p^{-2})^3$. For attainment, let $h_p = (K_{r_p} + K_{-r_p})/(2L(r_p)^{1/2})$ with $r_p = p^{-1/2}$, so that $\|h_p\|_2 = 1$, $\int h_p \,\mathrm{d}\lambda = L(r_p)^{-1/2}$ and $h_p(\pm r_p)^2 = L(r_p)$. The products $H_N = \prod_{j\le N} h_{p_j}$ satisfy $\|H_N - H_M\|_2^2 = 2 - 2\prod_{M<j\le N} L(r_{p_j})^{-1/2} \to 0$, so H_N^2 converges in L^1 to some F with $\|F\|_1 = 1$, and the sign average of H_N^2 is the constant $\prod_{j\le N} L(r_{p_j})$. In the Hardy case $c(x) = (1-|x|^4)^{-1}$ for every x, and the same argument gives $\prod_p (1-p^{-2})^{-1} = \zeta(2)$. □

For larger β the finitely many primes $p < \cot^2(\pi/(2\beta))$ fall outside Proposition 6.8(3), and their factors exceed $L(p^{-1/2})$ by item (4).

6.4. Dirichlet kernels

The contractive embedding also measures cancellation. Let $D_N = \sum_{n\le N} n^{-s}$ and $d_\beta(n)$ the coefficients of ζ^β, so that $\|D_N\|_{\mathscr{A}^2_\beta}^2 = \sum_{n\le N} 1/d_\beta(n)$.

PROPOSITION 6.13. *Fix $\beta \ge 1$ and put*

$$G_\beta(1) = \prod_p (1-1/p)^{1/\beta} \sum_{k\ge 0} \frac{k!\, p^{-k}}{(\beta)_k}.$$

Then

$$\|D_N\|_{\mathscr{A}^2_\beta}^2 \sim \frac{G_\beta(1)}{\Gamma(1/\beta)}\, N (\log N)^{1/\beta - 1},$$

and

$$(c_\beta + o(1))(\log N)^{-1/(4\beta)} \le \frac{\|D_N\|_{\mathscr{A}^1_\beta}}{\|D_N\|_{\mathscr{A}^2_\beta}} \le 1, \qquad c_\beta^2 = \frac{G_{2\beta}(1)\Gamma(1/\beta)}{G_\beta(1)\Gamma(1/(2\beta))}.$$

PROOF. The Dirichlet series of $1/d_\beta$ is $\zeta(s)^{1/\beta} G_\beta(s)$ with G_β analytic and nonzero near 1, and Selberg–Delange gives the asymptotic. The lower bound is $\|D_N\|_{\mathscr{A}^1_\beta} \ge \|D_N\|_{\mathscr{A}^2_{2\beta}}$, by tensorized Kulikov, with the asymptotic at 2β. □

For $\beta = 2$ this is [10, Theorem 10]. At $\beta = 1$ Harper's theorem [25] gives the true ratio $(\log\log N)^{-1/4}$, so the lower bound is far from sharp there. For $\beta > 1$ we do not know whether the ratio tends to 0; the heuristic chaos exponent $2/\sqrt{\beta}$

is subcritical, which suggests it does not. It cannot tend to 0 uniformly in β: the ratio tends to 1 when $\beta/N \to \infty$.

6.5. Positive measures in three geometries

With R, Q, ν and the measure $\mu_{\nu,R}$ as in section 4.2, let $\alpha_{\nu,R}(n) = \mathbf{1}[n_Q \in \square] \int n^{-s}\,\mathrm{d}\nu(s)$ and $[\nu]_\kappa = \sup_{0<t\le 1} \nu((1/2, 1/2+t])t^{-\kappa}$.

THEOREM 6.14. *Let R have density $\theta \in [0,1]$ with constant B, let κ be the kernel exponent of the geometry, and let ν be a finite positive Borel measure on $(1/2,\infty)$. The following are equivalent:*

(1) $H_{\alpha_{\nu,R}}$ is bounded on $X^2 \times X^2$;
(2) $[\nu]_\kappa < \infty$;
(3) $\int |F|\,\mathrm{d}\mu_{\nu,R} \lesssim \|F\|_{X^1}$;
(4) $H_{\alpha_{\nu,R}}$ has a bounded symbol.

The ratio between the smallest symbol norm and $\|H_{\alpha_{\nu,R}}\|$ is bounded by a constant depending only on θ, B and the geometry. If $\theta = 0$, all four hold for every finite ν. Moreover $H_{\alpha_{\nu,R}}$ is compact if and only if $\nu((1/2,1/2+t]) = o(t^\kappa)$, which is automatic when $\theta = 0$, and then its bounded symbols lie in the $L^\infty(\lambda)$-closure of $C + N_\lambda$, where N_λ is the annihilator of the analytic polynomials and C is $C(\overline{\mathbb{D}}^\infty)$ in the Bergman geometry and the bounded continuous functions in the Gauss geometry.

PROOF. The structure is that of Theorems 4.2 and 4.10. We record the three geometry-dependent inputs.

Kernel test. Let K be the reproducing kernel of X^2 at $\gamma_R(1/2+\delta)$, restricted to the coordinates of R. Its coefficients are positive and supported on integers with $n_Q = 1$, so Tonelli's theorem gives $H_{\alpha_{\nu,R}}(K,K) = \int K(\gamma(s))^2\,\mathrm{d}\nu(s)$. Also $K(\gamma(s)) \ge \|K\|_2^2$ for $s \le 1/2+\delta$. Hence

$$\nu((1/2,1/2+\delta]) \le \|H_{\alpha_{\nu,R}}\|\,\|K\|_2^{-2} \le e^{\kappa' B}(2\delta)^\kappa\,\|H_{\alpha_{\nu,R}}\|,$$

with $\kappa' = \beta$ in the Bergman case and $\kappa' = 1$ otherwise, by the lower bound $\|K\|_2^2 \ge e^{-\kappa' B}(2\delta)^{-\kappa}$ that follows from (2.1). The same computation with normalized kernels gives $\nu = o(t^\kappa)$ when $H_{\alpha_{\nu,R}}$ is compact.

Sufficiency. Average the Q-coordinates with Proposition 6.11. Then treat the remaining R-coordinates with Theorem 2.10 (2), with $p = 1$, if $\theta > 0$, or with the uniform bound $\|K_{\gamma_R(s)}\|_2^2 \le e^{\kappa'(1+B)}$ if $\theta = 0$. Treat $\Re s > 3/2$ with the point estimate. This gives $\int |F|\,\mathrm{d}\mu_{\nu,R} \le C_{\mathrm{sign}}\big(C[\nu]_\kappa + C'\nu((1/2,\infty))\big)\|F\|_{X^1}$. Symbols then come from Lemma 6.1.

Compactness. For the part of ν above $1/2+\varepsilon$, the moments satisfy $\alpha(n) \le \nu((1/2,\infty))n^{-1/2-\varepsilon}$, and Corollary 6.4 bounds the tail of the functional beyond $n > K$ by the square root of

$$\sum_{n>K} d(n)\,n^{-1-2\varepsilon}, \qquad \sum_{n>K} d_{2\beta}(n)\,n^{-1-2\varepsilon}, \qquad \sum_{n>K} \frac{2^{\Omega(n)}}{[n]!}\,n^{-1-2\varepsilon}.$$

These are tails of convergent series, the last being $\prod_p \exp(2p^{-1-2\varepsilon})$. The part below $1/2+\varepsilon$ is small by the sufficiency estimate. The rest of the argument is that of Theorem 4.10. □

In the Bergman case with $R = \emptyset$, every $H_{\alpha_{\nu,\emptyset}}$ is bounded, compact, and has a bounded symbol of norm at most $C_\beta\, \nu((1/2,\infty))$. Without the sign average, Ma, Lu and Zu show that boundedness requires $\nu \lesssim t^\beta$ and compactness requires $\nu = o(t^\beta)$ [38].

6.6. The Bergman multiplicative Hilbert form

Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [10] introduced the form on $\mathscr{A}^2 \times \mathscr{A}^2$, $\mathscr{A} = \mathscr{A}_2$, with analytic symbol

$$\varphi(s) = \sum_{n\geq 2} \frac{d(n)}{\sqrt{n}\log^2 n}\, n^{-s},$$

the Bergman analogue of the multiplicative Hilbert matrix. They proved that it is bounded and not compact, and that φ defines a bounded functional on $\mathscr{A}^p$ for $1 < p < \infty$. Ma, Lu and Zu [38, Theorem 1.1] have since determined its spectrum on the subspace $\mathscr{A}^2_0$ of series with $a_1 = 0$: the essential spectrum is $[0,1]$, absolutely continuous of multiplicity one, and above it there are finitely many eigenvalues, at least one, all simple. Its moments are $\alpha(n) = 1/(\sqrt{n}\log^2 n) = \int_0^\infty n^{-1/2-u}u\,\mathrm{d}u$, so it is the form of the measure $u\,\mathrm{d}u$ on the curve: the Bergman counterpart of Lebesgue measure.

The Bergman analogue of Problem 3 asks whether this form has a bounded symbol. It does.

THEOREM 6.15. *There is C such that $|\sum_{n\geq 2} a_n/(\sqrt{n}\log^2 n)| \leq C\|f\|_{\mathscr{A}^1}$ for every $f \in \mathscr{A}^1$, where the series converges in its natural order. Consequently, the Bergman multiplicative Hilbert form has a bounded symbol in $L^\infty(m_2^\infty)$.*

PROOF. For polynomials, the Gamma integral gives $\sum_{n\geq 2} a_n/(\sqrt{n}\log^2 n) = \int_0^\infty (f(1/2+u) - a_1)\,u\,\mathrm{d}u$. On $(0,1)$ apply Theorem 2.10 (2) in the Bergman geometry $\beta = 2$ with all primes, where $\kappa = 2$ and $\nu = u\,\mathrm{d}u$ has $[\nu]_2 = 1/2$. For $u \geq 1$ use $|a_n| \leq d(n)\|f\|_{\mathscr{A}^1}$, read off from the reproducing property. The bounded symbol then comes from Lemma 6.1.

For natural-order convergence, compare the partial sum up to N with the Abel sum at $\varepsilon = 1/\log N$. The Bergman Helson inequality with d_4 (Corollary 6.4) and the dyadic estimate $\sum_{e^u \leq n < e^{2u}} d_4(n)/(n\log^4 n) \leq 16e^2$ bound the contribution of the terms with $n > M$ by a constant times $(\sum_{n>M}|a_n|^2/d_4(n))^{1/2}$, which is small for fixed f and large M, while the terms with $n \leq M$ tend to 0. So the difference tends to 0. Translations $f \mapsto f(\cdot + \varepsilon)$ are contractions of $\mathscr{A}^1$ [10, Theorem 11], so the Abel sums converge to the value of the bounded functional. □

The same subtraction as in Corollary 4.19 also handles the finite measure $u\mathbf{1}_{(0,1]}(u)\,\mathrm{d}u$, with correction terms controlled by the d_4 Helson inequality. Combined with Theorem 6.14, it gives bounded symbols for the sign-averaged variants: for instance, the Bergman Hilbert form restricted to pairs with $(mn)_Q$ a square.

The Gauss geometry has its own multiplicative Hilbert form, with moments $\alpha(n) = 1/(\sqrt{n}\log n)$, $n \geq 2$, on $\mathscr{F}^2 \times \mathscr{F}^2$: the form of Lebesgue measure on the

curve, as in the Hardy case. Let M_0 be the Meissel–Mertens constant and γ_{E} Euler's constant.

PROPOSITION 6.16. *The Gauss multiplicative Hilbert form has a bounded symbol. On series with $a_1 = 0$ its essential spectrum is $[0, \pi e^{M_0-\gamma_{\mathrm{E}}}]$, and $\pi e^{M_0-\gamma_{\mathrm{E}}} = 2.291\ldots$.*

PROOF. Split Lebesgue measure at $u = 1$, in the shifted variable $u = s - 1/2$. The part on $(0, 1]$ has $[\nu]_1 = 1$, and Theorem 6.14 gives a bounded symbol. The rest has moments $n^{-3/2}/\log n$, and by the Gauss Helson inequality of Corollary 6.4 and Cauchy–Schwarz its functional is bounded on $\mathscr{F}^1$, since $\sum_n n^{-3}2^{\Omega(n)}/[n]! \le \prod_p e^{2p^{-3}} < \infty$. In the orthonormal basis $n^{-s}/\sqrt{[n]!}$ the form is $\mathcal{N}^*\mathcal{N}$ as in Lemma 5.1, with companion kernel

$$h(u+v) = \sum_{n\ge 2} \frac{n^{-1-u-v}}{[n]!} = \exp\Bigl(\sum_p p^{-1-u-v}\Bigr) - 1.$$

By Mertens' theorem $\sum_p p^{-1-x} = \log(1/x) + M_0 - \gamma_{\mathrm{E}} + o(1)$, so $xh(x) \to e^{M_0-\gamma_{\mathrm{E}}}$, and $h(x) = O(2^{-x})$ for $x \ge 1$. Theorem 5.4 with $\kappa = 1$ and $b = 1$ gives the essential spectrum $[0, e^{M_0-\gamma_{\mathrm{E}}}\Gamma(1/2)^2]$. □

The spectral dictionary of chapter 5 carries over. For a finite positive ν on $(0,\infty)$, in the shifted variable, let $M^X_{\nu,R}$ be the matrix of the form with moments $\alpha_\nu(n) = \int n^{-1/2-u}\,\mathrm{d}\nu(u)$ on the integers composed of primes in R, in the orthonormal basis $\sqrt{d_\beta(n)}\,n^{-s}$ (Bergman) or $n^{-s}/\sqrt{[n]!}$ (Gauss).

THEOREM 6.17. *In the Bergman geometry with $\beta > 1$ and in the Gauss geometry, let R have density $\theta \in (0, 1]$ and let κ be the kernel exponent. For $0 < p < \infty$ and $0 < q \le \infty$,*

$$\|M^X_{\nu,R}\|_{S_{p,q}} \asymp \|(e^{j\kappa}\nu(I_j))_j\|_{\ell^{p,q}} + \nu((1,\infty)),$$

with constants depending only on θ, B, p, q and the geometry.

PROOF. The proofs of Theorems 5.7 and 5.13 use five inputs, and each holds here with κ in place of θ. The matrix is $\mathcal{N}^*\mathcal{N}$ for the synthesis by $k(u) = (w_n n^{-1/2-u})_n$, with w_n the basis weights, and k is analytic in $\Re u > 0$ with $\|k(u)\|^2$ equal to $\zeta_R(1 + 2\Re u)^\beta$ or $\exp(\sum_{p\in R} p^{-1-2\Re u})$, which is $\asymp (\Re u)^{-\kappa}$ on $(0, 1]$ by Lemma 2.9; this gives the trace formula and the block estimate (5.3). The operator norm bound is Theorem 2.10 with $p = 2$. The normalized kernels at $\delta_j = e^{-j}$ have inner products decaying like $e^{-\kappa|j-k|/2}$, the βth power of the Hardy ones in the Bergman case and $\exp(-\sum_{p\in R} p^{-1}(p^{-\delta_j} - p^{-\delta_k})^2/2)$ in the Gauss case. Finally $\|k(w) - e_1\|^2 \le C4^{-\Re w}$ for $\Re w \ge 1$. □

PROPOSITION 6.18 (The Hardy pairing is unbounded). *The functional $L(f) = \sum_{n\ge 2} a_n d(n)/(\sqrt{n}\log^2 n)$, the pairing of f with φ in $\mathscr{H}^2$, is unbounded on $\mathscr{H}^1$. With $f_\delta = \zeta(s + 1/2 + \delta)^2/\zeta(1 + 2\delta)$, one has $\|f_\delta\|_{\mathscr{H}^1} = 1$ and $L(f_\delta) \sim 1/(3\zeta(2)\delta)$.*

PROOF. $\|f_\delta\|_{\mathscr{H}^1} = \|\zeta(s+1/2+\delta)\|^2_{\mathscr{H}^2}/\zeta(1+2\delta) = 1$. By $1/\log^2 n = \int_0^\infty t n^{-t}\,\mathrm{d}t$ and $\sum d(n)^2 n^{-s} = \zeta(s)^4/\zeta(2s) \sim \zeta(2)^{-1}(s-1)^{-4}$,

$$\sum_{n\geq 2} \frac{d(n)^2}{n^{1+\delta}\log^2 n} \sim \frac{1}{\zeta(2)} \int_0^\infty \frac{t\,\mathrm{d}t}{(\delta+t)^4} = \frac{1}{6\zeta(2)\delta^2},$$

and $\zeta(1+2\delta) \sim 1/(2\delta)$. Truncated Euler products give polynomial witnesses. □

So the Bergman symbol is bounded against the Bergman pairing, where the weight $d(n)$ cancels, and the natural Hardy analogue fails by a power of δ; the pointwise count u^{-3} against $u\,\mathrm{d}u$ was already the right size.

6.7. Comparison

	Hardy	Bergman (β)	Gauss
transition maps	Möbius	Möbius	translations
energy $E(r,s)$	$\sum \rho_p^2$	$\sum \rho_p^2$	$\sum \lvert r_p - s_p\rvert^2$
Hellinger affinity	$\le e^{-E/4}$	$\le e^{-\beta E/4}$	$= e^{-E/4}$
kernel exponent κ	θ	$\beta\theta$	θ
contractive embedding	Carleman	Kulikov	Proposition 6.3
Helson weight	$1/d(n)$	$1/d_{2\beta}(n)$	$[n]!/2^{\Omega(n)}$
two-point constant	$(1-\lvert x\rvert^4)^{-1}$ (sharp)	$\Lambda_\beta(\lvert x\rvert^2)^{1/2}$	$\cosh(2\lvert x\rvert^2)^{1/2}$
sign-average constant	$\zeta(2)$	C_β	< 1.58

The Gauss column is the cleanest: its Hellinger affinity is exact, and its sign-average constant is the smallest. The Hardy column is the only one with a sharp two-point constant for every x. In the Bergman column the sharp constant is the square bound exactly up to the first zero of $K_x + K_{-x}$, and not beyond it; in the Gauss column only the leading term is known (Proposition 6.8).

Open questions

QUESTION 5. What are the sharp two-point constants in the Gauss geometry, where the square bound always fails (Proposition 6.8(4)), and in the Bergman geometry beyond the first zero of $K_x + K_{-x}$?

Part 3

Where Nehari's theorem fails: Schatten classes

CHAPTER 7

Schatten-class forms without bounded symbols

The question of this chapter was posed by Brevig and Perfekt.

PROBLEM 5 (Brevig and Perfekt [14, Question 3]). *Is there a multiplicative Hankel form in S_p, $2 < p < \infty$, without a bounded symbol, and if so, for which p? Brevig and Perfekt remark that it is tempting to conjecture that such forms exist for every $p > 2$.*

Let p_* be the infimum of such p. Helson's theorem gives $p_* \geq 2$ in the sense that S_2 forms always have bounded symbols, and Brevig and Perfekt [14] proved $p_* \leq (1 - \log \pi / \log 4)^{-1} = 5.7388\ldots$. This chapter proves $p_* \leq 4$, which settles Problem 5 for every $p > 4$, by a construction in the Gaussian model, and shows that the threshold 4 is built into the method: for the natural blocks it is the identity $\|\varphi\|_{L^4} = \|A(\varphi)\|_{S_4}$ of Janson, Peetre and Rochberg.

Bampouras [4] was the first to prove $p_* \leq 4$. His symbols are $1 - t^2 + tF$, with F a product of normalized linear forms on finite tori, tested against perturbed squares $(1+tF)^2 - \eta t^2 F^2$; integrating out the scale t against $t^{-\kappa-1}\,\mathrm{d}t$ reduces everything to the moments $\mathbb{E}|F|^\kappa$. We found the same construction independently, in its Gaussian limit $1 + cP_n$, with a longer proof that fixes the scale and optimizes it by large deviations, at rates that are Legendre transforms of his moments. Section 7.2 gives our proof, which has explicit parameters; section 7.3 gives his, for a general block, and its Gaussian form (remark 7.13), the shortest proof we know. New here are the identity of Proposition 7.11, which shows that his criterion cannot reach $p \leq 4$ on any finite torus, and the barriers of chapter 8.

7.1. Blocks and the Gaussian model

For an analytic polynomial φ on $\mathbb{T}^d$ write $H_\varphi(f, g) = \langle fg, \varphi \rangle$, with matrix $(\hat{\varphi}(\alpha + \beta))_{\alpha,\beta}$.

LEMMA 7.1 (Brevig–Perfekt [14]). *If $p \geq 1$ and polynomials f, φ in finitely many variables satisfy $|\langle f, \varphi \rangle| > \|H_\varphi\|_{S_p} \|f\|_1$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. The symbol can be written down. Normalize $\|H_\varphi\|_{S_p} = 1$ and put $R = |\langle f, \varphi \rangle| / \|f\|_1 > 1$. Let φ_m and f_m be the mth tensor powers of φ and f in a block of md new variables, disjoint for different m, and put $\Phi = \sum_m m^{-2} \varphi_m$. Pairing, L^1 norm and Schatten norm are multiplicative over products in disjoint variables, so $\|H_\Phi\|_{S_p} \leq \sum_m m^{-2}$ and $|\langle f_m, \varphi_m \rangle| = R^m \|f_m\|_1$. For $k \neq m$ the blocks are disjoint, so $\langle f_m, \varphi_k \rangle = f_m(0)\overline{\varphi_k(0)}$, with $|f_m(0)| \leq \|f_m\|_1$ and $|\varphi_k(0)| \leq 1$.

Hence $|\langle f_m, \Phi\rangle| \geq (m^{-2}R^m - \pi^2/6)\|f_m\|_1$, which is unbounded in m, while a bounded symbol ψ would give $|\langle f_m, \Phi\rangle| \leq \|\psi\|_\infty \|f_m\|_1$. □

Let μ_d be the law of d independent standard complex Gaussians on $\mathbb{C}^d$. For $\varphi = \sum c_\alpha z^\alpha$ the Gaussian Hankel form $(f, g) \mapsto \langle fg, \varphi\rangle_{L^2(\mu_d)}$ has, in the basis $z^\alpha/\sqrt{\alpha!}$, the matrix $A(\varphi)_{\alpha\beta} = c_{\alpha+\beta}(\alpha+\beta)!/\sqrt{\alpha!\beta!}$. Analytic $L^2(\mu_d)$ is the Fock space, and $A(\varphi)$ is the matrix of the small Hankel form studied by Janson, Peetre and Rochberg [30]. Let $D(\varphi)$, the dual norm, be the norm of $f \mapsto \langle f, \varphi\rangle_{L^2(\mu_d)}$ on analytic polynomials with the $L^1(\mu_d)$ norm; by the Hahn–Banach theorem it is the least sup norm of a bounded representative of φ. If $D(\varphi) > \|A(\varphi)\|_{S_p}$ we call φ a *strict block* at p.

PROPOSITION 7.2 (Transfer). *Let $p \geq 2$. If $D(\varphi) > \|A(\varphi)\|_{S_p}$ for a polynomial φ on $\mathbb{C}^d$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. Replace z_j by $L_{N,j} = N^{-1/2}\sum_{\ell \leq N} z_{j,\ell}$ on disjoint blocks of Steinhaus variables. For fixed k, $\|L_N^k\|_2^2 \to k!$, $\|H_{L_N^k}\|_{S_2}^2 \to 2^k k!$ (a coefficient of index ν occurs $\prod(\nu_\ell + 1)$ times, and only squarefree indices survive), and $\mathbb{E}|L_N|^{2k} \leq k!$. These multiply over blocks, so $\|H_{\varphi_N}\|_{S_2} \to \|A(\varphi)\|_{S_2}$. The compression of H_{φ_N} to the normalized monomials of degree at most K converges entrywise to $A(\varphi)$. Since compression is an orthogonal projection in Hilbert–Schmidt norm, the remainder tends to 0 in S_2, hence in S_p. The central limit theorem and uniformly bounded second moments give $\|f(L_N)\|_1 \to \|f\|_{L^1(\mu_d)}$, and the pairings converge by orthogonality. Apply Lemma 7.1. □

7.2. Every $p > 4$

Let $Z_1, Z_2, \ldots$ be independent standard complex Gaussians and $P_n = Z_1 \cdots Z_n$, so $\mathbb{E}|P_n|^2 = 1$ and $\mathbb{E}|P_n|^4 = 2^n$.

Brevig and Perfekt's block is the linear symbol z, for which $D(z) = 2/\sqrt{\pi}$ and $\|A(z)\|_{S_p} = 2^{1/p}$; this gives exactly $(1 - \log\pi/\log 4)^{-1}$, and by the monomial computation in chapter 8 no one-variable homogeneous symbol does better. Going below needs two new ingredients: a constant term in the symbol, and a perturbed square as test.

The budget. Take the symbol $\varphi = 1 + cP_n$ with $c = r/(r^2-1) \approx 1/r$, so that the top singular value of $A(\varphi)$ is $\lambda = r^2/(r^2-1)$ (Lemma 7.3), and test with the square $f = (1+P_n/r)^2$. Then $\langle f, \varphi\rangle = \lambda(1+r^{-2})$ and $\|f\|_1 = \mathbb{E}|1+P_n/r|^2 = 1+r^{-2}$, so the square alone ties with the operator norm. Subtracting $t(P_n/r)^2$ leaves the pairing unchanged, since $P_n^2 \perp \varphi$, and changes $\|f\|_1$ by $-tG$, with $G = \mathbb{E}(|P_n|^2/r^2 - 1)_+$, plus a second-order cost of order $t^2\,\mathbb{E}|P_n|^4/r^4 = t^2 2^n r^{-4}$ (Lemma 7.4). Optimizing t, the gain is of order $G^2 r^4/2^n$. Against it stands the Schatten excess of $A(\varphi)$ over λ, which comes from the $2^n - 2$ residual singular values c and is of order $2^n r^{-p}$. At $p = 4$ the gain cannot win: $(x-1)_+ \leq x^2/4$ gives $G \leq 2^n r^{-4}/4$, so the gain is at most of the order of the excess, and Theorem 8.1 shows that it never wins. For $p > 4$ the excess is smaller by the factor r^{4-p}, while G, the large-deviation cost of $|P_n| > r = e^{\beta n}$, is not. The rates I and $J = 4\beta - \log 2$ in the proof of Theorem 7.6 make this precise.

LEMMA 7.3 (Spectrum). *For $c > 0$ let $\lambda = (1 + \sqrt{1 + 4c^2})/2$. The nonzero singular values of $A(1 + cP_n)$ are λ, $\lambda - 1$, and c with multiplicity $2^n - 2$. Hence $\|A(1 + cP_n)\|_{S_p} - \lambda \le 2^n c^p/p$ for $p \ge 2$.*

PROOF. Only squarefree monomials contribute, each paired with its complement. The pair $\{1, P_n\}$ carries $\left(\begin{smallmatrix} 1 & c \\ c & 0 \end{smallmatrix}\right)$, and the other $2^{n-1} - 1$ pairs carry $\left(\begin{smallmatrix} 0 & c \\ c & 0 \end{smallmatrix}\right)$. Since $\lambda - 1 \le c$ and $\lambda \ge 1$, concavity of $x^{1/p}$ gives the estimate. □

LEMMA 7.4 (First variation of L^1). *Let W be rotation invariant with $\mathbb{E}|W|^2 = 1$, $M = \mathbb{E}|W|^4$, and let $u = W/r$, $G = \mathbb{E}(|u|^2 - 1)_+$, $T = \mathbb{E}(|u|^2; |u| > 1/2)$. For $t \ge 0$,*

$$\mathbb{E}\big|(1 + u)^2 - tu^2\big| \le 1 + r^{-2} - tG + 16Mt^2r^{-4} + 8t^{5/4}T.$$

PROOF. For $w \ne 0$,

$$|w - tv| = |w| - t\Re(v\overline{w})/|w| + R, \qquad 0 \le R \le \min(2t|v|, 4t^2|v|^2/|w|).$$

Take $w = (1+u)^2$, $v = u^2$. The main term has mean $\mathbb{E}|1+u|^2 = 1+r^{-2}$. Averaging the linear term over the angle,

$$\int_{\mathbb{T}} \rho^2\zeta^2 \frac{1 + \rho\overline{\zeta}}{1 + \rho\zeta}\, \mathrm{d}m(\zeta) = (\rho^2 - 1)_+,$$

by expanding in powers of $\rho\zeta$ for $\rho < 1$ and of $1/(\rho\zeta)$ for $\rho > 1$; so it has mean $-tG$. On $|u| \le 1/2$, $R \le 16t^2|u|^4$. On $|u| > 1/2$, $R \le 2^{5/4}t^{5/4}|u|^{5/2}|1 + u|^{-1/2}$, and the conditional mean of $\rho^{1/2}|1 + u|^{-1/2}$ given $|u| = \rho \ge 1/2$ is at most $2^{5/4}$. □

LEMMA 7.5 (Large deviations). *Let $\Psi(s) = \log \mathbb{E}|Z_1|^s = \log \Gamma(1 + s/2)$. For $s > 2$ put $\beta = \Psi'(s)$, $I = s\beta - \Psi(s)$, $r_n = e^{\beta n}$, and let G_n, T_n be as in Lemma 7.4 for $W = P_n$, $r = r_n$. Then $\beta > 0$, $T_n \le 2^{s-2}e^{-nI}$ and $n^{-1} \log G_n \to -I$.*

PROOF. Ψ is strictly convex with $\Psi(0) = \Psi(2) = 0$, so $\beta > 0$. With $x = |P_n|/r_n$, $\mathbb{E}x^s = e^{-nI}$, and $(x^2 - 1)_+ \le x^s$, $x^2\mathbf{1}_{x>1/2} \le 2^{s-2}x^s$ give the upper bounds. For the lower bound, tilt by $e^{u \log|P_n| - n\Psi(u)}$ with $u > s$: under the tilted law $\log|P_n|/n$ concentrates at $\Psi'(u) > \beta$, and converting back costs $e^{n(\Psi(u) - u\Psi'(u) + 2(\Psi'(u) - \beta)) + o(n)}$. Let $u \downarrow s$. □

THEOREM 7.6. *For every $p > 4$ and all large n there are $c_n, r_n, t_n > 0$ such that $\varphi_n = 1 + c_nP_n$ and $f_n = (1 + P_n/r_n)^2 - t_n(P_n/r_n)^2$ satisfy $|\langle f_n, \varphi_n\rangle| > \|A(\varphi_n)\|_{S_p}\|f_n\|_1$. Consequently $p_* \le 4$.*

PROOF. Take $s > 4$ and the data of Lemma 7.5, and put $J = 4\beta - \log 2$, so $2^n/r_n^4 = e^{-nJ}$. Strict convexity and $\Psi(4) = \log 2$ give $I > J$, and $(p\beta - \log 2) - (2I - J) \to (p - 4)\Psi'(4) > 0$ as $s \downarrow 4$; fix s and $\varepsilon > 0$ with $2I - J + \varepsilon < p\beta - \log 2$. Let $c_n = r_n/(r_n^2 - 1)$, so that $\lambda_n = r_n^2/(r_n^2 - 1)$, and $t_n = e^{-n(I - J + \varepsilon)}$.

By Lemma 7.3, the Schatten excess is $\|A(\varphi_n)\|_{S_p} - \lambda_n \le (2^p/p)e^{-n(p\beta - \log 2)}$. The pairing is $\langle f_n, \varphi_n\rangle = 1 + 2c_n/r_n = \lambda_n(1 + r_n^{-2})$, because $P_n^2 \perp \varphi_n$. By Lemmas 7.4 and 7.5, $(1 + r_n^{-2}) - \|f_n\|_1 \ge t_nG_n(1 - o(1))$, the two error terms being $e^{-n(\varepsilon + o(1))}$ and $e^{-n((I - J + \varepsilon)/4 + o(1))}$ relative to t_nG_n. Hence $\langle f_n, \varphi_n\rangle/\|f_n\|_1 - \lambda_n \ge t_nG_n/4 = e^{-n(2I - J + \varepsilon + o(1))}$, which beats the excess. Apply Proposition 7.2. □

COROLLARY 7.7 (Dimension growth). *For $p > 4$ let $C_p(d)$ be the best constant in $|\langle f, \varphi\rangle| \le C\|H_\varphi\|_{S_p}\|f\|_{L^1(\mathbb{T}^d)}$, which is the minimal-symbol constant on $\mathbb{T}^d$. Then $C_p(d) \ge A_p e^{c_p d}$ with $c_p > 0$.*

PROOF. A finite pair with ratio $R_p > 1$ in d_p variables exists by Theorem 7.6 and Proposition 7.2; k disjoint copies give $C_p(kd_p) \ge R_p^k$, and $C_p(d)$ is nondecreasing in d. □

The rate degenerates as $p \downarrow 4$. For $2 < p \le 4$, Proposition 7.8 below shows that $C_p(d)$ either equals 1 for every d or grows exponentially; which of the two happens is open. In a fixed number of variables the constant is finite for every $p \le \infty$: Nehari's theorem holds on $\mathbb{D}^d$, by Ferguson and Lacey [23] for $d = 2$ and Lacey and Terwilleger [36] for $d > 2$, and $\|A\| \le \|A\|_{S_p}$ gives $1 \le C_p(d) \le C_\infty(d) < \infty$. Ortega-Cerdà and Seip [44] show that $C_\infty(d)$ grows at least exponentially in d, and Brevig [13] improved their rate. For $d = 2$, Bampouras and Perfekt [8] give a direct proof in the product Hardy space of two half-planes that Hankel operators in S_p have symbols in product BMO. So the counterexamples of this chapter live in the limit $d \to \infty$, and for $p > 4$ the corollary gives the exponential rate.

The constants $C_p(d)$, defined in the same way for every $p \ge 1$, see the threshold exactly.

PROPOSITION 7.8 (Dichotomy). *Let $p \ge 1$. Either $C_p(d) = 1$ for every d, or $C_p(d)$ grows exponentially in d; and $\sup_d C_p(d) < \infty$ if and only if every multiplicative Hankel form in S_p has a bounded symbol. Consequently the forms in S_p all have bounded symbols if and only if $p \le p_*$, the endpoint included, and $p_* = 4$ if and only if every multiplicative Hankel form in S_4 has a bounded symbol.*

PROOF. If $C_p(d_0) > 1$, disjoint copies give $C_p(kd_0) \ge C_p(d_0)^k$, and Lemma 7.1 gives a form in S_p without bounded symbol. If $\sup_d C_p(d) = C < \infty$ and $H_\varphi \in S_p$, then for a polynomial f in d variables, $\langle f, \varphi\rangle = \langle f, \varphi_d\rangle$ with φ_d the restriction to those variables; Fejér means of φ_d are averages of unitary conjugates of a compression of H_φ, so $|\langle f, \varphi\rangle| \le C\|H_\varphi\|_{S_p}\|f\|_1$, and the Hahn–Banach theorem gives a bounded symbol. For the endpoint: if $p < p_*$ then $C_p(d) = 1$, so $|\langle f, \varphi\rangle| \le \|H_\varphi\|_{S_p}\|f\|_1$ for polynomials; H_φ has finite rank, so $p \mapsto \|H_\varphi\|_{S_p}$ is continuous, and the inequality passes to $p = p_*$. The last claim uses $p_* \le 4$ from Theorem 7.6. □

7.3. The moment criterion and why it stops at 4

Bampouras's proof [4] integrates over the scale instead of fixing it. We record it for a general block, which shows where it stops. Call a polynomial W on $\mathbb{T}^N$ *admissible* if $\|W\|_2 = 1$, $W(0) = 0$, $\int W^2\overline{W}\,\mathrm{d}m = 0$, $\int g|W|^2\,\mathrm{d}m = \hat{g}(0)$ for analytic g, and the law of W is rotation invariant. Every normalized homogeneous polynomial is admissible, and so are products of admissible blocks in disjoint variables.

LEMMA 7.9. *For admissible F and $\varphi_t = 1 - t^2 + tF$, the span of 1 and F reduces H_{φ_t}, with compression $\left(\begin{smallmatrix} 1-t^2 & t \\ t & 0 \end{smallmatrix}\right)$, and $\|H_{\varphi_t}\|_{S_p}^p = 1 + t^{2p} + t^p(\|H_F\|_{S_p}^p - 2)$.*

PROOF. $H_F(F, g) = \int g|F|^2 = \hat{g}(0)$ and $H_F(F, F) = \int F^2\overline{F} = 0$, so H_F swaps 1 and F and maps their orthogonal complement into itself. □

THEOREM 7.10 (Bampouras's moment criterion). *Let W be admissible and $p > 2$. If $\|W\|_{L^4} > \|H_W\|_{S_p}$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. Let

$$D_\eta(u) = 1 + u^2 - \int_{\mathbb{T}} |(1+u\zeta)^2 - \eta u^2 \zeta^2| \, \mathrm{d}m(\zeta), \qquad I_\kappa(\eta) = \int_0^\infty D_\eta(u) u^{-\kappa-1} \, \mathrm{d}u;$$

by [4], $I_\kappa(\eta) > 0$ for $\kappa \in (2,4)$ and small η. By continuity choose $\kappa \in (2, \min(p,4))$ with $\|W\|_\kappa > \|H_W\|_{S_p}$. Suppose the conclusion fails. For $F = \prod_{j \le m} W(z^{(j)})$, with $A = \|H_W\|_{S_p}^p$, the test $F_t = (1+tF)^2 - \eta t^2 F^2$ and Lemma 7.9 give

$$\|F_t\|_1 \ge (1+t^2)(1 + t^{2p} + (A^m - 2)t^p)^{-1/p},$$

while rotation invariance gives $\|F_t\|_1 = 1 + t^2 - \int D_\eta(t|F|) \, \mathrm{d}m$. Integrate against $t^{-\kappa-1} \, \mathrm{d}t$ and substitute $u = t|F|$: $I_\kappa(\eta)(\mathbb{E}|W|^\kappa)^m \le C_{p,\kappa} A^{m\kappa/p}$. Taking mth roots gives $\|W\|_\kappa \le \|H_W\|_{S_p}$, a contradiction. □

PROPOSITION 7.11 (L^4 is dominated by S_4). *Let $\varphi = \sum_\alpha c_\alpha z^\alpha$ be a polynomial on $\mathbb{T}^d$, and for $\gamma \ge 0$ let $B_\gamma \varphi = \sum_\alpha c_{\alpha+\gamma} z^\alpha$ be its backward shift. Then, in the spirit of the one-variable integral identities of Janson, Upmeier and Wallstén [31],*

$$\|H_\varphi\|_{S_4}^4 = \sum_{\gamma \ge 0} \|B_\gamma \varphi\|_{L^4}^4.$$

In particular $\|\varphi\|_{L^4} \le \|H_\varphi\|_{S_4}$, with equality only for constants, and the defect satisfies

$$\|H_\varphi\|_{S_4}^4 - \|\varphi\|_{L^4}^4 \ge \sum_{\alpha,\alpha'} \Big(\prod_{j \le d} (1 + \min(\alpha_j, \alpha_j')) - 1 \Big) |c_\alpha|^2 |c_{\alpha'}|^2 \ge \frac{1}{d} \|\varphi - \varphi(0)\|_2^4.$$

In the Gaussian model $\|\varphi\|_{L^4(\mu)} = \|A(\varphi)\|_{S_4}$ [30].

PROOF. $\|H_\varphi\|_{S_4}^4 = \sum_{\beta,\beta'} |\sum_\alpha c_{\alpha+\beta} \overline{c_{\alpha+\beta'}}|^2$. Every pair (β, β') is uniquely $(\gamma + \delta^+, \gamma + \delta^-)$ with $\gamma = \min(\beta, \beta')$ coordinatewise and $\delta^\pm$ of disjoint supports, and for fixed γ the inner sums are the Fourier coefficients of $|B_\gamma \varphi|^2$ at $\delta^+ - \delta^-$; each frequency occurs once. Parseval gives the identity, and $\gamma = 0$ is $\|\varphi\|_4^4$. For the defect, use $\|B_\gamma \varphi\|_4 \ge \|B_\gamma \varphi\|_2$ and count the $\gamma \ne 0$ with $\gamma \le \min(\alpha, \alpha')$. Keeping only $\gamma = e_j$ gives $\sum_j I_j^2$ with $I_j = \sum_{\alpha_j \ge 1} |c_\alpha|^2$, and $\sum_j I_j \ge \|\varphi - \varphi(0)\|_2^2$. □

For a Dirichlet polynomial $\sum a_n n^{-s}$ the product in the defect is $d(\gcd(m,n))$: approaching the Gaussian equality $L^4 = S_4$ requires the coefficients to have negligible common divisors, which is why spreading a block over many independent primes helps.

COROLLARY 7.12 (The method's threshold). *(1) For $p \le 4$ the criterion fails for every admissible block on every finite torus, since $\|W\|_{L^\kappa} \le \|W\|_{L^4} \le \|H_W\|_{S_4} \le \|H_W\|_{S_p}$.*

(2) *(Bampouras [4].) For a normalized linear form $L = \sum_{i\le N} c_i z_i$ it succeeds exactly for $p > 4\log 2/\log(2 - \sum|c_i|^4)$; with equal coefficients this is* 6.84, 5.43, 4.95, 4.32 *for* $N = 2, 3, 4, 10$, *and* $4 + 2/(N\log 2) + O(N^{-2})$ *in general.*

(3) *If H_W has rank two, then W is linear: a monomial of top degree k and a nested chain of divisors give a triangular $(k+1)\times(k+1)$ submatrix. So the rank-two version of the criterion is exactly the linear case.*

PROOF. (2) $\|H_L\|_{S_p}^p = 2$ and $\|L\|_4^4 = 2 - \sum|c_i|^4$. □

So three variables already beat the Brevig–Perfekt exponent. For every nonconstant Gaussian polynomial W the crossing $\|W\|_{L^p} = \|A(W)\|_{S_p}$ happens exactly at $p = 4$, since $\|W\|_{L^p(\mu)}$ increases and $\|A(W)\|_{S_p}$ decreases in p, with equality at 4 by Janson–Peetre–Rochberg.

REMARK 7.13 (The criterion in the Gaussian model). Bampouras's argument runs verbatim in the Gaussian model, with Proposition 7.2 in place of Lemma 7.1: circle invariance gives the structure of Lemma 7.9. The criterion then reads: if $\|W\|_{L^\kappa(\mu)} > \|A(W)\|_{S_p}$ for some $\kappa < 4$, there is a form in S_p without bounded symbol. For $W = Z_1$ one has $\|A(Z_1)\|_{S_p}^p = 2$ and $\mathbb{E}|Z_1|^\kappa = \Gamma(1+\kappa/2)$, and letting $\kappa \to 4$ gives every $p > 4$. This is Bampouras's argument with the limit $N \to \infty$ taken inside the criterion; it is the shortest proof of $p_* \le 4$ we know.

Going below 4 with Lemma 7.1 requires a block outside the admissible class, so that the $\{1, F\}$ splitting fails, or a test family not of the form $(1+tF)^2 - \eta t^2 F^2$.

CHAPTER 8

Barriers: why 4, and why not 2

Question. Can the Gaussian-block method of chapter 7 reach any $p \le 4$? And what keeps every construction away from $p = 2$?

Throughout, μ is Gaussian probability measure, $A(\varphi)$ the Gaussian Hankel matrix and $D(\varphi)$ the norm of $f \mapsto \langle f, \varphi\rangle$ on analytic polynomials in $L^1(\mu)$, as in section 7.1. A symbol gives an S_p counterexample through Proposition 7.2 when $D(\varphi) > \|A(\varphi)\|_{S_p}$; we call this a *strict block* at p. The Gaussian Carleman inequality $\sum_\alpha \alpha! 2^{-|\alpha|}|a_\alpha|^2 \le \|f\|^2_{L^1(\mu)}$ (Proposition 6.3 with Minkowski's inequality) and Cauchy–Schwarz give $D(\varphi) \le \|A(\varphi)\|_{S_2}$, the Gaussian form of Helson's inequality, and $\|A(\varphi)\|_{S_4} = \|\varphi\|_{L^4(\mu)}$ [30].

The answers: the constant-plus-product family that reaches every $p > 4$ has no strict block at $p \le 4$, for every n and every test, so 4 is exactly its threshold. Bampouras [4] singles out the endpoint $p = 4$ as especially interesting; for the Gaussian limit of his family the answer is negative. Affine symbols, even quadratic symbols of rank at most two, and all zero-free Hilbert–Schmidt symbols, are equally stuck; and every family of bounded degree in a bounded number of variables is excluded on an interval $(2, 2+\varepsilon)$. Whether *any* Gaussian symbol beats S_4 is the L^1–L^4 question of section 8.3. The Paley–Wiener threshold 4 of Bampouras has a different origin (section 8.5). The sharpness of the product family at 4, the barriers for affine, even quadratic and zero-free symbols, the interval $(2, 2+\varepsilon)$, the disc blocks and the polydisc growth are, to our knowledge, new.

8.1. The product family is sharp at 4

Let $P_n = Z_1 \cdots Z_n$. By Lemma 7.3, $\|A(c_0 + c_1 P_n)\|^4_{S_4} = |c_0|^4 + 4|c_0 c_1|^2 + 2^n |c_1|^4$.

THEOREM 8.1. *For every $n \ge 1$ and all $c_0, c_1 \in \mathbb{C}$, $D(c_0 + c_1 P_n) \le \|A(c_0 + c_1 P_n)\|_{S_4}$. Consequently the infimum of exponents with a strict block in this family is exactly 4.*

The cases $n = 1$ and $n \ge 2$ are proved differently: for $n = 1$ there are no residual singular values, $M = 2$, and the envelope argument below, which needs $M \ge 4$, does not apply; that case is handled by an explicit bounded representative.

The affine case

For $k > 0$ and $R = |Z|$ put $u_0 = \mathbb{E}(1 - k^2 R^2)_+$ and $u_1 = \mathbb{E}[R \min(kR, 1)]$.

LEMMA 8.2. *$Q(k) = u_0^4 + 4u_0^2 u_1^2 + 2u_1^4$ increases strictly on $(0, \infty)$ from the limit 1 at $k = 0$.*

PROOF. With $B = \mathbb{E}[R^2\mathbf{1}_{kR<1}]$, $u_0' = -2kB$ and $u_1' = B$, so $Q' = 8B\Phi$ with $\Phi = u_1^3 + u_0^2u_1 - ku_0^3 - 2ku_0u_1^2$. Put $t = 1/k$, $x = t^2$, $e = e^{-x}$. Then $u_0 = 1 - (1-e)/x$, and the Mills bound $\int_t^\infty e^{-s^2}\,\mathrm{d}s \geq te^{-t^2}/(2t^2+1)$ gives $u_1 \geq u_e = t^{-1}(1 - e(x+1)/(2x+1))$. For $x \leq 1$, $u_e > 2ku_0$ reduces to $2 + 3x - 2x^2 > e(x^2 + 5x + 2)$, which follows from the fourth-order Taylor bound for e^{-x} and the identity $(2 + 3x - 2x^2) - (x^2 + 5x + 2)(1 - x + x^2/2 - x^3/6 + x^4/24) = x^2(1-x)(x^3 + 2x^2 - 4x + 24)/24$; then $\Phi = u_1^2(u_1 - 2ku_0) + u_0^2(u_1 - ku_0) > 0$. For $x \geq 1$, write $u_1 = \rho u_0$. Then $\partial_{u_1}\Phi = u_0^2(3\rho^2 - 4k\rho + 1)$, which is nonnegative for $\rho \geq k$ (its larger root $(2k + \sqrt{4k^2 - 3})/3$ is at most k, and there is no real root once $x > 4/3$), and $\partial_{u_0}\Phi = u_0^2(2\rho - 3k - 2k\rho^2)$. For $x < 6$ the last quadratic has no real root, so $\partial_{u_0}\Phi < 0$. For $x \geq 6$, $u_1 \leq k\mathbb{E}R^2 = k$ and $u_0 \geq 1 - k^2 \geq 5/6$ give $\rho \leq 6k/5$, while the smaller root satisfies $(1 - \sqrt{1 - 6k^2})/(2k) \geq 3k/2$, so again $\partial_{u_0}\Phi \leq 0$. Replacing e by the larger $1/T(x)$, $T(x) = \sum_{j\leq 4} x^j/j!$, raises u_0 and lowers u_e, so ρ only decreases along the way; at the end $\rho \geq k$ is $T(x)(2x+1) \geq x^2 + 3x + 1$, which holds. So both signs persist, and the replacement only lowers Φ. The resulting expression is $x^{3/2}(x+2)^3P(x)/(13824(2x+1)^3T(x)^3)$ with $P(x) = x^3[8x^2(x - 9/4)^2 + 75x^2/2 - 163x + 1326] + 2268x^2 + 6408x + 6912 > 0$. □

PROOF OF THEOREM 8.1 FOR $n = 1$. For $b = \min(kR, 1)$ the unimodular function

$$\psi_k(R\zeta) = \frac{1 + b\zeta}{1 + b\overline{\zeta}}$$

has analytic coefficients $1-b^2, b, 0, \dots$ on each circle, so it represents the functional of u_0+u_1z, and $D(1+(u_1/u_0)z) \leq 1/u_0 < (1+4(u_1/u_0)^2+2(u_1/u_0)^4)^{1/4}$ by Lemma 8.2. The ratio u_1/u_0 increases continuously from 0 to ∞, so scaling and rotation cover every symbol with $c_0c_1 \neq 0$; for $c_0 = 0$, $D(c_1z) = 2|c_1|/\sqrt{\pi} < 2^{1/4}|c_1|$. □

Products of at least two factors

For $M \geq 4$ put $S_M(c) = (1 + 4c^2 + Mc^4)^{1/4}$ and $L_M(b) = \sup_{c\geq 0}(1 + bc)/S_M(c)$.

Here $S_M(c)$ is the S_4 norm of $A(1 + cP_n)$ when $M = \mathbb{E}|P_n|^4 = 2^n$, and $L_M(b)$ is the smallest L^1 norm, relative to $|a_0|$, that a test function with coefficient ratio $|a_1/a_0| = b$ must have if it is to lose against every symbol $c_0 + c_1P_n$. The proof is an induction on n. Multiplying by one more Gaussian Z does two things at once: it replaces the coefficient ratio b by the random ratio $b|Z|$, and it doubles M. The envelope inequality (8.1) below says that averaging L_M over the random ratio $b|Z|$ costs at least as much as doubling M: these two changes are compatible. To prove it, the single multiplication by $|Z|^2$ is split into a continuum of fractional steps, the log-Gamma semigroup of Lemma 8.5, while M moves continuously from M to $2M$. The discrete step then becomes a comparison between the generator of the semigroup and the speed at which M moves (Theorem 8.6), and that comparison is a one-variable inequality, checked below.

LEMMA 8.3 (Reduction). *The theorem holds for $n \geq 2$ if every polynomial $q(w) = a_0 + a_1w + \cdots$ with $a_0 \neq 0$ satisfies $\mathbb{E}|q(P_n)| \geq |a_0|L_{2^n}(|a_1/a_0|)$.*

PROOF. Averaging a test over the rotations with $\zeta_1 \cdots \zeta_n = 1$ keeps exactly the monomials with equal exponents, preserves the pairing and contracts L^1; so tests

may be taken of the form $q(P_n)$. Then $|a_0|+c|a_1|\le S_M(c)\mathbb{E}|q(P_n)|$ is the claim. If $a_0=0$, coefficient extraction gives $\mathbb{E}|q(P_n)|\ge|a_1|(\sqrt{\pi}/2)^n\ge|a_1|M^{-1/4}$. □

LEMMA 8.4 (Circle estimate). *If G is analytic on a neighbourhood of $\overline{\mathbb{D}}$ with $G(0)=1$ and $|G'(0)|=s$, then $\int_{\mathbb{T}}|G|\,\mathrm{d}m\ge h(s)$, where $h(s)=1+s^2/4$ for $s\le 2$ and $h(s)=s$ for $s\ge 2$.*

PROOF. Factor $G=uv$ with $\|u\|_2\|v\|_2=\|G\|_1$ and pair with $1+\lambda z$: $1+\lambda s\le\|G\|_1(1+\sqrt{1+4\lambda^2})/2$. Take $\lambda=2s/(4-s^2)$ for $s<2$; for $s\ge 2$ extract the coefficient. □

For $n=2$: with $\beta=|a_1/a_0|$, the circle estimate and rotation invariance give $\mathbb{E}|q(Z_1Z_2)|\ge|a_0|\mathbb{E}h(\beta|Z_1Z_2|)$. Since $x\mapsto h(\beta\sqrt{x})$ is concave, Jensen gives $\mathbb{E}h(\beta|W|)\ge\mathbb{E}h(\beta\sqrt{2}|\Re W|)$ for $W=Z_1Z_2$, and $\sqrt{2}|\Re W|$ is distributed as $X/\sqrt{2}$ with X exponential. Then $\mathbb{E}h(bX)=1+(b^2/2)(1-e^{-2/b})\ge\sqrt{1+b^2}$, because $\sinh(1/b)\ge 1/b$. This is $L_4(b)=\sqrt{1+b^2/2}$ at $b=\beta/\sqrt{2}$.

The induction from n to $n+1$ conditions on Z_{n+1}: the polynomial $w\mapsto q(wZ_{n+1})$ has coefficients a_0 and a_1Z_{n+1}, so it suffices to prove the envelope inequality

$$\mathbb{E}L_M(b|Z|)\ge L_{2M}(b)\qquad(M\ge 4,\ b\ge 0).\tag{8.1}$$

LEMMA 8.5 (The log-Gamma semigroup). *There are positive random variables X_a, $a\ge 0$, with $X_0=1$, $X_1\overset{d}{=}|Z|^2$, $\mathbb{E}X_a^s=\Gamma(1+s)^a$ and $X_{a+a'}\overset{d}{=}X_aX'_{a'}$. The semigroup $P_ag(t)=\mathbb{E}g(tX_a)$ has generator*

$$\mathcal{A}g(t)=\int_0^1\frac{g(tu)-g(t)+(1-u)tg'(t)}{(-\log u)(1-u)}\,\mathrm{d}u,$$

with positive kernel and $\mathcal{A}(t^s)=\log\Gamma(1+s)\,t^s$.

PROOF. $\log|Z|^2$ is infinitely divisible [40]: the Lévy measure $e^y\,\mathrm{d}y/(|y|(1-e^y))$ on $(-\infty,0)$ has Laplace exponent $\int_0^1(u^s-1+s(1-u))/((-\log u)(1-u))\,\mathrm{d}u$, whose second derivative is $\sum_{k\ge1}(k+s)^{-2}$ and which vanishes at $s=0,1$, so it equals $\log\Gamma(1+s)$. Put $X_a=e^{\xi_a}$. □

Write $f_M(t)=L_M(\sqrt{t})$; then (8.1) is $P_1f_M\ge f_{2M}$. The family $g_a=f_{2^aM}$ moves with speed $(\log 2)N\partial_Nf_N|_{N=2^aM}$, and a comparison principle for $\mathcal{A}$ on functions of square-root growth (Dynkin's formula, then differentiating $s\mapsto P_sg_{1-s}$) shows $g_1\le P_1g_0$ as soon as g_a is a subsolution:

THEOREM 8.6. *For $M\ge 4$ and $t>0$, $\mathcal{A}f_M(t)>(\log 2)M\partial_Mf_M(t)$.*

PROOF. Let c be the maximizer in $L_M(\sqrt{t})$, so $\sqrt{t}=(2c+Mc^3)/(1+2c^2)$, and put $y=Mc^2$. The envelope theorem gives

$$tf'_M=\frac{c^2(2+y)}{2(1+2c^2)S_M(c)},\qquad M\partial_Mf_M=-\frac{y}{2(2+y)}\,tf'_M,$$

where we used $1+\sqrt{t}c=S_M(c)^4/(1+2c^2)$, so that $f_M=S_M(c)^3/(1+2c^2)$. Since $d\sqrt{t}/dc=(2-4c^2+3Mc^2+2Mc^4)/(1+2c^2)^2>0$ for $M\ge4$, the maximizer c,

and with it y, increases with t, and f_M is increasing. Differentiating once more in c,

$$-\frac{tf_M''}{f_M'} = \frac{c^2(2M^2c^6 + 3M^2c^4 + 12Mc^2 + 2M - 24c^2 - 4)}{2S_M(c)^4(2Mc^4 + 3Mc^2 - 4c^2 + 2)},$$

which is positive for $M \geq 4$ (as $12M \geq 48$), so f_M is concave; and

$$\frac{y}{2(1+y)} + \frac{tf_M''}{f_M'} = \frac{c^2\big(2M^2c^6 + M^2c^4 + 8Mc^4 + Mc^2(M-4) + 24c^2 + 4\big)}{2(1+y)S_M(c)^4(2Mc^4 + 3Mc^2 - 4c^2 + 2)} > 0.$$

Differentiating a third time, $f_M^{(3)}$ is a positive multiple of a polynomial in c and M of degree 16 in c, all of whose coefficients become nonnegative after the substitution $M = 4 + m$; so $f_M^{(3)} > 0$.

If $y \leq 4/3$: since $f_M^{(3)} > 0$, the divided difference $(f_M(tu) - f_M(t) + (1-u)tf_M'(t))/(1-u)^2$ increases in u, so its value $1 - f_M(t) + tf_M'(t)$ at $u = 0$ bounds it below. Since $\int_0^1 (1-u)/(-\log u)\,\mathrm{d}u = \log 2$, this gives $\mathcal{A}f_M \geq (\log 2)(1 - f_M + tf_M')$. With the formulas above, the inequality $1 - f_M + tf_M' > M\partial_M f_M$ becomes $(1+2c^2)S_M(c) > 1 + 3c^2 + Mc^4/4$. Put $c^2 = y/M$ and raise to the fourth power; M^5 times the difference is

$$\begin{aligned}(M+2y)^4(M+4y+y^2) - M(M+3y+y^2/4)^4 = {}& \frac{y^2(y+4)(4-3y)}{8}M^3 \\ &+ \frac{y^3(y+4)(80-32y-y^2)}{16}M^2 \\ &+ \frac{y^4(y+4)(4032-688y-44y^2-y^3)}{256}M + 16y^5(y+4),\end{aligned}$$

and every bracket is nonnegative for $0 < y \leq 4/3$, the second one strictly.

If $y \geq 4/3$: let $\alpha = (2+y)/(2+2y) \in (1/2, 5/7]$. The curvature bound integrates to the minorant $f_M(tu) \geq f_M(t) + tf_M'(t)(u^\alpha - 1)/\alpha$, which touches at $u = 1$, so positivity of the kernel gives $\mathcal{A}f_M(t) \geq tf_M'(t)\log\Gamma(1+\alpha)/\alpha$. Finally $\log\Gamma(1+\alpha) > -(\log 2)(1-\alpha)/2$ on $[1/2, 5/7]$. The difference $g(\alpha)$ of the two sides is convex. Since ψ is concave, $\psi(12/7) \leq \psi(2) - (2/7)\psi'(2) = 1 - \gamma_{\mathrm{E}} - (2/7)(\zeta(2)-1) < 0.24 < (\log 2)/2$, so $g'(5/7) < 0$, and g is smallest at $5/7$. Taylor's formula at 2, whose fourth-order remainder is positive, gives $\log\Gamma(12/7) \geq -(2/7)(1-\gamma_{\mathrm{E}}) + (2/49)(\zeta(2)-1) + (8/1029)(\zeta(3)-1) > -0.0931 > -(\log 2)/7$, so $g(5/7) > 0$. Since $(1-\alpha)/\alpha = y/(2+y)$, this is the claim. □

This completes the proof of Theorem 8.1.

8.2. Other symbol families

PROPOSITION 8.7 (Even quadratic symbols). *Every analytic polynomial $f = \sum a_n z^n$ in one Gaussian variable has $\|f\|_1^2 \geq |a_0|^2 + |a_2|^2$. Hence $D(c_0 + c_2 z^2) \leq (|c_0|^2 + 4|c_2|^2)^{1/2} < \|A(c_0 + c_2 z^2)\|_{S_4}$ for $c_2 \neq 0$.*

PROOF. If $a_0 = 0$, extracting the coefficient gives $\|f\|_1 \geq |a_2|$. Otherwise pass to the even part $g(z^2)$. With $\beta = |a_2/a_0|$, Lemma 8.4 on each circle and $x = r^2$ give $\|f\|_1/|a_0| \geq \int_0^\infty h(\beta x)e^{-x}\,\mathrm{d}x = 1 + (\beta^2/2)(1 - e^{-2/\beta}) \geq \sqrt{1+\beta^2}$, as in the case

$n = 2$ above. Then $\langle f, \varphi\rangle = a_0\overline{c_0} + 2a_2\overline{c_2}$ and Cauchy–Schwarz. The Hankel block has $\|A\|_{S_4}^4 = |c_0|^4 + 8|c_0c_2|^2 + 24|c_2|^4$, which exceeds $(|c_0|^2 + 4|c_2|^2)^2$ by $8|c_2|^4$. □

The even case extends to every number of variables. Let $\varphi = c + z^T Cz$ with C complex symmetric, let $\lambda_j \geq 0$ be the Takagi singular values of C, and put $r^2 = \sum_j \lambda_j^2$.

PROPOSITION 8.8. *$D(\varphi) \leq (|c|^2 + 4r^2)^{1/2}$ and $\|A(\varphi)\|_{S_4}^4 = |c|^4 + 8|c|^2r^2 + 8r^4 + 16\sum_j \lambda_j^4$. Hence φ has no strict block at any $p \leq 4$ whenever $(\sum_j \lambda_j^2)^2 \leq 2\sum_j \lambda_j^4$, in particular whenever C has rank at most two.*

PROOF. A unitary change of variables preserves μ, analytic L^1 norms and the pairing, so we may take $\varphi = c + \sum_j \lambda_j z_j^2$. Applying Proposition 8.7 in one variable at a time and Minkowski's inequality in the others gives $\|f\|_1^2 \geq \sum_{\alpha\in\{0,2\}^d} |a_\alpha|^2$. Since $\langle f, \varphi\rangle = a_0\overline{c} + 2\sum_j a_{2e_j}\lambda_j$, Cauchy–Schwarz gives the bound for D. For the norm, $\|A(\varphi)\|_{S_4}^4 = \mathbb{E}|\varphi|^4$ by Proposition 7.11; with $X = \sum_j \lambda_j z_j^2$ one has $\mathbb{E}X = \mathbb{E}X^2 = 0$, $\mathbb{E}|X|^2 = 2r^2$ and $\mathbb{E}|X|^4 = 24\sum_j \lambda_j^4 + 8\sum_{j\neq k} \lambda_j^2\lambda_k^2$. The condition is $(|c|^2 + 4r^2)^2 \leq \|A\|_{S_4}^4$, and $(a^2 + b^2)^2 \leq 2(a^4 + b^4)$. □

So an even quadratic counterexample must have effective rank above two.

PROPOSITION 8.9 (Zero-free symbols). *Suppose that φ is entire without zeros and that $\|A(\varphi)\|_{S_2} < \infty$. Then $D(\varphi)$ equals the operator norm of $A(\varphi)$, so φ has no strict block at any p.*

PROOF. The growth bound $|\varphi(z)| \leq \|A\|_{S_2} e^{|z|^2/4}$ and Borel–Carathéodory make $\log\varphi$ a quadratic polynomial, and the Hilbert–Schmidt condition forces $\varphi = e^{az^2+bz+c}$ with $|a| < 1/4$. A Weyl translation, unitary on analytic L^2, isometric on analytic L^1 and compatible with products, removes b, and a rotation makes $a \geq 0$. Write $a = q/(2(1 + q^2))$ with $0 \leq q < 1$. The bounded function $\sqrt{1 + q^2}\exp(iq\Im z^2)$ represents e^{az^2}, so $D \leq \sqrt{1 + q^2}$; and $f = g^2$ with $g = e^{qz^2/2}$ gives $\langle f, \varphi\rangle/\|g\|_2^2 = \sqrt{1 + q^2}$, so $\|A\| \geq \sqrt{1 + q^2}$. □

For monomials the computation is explicit: $D(z^m) = m!/\Gamma(1+m/2)$ by circular averaging, and $A(z^m)$ is anti-diagonal. For even m this equals $\|A\|_{S_\infty}$; for $m = 2k+1$ a strict block needs $p > \log 2/\log(\sqrt{k!(k+1)!}/\Gamma(k+3/2)) \geq (1 - \log\pi/\log 4)^{-1}$. So no one-variable homogeneous symbol improves on Brevig and Perfekt.

Every barrier in this chapter is invariant under a group. Unitary changes of variables preserve μ. The Weyl operator $W_a f(z) = f(z - a)e^{\overline{a}\cdot z - |a|^2/2}$ is unitary on $L^2(\mu)$, and $W_a f\, W_a g = S(fg)$ with $Sh(z) = h(z - a)e^{2\overline{a}\cdot z - |a|^2}$, which the substitution $z = w + a$ shows to be an isometry of analytic $L^1(\mu)$ onto itself. Hence $H_\varphi(W_a f, W_a g) = \langle fg, S^*\varphi\rangle$, where moving the factor $e^{\overline{a}\cdot w}$ across the pairing as a translation gives

$$S^*\varphi(w) = e^{-|a|^2}e^{-\overline{a}\cdot w}\varphi(w + 2a).$$

So $A(S^*\varphi)$ is unitarily equivalent to $A(\varphi)$ and $D(S^*\varphi) = D(\varphi)$. Consequently Theorem 8.1, Proposition 8.8 and Theorem 8.10 hold for every symbol $ce^{-\overline{b}\cdot z}\psi(z+2b)$

with $\psi(z) = \varphi(Uz)$ obtained from theirs, U unitary, $b \in \mathbb{C}^d$ and $c \neq 0$; for instance $D \leq \|A\|_{S_4}$ for $e^{-\overline{b}\cdot z}(c_0 + c_1 \prod_j (z_j + 2b_j))$.

8.3. The L^1–L^4 question

PROBLEM 6 (The L^1–L^4 question). *Is $|\int f\overline{\varphi}\,\mathrm{d}\mu_d| \leq \|f\|_{L^1(\mu_d)}\|\varphi\|_{L^4(\mu_d)}$ for all analytic polynomials f, φ on $\mathbb{C}^d$ and all d? Equivalently, does every polynomial φ have a bounded representative ψ with $\|\psi\|_\infty \leq \|\varphi\|_{L^4}$?*

Since $\|\varphi\|_{L^4} = \|A(\varphi)\|_{S_4}$, a counterexample gives a strict block at every p slightly below 4, hence $p_* < 4$. A positive answer shows that 4 is the limit of the Gaussian method: it holds exactly when no Gaussian block is strict at any $p \leq 4$. It would not decide p_*. Nothing transfers back from the Gaussian model to the torus, and the torus analogue of the inequality is false already on $\mathbb{T}$: $\varphi = (1-z)^{-1/5}$ lies in $L^4(\mathbb{T})$ but not in BMOA, so its dilates have $D(\varphi_r)/\|\varphi_r\|_4 \to \infty$. Blocks on a fixed finite torus that are far from Gaussian, like the Ortega-Cerdà–Seip examples for bounded forms, are therefore not covered. The product, affine, zero-free and one-variable even quadratic symbols, and the even quadratic symbols of rank at most two, all satisfy the inequality. The natural remaining candidates are cumulative products $c_0 + \sum_j c_j Z_1 \cdots Z_j$ and quadratic symbols of large rank.

Near constants the inequality is sharp to second order: for $\varphi = 1 + \varepsilon b$ and $f = 1 + \varepsilon h$ with $b(0) = h(0) = 0$,

$$\log \frac{|\int f\overline{\varphi}\,\mathrm{d}\mu|}{\|f\|_{L^1}\|\varphi\|_{L^4}} = -\frac{\varepsilon^2}{4}\|h - 2b\|_2^2 + O(\varepsilon^3),$$

optimal but not positive at $h = 2b$.

8.4. Near $p = 2$

THEOREM 8.10 (Bounded degree and dimension). *For all $d, K \geq 1$ there is $\varepsilon_{d,K} > 0$ such that $D(\varphi) \leq \|A(\varphi)\|_{S_p}$ for $2 \leq p \leq 2 + \varepsilon_{d,K}$ and every polynomial φ on $\mathbb{C}^d$ of total degree at most K, against tests of any degree.*

PROOF. *Near constants.* Normalize $c_0 = 1$ and let $\tau = \max_{|\alpha|\geq 1}|c_\alpha|$. A test with quotient above 1 and $f(0) = 1$ has, by the Gaussian Carleman inequality $\sum \alpha! 2^{-|\alpha|}|a_\alpha|^2 \leq \|f\|_1^2$, all coefficients of degree at most K of size $O(\tau)$. Let g be the square root of f truncated at degree K. On tori of radius r with $r_j \leq R = c\tau^{-1/K}$, $|g - 1| \leq 1/2$, and the unimodular function $g/\overline{g}$ has analytic projection of degree at most K, so $\int_{\mathbb{T}^d}|f(r\zeta)|\,\mathrm{d}m \geq \int_{\mathbb{T}^d}|g(r\zeta)|^2\,\mathrm{d}m$. Integrating radially, the region outside the polydisc of radius R costs $O(e^{-R^2})$, so $D(\varphi) \leq \|A(\varphi)\|_{S_\infty} + Ce^{-b\tau^{-2/K}}$. On the other hand, put $m_\alpha = \alpha! c_\alpha$. The 2×2 submatrix of A in rows $0, e_j$ and columns $0, \alpha$ has determinant $(m_{\alpha+e_j} - m_{e_j} m_\alpha)/\sqrt{\alpha!}$ and norm at most 2, so $s_2(A) \geq \varepsilon/2$, where ε is the largest of these determinants over j and $|\alpha| \leq K$. With $\mu_j = m_{e_j}$, induction on $|\alpha|$ gives $m_\alpha = \mu^\alpha + O_K(\varepsilon)$ for $|\alpha| \leq K + 1$. At $\alpha = (K+1)e_j$ the left side vanishes, so $|\mu_j| \lesssim \varepsilon^{1/(K+1)}$, and then $\tau \lesssim \varepsilon^{1/(K+1)}$. Thus $s_2(A) \gtrsim \tau^{K+1}$, so $\|A\|_{S_p} - \|A\|_{S_\infty} \gtrsim \tau^{p(K+1)}$, which wins for small τ.

Away from constants. If $D(\varphi) = \|A(\varphi)\|_{S_2}$ for a nonconstant φ, a maximizing sequence would converge to $\varphi(2z)$ up to normalization and force equality in

Gaussian Carleman, which holds only for exponentials [52]. So $D < \|A\|_{S_2}$, and by compactness of the normalized symbol sphere outside the constant neighbourhood, $D \le (1-\eta_0)\|A\|_{S_2}$ there. Since $\operatorname{rank} A \le \binom{d+K}{K} = r$, $\|A\|_{S_p} \ge r^{1/p-1/2}\|A\|_{S_2}$, which exceeds $(1-\eta_0)\|A\|_{S_2}$ for p close to 2. □

PROPOSITION 8.11 (Quadratic compression). *For $\varphi = c + b^T Z + Z^T C Z$ on $\mathbb{C}^d$ with $C = C^T$, the nonzero singular values of $A(\varphi)$ are those of*

$$\begin{pmatrix} c & b^T & \sqrt{2}\|C\|_F \\ b & 2C & 0 \\ \sqrt{2}\|C\|_F & 0 & 0 \end{pmatrix}.$$

If $\operatorname{rank} C = r$, a unitary change of variables reduces φ to $r+1$ Gaussians, preserving D and all singular values. So quadratic symbols approaching $p = 2$ need $\operatorname{rank} C \to \infty$.

PROOF. The quadratic row of the full matrix is a vector q with $\|q\|^2 = 2\|C\|_F^2$; a unitary congruence in the quadratic block rotates it to one coordinate. Takagi factorization diagonalizes C, and Theorem 8.10 applies with $d = r+1$, $K = 2$. □

REMARK 8.12 (Entropy form of a strict block). For a finite-rank block, put $q_j = s_j^2/\|A\|_{S_2}^2$ and $\Delta = \log(\|A\|_{S_2}/D) \ge 0$. Then $D > \|A\|_{S_p}$ exactly when $\Delta < (1/2 - 1/p)H_{p/2}(q)$, with H_α the Rényi entropy; in particular rank r forces $\Delta < (1/2 - 1/p)\log r$. Both Δ and H_α add under tensor products. So a sequence of blocks succeeding at exponents $p \downarrow 2$ needs either Helson deficits $\Delta \to 0$ at bounded rank, which Theorem 8.10 rules out at bounded degree and dimension, or entropy growing faster than the deficit.

8.5. The Paley–Wiener threshold

Bampouras [6] proved that Nehari's theorem fails in S_p, $p > 4$, for Hankel operators on Paley–Wiener spaces $PW(\Omega)$ of convex $\Omega \subset \mathbb{R}^n$ with a curved boundary piece. The cap counting of Proposition 8.14 reorganizes his proof around a scale-uniform L^1 bound, which suggests that the coincidence with the $\mathbb{T}^\infty$ threshold comes from a different mechanism.

LEMMA 8.13 (Lossless cap sums). *Let $\hat{b}$ be supported in the unit ball, $\rho > 0$, and $|\xi_i - \xi_j| \ge 2\rho$ for $i \ne j$. Then for every integer $k > n/4$,*

$$\Big\| \sum_j c_j e^{2\pi i \langle \xi_j, x\rangle} \rho^n b(\rho x) \Big\|_{L^1(\mathbb{R}^n)} \le \|(1+|x|^2)^k b\|_2 \, \|(1+|x|^2)^{-k}\|_2 \, \|c\|_{\ell^2}.$$

PROOF. Let f be the sum. By Cauchy–Schwarz with the weight $w(x) = (1+|\rho x|^2)^k$, $\|f\|_1 \le \|wf\|_2 \|1/w\|_2$, and $\|1/w\|_2 = \rho^{-n/2}\|(1+|x|^2)^{-k}\|_2$, finite since $4k > n$. Multiplication by w is the operator $(1 - \rho^2\Delta/4\pi^2)^k$ on the Fourier side, which preserves supports, so the terms of wf have the disjoint Fourier supports $B(\xi_j, \rho)$ and are orthogonal. Each has L^2 norm $\rho^{n/2}\|(1+|x|^2)^k b\|_2$, by scaling, and Plancherel finishes. □

PROPOSITION 8.14. *Let N caps $\hat\varphi_i = \beta((\cdot - z_i)/\rho)$ in 2Ω satisfy the hypotheses of the lemma and give orthogonal Hankel blocks, and let a bound $\omega_\Omega = \chi_\Omega * \chi_\Omega$ on them. For $\psi = \sum \varphi_i$ and $p \ge 2$,*

$$\frac{\langle \psi, \psi\rangle}{\|\psi\|_1 \|H_\psi\|_{S_p}} \ge c\, N^{1/2-1/p} (\rho^n/a)^{1/p}.$$

PROOF. The caps have disjoint supports, so the pairing is $\sum_i \|\hat\varphi_i\|_2^2 = N\rho^n \|\beta\|_2^2$, up to the normalization of the Fourier transform. Lemma 8.13, with $c_j = 1$ and $\rho^n b(\rho\,\cdot) = \varphi_i$ translated in frequency, gives $\|\psi\|_1 \lesssim \sqrt{N}$, independently of ρ: the L^1 norm of a sum of separated caps grows only like the ℓ^2 norm of the coefficients. Bampouras's bound

$$\|H_{\varphi_i}\|_{S_p} \le \|\hat\varphi_i\, \omega^{1/p}\|_{L^{p'}},$$

with orthogonality, gives $\|H_\psi\|_{S_p} \lesssim N^{1/p} \rho^{n/p'} a^{1/p}$. □

With $\rho^n/a \approx N^{-\gamma}$ the ratio grows like $N^{1/2-1/p-\gamma/p}$, so the threshold is $p = 2(1+\gamma)$. The factor 2 is square-root cancellation in L^1, and γ is geometric. For caps of radius ε^2 along a fully curved boundary, $N \approx \varepsilon^{-(n-1)}$ caps fit, ω is of size ε^{n+1} on them, and $\gamma = 1$: this is Bampouras's threshold 4, in every dimension. Heuristically, partial curvature, with $k < n-1$ curved directions, gives $\gamma = (n-1)/k$ and larger thresholds, for instance 6 for $n = 3$, $k = 1$; this assumes the corresponding vanishing rate of ω, which we have not proved. The mechanism has no analogue on $\mathbb{T}^\infty$: there $\|H_\psi\| \ge \|H_\psi 1\| = \|\psi\|_2$ for every symbol, so a sum of symbols cannot split into orthogonal blocks, and decoupling must come from tensor products, where 4 is $\mathbb{E}|Z|^4 = 2$. Accordingly the $\mathbb{T}^\infty$ barriers do not transfer verbatim. The Paley–Wiener analogue of Problem 6 asks whether $|\langle h, \varphi\rangle| \le C\|h\|_1 \|H_\varphi\|_{S_4}$; at $p = 4$ the counting bound above stops growing.

At $p = 2$ the Paley–Wiener question goes back to Brevig and Perfekt.

PROBLEM 7 (Brevig and Perfekt [15]). *Does every Hankel operator on $PW(D)$, D the unit disc, that belongs to S_2 have a bounded symbol?*

The answer is yes. Bampouras and Perfekt [7] proved it for balls in every dimension, and Bampouras [5] then for every convex set free of affine lines. Both go through a duality that turns the problem into a Paley–Wiener form of Helson's inequality. Put $\omega(\zeta) = |D \cap (\zeta - D)|$, the density of $\xi + \eta$ for ξ, η uniform in D. Then $\|H_\varphi\|_{S_2}^2 = \int_{2D} |\hat\varphi|^2 \omega$, and Cauchy–Schwarz, with equality for $\hat\varphi = \hat h/\omega$, shows that the best constant C in $|\langle h, \varphi\rangle| \le C\|h\|_1 \|H_\varphi\|_{S_2}$ is given by

$$C^2 = \sup_h \frac{1}{\|h\|_1^2} \int_{2D} \frac{|\hat h|^2}{\omega}, \tag{8.2}$$

the supremum taken over nonzero $h \in L^1(\mathbb{R}^2)$ with $\hat h$ supported in $2\overline{D}$. By the Hahn–Banach theorem, $C < \infty$ is exactly a positive answer to Problem 7. Near $\partial(2D)$ the weight ω vanishes like $\operatorname{dist}(\zeta, \partial(2D))^{3/2}$, so $1/\omega$ is not integrable and (8.2) is not trivial. Bampouras and Perfekt construct $\psi \in L^\infty(\mathbb{R}^2)$ whose Fourier transform equals $1/\omega$ on $2D$, as a distribution, from the boundary values of $(1 - |x|^2 \pm i0)^{-3/2}$;

then, for $\hat{h} \in C_c^\infty(2D)$,

$$\int_{2D} \frac{|\hat{h}|^2}{\omega} = \int |\hat{h}|^2 \hat{\psi} = \langle h * \psi, h \rangle \le \|\psi\|_\infty \|h\|_1^2,$$

so $C^2 \le \|\psi\|_\infty$. Bampouras's theorem for general convex $\Omega \subset \mathbb{R}^n$ comes with the explicit constant $2^n (2n)!\,\Gamma(n/2+1)^2/(n!)^3 \le 8^n$ in Helson's inequality, for the weight $|\Omega \cap (2x - \Omega)|$. The constant cannot be 1, already at $p = 2$; this is the first assertion below, and with the symbol ψ it pins down the growth of the constant on products of discs. On $\mathbb{T}^\infty$, by contrast, Helson's inequality gives the constant 1 in S_2 (Theorem 5.19).

PROPOSITION 8.15 (Disc blocks). *Let $D \subset \mathbb{R}^2$ be the unit disc. For every $p \ge 2$ there are Schwartz functions h, φ with Fourier supports in $2\overline{D}$ and*

$$|\langle h, \varphi \rangle| > \|h\|_1 \|H_\varphi\|_{S_p}.$$

Consequently, for every $p \ge 2$, the best constant in $|\langle h, \varphi \rangle| \le C \|h\|_1 \|H_\varphi\|_{S_p}$ on $PW(D^m)$ is at least exponential in m (for $p > 4$ it is already infinite, by Bampouras [6]).

PROOF. Let $v = |D|$ and $k = \check{\chi}_D$, and put $h_0 = k^2$. Then $\widehat{h_0} = \chi_D * \chi_D$ is supported in $2\overline{D}$, $h_0 \ge 0$, $h_0(0) = v^2$ and $\|h_0\|_1 = \|\chi_D\|_2^2 = v$; moreover $h_0 > 0$ almost everywhere, since k is real analytic and not identically zero. Let $\beta \ge 0$ be smooth, even and nonzero, with support in $2D$ away from the origin, and $h_1 = \check{\beta}$. Then h_1 is real, $\int h_1 = \beta(0) = 0$ and $h_1(0) = \int \beta > 0$. By dominated convergence, $\|h_0 + \varepsilon h_1\|_1 = v + \varepsilon \int h_1 + o(\varepsilon) = v + o(\varepsilon)$, while $(h_0 + \varepsilon h_1)(0) = v^2 + \varepsilon h_1(0)$. So $h = h_0 + \varepsilon h_1$ satisfies $h(0) > v\|h\|_1$ for small $\varepsilon > 0$. Replacing h by $h(rx)\eta(\delta x)$, with $\hat{\eta}$ supported in D, $\eta(0) = 1$, $r \uparrow 1$ and $\delta < 2(1 - r)$, makes it Schwartz with the same Fourier support condition and preserves the strict inequality, since these functions converge to h in L^1 and have the same value at 0.

For the symbol take $\hat{\varphi}_\ell = a_\ell \in C_c^\infty(2D)$ with $0 \le a_\ell \le 1$ and $a_\ell \to 1$ on $2D$. The kernels $a_\ell(\xi + \zeta)$ converge in $L^2(D \times D)$ to the constant 1, whose operator has rank one and norm v; hence $\|H_{\varphi_\ell}\|_{S_p} \to v$ for every $p \ge 2$, while $\langle h, \varphi_\ell \rangle = \int \hat{h}\, a_\ell \to h(0)$. For large ℓ the quotient exceeds 1. As in Ortega-Cerdà and Seip [44], on D^m the tensor powers $h^{\otimes m}$ and $\varphi^{\otimes m}$ have pairing, L^1 norm and S_p norm equal to the mth powers of those on D. □

The proof uses only that $\check{\chi}_D$ is real, so that $h_0 \ge 0$, and that $2D \setminus \{0\}$ contains the support of a nonnegative bump. It therefore applies verbatim to every convex body symmetric about a point, after a modulation that moves the center to 0; polygons with this symmetry included. For such bodies the S_p Nehari constant on $PW(\Omega^m)$ is at least exponential in m for every $p \ge 2$, although for a polygon it is finite when $m = 1$, by the theorem of Bampouras and Perfekt below.

COROLLARY 8.16 (Polydiscs). *Let ψ be the symbol of Bampouras and Perfekt [7], and let C_m be the best constant in $|\langle h, \varphi \rangle| \le C_m \|h\|_1 \|H_\varphi\|_{S_2}$ on $PW(D^m)$. Then*

$$C_1^m \le C_m \le \|\psi\|_\infty^{m/2}, \qquad C_1 > 1.$$

PROOF. The lower bound is the tensor argument of Proposition 8.15 applied to near extremals for C_1, and $C_1 > 1$ is that proposition at $p = 2$. For the upper bound, the density of $\xi + \eta$ for ξ, η uniform in D^m is $\omega^{\otimes m}$, so (8.2) holds on D^m with this weight. The Fourier transform of $\psi^{\otimes m}$ is $\hat{\psi}^{\otimes m}$, which equals $(1/\omega)^{\otimes m}$ on $(2D)^m$, since tensor products of test functions are dense there, and $\|\psi^{\otimes m}\|_\infty = \|\psi\|_\infty^m$. The computation above, with h ranging over the dense set of $h \in L^1$ with $\hat{h} \in C_c^\infty((2D)^m)$ and Fatou's lemma for the rest, gives $C_m^2 \le \|\psi\|_\infty^m$. Bampouras's constant, with $n = 2m$, gives an upper bound of the same exponential form. □

So there is no Paley–Wiener analogue of Helson's dimension-free inequality: on D^m the constant grows exactly exponentially, and on D^∞ it is infinite. For balls the tensor argument is not available, and the argument of Proposition 8.15 gives only a constant larger than 1; whether the constant for the ball B^n grows with n is open.

For polygons there is no threshold at all. Bampouras and Perfekt [8, Corollary 1.2] prove that every Hankel operator on $PW(P)$, P a polygon, that lies in some S_p, $p < \infty$, has a bounded symbol, by lifting a theorem on the product Hardy space of two half-planes. This is the limit $k = 0$ of the counting above, where the threshold $2(1 + (n-1)/k)$ is infinite, which suggests that the obstruction above S_2 is curvature.

For $2 < p \le 4$ the disc remains open. Bampouras and Perfekt show that for the disc and $2 \le p < 6$, where the Besov description of S_p is available, Nehari's theorem for S_p implies $\int_{2D} |\hat{h}|^{p'}/\omega \lesssim \|h\|_1^{p'}$. A function h violating this inequality for some $p' \in [4/3, 2)$ would therefore show that Nehari's theorem fails for $S_p(PW(D))$ at the dual exponent p. Their proof of the case $p' = 2$ uses L^2 through the identity $\int |\hat{h}|^2 \hat{\psi} = \langle h * \psi, h \rangle$, which has no analogue for $p' < 2$.

Open

(1) Is $p_* = 4$? By Proposition 7.8 this is the same as asking whether multiplicative Hankel forms in S_4 have bounded symbols.
(2) Problem 6, which decides whether Gaussian methods can go below 4.
(3) A dimension- and degree-free interval $(2, 2 + \varepsilon)$ with no strict blocks, or a family of blocks with exponents tending to 2.
(4) Nehari's theorem for $S_p(PW(D))$, $2 < p \le 4$; by Bampouras and Perfekt it would give $\int_{2D} |\hat{h}|^{p'}/\omega \lesssim \|h\|_1^{p'}$.
(5) Does the S_2 constant for the ball B^n grow with n?

Part 4

The borderline: the multiplicative Hilbert matrix

CHAPTER 9

Factorization and π

For a Dirichlet series $f = \sum a_n n^{-s}$ write $c_n = 1/(\sqrt{n}\log n)$ and

$$\ell(f) = \sum_{n\geq 2} \frac{a_n}{\sqrt{n}\log n} = \int_0^\infty \big(f(1/2+\sigma) - a_1\big)\,\mathrm{d}\sigma.$$

Let C be the norm of ℓ on $\mathscr{H}^1_0$, the series with $a_1 = 0$. Write $r(\sigma) = (p^{-1/2-\sigma})_p$ for the prime curve, F for the Bohr lift of f, and $J(F) = \int_0^\infty |F(r(\sigma))|\,\mathrm{d}\sigma$. Let C_J be the least constant in $J(F) \leq C_J \|F\|_1$ on H^1_0. Since $|\ell(f)| \leq J(F)$, we have $C \leq C_J$.

Part IV is about four inequalities that share the constant π in one variable (Proposition 9.1).

	One variable	On $\mathbb{T}^\infty$
Hilbert form on $H^2 \times H^2$	π	π [16]; also with the row $m = 1$ adjoined (Theorem 9.2)
Signed functional ℓ on H^1_0	π	finite [45]; $\pi \leq C \leq 5.515$ (Proposition 9.14)
Fejér–Riesz functional J on H^1_0	π	finite [45]; $C \leq C_J \leq 5.515$
Hardy: $\sum \lvert a_n\rvert c_n$ on H^1	π	open (Problem 9); below π on primes up to 10^{216}

PROBLEM 8 (The sharp constant). *Is $C = \pi$, the norm of the multiplicative Hilbert matrix on $\mathscr{H}^2$?*

PROBLEM 9 (Perfekt [45]). *Is $|a_1| + \sum_{n\geq 2} |a_n|/(\sqrt{n}\log n) \lesssim \|f\|_{\mathscr{H}^1}$ for every Dirichlet polynomial f?*

Several different gaps involving π appear in Part IV, and they answer different questions.

- *Operator norm against symbol norm.* The Hilbert matrix has norm π; its smallest symbol has norm $C \geq \pi$. Problem 8 asks whether these agree.
- *Finitely many primes.* On d primes the constant C_d exceeds the H^2 benchmark π^*_d by about six per cent, and for two large primes by $\pi/(2\sqrt{2}) - 1 \approx 11$ per cent (Proposition 9.10). This is an effect of linear symbols at a scale where both constants are far below π. It says nothing about C: for functions of the primes below 10^{216} even the absolute sum is below π.
- *The Gram constant.* The Hellinger–Bessel method gives $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ near the critical line. That is a limitation of the method, which discards phases, not an excess of C.

- *The Gaussian constant.* $C \geq C_{\mathrm{Gauss}} \geq \pi$ (chapter 11). Here a strict inequality $C_{\mathrm{Gauss}} > \pi$ would be an actual excess, and it would survive the removal of every continuous symbol.

In one variable the four constants coincide for one reason: every H^1 function is a product of two H^2 functions with the same norm, and Hilbert's inequality bounds the product. On $\mathbb{T}^\infty$ the first row survives with the same constant, while factorization fails, even in its weak form, because Nehari's theorem does [44]. Part IV measures what that failure costs, row by row. This chapter treats the H^2 side: the multiplicative Hilbert matrix and its near-extremal vectors (section 9.2), and the equivalent forms of $C = \pi$, each with the reason it is not automatic (section 9.3), and it ends with explicit bounds: $\pi \leq C \leq 5.515$, and a constant below π for functions of the primes below 10^{216} (section 9.4). Chapter 10 treats the relatives of ℓ: absolute values, other weights and exponents, and partial sums; it shows in passing that a counterexample must mix degrees. Chapter 11 then studies the regime where C is decided, a Gaussian scaling limit in which π is very probably exceeded.

Brevig, Perfekt, Seip, Siskakis and Vukotić introduced the multiplicative Hilbert matrix, showed that its norm on $\mathscr{H}^2$ is π, and observed that it has a bounded symbol exactly when $C < \infty$ [16]; the question became Problem 3.2 of Saksman and Seip [49]. Partial answers replaced $\mathscr{H}^1$ by $\mathscr{H}^p$, $p > 1$, by Bayart and Brevig [9], and $\log n$ by $(\log n)^\beta$, $\beta > 1$, by Bondarenko, Brevig, Saksman and Seip [11]. Perfekt [45, Theorem 1.1 and Corollary 1.2] proved $C_J < \infty$, with a constant of order $3 \cdot 10^5$; so $C < \infty$ and the multiplicative Hilbert matrix has a bounded symbol. He also noted that Problem 9. Perfekt and Pushnitski proved that the multiplicative Hilbert matrix has purely absolutely continuous spectrum $[0, \pi]$ [47]. Results below that carry no other attribution are, to our knowledge, new.

The *sharp Fejér–Riesz conjecture* $C_J = \pi$ implies $C = \pi$. Both are very probably false (remark 11.10).

9.1. One variable

PROPOSITION 9.1. *Let* $f = \sum_{k\geq 0} a_k z^k \in H^1(\mathbb{T})$, *and let* $g = \sum_j b_j z^j$ *and* $h = \sum_k d_k z^k$ *belong to* $H^2(\mathbb{T})$.

(1) (Hilbert) $|\sum_{j,k} b_j d_k/(j+k+1)| \leq \pi \|g\|_2 \|h\|_2$.
(2) (Fejér–Riesz) $\int_0^1 |f(x)|\,\mathrm{d}x \leq \pi \|f\|_1$.
(3) (Hardy) $\sum_k |a_k|/(k+1) \leq \pi \|f\|_1$.
(4) $|\sum_k a_k/(k+1)| \leq \pi \|f\|_1$.

In each, π *is best possible.*

PROOF. (1) The entries are positive, so we may take $b, d \geq 0$. Put $A = j + 1/2$. The function $y \mapsto 1/((A+y)\sqrt{y})$ is convex on $(0,\infty)$, so its value at $k+1/2$ is at most its integral over $(k, k+1)$, and

$$\sum_{k\geq 0} \frac{(k+1/2)^{-1/2}}{j+k+1} \leq \int_0^\infty \frac{\mathrm{d}y}{(A+y)\sqrt{y}} = \pi (j+1/2)^{-1/2}.$$

Schur's test with the weight $(k+1/2)^{-1/2}$ on rows and columns gives the bound. The same test applies to the smaller entries $\rho^{j+k+1}/(j+k+1)$, $0<\rho<1$.

(2) Write $f=IO$ with I inner and O outer, and put $g=IO^{1/2}$, $h=O^{1/2}$. Then $f=gh$ and $\|g\|_2^2=\|h\|_2^2=\|f\|_1$. For $0<\rho<1$, $\int_0^\rho |g(x)|^2\,\mathrm{d}x = \sum_{j,k} b_j\overline{b_k}\rho^{j+k+1}/(j+k+1) \le \pi\|g\|_2^2$ by (1), and similarly for h. Cauchy–Schwarz and $\rho\to 1$ give (2).

(3) With $f=gh$ as in (2), $|a_k|\le\sum_{i+j=k}|b_i||d_j|$, so (1) applied to $(|b_i|)$ and $(|d_j|)$ gives (3).

(4) For polynomials $\sum_k a_k/(k+1)=\int_0^1 f(x)\,\mathrm{d}x$; both sides are continuous on H^1 by (2) and (3), so (4) follows from either.

Sharpness. Let $b_k=(k+1)^{-1/2-\varepsilon}$, $g=\sum b_k z^k$ and $f=g^2$. The coefficients of f are positive, $\|f\|_1=\|g\|_2^2$, and each left side in (1)–(4), with $h=g$, equals $Q=\sum_{j,k} b_jb_k/(j+k+1)$. Since $j+k+1\le(j+1)+(k+1)$ and $y\mapsto y^{-1/2-\varepsilon}/(J+y)$ decreases, with $J=j+1$,

$$\sum_k \frac{b_k}{j+k+1} \ge \int_1^\infty \frac{y^{-1/2-\varepsilon}\,\mathrm{d}y}{J+y} \ge J^{-1/2-\varepsilon}\Big(I_\varepsilon - \frac{J^{-1/2+\varepsilon}}{1/2-\varepsilon}\Big),$$

where $I_\varepsilon=\int_0^\infty u^{-1/2-\varepsilon}(1+u)^{-1}\,\mathrm{d}u$. Summing against $b_j=J^{-1/2-\varepsilon}$ gives $Q\ge I_\varepsilon\|g\|_2^2-O(1)$. As $\varepsilon\to 0$, $\|g\|_2^2=\sum_J J^{-1-2\varepsilon}\to\infty$ and $I_\varepsilon=\pi/\cos(\pi\varepsilon)\to\pi$, so $Q/\|f\|_1\to\pi$. □

The dictionary to Dirichlet series is $z^k\leftrightarrow n^{-s}$, $\int_0^1 x^k\,\mathrm{d}x=1/(k+1)\leftrightarrow\int_0^\infty n^{-1/2-\sigma}\,\mathrm{d}\sigma=c_n$, and $1/(j+k+1)\leftrightarrow c_{mn}$. Only the first step of the proof, the factorization $f=gh$, has no counterpart on $\mathbb{T}^\infty$.

9.2. The multiplicative Hilbert matrix

Let $\mathscr{M}=(c_{mn})_{m,n\ge2}$, of norm π [16], and let $\mathscr{R}=(c_{mn})_{m\ge1,n\ge2}$ be $\mathscr{M}$ with the row $m=1$ adjoined. For $g\in H^2$, $h\in H^2_0$ one has $\ell(gh)=\langle\mathscr{R}\hat{h},\bar{\hat{g}}\rangle$, so $C\ge\|\mathscr{R}\|$.

THEOREM 9.2. *$\|\mathscr{R}\|=\pi$. If π_d^* is the norm of $\mathscr{R}$ restricted to the integers $\mathcal{S}_d$ generated by the first d primes, then $\pi_d^*\uparrow\pi$, and*

$$\frac{C}{\pi}=\lim_{d\to\infty}\frac{C_d}{\pi_d^*},$$

where C_d is the norm of ℓ on the same restricted $\mathscr{H}_0^1$.

PROOF. We apply Schur's test with weights $u_n=(n\log n)^{-1/2}$ on columns and $v_1=1$, $v_m=(m\log m)^{-1/2}$ on rows; it suffices that $\sum_n c_{mn}u_n\le\pi v_m$ and $\sum_m c_{mn}v_m\le\pi u_n$.

Rows $m\ge2$. With $x=\log m$, the summand $1/(t\sqrt{\log t}(x+\log t))$ decreases in t, so $v_m^{-1}\sum_n c_{mn}u_n\le\sqrt{x}\int_1^\infty dt/(t\sqrt{\log t}(x+\log t))=2\sqrt{x}\int_0^\infty dr/(x+r^2)=\pi$.

Row $m=1$. With $g(t)=1/(t\log^{3/2}t)$ decreasing, $\sum_n g(n)\le g(2)+g(3)+2/\sqrt{\log 3}<3.091<\pi$, using $\log 2>0.69$ and $\log 3>1.09$.

Columns. With $y = \log n$ and $f_y(t) = 1/(t\sqrt{\log t}(\log t + y))$,

$$u_n^{-1}\sum_m c_{mn}v_m = \frac{1}{\sqrt{y}} + \sqrt{y}\sum_{m\ge 2} f_y(m) \le \frac{1}{\sqrt{y}} + \sqrt{y}\Big(\frac{\pi}{\sqrt{y}} - \int_1^2 f_y + f_y(2)\Big),$$

since $\int_1^\infty f_y = \pi/\sqrt{y}$. So it suffices that $\int_1^2 f_y - f_y(2) \ge 1/y$ for $y \ge \log 2$. With $a = \log 2$, $A = \sqrt{a}$ and $u = \sqrt{a/y} \in (0,1]$, this reads

$$\Psi(u) = 2A\frac{\arctan u}{u} - \frac{1}{2A(1+u^2)} - 1 \ge 0.$$

Put $t = u^2$. Since $\arctan u/u = \int_0^1 (1+tx^2)^{-1}\,\mathrm{d}x$, completing the square gives $\arctan u/u \ge 2\int_0^1 P - \int_0^1 (1+tx^2)P^2$ for every polynomial P, and the best P in the span of 1 and x^2 gives

$$\frac{\arctan u}{u} \ge \frac{105 + 55t}{105 + 90t + 9t^2}.$$

Substituting, $\Psi \ge 0$ follows from the positivity on $[0,1]$ of

$$(420A^2 - 210A - 105) + (640A^2 - 390A - 90)t + (220A^2 - 198A - 9)t^2 - 18At^3,$$

which is concave there, since $220A^2 - 198A - 9 < 0$, and positive at $t = 0$ and $t = 1$; the bounds $0.8325 < A < 0.8326$ suffice for all three signs.

Limits. Monotonicity in d is clear, and every finitely supported test vector lives in some $\mathcal{S}_d$. The same density argument gives $C_d \uparrow C$. □

Equivalently $\mathscr{R}^*\mathscr{R} = \mathscr{M}^2 + c\otimes c \le \pi^2$: the rank-one term is absorbed although it is not zero. Since $\pi_d^* \le C_d < 1.91$ for $d \le 7$ (Proposition 9.14), finite-prime numerics are far from the regime in which C/π is decided.

9.2.1. Near-extremal vectors

Let $N : \ell^2(\{2,3,\dots\}) \to L^2(0,\infty)$, $Nb(\sigma) = \sum b_n n^{-1/2-\sigma}$, so $\mathscr{M} = N^*N$, and let S_m be the isometry $e_n \mapsto e_{mn}$. Then $NS_m = w_m N$ with $w_m(\sigma) = m^{-1/2-\sigma}$.

THEOREM 9.3 (Slow variation). *If b_k are unit vectors with $\langle \mathscr{M}b_k, b_k\rangle \to \pi$, then for all positive integers m, m'*

$$\sum_n b_k(mn)\overline{b_k(m'n)} \to (mm')^{-1/2}.$$

PROOF. Put $g_k = Nb_k$. From $\|(\pi - \mathscr{M})b\|^2 \le \pi\langle(\pi-\mathscr{M})b, b\rangle$, $b_k = \pi^{-1}N^*g_k + o(1)$. Weak limits are eigenvectors, and $\mathscr{M}$ has none [47], so $b_k \rightharpoonup 0$. As N restricted to (δ,∞) is Hilbert–Schmidt, $\int_\delta^\infty |g_k|^2 \to 0$, hence $\|w_m g_k - m^{-1/2}g_k\| \to 0$. Therefore $S_m^* b_k = \pi^{-1}N^* w_m g_k + o(1) = m^{-1/2}\pi^{-1}\mathscr{M}b_k + o(1) = m^{-1/2}b_k + o(1)$. Hence $\sum_n b_k(mn)\overline{b_k(m'n)} = \langle S_m^* b_k, S_{m'}^* b_k\rangle \to (mm')^{-1/2}$. □

Since $\mathscr{M}$ is real and positive, $|\ell(h^2)| = |\langle \mathscr{M}b, \bar b\rangle| \le \langle \mathscr{M}b, b\rangle$ for $h = \sum b_n n^{-s}$, so the theorem applies to every sequence with $\|h_k\|_2 = 1$ and $|\ell(h_k^2)| \to \pi$.

COROLLARY 9.4 (Marginals and fixed factors). *Let S be a finite set of primes and $r_0 = (p^{-1/2})_{p\in S}$. For such h_k:*

(1) *the marginal of* $|h_k|^2 dm$ *on* $\mathbb{T}^S$ *tends weakly to the Poisson measure* $P_{r_0}\,\mathrm{d}m_S$*;*
(2) *for every polynomial* $u \neq 0$ *in the variables of* S*, overlapping those of* h_k *or not,*

$$\limsup_k \frac{|\ell(uh_k^2)|}{\|uh_k^2\|_1} \le \pi\,\frac{|u(r_0)|}{\int |u| P_{r_0}\,\mathrm{d}m_S} \le \pi.$$

Proof. (1) For coprime S-integers m, m', the Fourier coefficient at $\kappa(m)-\kappa(m')$ is $\sum_t b_k(mt)\overline{b_k(m't)}$, and every character of $\mathbb{T}^S$ arises from exactly one such reduced pair. (Without coprimality the identity is false: $m = m' = 2$ gives the constant character.) (2) $\ell(uh_k^2) = \int_0^\infty u(r(\sigma))g_k(\sigma)^2\,\mathrm{d}\sigma$ and $|g_k|^2$ concentrates at $\sigma = 0$, so this is $u(r_0)\ell(h_k^2) + o(1)$; the denominator is $\int |u||h_k|^2\,\mathrm{d}m \to \int |u| P_{r_0}\,\mathrm{d}m_S$ by (1). □

For a single low prime no extremality is needed: if $g \in H^1$ depends on one prime and $h \in H^2_0$ on the others, then $|\ell(gh^2)| \le \pi\|gh^2\|_1$, by inner-outer factorization of g and Theorem 9.2.

9.3. Equivalent forms and obstructions

Let $G = \widehat{\mathbb{Q}_{>0}} \cong \mathbb{T}^\infty$ and $P = \{q \in \mathbb{Q}_{>0} : q \ge 1\}$, a total positive cone containing $\mathbb{N}$. For $b : P \to \mathbb{C}$ let $\mathcal{H}_b(u,v) = \sum_{q,r\in P} \hat u(q)\hat v(r) b(qr)$ on $H^2_P(G)$.

Theorem 9.5 (Completion principle).

$$C = \min\bigl\{\|\mathcal{H}_b\| : b(n) = c_n \text{ for all integers } n \ge 2\bigr\},$$

with $b(1)$ *free. So* $C = \pi$ *if and only if the multiplicative Hilbert coefficients extend to the ordered rationals with Hankel norm* π*.*

Proof. A Hahn–Banach extension Λ of ℓ to $L^1(G)$ gives $b(q) = \Lambda(\chi_q)$ with $\|\mathcal{H}_b\| \le C$. Conversely, the Nehari theorem for ordered groups [53] gives a symbol of norm $\|\mathcal{H}_b\|$ that represents ℓ on $\mathscr{H}^1_0$. □

Remark 9.6 (What the reformulation does and does not give). By compactness, $C > \pi$ would be witnessed by one finite set $F \subset P$ and one trace-class matrix X on F with $\sum_{qr=t} X(q,r) = 0$ for $t \notin \mathbb{N}_{\ge 2}$; finitely supported rank-one $X = u \otimes v$ never works: by Gauss's lemma in the Laurent ring $u = z^\gamma u_0$ and $v = z^\delta v_0$ with $\gamma + \delta \ge 0$, so moving $z^{\gamma+\delta}$ onto v gives integer spectra, and the form is a compression of $\mathscr{R}$. By Helson–Lowdenslager factorization [28], $C = \pi$ is equivalent to $|\sum_n c_n \sum_{qr=n} \hat u(q)\hat v(r)| \le \pi\|u\|_2\|v\|_2$ for $u, v \in H^2_P(G)$ whose product is an integer Dirichlet polynomial; the factors need not have spectrum in $\mathbb{N}$, already for $z_1 + z_2$. The natural completions all fail: continuous interpolation $b(q) = F(\log q)$, the integrated flat representatives, and $b = 0$ off $\mathbb{N}$ are unbounded, and no completely positive Schur damping fixes the integer entries non-trivially. A bounded completion must cancel the non-square-summable entries $\sum_m c_{Am}c_{Bm}$, $2 \le A < B$ coprime, through its non-integer layers.

9.3.1. Curve factorization

PROPOSITION 9.7 (Curve factorization). *For $F \in H^1_0$ let $\Phi(F)$ be the infimum of $\|G\|_2\|H\|_2/\|F\|_1$ over $G \in H^2_0$, $H \in H^2$ with $|F(r(\sigma))| \le |G(r(\sigma))H(r(\sigma))|$ for all $\sigma > 0$. Then $J(F) \le \pi\Phi(F)\|F\|_1$.*

PROOF. $J(F) \le \sum_{m\ge1,n\ge2} c_{mn}|\hat H(m)||\hat G(n)| \le \pi\|G\|_2\|H\|_2$ by Theorem 9.2, which applies to absolute values since the matrix is positive. □

In one variable inner-outer factorization gives $\Phi \le 1$. On $\mathbb{T}^\infty$ global weak factorization fails [44], but Φ asks only for a majorant on the curve, which is far weaker.

PROPOSITION 9.8 (Curve factorization fails). *For every $F \in H^1_0$ and $\sigma > 0$, with $\zeta_\sigma = \zeta(1+2\sigma)$,*

$$\Phi(F)\|F\|_1 \ge \frac{|F(r(\sigma))|}{\sqrt{\zeta_\sigma(\zeta_\sigma-1)}}.$$

Consequently $\Phi(K_{r(\sigma)}(K_{r(\sigma)}-1)) > 1$ for every $\sigma > 0$, and $\Phi(F) > 1.0151$ for $f = 2^{-s} + (2/5)3^{-s}$.

PROOF. For admissible $G \in H^2_0$, $H \in H^2$, $|G(r)H(r)| \le \|G\|_2\sqrt{\zeta_\sigma - 1}\cdot\|H\|_2\sqrt{\zeta_\sigma}$. For $F = K(K-1)$ with $K = K_{r(\sigma)}$, $F(r(\sigma)) = \zeta_\sigma(\zeta_\sigma-1)$ while $\|F\|_1 < \|K\|_2\|K-1\|_2 = \sqrt{\zeta_\sigma(\zeta_\sigma-1)}$, strictly, because $|1-1/K|$ is not constant. For the binomial, $\|F\|_1 = \int|1+2z/5|\,\mathrm{d}m \le 1.04048$, and at $\sigma = 3$, $|f(7/2)| = 0.0969\ldots$ against $\sqrt{\zeta(7)(\zeta(7)-1)} \le 0.09179$. □

The mechanism is again Ortega-Cerdà–Seip: for $2^{-s} + \varepsilon 3^{-s}$ the evaluation gains to first order in ε at a suitable point while the L^1 norm pays only to second order. Near extremizers the picture is different: for the two-prime near-extremizer of Proposition 9.14, the convex program for Φ and its dual agree at $\Phi \approx 0.994$. So the reduction survives in the weaker form that matters: $C = \pi$ if $\Phi(F_k) \to 1$ along some extremizing sequence.

REMARK 9.9 (A warning from one variable). Building a symbol from the phases of extremal squares, $\psi = \pi F/|F|$, does not work directly. In one variable, with $F = (1-rz)^{-1}$ and $r \to 1$, the limit symbol has analytic coefficients $1/(k+1/2)$, so it represents $\sum a_k/(k+1/2)$ rather than $\sum a_k/(k+1)$. The multiplicative analogue is the family $\ell_c(f) = \sum a_n/(\sqrt{n}(\log n + c)) = \int_0^\infty e^{-c\sigma} f(1/2+\sigma)\,\mathrm{d}\sigma$, all of H^2 norm π. A phase-limit symbol would settle one member ℓ_{c_*}; passing to $c = 0$ is the completion problem of Theorem 9.5.

9.3.2. Small primes: a linear-symbol effect

Numerically $C_d/\pi^*_d \approx 1.06$ for $d = 2, 3, 4$. For primes $p < q$ let $C_{p,q}$ and $\pi^*_{p,q}$ be the two-prime analogues.

PROPOSITION 9.10. *As $p, q \to \infty$ with $q/p \to 1$, $C_{p,q}/\pi^*_{p,q} \to \pi/(2\sqrt{2}) = 1.1107\ldots$.*

PROOF. With $a = c_p$, $b = c_q$ the functional is $a\hat f(p)+b\hat f(q)+R$, where R has square-summable coefficients of size $o(a)$ and so dual norm $o(a)$. Hence $C_{p,q} = D(az_1 + bz_2) + o(a)$, where D is the dual norm, and similarly $\pi^*_{p,q} = \sqrt{a^2+b^2} + o(a)$. For $a = b$, $D(z_1+z_2) = 2/\|z_1+z_2\|_1 = \pi/2$, with extremal symbol $(z_1+z_2)/|z_1+z_2|$. □

(p,q)	$(2,3)$	$(5,7)$	$(11,13)$	$(101,103)$	$(1009,1013)$
$C_{p,q}/\pi^*_{p,q} \ge$	1.0617	1.0905	1.1038	1.1101	1.1107

The same ratio appears exactly as a factorization gap: for $q_d = \prod_{j\le d}(z_{2j-1} + z_{2j})$,

$$\|q_d\|_1 = (4/\pi)^d, \qquad \|q_d\|_{H^2\odot H^2_0} = 2^{d/2}, \qquad \frac{\|q_d\|_{H^2\odot H^2_0}}{\|q_d\|_1} = \Big(\frac{\pi}{2\sqrt{2}}\Big)^d.$$

Since $|\ell(f)| \le \pi\|f\|_{H^2\odot H^2_0}$ by Theorem 9.2, this shows that the rectangular norm alone can never prove $C = \pi$: the projective norm and the H^1 norm are not equivalent on H^1_0.

9.4. Perfekt's theorem and explicit bounds

Corollary 2.6 reproves Perfekt's theorem with the explicit bound $\pi \le C < 9$; this section lowers 9 to 5.515, and below π on primes up to 10^{216}.

REMARK 9.11 (The limit of $|F|$-only methods). For Lebesgue measure on $(0,\delta)$ along the curve, the Gram kernel of Theorem 2.1 is asymptotically $2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$, homogeneous of degree -1, whose Mellin multiplier is $\Gamma(1/4)^2/\sqrt{2\pi} = 5.2441\ldots$. The H^2 analogue gives the Carleman kernel $1/(\sigma+\tau)$, of norm π. So, up to uniformity of the asymptotics, the unrestricted Gram estimate of Theorem 2.1 stops at 5.244. This does not rule out arguments that use the special structure of $u = |F|^{1/2}$. Since J is subadditive and $\|\sum F_i\|_1 \ge \sum\|F_i\|_1 - 2\sum_{i<j}(\|F_i\|_1\|F_j\|_1)^{1/2}H(\nu_i,\nu_j)$, profiles at well-separated scales cannot create an excess either; extremizing sequences for $\mathscr{M}$ spread over a long range of $\log(1/\sigma)$, so the sharp constant is a per-unit-scale quantity.

The local constant of Theorem 2.8 handles $\sigma \le 2$. Beyond that, a second refinement uses Helson's inequality in the odd primes together with an exact factorization in z_2.

LEMMA 9.12 (Vector-valued outer factorization). *If V is a nonzero $\mathbb{C}^N$-valued analytic polynomial on $\mathbb{T}$, there are an outer $g \in H^2(\mathbb{T})$ and $W \in H^2(\mathbb{T};\mathbb{C}^N)$ with $V = gW$, $\|g\|_2^2 = \int\|V\|\,\mathrm{d}m$ and $\|W\|_2^2 \le \int\|V\|\,\mathrm{d}m$.*

PROOF. Let g be outer with $|g|^2 = \|V\|$ on $\mathbb{T}$. Since $\log\|V\|$ is subharmonic, $\|V\| \le |g|^2$ in $\mathbb{D}$, so $W = V/g$ is holomorphic with $\|W\| \le |g|$. □

For a positive measure ν on $(0,\infty)$ put $c_N(\nu) = \int N^{-1/2-\sigma}\,\mathrm{d}\nu$ and let A_ν be the nonnegative matrix with rows $k \ge 0$, columns (j,m) with $j \ge 0$, m odd and $(j,m) \ne (0,1)$, and entries $\sqrt{d(m)}\,c_{2^{k+j}m}(\nu)$.

THEOREM 9.13 (Tail bound). *For $F \in H^1_0(\mathbb{T}^\infty)$, $\int|F(r(\sigma))|\,\mathrm{d}\nu(\sigma) \le \|A_\nu\|\,\|F\|_1$, and $\|A_\nu\|^2 \le \sum_{N\ge2} d(N)c_N(\nu)^2 - \sum_{k\ge1} c_{2^k}(\nu)^2$. The same holds on $\mathcal{P}$-smooth functions with the sums restricted accordingly.*

PROOF. Write $F = \sum_{m \text{ odd}} F_m(z_2) w^{\kappa(m)}$. For each z_2, Helson's inequality in w gives $(\sum_m |F_m(z_2)|^2/d(m))^{1/2} \le \int |F(z_2, w)|\, \mathrm{d}m(w)$. Apply Lemma 9.12 to $V = (F_m/\sqrt{d(m)})_m$: $F_m = \sqrt{d(m)}\, g W_m$ with $\|g\|_2 \|W\|_2 \le \|F\|_1$ and $\hat{W}_1(0) = 0$. Expand $F(r(\sigma))$ in powers of $a_\sigma = 2^{-1/2-\sigma}$ and integrate in ν. For the Hilbert–Schmidt bound, the entries with $k + j = e$ for $N = 2^e m$ number $e + 1$, and $(e+1)d(m) = d(N)$. □

PROPOSITION 9.14 (Computer-assisted bounds). *The following bounds hold.*

(1) For every $F \in H^1_0$, $J(F) \le 5.5150\, \|F\|_1$. Hence $\pi \le C \le C_J \le 5.5150$.

(2) If $F \in H^1_0$ uses only primes $p \le P$ with $\prod_{p \le P} (1 - 1/p)^{-1} \le 890$, in particular $P \le 10^{216}$, then $J(F) \le \sum_{n \ge 2} |a_n|/(\sqrt{n} \log n) \le 3.1407 \|F\|_1 < \pi \|F\|_1$.

(3) For the first d primes,

d	1	2	3	4	5	6	7
$C_d \le$	1.19914	1.58582	1.72142	1.80494	1.84758	1.88301	1.90873

Moreover $C_1 = \pi_1^*$, $1.54792 \le C_2 \le 1.58582$ and $1.0617 \le C_2/\pi_2^* \le 1.0878$.

PROOF. (1) On $(0, 2)$ use Theorem 2.8. On $(2, \infty)$ use Theorem 9.13 with Lebesgue measure: the finite part of A_ν with $N \le 2000$ has norm at most 0.27086 by Schur's test with weights verified in ball arithmetic, and the rest has Hilbert–Schmidt norm below 10^{-6}.

(2) The proof of Theorem 9.13 bounds $\sum |a_n| c_n(\nu)$, not only $\int |F(r(\sigma))|\, \mathrm{d}\nu$: it estimates each coefficient $a_{2^e m} = \sqrt{d(m)} \sum_{k+j=e} \hat{g}(k) \widehat{W_m}(j)$ in absolute value, so that $\sum |a_n| c_n(\nu) \le \langle A_\nu |\hat{g}|, |\widehat{W}| \rangle$. With Lebesgue measure it gives $J(F)^2/\|F\|_1^2 \le \int_0^\infty \sigma(\Pi_P(\sigma)^2 - 1)\, \mathrm{d}\sigma - \lambda_0$, where $\Pi_P(\sigma) = \prod_{p \le P} (1 - p^{-1-\sigma})^{-1} \le \min\{\zeta(1 + \sigma), \Pi_P(0)\}$ and $\lambda_0 = \mathrm{Li}_2(1/2)/\log^2 2$. With $\Pi_P(0) \le 890$ the integral is at most 11.07528 (ball arithmetic), so the bound is $\sqrt{9.86343} < 3.1407$. The bound $\prod_{p \le x} (1 - 1/p)^{-1} < e^{\gamma_{\mathrm{E}}} \log x\, (1 + 1/(2 \log^2 x))$ of Rosser and Schoenfeld [48] gives 885.9 at $x = 10^{216}$.

(3) Upper bounds: Theorem 9.13 with Schur weights verified in ball arithmetic and a Hilbert–Schmidt tail. $d = 1$: Nehari's theorem in one variable identifies C_1 with the Hankel matrix that defines π_1^*. Lower bound for C_2: an explicit polynomial in z_2, z_3 of degree at most 11 in each variable, found by second-order cone programming, with $\ell(q) = 1.5487079\ldots$ enclosed in ball arithmetic and $\|q\|_1 \le 1.0005030$ by a midpoint rule with explicit first- and second-order Taylor errors on an 8192^2 grid. The grid values are computed in binary64 as products of small matrices of exponentials enclosed in ball arithmetic, with an a priori bound for the rounding error that uses only the IEEE model for addition and multiplication. □

The tail bound is nearly tight for its part: $F = z_2$ has $\int_2^\infty |F(r)|\, \mathrm{d}\sigma = 2^{-5/2}/\log 2 \approx 0.255$. No refinement of this scheme reaches $\Gamma(1/4)^2/\sqrt{2\pi}$ itself, and making finitely many primes exact does not lower the boundary constant: the modified kernel is still asymptotic to the kernel L_0 of Theorem 2.8 near 0.

By (2), a counterexample to $C = \pi$ must involve a prime above 10^{216}. The finite-prime constants themselves are much smaller, and by (3) the two-prime excess of section 9.3.2 is real and capped, between six and nine per cent.

Open

(1) Is there a bounded completion of (c_n) to the ordered rationals with norm π (Theorem 9.5)?
(2) Does $\Phi(F_k) \to 1$ along some extremizing sequence? That would give $C = \pi$; the certificate of remark 11.10 predicts not, and a direct proof that $\Phi(F_k) \not\to 1$ would be a first non-probabilistic sign of $C > \pi$. ($\Phi \le 1$ fails in general, Proposition 9.8.)
(3) Close $1.0617 \le C_2/\pi_2^* \le 1.0878$, and certify a lower bound for C_3.
(4) Improve $C \le 5.515$. Arguments that use only $|F|$ cannot go below $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$; going further needs joint phase information about infinitely many large primes.

CHAPTER 10

Coefficient inequalities

The functional ℓ has relatives that ask for more: absolute values on the coefficients (Problem 9), other logarithmic weights, other $\mathscr{H}^p$, and partial sums. One arithmetic test runs through the first part of the chapter: by Helson's inequality, the absolute inequality holds on a set S of integers as soon as $B(S) = \sum_{n\in S} d(n)/(n\log^2 n)$ is finite, and each support restriction below is a way of taming the same Euler-product divergence $\zeta(1+\sigma)^2 \asymp \sigma^{-2}$. The second part replaces absolute values by cancellation: on prime sets of density θ the signed sum is bounded down to the weight $(\log n)^{-\theta}$, and the fractional primitives $\sum a_n/(\sqrt{n}(\log n)^{\beta})$ are classified on every $\mathscr{H}^p$, exactly as on the disc. Partial sums close the chapter. This chapter does not use chapter 11; it needs Helson's inequality, chapter 2, and, for the signed results, Theorem 3.1.

10.1. The absolute inequality

In one variable the absolute inequality is part (3) of Proposition 9.1; the same argument works for Paley–Wiener spaces whenever Nehari's theorem, equivalently weak factorization, holds [7, Lemma 7.5]. If $f = \sum_j g_j h_j$ with $\sum_j \|g_j\|_{\mathscr{H}^2}\|h_j\|_{\mathscr{H}^2} \lesssim \|f\|_{\mathscr{H}^1}$, then $|a_n|$ is dominated by the Dirichlet convolutions of the coefficient moduli, and the multiplicative Hilbert matrix, of norm π, bounds the weighted sum. On $\mathbb{T}^\infty$ this weak factorization fails, since Nehari's theorem does [44], and the S_2 substitute of chapter 5 controls only $\sum_j (\|g_j\|_{\mathscr{H}^2}\|h_j\|_{\mathscr{H}^2})^2$.

Problem 9 is open, although it holds below π on primes up to 10^{216} (Proposition 9.14). This section proves the inequality on every piece of $\mathscr{H}^1$ cut out by a degree constraint that the Euler product can see: one homogeneous degree, one degree in a subset of the primes, integers composed of a thin set of primes, and the balanced four-class diagonals together with summable unions of them. It also proves it for products across two complementary prime sets, and for sums of r such products at the sharp cost $\sqrt{r}$. All of these rest on Helson's inequality. It ends with the homogeneous dual constants.

10.1.1. An arithmetic criterion

LEMMA 10.1. *Let $S \subseteq \mathbb{N}$ and $B(S) = \sum_{n\in S,\, n\geq 2} d(n)/(n\log^2 n)$. Every $f \in \mathscr{H}^1$ supported on S satisfies*

$$|a_1| + \sum_{n\geq 2} \frac{|a_n|}{\sqrt{n}\log n} \leq \bigl(1 + B(S)^{1/2}\bigr)\|f\|_{\mathscr{H}^1}.$$

Moreover $B(S) = \int_0^\infty \sigma \sum_{n\in S,\, n\ge 2} d(n) n^{-1-\sigma}\, \mathrm{d}\sigma$.

PROOF. Use Cauchy–Schwarz with weights $d(n)$ and Helson's inequality [27], which states that $\sum |a_n|^2/d(n) \le \|f\|^2_{\mathscr{H}^1}$. For the formula, $\int_0^\infty \sigma n^{-\sigma}\, \mathrm{d}\sigma = 1/\log^2 n$. □

So every result in the next subsection is a bound on a Dirichlet series of $d(n)$ restricted to S, near $\sigma = 0$. The weight $\sigma\, \mathrm{d}\sigma$ is exactly what makes $\sigma^{-2+\varepsilon}$ integrable and σ^{-2} not: the full set $S = \mathbb{N}$ gives $\zeta(1+\sigma)^2 \asymp \sigma^{-2}$ and $B = \infty$, which is why the method cannot settle the full inequality.

10.1.2. Pieces

For a set $\mathcal{P}$ of primes let $\Omega_{\mathcal{P}}(n) = \sum_{p\in\mathcal{P}} v_p(n)$. Enumerate the primes $p_1 < p_2 < \cdots$ and let $\mathcal{P}_j$, $1 \le j \le 4$, be the residue classes of indices modulo 4.

THEOREM 10.2. *The absolute inequality* $|a_1| + \sum_{n\ge 2} |a_n|/(\sqrt{n}\log n) \le C\|f\|_{\mathscr{H}^1}$ *holds for* f *supported on each of the following sets, with* C *as stated.*

1. $\{\Omega(n) = m\}$*, with* C *absolute and independent of* m*. For* $m \ge 1$*, and also for* $\{\Omega(n) \in \{1,2\}\}$*, the sum alone satisfies* $\sum_{n\ge 2} |a_n|/(\sqrt{n}\log n) \le (2389/256)^{1/2}\|f\|_{\mathscr{H}^1} < \pi\|f\|_{\mathscr{H}^1}$*, uniformly in* m *and in the number of primes.*
2. $\{\Omega_{\mathcal{P}}(n) = k\}$*, where* $\sum_{p\in\mathcal{P}} p^{-1-\sigma} \le \delta\log(1/\sigma) + C_0$ *and* $\sum_{p\notin\mathcal{P}} p^{-1-\sigma} \le (1-\delta)\log(1/\sigma) + C_0$ *on* $(0,1]$*,* $0 < \delta \le 1$*;* C *depends on* δ, C_0 *only, not on* k*. The degrees in the primes outside* $\mathcal{P}$ *are unrestricted.*
3. *The integers composed of primes in* $\mathcal{P}$*, where* $\sum_{p\in\mathcal{P}} p^{-1-\sigma} \le \delta\log(1/\sigma) + C_0$ *with* $\delta < 1$*; for instance either alternating half of the primes. More generally,* $\sum_{n\ge 2} |a_n|/(\sqrt{n}(\log n)^\beta) \le C\|f\|_{\mathscr{H}^1}$ *on these integers for every* $\beta > \delta$*, with* C *depending on* β, δ, C_0*.*
4. *The union over* $a \in \mathcal{A} \subseteq \mathbb{Z}^3$ *of the diagonals* $\{\Omega_{\mathcal{P}_j}(n) - \Omega_{\mathcal{P}_4}(n) = a_j,\ j \le 3\}$*, with* $C \lesssim (\sum_{a\in\mathcal{A}} (1+|a|_\infty)^{-1})^{1/2}$*; for a single diagonal* C *is absolute.*

PROOF. In each case we bound $B(S)$ through Lemma 10.1, by marking degrees in the Euler product $\prod_p (1 - u_p p^{-1-\sigma})^{-2}$ and comparing coefficientwise. For $0 < \sigma \le 1$, expanding the logarithm gives the majorant $\exp(2\sum_p u_p p^{-1-\sigma}) E(u)$, where E collects the terms $u^\ell/(\ell p^\ell)$, $\ell \ge 2$, and has nonnegative coefficients and $E(1) < \infty$; for $\sigma \ge 1$, every coefficient is at most $\zeta(1+\sigma)^2 - 1 \le C 2^{-\sigma}$.

(1) With one marker u, $\sum_p p^{-1-\sigma} \le \log(2/\sigma)$ gives $[u^m] \le [u^m](2/\sigma)^{2u} E(u)$. Integrating σ^{1-2u} over $(0,1)$ gives $4^u/(2(1-u))$, and the coefficient of u^m in $4^u E(u)/(1-u)$ is a partial sum of nonnegative coefficients, hence at most $4E(1)$. For the explicit constant, $a_1 = 0$ when $m \ge 1$, and the proof of Lemma 10.1 bounds the sum by $B_m^{1/2}\|f\|_{\mathscr{H}^1}$, where $B_m = \sum_{\Omega(n)=m} d(n)/(n\log^2 n)$. The argument just given bounds the part $\sigma \le 1$ of B_m by $2E(1)$, and $E(1) \le e$ because $\sum_p \sum_{\ell\ge 2} 1/(\ell p^\ell) \le \sum_{n\ge 2} 1/(2n(n-1)) = 1/2$. For $\sigma \ge 1$ the coefficient is at most $\zeta(1+\sigma)^2 - 1$, and with $x = \zeta(1+\sigma) - 1 \le 2^{-\sigma}(1/2 + 1/\sigma)$ the integrand $\sigma(2x + x^2)$ is at most $2^{-\sigma}(\sigma+2) + 4^{-\sigma}(\sigma/4 + 2)$, whose integral over $[1,\infty)$ is below $981/256$ since $\log 2 > 2/3$. Hence $B_m < 2e + 981/256 < 2389/256 < \pi^2$,

using $e < 11/4$ and $\pi^2 > 72 - 36\sqrt{3} > 48/5$. For the two layers $\{1,2\}$ the same estimates give $B_1 + B_2 < (P_1 + P_2)/2 + 981/256 < 141/16$, where $P_1 = 1 + \log 4$ and $P_2 = P_1 + (\log 4)^2/2 + \sum_p p^{-2}$ are partial sums of the coefficients of $4^z E(z)$.

(2) Mark only the primes of $\mathcal{P}$. The majorant is $C\sigma^{-2(1-\delta)}\sigma^{-2\delta u}e^{2C_0 u}E(u)$, and $\int_0^1 \sigma^{-1+2\delta(1-u)}\,\mathrm{d}\sigma = 1/(2\delta(1-u))$; conclude as in (1).

(3) $\prod_{p\in\mathcal{P}}(1-p^{-1-\sigma})^{-2} \le C\sigma^{-2\delta}$ and $\int_0^1 \sigma^{1-2\delta}\,\mathrm{d}\sigma < \infty$. For the alternating halves, $0 \le \sum_j (-1)^{j-1} p_j^{-s} \le 2^{-s}$ gives $\delta = 1/2$. For the weight $(\log n)^{-\beta}$, run the proof of Lemma 10.1 with

$$(\log n)^{-2\beta} = \frac{1}{\Gamma(2\beta)} \int_0^\infty \sigma^{2\beta-1} n^{-\sigma}\,\mathrm{d}\sigma;$$

the bound becomes $\int_0^1 \sigma^{2\beta-1-2\delta}\,\mathrm{d}\sigma < \infty$.

(4) Each class $\mathcal{P}_j$ satisfies $\sum_{p\in\mathcal{P}_j} p^{-1-\sigma} \le \log(1/\sigma)/4 + C_0$, by the same alternating comparison. With $T = \log(1/\sigma)$ and $\mu = T/2$, the coefficients of $\exp(\mu\sum_j u_j)$ are $e^{4\mu}$ times the joint law of four independent Poisson variables of mean μ, and the factor $e^{4\mu} = \sigma^{-2}$ cancels against $\sigma\,\mathrm{d}\sigma = e^{-2T}\,\mathrm{d}T$. So the contribution of a diagonal is $\int_0^\infty \mathbb{P}(N_j - N_4 = a_j,\ j \le 3)\,\mathrm{d}\mu$, convolved with the coefficients of E. The integral is at most $C(1+|a|_\infty)^{-1}$: conditioning on N_j, N_4 for the largest $|a_j|$ gives the factor $(1+\mu)^{-1}$ from the other two classes, and a tilted Fourier bound gives $\mathbb{P}(N_j - N_4 = A) \lesssim (1+\mu)^{-1/2} e^{-\min(A, A^2/\mu)/8}$. The shifts from E are absorbed using $E(3/2,\dots,3/2) < \infty$, and for $\sigma \ge 1$ every selected n has $\Omega(n) \ge |a|_\infty$, which costs $(3/2)^{-|a|_\infty}$. Different diagonals are disjoint; sum over $\mathcal{A}$. □

Since $J(F) \le \sum_n |a_n| c_n$, part (1) proves the sharp Fejér–Riesz conjecture on every homogeneous layer: any counterexample mixes degrees in an essential way. Mixing a bounded number of degrees costs little: for $S = \{\Omega(n) \le D\}$, Lemma 10.1 and the uniform bound $B_m < 2389/256$ give $|a_1| + \sum_{n\ge 2} |a_n|/(\sqrt{n}\log n) \le (1 + 3.06\sqrt{D})\|f\|_{\mathscr{H}^1}$.

The layer constants improve with the degree.

PROPOSITION 10.3. *With $B_m = \sum_{\Omega(n)=m} d(n)/(n\log^2 n)$, $B_m = 1/2 + O(\rho^m)$ for every $\rho > 2/3$. Hence the absolute inequality on the layer $\{\Omega(n) = m\}$ holds with constant at most $2^{-1/2} + o(1)$ as $m \to \infty$.*

PROOF. Since $\sum_{\Omega(n)=m} d(n) n^{-1-\sigma}$ is the coefficient of u^m in $\prod_p (1 - up^{-1-\sigma})^{-2}$, Lemma 10.1 gives $\Phi(u) = \sum_m B_m u^m = \int_0^\infty \sigma\big(\prod_p (1-up^{-1-\sigma})^{-2} - 1\big)\,\mathrm{d}\sigma$. On $(0,1]$ write the product as $\sigma^{-2u} H(u,\sigma)$, with

$$H(u,\sigma) = \big(\sigma\zeta(1+\sigma)\big)^{2u} \prod_p e^{-2u\sum_{k\ge 2} p^{-k(1+\sigma)}/k} \big(1 - up^{-1-\sigma}\big)^{-2} e^{-2up^{-1-\sigma}},$$

which is analytic in $|u| < 2$ and real analytic in $\sigma \in [0,1]$; the part $\sigma \ge 1$ of the integral is analytic in $|u| < 2$. Expanding H in σ, $\int_0^1 \sigma^{1-2u+k}\,\mathrm{d}\sigma = 1/(2-2u+k)$, so Φ is meromorphic in $|u| < 3/2$ with a single simple pole at $u = 1$, of residue

$-H(1,0)/2 = -1/2$, since $H(1,0) = \lim_{\sigma\to 0} \sigma^2\zeta(1+\sigma)^2 = 1$. The layer constant is $B_m^{1/2}$ by the proof of Theorem 10.2. □

Part (4) needs four classes for a reason that is easiest to see probabilistically. Along σ, the degrees in the four classes behave like independent Poisson variables of mean $\mu = \log(1/\sigma)/2$, and the weight $\sigma\,\mathrm{d}\sigma$ turns the contribution of a diagonal into the expected time $\int \mathbb{P}(N_j - N_4 = a_j,\ j \le 3)\,\mathrm{d}\mu$ that the three differences spend at one point. That is a return probability of order $\mu^{-3/2}$, which is integrable. With three classes there are two differences, the return probability is of order μ^{-1}, and the integral diverges, as for recurrent walks in two dimensions. That is a limit of the method, not a counterexample.

PROPOSITION 10.4 (Products across complementary prime sets). *Let $\mathcal{P}_0, \mathcal{P}_1$ be the primes of odd and even index.*

1. *If $g, h \in \mathscr{H}^1$ are supported on integers composed of primes in $\mathcal{P}_0$ and $\mathcal{P}_1$ respectively, then $f = gh$ satisfies the absolute inequality with an absolute constant, and $\|f\|_{\mathscr{H}^1} = \|g\|_{\mathscr{H}^1}\|h\|_{\mathscr{H}^1}$.*
2. *If $f = \sum_{j\le r} g_j h_j$ with such factors, then the absolute inequality holds with constant $C\sqrt{r}$.*
3. *The cost $\sqrt{r}$ of separated representations is sharp. For*

$$f_N = \sum_{j\le N} (p_{2j-1}p_{2j})^{-s},$$

every representation of f_N as a sum of products $u_k v_k$ with separated supports costs at least $N/2$, while $\|f_N\|_{\mathscr{H}^1} \le \sqrt{N}$.

PROOF. (1) The lifts use disjoint variables. Group m, n into blocks $2^j \le \log n < 2^{j+1}$. By the bound $\prod_{p\in\mathcal{P}_0}(1-p^{-1-\sigma})^{-2} \le K\sigma^{-1}$ from the proof of part (3) of Theorem 10.2, with $\sigma = 2^{-j}$, each block has $\sum d(n)/(n\log n) \le C$, and $\sqrt{\log m \log n}/(\log m + \log n) \le \sqrt{2}\,2^{-|j-k|/2}$ across blocks. Cauchy–Schwarz per pair of blocks, a sum over diagonals $j-k$, and Helson's inequality for g and h finish.

(2) The span E of the lifts of the g_j in L^1 has dimension at most r, so the Kadets–Snobar theorem [32] gives a projection onto E of norm at most $\sqrt{2r}$, represented by a kernel $K(x) \in E$ with $\operatorname{ess\,sup}\|K(x)\|_1 \le \sqrt{2r}$. Since every w-section of the lift of f lies in E, $f(z,w) = \int K(x)(z) f(x,w)\,\mathrm{d}m(x)$, which is a representation with projective cost at most $\sqrt{2r}\|f\|_{\mathscr{H}^1}$. Apply (1) to each term.

(3) Projecting each u_k, v_k to degree one and Helson's inequality with $d(p) = 2$ give $\sum_j |\hat{u}_k(p_{2j-1})|^2 \le 2\|u_k\|^2$ and similarly for v_k; summing the coefficients at $p_{2j-1}p_{2j}$ gives $N \le 2\sum_k \|u_k\|\|v_k\|$, while $\|f_N\|_{\mathscr{H}^1} \le \|f_N\|_{\mathscr{H}^2} = \sqrt{N}$. □

REMARK 10.5 (Product Hardy inequality). The tensorized Hardy inequality

$$\sum_{\alpha} \frac{|a_\alpha|}{\prod_j(\alpha_j+1)} \le \pi^d \|f\|_{H^1(\mathbb{T}^d)},$$

sharp by tensoring near-extremal squares for the Hilbert matrix. On the integers generated by d primes it reads $\sum_n |a_n|/d(n) \le \pi^d\|f\|_{\mathscr{H}^1}$. So no dimension-free

inequality holds for the weight $1/d(n)$; the question is only about smaller weights such as $1/(\sqrt{n}\log n)$.

10.1.3. Homogeneous dual constants

Let c_m be the norm of $\varphi \mapsto \Lambda_\varphi$ from homogeneous degree-m symbols in $H^2(\mathbb{T}^\infty)$ to $(H^1)^*$; equivalently, the best constant in $\|f\|_2 \le c_m \|f\|_1$ for homogeneous f of degree m.

PROPOSITION 10.6. *$\sqrt{m!}/\Gamma(1+m/2) \le c_m \le 2^{m/2}$, so $c_m^{1/m} \to \sqrt{2}$; moreover $c_{m+n} \ge c_m c_n$. Consequently any inequality $\|\Lambda_\varphi\| \le C(\sum_m w_m \|\varphi_m\|_2^2)^{1/2}$ over homogeneous components needs $w_m \ge C^{-2} m!/\Gamma(1+m/2)^2$, so $\liminf w_m^{1/m} \ge 2$.*

PROOF. Projection onto degree m is an H^1 contraction, which gives the equivalence. The upper bound is Helson's inequality with $\prod(\nu_j+1) \le 2^m$. For the lower bound take $((z_1+\cdots+z_d)/\sqrt{d})^m$ and let $d \to \infty$: the moments converge to those of a complex Gaussian, $\mathbb{E}|Z|^q = \Gamma(1+q/2)$. Supermultiplicativity comes from disjoint variables, and $c_1 = 2/\sqrt{\pi}$ is Brevig's value [12]. □

10.2. Signed functionals and fractional primitives

10.2.1. Signed functionals on prime sets

COROLLARY 10.7. *Let R have density $\theta \in (0,1]$. For $f \in \mathscr{H}^1$ supported on integers composed of primes in R,*

$$\Big| a_1 + \sum_{n \ge 2} \frac{a_n}{\sqrt{n}(\log n)^\theta} \Big| \le C \|f\|_{\mathscr{H}^1},$$

and the series converges in its natural order.

PROOF. For polynomials, $\sum_{n\ge2} a_n/(\sqrt{n}(\log n)^\theta) = \Gamma(\theta)^{-1}\int_0^\infty (f(1/2+\sigma) - a_1)\sigma^{\theta-1}\,\mathrm{d}\sigma$. On $(0,1)$ use Theorem 3.1 with $p=1$; on $(1,\infty)$ use $|f(1/2+\sigma)-a_1| \le C2^{-\sigma}\|f\|_{\mathscr{H}^1}$. For natural-order convergence we follow Perfekt's Tauberian argument for $\theta = 1$ [45], comparing partial sums with Abel sums at $1/\log N$: the upper density bound gives $\sum_{e^u \le n < e^{2u}} d(n)/(n(\log n)^{2\theta}) \le C$ on R-integers, which with Helson's inequality makes the difference tend to 0, and horizontal translation is contractive on $\mathscr{H}^1$. □

Compare part (3) of Theorem 10.2: on a prime set of density $\theta < 1$ the *absolute* sum with weight $(\log n)^{-\beta}$ is bounded for every $\beta > \theta$, and the *signed* sum is bounded at the weight $(\log n)^{-\theta}$, which is critical by Corollary 5.3. So cancellation can matter only at the endpoint $\beta = \theta$, and whether the absolute inequality holds there is open. For $\theta = 1$ this endpoint question is exactly Perfekt's.

10.2.2. Fractional primitives of the zeta function

Bondarenko, Brevig, Saksman and Seip [11] considered, for $\beta > 0$, the functionals

$$\varphi_\beta(f) = a_1 + \sum_{n\ge2} \frac{a_n}{\sqrt{n}(\log n)^\beta} = a_1 + \frac{1}{\Gamma(\beta)} \int_0^\infty \big(f(1/2+\sigma) - a_1\big)\sigma^{\beta-1}\,\mathrm{d}\sigma.$$

In one variable the analogue is $f \mapsto \Gamma(\beta)^{-1}\int_0^1 f(x)(1-x)^{\beta-1}\,\mathrm{d}x$, with coefficients $\Gamma(j+1)/\Gamma(j+1+\beta)$. It is bounded on $H^p(\mathbb{D})$ for $p>1$ exactly when $\beta > 1/p$, and for $p \le 1$ exactly when $\beta \ge 1/p$ [11, Theorem 3.3].

PROBLEM 10 (Conjecture of Bondarenko, Brevig, Saksman and Seip [11, Section 3]). *Let $0 < p \le 2$. Is $\varphi_{1/p}$ bounded on $\mathscr{H}^p$ if and only if $p \le 1$?*

The case $p = 1$ is Perfekt's theorem, and $p = 2$ is trivial. The answer for every p and β, and on prime sets, needs one more lemma.

LEMMA 10.8 (Synthesis). *Let $1 \le p < \infty$, and let $P_j \ge 0$ be integrable functions with $\int \sqrt{P_i P_j} \le A e^{-b|i-j|}$. Then for finitely many $c_j \ge 0$,*

$$\int \Big(\sum_j c_j P_j^{1/p}\Big)^p \le C(A,b,p)\sum_j c_j^p.$$

PROOF. Fix a point, and let j be an index with $c_j^p P_j$ largest there. Choose $a > 0$ with $a(p-1) < b$. Hölder's inequality with the weights $e^{-a|i-j|}$ gives $(\sum_i c_i P_i^{1/p})^p \le C_a \sum_i e^{a(p-1)|i-j|} c_i^p P_i$, and $c_i^p P_i \le (c_i^p P_i)^{1/2}(c_j^p P_j)^{1/2}$ by the choice of j. Bounding the term of this j by the sum over all j and integrating,

$$\int \Big(\sum_i c_i P_i^{1/p}\Big)^p \le C_a A \sum_{i,j} e^{(a(p-1)-b)|i-j|} c_i^{p/2} c_j^{p/2},$$

and Schur's test bounds the right side by a constant times $\sum_j c_j^p$. □

For a set R of primes let $\mathscr{H}^p_R$ be the closure in $\mathscr{H}^p$ of the Dirichlet polynomials supported on integers composed of primes in R.

THEOREM 10.9. *Let R have density $\theta \in (0,1]$ with constant B, let $0 < p < \infty$ and $\beta > 0$. Then φ_β is bounded on $\mathscr{H}^p_R$ if and only if $\beta > \theta/p$, or $\beta = \theta/p$ and $p \le 1$. For R all primes, φ_β is bounded on $\mathscr{H}^p$ if and only if $\beta > 1/p$, or $\beta = 1/p$ and $p \le 1$, exactly as on the disc.*

PROOF. By Theorem 1.7 and Cauchy–Schwarz, $|f(1/2+\sigma)|^p \le \zeta(1+2\sigma)\|f\|^p$, and $\sigma \to \infty$ gives $|a_1| \le \|f\|$. For $\sigma \ge 1$ let F be the Bohr lift of a Dirichlet polynomial f, put $\rho = 2^{\sigma-1}r(\sigma)$, so that $\rho_q \le q^{-3/2}$, and $G(\lambda) = F(\lambda\rho)$. For $|\lambda| = 1$, plurisubharmonicity of $|F|^p$ gives $|G(\lambda)|^p \le \|f\|^p \sup P_{\lambda\rho} < 6\|f\|^p$, and the Schwarz lemma applied to $G - a_1$ gives $|f(1/2+\sigma) - a_1| \le 2\cdot 6^{1/p}2^{1-\sigma}\|f\|$; no triangle inequality is used, so this holds for $p < 1$ too. So the part $\sigma \ge 1$ of the integral and a_1 are bounded. By (2.1), $\zeta_R(1+2\sigma) \le C\sigma^{-\theta}$ on $(0,1]$, and $\nu = \sigma^{\theta-1}\,\mathrm{d}\sigma$ has $[\nu]_\theta = 1/\theta$, so part (2) of Theorem 2.10 gives $\int_0^1 |f(1/2+\sigma)|^p \sigma^{\theta-1}\,\mathrm{d}\sigma \le C\|f\|^p$.

Boundedness. If $p \le 1$ and $\beta \ge \theta/p$, write $|f| = |f|^p|f|^{1-p}$ and use $|f(1/2+\sigma)|^p \le \zeta_R(1+2\sigma)\|f\|^p \le C\sigma^{-\theta}\|f\|^p$ in the second factor:

$$|f|\sigma^{\beta-1} \le C|f|^p \sigma^{-\theta(1-p)/p+\beta-1}\|f\|^{1-p},$$

and the exponent is at least $\theta - 1$. If $p > 1$ and $\beta > \theta/p$, Hölder's inequality gives

$$\int_0^1 |f|\sigma^{\beta-1}\,\mathrm{d}\sigma \le \Big(\int_0^1 |f|^p\sigma^{\theta-1}\,\mathrm{d}\sigma\Big)^{1/p}\Big(\int_0^1 \sigma^{(\beta-\theta/p)p'-1}\,\mathrm{d}\sigma\Big)^{1/p'},$$

with $p' = p/(p-1)$, and the last integral is finite.

Unboundedness. Let K_δ be the Bohr lift of $\sum n^{-1/2-\delta}n^{-s}$ over integers composed of primes in R, and $F_\delta = K_\delta^{2/p}\zeta_R(1+2\delta)^{-1/p}$, so that $|F_\delta|^p$ is the Poisson kernel P_δ of $r_R(\delta)$, $\|F_\delta\|_p = 1$, and F_δ has positive coefficients with constant term at most 1. For $\sigma \le \delta$, $F_\delta(1/2+\sigma) \ge \zeta_R(1+2\delta)^{1/p} \ge c\delta^{-\theta/p}$, so

$$\varphi_\beta(F_\delta) \ge \frac{1}{\Gamma(\beta)}\int_0^\delta \big(F_\delta(1/2+\sigma)-1\big)\sigma^{\beta-1}\,\mathrm{d}\sigma \ge c'\delta^{\beta-\theta/p}$$

for small δ. If $\beta < \theta/p$ this is unbounded. If $\beta = \theta/p$ and $p > 1$, put $\delta_j = 2^{-j}$ and $g = \sum_{j=j_0}^{j_0+M} F_{\delta_j}$, so that $\varphi_\beta(g) \ge c'M$. By part (4) of Proposition 1.3 in each coordinate and (2.1),

$$\int \sqrt{P_{\delta_i}P_{\delta_j}} \le \Big(\frac{\zeta_R(1+\delta_i+\delta_j)^2}{\zeta_R(1+2\delta_i)\zeta_R(1+2\delta_j)}\Big)^{1/4} \le Ce^{-b|i-j|}, \qquad b = \theta(\log 2)/4,$$

so Lemma 10.8 gives $\|g\|_p^p \le CM$ and $\varphi_\beta(g)/\|g\|_p \ge cM^{1-1/p} \to \infty$. The functions F_δ involve infinitely many primes; truncating to the primes $q \le N$, taking Taylor sections, and letting $N \to \infty$ gives the same conclusion for Dirichlet polynomials, by monotone convergence in the positive coefficients. □

COROLLARY 10.10. *Let $0 < p \le 2$. The functional $\varphi_{1/p}$ is bounded on $\mathscr{H}^p$ if and only if $p \le 1$. This answers Problem 10.*

PROOF. The case R all primes and $\beta = 1/p$ of Theorem 10.9. □

For R all primes and $0 < p \le 2$, the cases $\beta \ne 1/p$ are [11, Theorem 3.1(b)]; what is new is the critical exponent, the range $p > 2$, and prime sets.

At the endpoint $\beta = 1/p$, $p > 1$, the growth on short polynomials is slow.

PROPOSITION 10.11. *Let $p > 1$. For Dirichlet polynomials f of length $N \ge 3$, $|\varphi_{1/p}(f)| \le C_p(\log\log N)^{1-1/p}\|f\|_{\mathscr{H}^p}$.*

PROOF. Put $\delta = 1/\log N$. The parts $\sigma \ge 1$ and a_1 are bounded as in the proof of Theorem 10.9. On $(\delta,1)$, Hölder's inequality and Theorem 2.5 give $\int_\delta^1 |f|\sigma^{1/p-1}\,\mathrm{d}\sigma \le C(\int_\delta^1 \sigma^{-1}\,\mathrm{d}\sigma)^{1-1/p}\|f\|$. On $(0,\delta)$, vertical translations act isometrically on $\mathscr{H}^p$ and f has frequencies in $[0,\log N]$, so Bernstein's inequality for vector-valued functions of exponential type gives $\|f'\|_{\mathscr{H}^p} \le \log N\|f\|_{\mathscr{H}^p}$. Hence, by Theorem 2.5 for f' and Hölder, $|f(1/2+\sigma)| \le |f(1/2+\delta)| + C\delta^{1-1/p}\log N\|f\|$, and $|f(1/2+\delta)| \le \zeta(1+2\delta)^{1/p}\|f\| \le C\delta^{-1/p}\|f\|$. Integrating against $\sigma^{1/p-1}$ over $(0,\delta)$ gives $O(\|f\|)$. □

The mechanism is the one-variable one: normalized kernels at dyadic scales are seen with weight of order one by $\varphi_{1/p}$, while their pth powers are almost orthogonal when $p > 1$. On $\mathbb{T}^\infty$ the almost orthogonality is the decay of the Hellinger affinity across scales.

10.3. Partial sums

Partial sums lie outside the story of π: the operators S_N are not coefficient functionals, and they share with the rest of Part IV only the tools, Helson's inequality and Harper's low moments. We include them because the $\mathscr{H}^1$ problem is open and those tools give something.

PROBLEM 11 (Saksman and Seip [49, Problem 4.5]). *What is the precise growth of* $\|S_N\|_{\mathscr{H}^1\to\mathscr{H}^1}$, *where* $S_N\sum a_n n^{-s}=\sum_{n\le N}a_n n^{-s}$?

For Problem 11, the order of growth is known for every p except $p=1$. For $1<p<\infty$ the operators S_N are uniformly bounded, by Helson's theorem on half-spaces of ordered groups applied to $\mathbb{Q}_+$ ordered by size. On $\mathscr{H}^\infty$, Balasubramanian, Calado and Queffélec [3] proved $\|S_N\|\lesssim\log N$, and Chen and Fu [20] have shown that $\|S_N\|\ge(1/(2\pi)+o(1))\log N$, so the order is $\log N$. For $0<p<1$, Bondarenko, Brevig, Saksman and Seip [11] showed that $\log\|S_N\|_{\mathscr{H}^p\to\mathscr{H}^p}$ is of order $\log N/\log\log N$. On $\mathscr{H}^1$ the lower bound $\log\log N\lesssim\|S_N\|$ was known earlier, and they proved the upper bound $\|S_N\|\lesssim\log N/\log\log N$. The lower bound is a one-variable fact: on series in the powers 2^{-ks}, S_N is the partial sum of degree $\log_2 N$ on $H^1(\mathbb{D})$. Any larger lower bound must therefore use many primes.

Optimizing ε in the inequality from the proof of [11, Theorem 5.2] gives an output-sensitive bound.

PROPOSITION 10.12. *There is an absolute* C *such that, for* $f\in\mathscr{H}^1$ *and* $g=S_Nf=\sum_{n\le N}a_nn^{-s}\neq0$,

$$\frac{\|g\|_{\mathscr{H}^1}}{\|f\|_{\mathscr{H}^1}}\le C\Big(1+\log\frac{\|g\|_{\mathscr{H}^2}}{\|g\|_{\mathscr{H}^1}}\Big).$$

PROOF. Bondarenko, Brevig, Saksman and Seip [11] prove that

$$\|S_Nf\|_{\mathscr{H}^{1-\varepsilon}}\le C_0\varepsilon^{-1}\|f\|_{\mathscr{H}^1}.$$

Hölder's inequality gives

$$\|g\|_1\le\|g\|_{1-\varepsilon}^{(1-\varepsilon)/(1+\varepsilon)}\|g\|_2^{2\varepsilon/(1+\varepsilon)},\qquad\text{hence}\qquad\frac{\|g\|_1}{\|f\|_1}\le\frac{C_0}{\varepsilon}\Big(\frac{\|g\|_2}{\|g\|_1}\Big)^{2\varepsilon/(1-\varepsilon)}.$$

Take $\varepsilon=1/(2(1+\log(\|g\|_2/\|g\|_1)))$. □

COROLLARY 10.13. *Uniformly over all tails:*

(1) if $S_Nf=\sum_{n\le N}n^{-s}$, *then* $\|S_Nf\|_{\mathscr{H}^1}\lesssim(1+\log\log\log N)\|f\|_{\mathscr{H}^1}$;
(2) if $S_Nf=\sum_{n\le N}n^{-1/2-s}$, *then* $\|S_Nf\|_{\mathscr{H}^1}\lesssim(1+\log\log N)\|f\|_{\mathscr{H}^1}$;
(3) if every n *in the support of* S_Nf *has* $\Omega(n)\le m$, *then* $\|S_Nf\|_{\mathscr{H}^1}\lesssim(m+1)\|f\|_{\mathscr{H}^1}$.

PROOF. (1) Harper's theorem [25] gives $\|\sum_{n\le N}n^{-s}\|_1\asymp\sqrt{N}(\log\log N)^{-1/4}$. (2) Gerspach's pseudomoment bound [24] gives $\|\sum_{n\le N}n^{-1/2-s}\|_1\asymp(\log N)^{1/4}$ against an $\mathscr{H}^2$ norm $\asymp(\log N)^{1/2}$. (3) $d(n)\le2^m$ on the support, so Helson gives $\|g\|_2\le2^{m/2}\|g\|_1$. □

The known lower bound for $\|S_N\|_{\mathscr{H}^1\to\mathscr{H}^1}$ is of order $\log\log N$. A large amplification therefore needs an output with an exponentially large ratio of $\mathscr{H}^2$ to $\mathscr{H}^1$ norms. The Dirichlet kernel is not such an output, and neither is the partial sum of $\zeta(1/2+s)$, which can at most match the known order. Vertical blow-up does not help either (section 4.4.2).

Open

(1) The absolute inequality $\sum |a_n|/(\sqrt{n}\log n) \lesssim \|f\|_{\mathscr{H}^1}$ (Problem 9). By Lemma 10.1, any proof must use more than Helson's inequality on the full set of integers.
(2) Is $\|S_N\|_{\mathscr{H}^1 \to \mathscr{H}^1}$ of order $\log\log N$, or larger?

CHAPTER 11

The Gaussian boundary model

Chapters 9 and 10 say where a counterexample to $C = \pi$ would have to live: it must use primes beyond 10^{216} (Proposition 9.14), mix degrees (Theorem 10.2), and gain nothing from a fixed finite set of primes (Corollary 9.4). So any excess comes from infinitely many primes near the critical line. This chapter finds the natural model for that regime. For quadratic combinations of Euler kernels at many scales near the critical line, the H^1 norm has an exact Gaussian limit (section 11.1), so C is at least a constant C_{Gauss} of a log-correlated Gaussian field, and a probabilistic certificate puts that constant above π. The same witnesses bound the essential size of the symbol and every boundary-weighted functional. Section 11.2 shows that extremizing sequences either concentrate at the critical line or produce an extremizer, and that above the essential threshold the norm is attained.

11.1. The Gaussian scaling limit

Products of normalized Euler kernels at many scales near the critical line do all three. Their H^1 norms have an exact Gaussian limit, and in that limit π is very probably exceeded.

Fix scales $a_1, \dots, a_m > 0$ and write $a_i = e^{t_i}$. For $\varepsilon > 0$ let

$$k_i^\varepsilon(z) = \zeta(1 + 2\varepsilon a_i)^{-1/2} \prod_p \bigl(1 - p^{-1/2-\varepsilon a_i} z_p\bigr)^{-1},$$

the normalized reproducing kernel at the point $r(\varepsilon a_i)$. For a real symmetric $m \times m$ matrix A put

$$F_\varepsilon = \sum_{i,j} A_{ij} k_i^\varepsilon k_j^\varepsilon, \qquad F_\varepsilon^0 = F_\varepsilon - F_\varepsilon(0) \in H_0^1.$$

PROPOSITION 11.1. *As $\varepsilon \to 0$, $\langle k_i^\varepsilon, k_j^\varepsilon \rangle \to G_{ij}$ and $\ell(F_\varepsilon^0) \to \sum_{i,j} A_{ij} H_{ij}$, where*

$$G_{ij} = \operatorname{sech} \frac{t_i - t_j}{2}, \qquad H_{ij} = \frac{t_i - t_j}{\sinh((t_i - t_j)/2)}, \quad H_{ii} = 2.$$

Moreover $H \preceq \pi G$ for every choice of scales.

PROOF. The Gram entry is $\zeta(1 + \varepsilon(a_i + a_j))/\sqrt{\zeta(1 + 2\varepsilon a_i)\zeta(1 + 2\varepsilon a_j)}$, and $\zeta(1 + t) = t^{-1} + O(1)$. The functional is $\ell(F) = \int_0^\infty (F(r(\sigma)) - F(0))\, \mathrm{d}\sigma$, and the entry for (i, j) equals $\int_0^\infty (\zeta(1+\varepsilon a_i+\sigma)\zeta(1+\varepsilon a_j+\sigma)-1)\, \mathrm{d}\sigma$ divided by the same normalization. Replacing both zeta factors by their poles costs $O(\log 1/\varepsilon)$, which is $o(1)$ after normalization. The substitution $\sigma = \varepsilon x$ then gives $2\sqrt{a_i a_j} \int_0^\infty dx/((a_i+x)(a_j+x)) = H_{ij}$.

For the last claim, $\pi\operatorname{sech}(x/2) - x/\sinh(x/2)$ is a positive definite function on $\mathbb{R}$: the Fourier transforms of $\operatorname{sech}(x/2)$ and $x/\sinh(x/2)$ are $2\pi\operatorname{sech}(\pi\xi)$ and $2\pi^2\operatorname{sech}^2(\pi\xi)$, and $2\pi^2\operatorname{sech}(\pi\xi)(1-\operatorname{sech}(\pi\xi)) \geq 0$. □

The last statement explains why squares never beat π here. If $A = uu^T$, then $F_\varepsilon = (\sum u_i k_i^\varepsilon)^2$, its norm tends to $u^T G u$, and $u^T H u \leq \pi u^T G u$. Any excess must come from quadratics that are not squares. These include positive semidefinite A of higher rank: for $A = \sum_\ell u_\ell u_\ell^T$ the squares $(\sum_i u_{\ell i} k_i^\varepsilon)^2$ are complex and can cancel, so the norm can fall below $\operatorname{tr}(AG)$. Their norms are not determined by G, and the next theorem computes them.

The question is under which measure the normalized Euler kernels have a useful limit. Under Haar measure they have none: each k_i^ε tends to zero weakly while its modulus concentrates. The proof below therefore writes $\|F_\varepsilon\|_1$ as an average against the mixture $m^{-1}\sum_j |k_j^\varepsilon|^2$ of the harmonic measures of the m points. Under each $|k_j^\varepsilon|^2\,\mathrm{d}m$ the coordinates z_p are independent with explicit Poisson laws, and after dividing by the mixture the integrand depends only on the ratios $k_i^\varepsilon/k_0^\varepsilon$. The ratios remove what all the kernels share, and their logarithms are sums over primes of small independent terms, to which the central limit theorem applies. The integrand is bounded, so convergence in law gives convergence of the norm.

Fix a reference scale a_0 among the a_i. Let $D = (D_i)$ be a centered circular complex Gaussian vector with $\mathbb{E} D_i \overline{D_k} = \Sigma_{ik}$, where

$$\Sigma_{ik} = \log\frac{(a_i + a_0)(a_k + a_0)}{2a_0(a_i + a_k)} = \int_0^\infty (e^{-a_i x} - e^{-a_0 x})(e^{-a_k x} - e^{-a_0 x})\,\frac{\mathrm{d}x}{x},$$

and put $R_i^{(j)} = \sqrt{a_i/a_0}\,(a_0 + a_j)(a_i + a_j)^{-1} e^{D_i}$.

Lemma 11.2 (The kernel ratios). *Fix j. Under the probability measure $|k_j^\varepsilon|^2\,\mathrm{d}m$, the vector $(k_i^\varepsilon/k_0^\varepsilon)_i$ converges in law to $R^{(j)}$ as $\varepsilon \to 0$.*

Proof. Under $|k_j^\varepsilon|^2\,\mathrm{d}m$ the coordinates z_p are independent and Poisson-kernel distributed, with $\mathbb{E} z_p^n = p^{-n(1/2 + \varepsilon a_j)}$. The logarithm of $r_i = k_i^\varepsilon/k_0^\varepsilon$ is $\log(a_i/a_0)/2 + o(1)$ plus the linear sum

$$\sum_p p^{-1/2}\bigl(p^{-\varepsilon a_i} - p^{-\varepsilon a_0}\bigr) z_p$$

plus a remainder of degree at least two, which tends to 0 in mean and in L^2, by domination with p^{-2}. The linear sum has mean

$$\sum_p p^{-1}\bigl(p^{-\varepsilon(a_i + a_j)} - p^{-\varepsilon(a_0 + a_j)}\bigr) \to \log\frac{a_0 + a_j}{a_i + a_j},$$

by $\sum_p p^{-1-t} = \log(1/t) + c + o(1)$. In the same way its covariances tend to Σ_{ik}, and its pseudo-covariances vanish because $\mathbb{E}(z_p - \mathbb{E} z_p)^2 = 0$. The Lindeberg theorem finishes the proof. □

Theorem 11.3.

$$\lim_{\varepsilon \to 0} \|F_\varepsilon^0\|_1 = \mathcal{N}(A) \coloneqq \frac{1}{m}\sum_{j=1}^m \mathbb{E}\,\frac{m\,|R^{(j)T} A R^{(j)}|}{|R^{(j)}|^2}.$$

Consequently $C \ge |\sum_{i,j} A_{ij}H_{ij}|/\mathcal{N}(A)$ *whenever* $\mathcal{N}(A) > 0$.

PROOF. With $q_\varepsilon = m^{-1}\sum_j |k_j^\varepsilon|^2$, a probability density, one has exactly

$$\|F_\varepsilon\|_1 = \frac{1}{m}\sum_j \int \frac{m\,|\sum A_{ik} r_i r_k|}{\sum_i |r_i|^2}\,|k_j^\varepsilon|^2\,\mathrm{d}m, \qquad r_i = k_i^\varepsilon/k_0^\varepsilon.$$

The integrand is a continuous function of r, bounded by $m\|A\|$, so Lemma 11.2 gives the convergence of each integral. The mixture is what makes the integrand bounded; a single reference measure would leave an unbounded integrand and a uniform integrability problem. Finally $F_\varepsilon(0) = O(\varepsilon)$. □

The mixture over j collapses to a single expectation. Put $g_i = G_{i0} = 2\sqrt{a_i a_0}/(a_i + a_0)$ and $R_i = g_i e^{D_i}$.

PROPOSITION 11.4. $\mathcal{N}(A) = \mathbb{E}\,|R^T A R|$. *Moreover* $\mathbb{E}[R^u \overline{R}^{\,v}] = g^{u+v}\exp(u^T\Sigma v)$ *for multi-indices* $u, v \ge 0$*; in particular* $\mathbb{E}[R_i\overline{R_k}] = G_{ik}$.

PROOF. Since D is circular, $\mathbb{E}[e^{u\cdot D + v\cdot\overline{D}}] = \exp(u^T\Sigma v)$, which gives the moments; with $e^{\Sigma_{ik}} = (a_i + a_0)(a_k + a_0)/(2a_0(a_i + a_k))$, the case $u = e_i$, $v = e_k$ is G_{ik}, and $\mathbb{E}|R_j|^2 = 1$. Under the probability measure $|R_j|^2\,\mathrm{d}\mathbb{P}$, the vector D is shifted by $\mathbb{E}[D\,2\Re D_j] = (\Sigma_{ij})_i$, so R_i becomes $g_i e^{\Sigma_{ij}} e^{D_i}$, and $g_i e^{\Sigma_{ij}} = \sqrt{a_i/a_0}\,(a_0 + a_j)(a_i + a_j)^{-1}$: this is the law of $R^{(j)}$. Hence $\mathcal{N}(A) = \sum_j \mathbb{E}[|R_j|^2 |R^T A R|/|R|^2] = \mathbb{E}|R^T A R|$. □

The single expectation decides whether a square can be improved. Let u be real with $u^T G u = 1$, so that $\mathcal{N}(uu^T) = 1$, put $\lambda = u^T H u$, and

$$\Delta_u = \Re\,\mathbb{E}\Big[RR^T\,\frac{\overline{(u^T R)^2}}{|u^T R|^2}\Big], \qquad K_u = H - \lambda\Delta_u.$$

PROPOSITION 11.5. *For real symmetric* B*, the derivative at* $t = 0$ *of* $\operatorname{tr}(H(uu^T + tB))/\mathcal{N}(uu^T + tB)$ *is* $\operatorname{tr}(K_u B)$*, and* $\Delta_u u = Gu$*. If* $K_u = 0$*, the square* uu^T *maximizes the ratio over all real symmetric* A *with these scales; if* $K_u \preceq 0$*, it maximizes it over positive semidefinite* A*.*

PROOF. For distinct scales $u^T R \neq 0$ almost surely, and $|R^T B R| \le \sum |B_{ik}||R_i R_k|$ is integrable, so $\mathcal{N}$ is differentiable at uu^T with derivative $B \mapsto \operatorname{tr}(\Delta_u B)$, and $\Delta_u u = \Re\,\mathbb{E}[R\overline{u^T R}] = Gu$ by Proposition 11.4. Since $\mathcal{N}$ is convex and positively homogeneous, $\mathcal{N}(A) \ge \operatorname{tr}(\Delta_u A)$ for every A. If $K_u = 0$ then $\operatorname{tr}(HA) = \lambda\operatorname{tr}(\Delta_u A) \le \lambda\mathcal{N}(A)$; if $K_u \preceq 0$ and $A \succeq 0$ then $\operatorname{tr}(HA) \le \lambda\operatorname{tr}(\Delta_u A)$ as well. □

So proving $C_{\mathrm{Gauss}} > \pi$, hence $C > \pi$, is reduced to bounding one finite-dimensional Gaussian expectation from above. The anchored form of D makes this practical: with $L(t) = \log\cosh(t/2)$, the increments $D(t_j + s) - D(t_j)$ form a stationary field with covariance $L(s) + L(s') - L(s - s')$. Squares give exactly $u^T G u$: the square $|u^T R|^2$ has exact mean $u^T G u$, which provides an exact control variate.

Let C_{Gauss} be the supremum of $|\sum_{i,j} A_{ij}H_{ij}|/\mathcal{N}(A)$ over all m, all scales $a_1, \dots, a_m > 0$, and all real symmetric A with $\mathcal{N}(A) > 0$. It is a constant of the log-correlated Gaussian field D, and involves no arithmetic. Squares alone give

$C_{\text{Gauss}} \geq \pi$: for $A = uu^T$ the ratio is u^THu/u^TGu, and on fine grids of scales with long span this approaches the supremum π of the ratio $\pi\operatorname{sech}(\pi\xi)$ of the Fourier transforms in Proposition 11.1.

COROLLARY 11.6. $C \geq C_{\text{Gauss}}$.

PROOF. This is the last statement of Theorem 11.3. □

The same witnesses see past continuous symbols. Let $\mathscr{M}$ be the multiplicative Hilbert matrix and $\mathcal{E}_{\text{sym}}$ the essential size of its symbols, as in section 4.6.

COROLLARY 11.7. $\mathcal{E}_{\text{sym}}(\mathscr{M}) \geq C_{\text{Gauss}}$.

PROOF. Let g be a bounded symbol of $\mathscr{M}$, so that $\int Fg\,\mathrm{d}m = \ell(F)$ for $F \in H^1_0$, and let $c \in C(\mathbb{T}^\infty)$ and $v \in N$. Since v annihilates analytic polynomials, $|\ell(F^0_\varepsilon)| \leq \|g - c - v\|_\infty \|F^0_\varepsilon\|_1 + |\int F^0_\varepsilon c\,\mathrm{d}m|$. The nth coefficient of $k^\varepsilon_i k^\varepsilon_j$ is at most $d(n)n^{-1/2}(\zeta(1+2\varepsilon a_i)\zeta(1+2\varepsilon a_j))^{-1/2}$, which tends to 0; so $\int F^0_\varepsilon P\,\mathrm{d}m \to 0$ for every trigonometric polynomial P, and, since $\|F^0_\varepsilon\|_1$ is bounded, also for every $c \in C(\mathbb{T}^\infty)$. Letting $\varepsilon \to 0$ in Proposition 11.1 and Theorem 11.3 gives $|\sum A_{ij}H_{ij}| \leq \|g - c - v\|_\infty \mathcal{N}(A)$. □

The witnesses also live at the critical line. For $\sigma \geq \delta$, $|k^\varepsilon_i k^\varepsilon_j(r(\sigma)) - k^\varepsilon_i k^\varepsilon_j(0)| \leq C\varepsilon(\zeta(1+\sigma)^2 - 1)$, so $\int_\delta^\infty |F^0_\varepsilon(r(\sigma))|\,\mathrm{d}\sigma = O(\varepsilon/\delta)$.

COROLLARY 11.8. *Let w be bounded on $[0,\infty)$, continuous at 0, with $w(0) = 1$, and $\ell_w(f) = \int_0^\infty w(\sigma)(f(1/2+\sigma) - a_1)\,\mathrm{d}\sigma$. Then $\|\ell_w\|_{(\mathscr{H}^1_0)^*} \geq C_{\text{Gauss}}$. In particular this holds for every ℓ_c of remark 9.9 and for the restriction of ℓ to any interval $(0,\delta)$.*

PROOF. $\ell_w(F^0_\varepsilon) - \ell(F^0_\varepsilon) = \int_0^\infty (w-1)F^0_\varepsilon$. On $(0,\delta)$ this is at most $\sup_{(0,\delta)}|w-1|$ times $\int_0^\delta |F^0_\varepsilon| \leq (5+2\sqrt{2})\|F^0_\varepsilon\|_1$ (Theorem 2.5), and on (δ,∞) it is $O(\varepsilon/\delta)$. Let $\varepsilon \to 0$, then $\delta \to 0$. □

Since $\|\mathscr{M}\|_{\text{e}} = \pi$ (Corollary 4.14), the inequality $C_{\text{Gauss}} > \pi$ suggested by remark 11.10 would give $\mathcal{E}_{\text{sym}}(\mathscr{M}) > \|\mathscr{M}\|_{\text{e}}$: the excess over π would survive removing every continuous symbol.

CONJECTURE 1. $C = C_{\text{Gauss}} > \pi$.

The evidence for the inequality is remark 11.10 below. The evidence for the equality is structural. Near-extremal vectors for the multiplicative Hilbert matrix tend weakly to zero, put asymptotically no mass at $\sigma > \delta$, and vary slowly under multiplication (Theorem 9.3). For ℓ on $\mathscr{H}^1_0$, section 11.2 proves that every extremizing sequence either concentrates at $\sigma = 0$ or produces an extremizer, and that the constant at $\sigma = 0$, suitably normalized, does not see any finite set of primes (Theorem 11.13 and Proposition 11.16). So the conjecture splits into two statements: that ℓ does not attain its norm, and that the concentrated problem is governed by the Gaussian limit of Theorem 11.3. If the second fails for quadratics, the natural replacement allows kernel polynomials of higher degree, whose norms we have not computed.

REMARK 11.9 (The excess, numerically). Take the scales $t_i = i$, $|i| \le T/2$, and let A range over quadratics in the top few generalized eigenvectors of (H, G). Then minimize $\mathcal{N}(A)$ for fixed $\sum A_{ij}H_{ij}$ on one sample and evaluate on an independent one. The best ratios are then $\sum A_{ij}H_{ij}/\mathcal{N}(A) \approx 3.1386$, 3.1422, 3.1442, 3.1450 for $T = 120$, 160, 200, 240, against 3.1326, 3.1363, 3.1381, 3.1392 for squares. Indefinite quadratics thus gain about 0.006 over squares at every span. The squares' deficit decays with the span, and the two cross between $T = 120$ and $T = 160$. A refinement of the grid changes nothing.

REMARK 11.10 (A probabilistic certificate). For $T = 240$ and one explicit quadratic in six modes, normalized so that $\sum A_{ij}H_{ij} = 1$ (to 10^{-15}, in interval arithmetic), 10^8 samples drawn from a seed fixed in advance give $\mathcal{N}(A) \le 0.3179567$. The tool is a one-sided Bernstein inequality for the control-variate residual, with explicit Lipschitz, coupling and rounding budgets. The threshold is $1/\pi = 0.3183099$, so, if the certificate is right, $C \ge C_{\text{Gauss}} \ge 3.1450 > \pi$. This is not a proof. It holds except on an event of probability at most $2 \cdot 10^{-30}$, assuming that the pseudo-random generator behaves like independent draws, that A was chosen without using the certification stream, and that floating-point arithmetic obeys its rounding model. A deterministic certificate needs a bound for one bounded Gaussian integral in 480 real dimensions, to about $4 \cdot 10^{-4}$. Grids are hopeless at that dimension, and quadratic majorants provably fail (their best bound is the nuclear norm of A).

The certificate is consistent with everything above. The kernels involve all primes, and the excess appears only as $\varepsilon \to 0$, so only primes far beyond 10^{216} matter. The quadratic is not a square, and it mixes all degrees. If it is right, then $C > \pi$, the sharp Fejér–Riesz conjecture fails, and $\Phi(F_k) \not\to 1$ along extremizing sequences.

REMARK 11.11 (Fewer scales). Splitting the primes into K interleaved classes and summing the squares of the top eigenvectors on each class gives ratios 1.29, 0.61, 0.47 for $K = 1, 2, 3$: the Ortega-Cerdà–Seip gain from independent blocks is real, about $\sqrt{K}$, but sparse classes give trace-class matrices (Corollary 5.3) whose top eigenvalues fall like $1/K$. None of these constructions exceeds the square bound, and the Euler-product family $\prod_{p \le y}(1 + 2z_p/\sqrt{p}) - 1$ stays below $e^{\gamma_{\text{E}}} < \pi$. All of them use a bounded number of scales; the excess of remark 11.9 appears numerically only from about one hundred and fifty scales.

11.2. Extremizing sequences

The obstruction to compactness is that mass can escape toward the critical line: an extremizing sequence may put almost all of $\int |f(1/2 + \sigma)| \, \mathrm{d}\sigma$ near $\sigma = 0$, where no fixed coefficient sees it. The proof of Theorem 11.13 handles the other alternative with three tools, each doing one job. Komlós's theorem replaces a subsequence by Cesàro means that converge almost everywhere; Lemma 11.12 shows that such a limit of a bounded sequence in H^1 is again analytic, with the limiting coefficients; and the Brezis–Lieb lemma splits the L^1 norm between the limit and the remainder, which is what lets the extremal ratio pass to the limit.

For the multiplicative Hilbert matrix, near-extremal vectors tend weakly to zero because $\mathscr{M}$ has no eigenvectors (Theorem 9.3). In H^1 weak-* limits do not split

the norm, since L^1 norms are not additive under weak convergence. The substitute is an averaging argument and the following lemma, in which analyticity is what matters. Write E_d for the conditional expectation onto the first d coordinates, and P_x for the Poisson kernel of $x \in \mathbb{D}^d$, viewed as a function on $\mathbb{T}^\infty$.

LEMMA 11.12. *Let $F_N \in H^1(\mathbb{T}^\infty)$ with $\sup_N \|F_N\|_1 < \infty$ and $F_N \to u$ almost everywhere. Then $u \in H^1(\mathbb{T}^\infty)$, and the Fourier coefficients of F_N converge to those of u.*

PROOF. Let $K = \sup_N \|F_N\|_1$. By Fatou, $u \in L^1$. Let μ be a weak-* cluster point of $F_N \,\mathrm{d}m$; passing to a subsequence, $F_N \,\mathrm{d}m \to \mu$ weak-*. Since $|G|^{1/2}$ is plurisubharmonic for G analytic, and both sides are continuous on H^1,

$$|E_d G(x)|^{1/2} \le \int |G|^{1/2} P_x \,\mathrm{d}m \qquad (G \in H^1,\ x \in \mathbb{D}^d).$$

Fix M and put $g = |u - F_M|^{1/2}$. Apply this to $G = F_N - F_M$ and let $N \to \infty$. On the left, $E_d F_N(x) \to \int P_x \,\mathrm{d}\mu$. On the right, $|F_N - F_M|^{1/2} \to g$ almost everywhere with $\||F_N - F_M|^{1/2}\|_2^2 \le 2K$, hence weakly in L^2. So the Poisson extension h_d of the analytic measure $\nu = E_d\mu - E_d F_M$ on $\mathbb{T}^d$ satisfies $|h_d(x)| \le (P_x[E_d g])^2$. As $\rho \to 1$ the functions $(P_{\rho\zeta}[E_d g])^2$ converge in $L^1(\mathbb{T}^d)$, so $\{h_d(\rho\,\cdot)\}$ is uniformly integrable; therefore ν is absolutely continuous, and letting $\rho \to 1$ along a suitable sequence gives $|E_d\mu - E_d F_M| \le (E_d g)^2$ almost everywhere on $\mathbb{T}^d$.

By Doob's maximal inequality $\sup_d (E_d g)^2 \in L^1$, and $\{E_d F_M\}_d$ is uniformly integrable, so the martingale $(E_d\mu)_d$ is uniformly integrable. Hence $\mu = F\,\mathrm{d}m$ with $F \in L^1$, and martingale convergence gives $|F - F_M| \le |u - F_M|$ almost everywhere, for every M. Letting $M \to \infty$ gives $F = u$. So $\mu = u\,\mathrm{d}m$, which has analytic spectrum, and the whole sequence converges. □

THEOREM 11.13 (Concentration or an extremizer). *Let (f_k) be an extremizing sequence: $f_k \in \mathscr{H}^1_0$, $\|f_k\|_{\mathscr{H}^1} = 1$ and $\ell(f_k) \to C$. Then either*

$$\int_\delta^\infty |f_k(1/2 + \sigma)| \,\mathrm{d}\sigma \to 0 \qquad \text{for every } \delta > 0,$$

or ℓ attains its norm: there is $u \in \mathscr{H}^1_0$, $u \ne 0$, with $|\ell(u)| = C\|u\|_{\mathscr{H}^1}$.

PROOF. For $\|f\|_1 \le 1$ and $a_1 = 0$ we have $|f(1/2+\sigma)| \le \zeta(1+2\sigma)$ by Proposition 1.2 and $|f(1/2+\sigma)| \le \zeta(1/2+\sigma) - 1$ for $\sigma \ge 1$, an integrable majorant on $[\delta, \infty)$. Helson's inequality $\sum |a_n|^2/d(n) \le \|f\|_1^2$ [27] makes the tails $\sum_{n>N} |a_n| n^{-1/2-\sigma}$ small uniformly on the unit ball. Hence, if the coefficients of f_k converge to those of u, then $f_k(1/2+\sigma) \to u(1/2+\sigma)$ for every $\sigma > 0$, and $\int_\delta^\infty |f_k| \to \int_\delta^\infty |u|$ by dominated convergence.

Suppose that $\int_\delta^\infty |f_k| \ge \eta > 0$ along a subsequence. Pass to a further subsequence whose coefficients converge, to those of a measure μ, and then, by Komlós's theorem [34] applied to the real and imaginary parts, to one whose Cesàro means F_N converge almost everywhere. Then $F_N \in \mathscr{H}^1_0$, $\|F_N\|_1 \le 1$ and $\ell(F_N) \to C$, so $\|F_N\|_1 \to 1$; the coefficients of F_N still converge to those of μ. By Lemma 11.12, $\mu = u\,\mathrm{d}m$ with $u \in \mathscr{H}^1_0$ and $F_N \to u$ almost everywhere, and $\int_\delta^\infty |u| \ge \eta$ (apply the first paragraph to the f_k), so $u \ne 0$. The Brezis–Lieb lemma [17] gives

$\|F_N - u\|_1 \to 1 - \|u\|_1$. Since $F_N - u \in \mathscr{H}^1_0$,

$$C = \lim|\ell(F_N)| \le |\ell(u)| + C\lim\|F_N - u\|_1 = |\ell(u)| + C(1 - \|u\|_1),$$

so $|\ell(u)| \ge C\|u\|_1$. □

Write $\mathcal{E} = \mathcal{E}_{\mathrm{sym}}(\mathscr{M})$. Since $(\mathscr{H}^1_0)^* = L^\infty/(N + \mathbb{C})$ and trigonometric polynomials are dense in $C(\mathbb{T}^\infty)$, $\mathcal{E}$ is the distance from ℓ to the closure of the functionals that depend on finitely many coefficients. Clearly $\mathcal{E} \le C$, and $\mathcal{E} \ge C_{\mathrm{Gauss}}$ by Corollary 11.7.

COROLLARY 11.14. *If $C > \mathcal{E}$, then ℓ attains its norm. Equivalently, if ℓ does not attain its norm, then $C = \mathcal{E}$, and moreover $C \le \Gamma(1/4)^2/\sqrt{2\pi} < 5.2442$.*

PROOF. Suppose ℓ does not attain its norm, and let (f_k) be extremizing; by Theorem 11.13 it concentrates. Then every coefficient of f_k tends to 0: otherwise a subsequence has coefficients converging to those of a nonzero Dirichlet series u, the first paragraph of the proof above gives $f_k(1/2+\sigma) \to u(1/2+\sigma)$ for $\sigma > 0$, and u, being nonzero, does not vanish identically on (δ, ∞), contradicting concentration. So $\lambda(f_k) \to 0$ for every functional λ in the closure, and $C = \lim|\ell(f_k)| \le \|\ell - \lambda\|$. Hence $C \le \mathcal{E}$. Finally $C = \lim|\int_0^2 f_k(1/2+\sigma)\,\mathrm{d}\sigma|$ by concentration, and Theorem 2.8 bounds this by $\Gamma(1/4)^2/\sqrt{2\pi}$. □

An extremizer, if there is one, is rigid.

PROPOSITION 11.15 (Extremizers). *Let $F \in H^1_0$ with $\|F\|_1 = 1$ and $\ell(F) = C$.*

(1) $\ell(QF) = C\int Q|F|\,\mathrm{d}m$ for every analytic polynomial Q.
(2) F depends on infinitely many variables.
(3) F is not of the form h^2 with $h \in H^2$.

PROOF. (1) For complex w, $|1+w| \le (1+|1+w|^2)/2 = 1 + \Re w + |w|^2/2$. So, for complex t,

$$\begin{aligned} C + \Re\big(t\,\ell(QF)\big) &\le |\ell(F(1+tQ))| \le C\|F(1+tQ)\|_1 \\ &\le C + C\Re\Big(t\int Q|F|\,\mathrm{d}m\Big) + \frac{C|t|^2}{2}\|Q\|_\infty^2, \end{aligned}$$

and letting $t \to 0$ in every direction gives (1).

(2) Suppose F depends only on the variables of a finite set S of primes. For $p \notin S$, $\int z_p|F|\,\mathrm{d}m = 0$, so by (1) $\int_0^\infty p^{-\sigma} g(\sigma)\,\mathrm{d}\sigma = 0$, where $g(\sigma) = F(r(\sigma))$. The function g is real analytic on $(-1/2, \infty)$, bounded on $[0,\infty)$, and not identically zero since $\ell(F) \ne 0$; if $g(\sigma) \sim c\sigma^m$ at 0, then Watson's lemma gives $\int_0^\infty e^{-x\sigma} g(\sigma)\,\mathrm{d}\sigma \sim c\,m!\,x^{-m-1} \ne 0$ as $x \to \infty$, which contradicts its vanishing at $x = \log p$ for every large prime p.

(3) If $F = h^2$ with $h = \sum b_n n^{-s}$, then $b_1 = 0$ and $C = |\langle \mathscr{M} b, \bar{b}\rangle| \le \langle \mathscr{M}|b|, |b|\rangle \le \pi\|b\|^2 = \pi$. Since $C \ge \pi$, $|b|$ would be an eigenvector of $\mathscr{M}$ with eigenvalue π, and there are none [47]. □

In the first alternative of Theorem 11.13 the constant is local. For $0 < \delta \le 1$ let $C^{(\delta)}$ be the norm of $f \mapsto \int_0^\delta f(1/2+\sigma)\,\mathrm{d}\sigma$ on $\mathscr{H}^1$, and $C^{(\delta)}_{>P}$ its norm on the series supported on integers with no prime factor $p \le P$. Put $c_P = \prod_{p \le P}(1 - 1/p)$.

PROPOSITION 11.16 (Small primes). *For every P there is K_P with $|C^{(\delta)} - C^{(\delta)}_{>P}/c_P| \le K_P\delta$ for $0 < \delta \le 1$. If ℓ does not attain its norm, then $C \le C^{(\delta)}$ for every δ, so*

$$C \le \liminf_{\delta\to 0} \frac{C^{(\delta)}_{>P}}{c_P} \qquad \text{for every } P.$$

PROOF. Let S be the primes up to P, write $z = (\zeta, w)$ and $r(\sigma) = (r_S(\sigma), r'(\sigma))$, and put $r_0 = r_S(0)$. For a polynomial F, $F(r(\sigma)) = \int_{\mathbb{T}^S} F(\zeta, r'(\sigma))P_{r_S(\sigma)}(\zeta)\,\mathrm{d}m(\zeta)$, and $|P_{r_S(\sigma)} - P_{r_0}| \le K_P\sigma$ for $\sigma \le 1$ because r_0 lies inside the polydisc. The function $H(w) = F(r_0, w)$ has $\|H\|_1 \le \|F\|_1/c_P$: apply $|g(a)| \le \|g\|_{H^1(\mathbb{T})}/(1-|a|^2)$ in each variable of S and integrate. Splitting $P_{r_S(\sigma)} = P_{r_0} + (P_{r_S(\sigma)} - P_{r_0})$,

$$\Big|\int_0^\delta F(r(\sigma))\,\mathrm{d}\sigma\Big| \le \frac{C^{(\delta)}_{>P}}{c_P}\|F\|_1 + K_P\delta \int_{\mathbb{T}^S}\int_0^1 |F(\zeta, r'(\sigma))|\,\mathrm{d}\sigma\,\mathrm{d}m(\zeta),$$

and the last integral is at most $(5+2\sqrt{2})\|F\|_1$ by Theorem 2.5, applied for each ζ to $w \mapsto F(\zeta, w)$. Conversely, for g supported on integers free of primes $p \le P$, put $f = ug$ with $u = \prod_{p\le P}(1-1/p)(1-p^{-1/2}z_p)^{-2}$. The variables are disjoint, so $\|f\|_1 = \|u\|_1\|g\|_1 = \|g\|_1$, and $u(r_S(\sigma)) = c_P\prod_{p\le P}(1-p^{-1-\sigma})^{-2} = c_P^{-1} + O_P(\sigma)$. Hence $\int_0^\delta f(r)\,\mathrm{d}\sigma = c_P^{-1}\int_0^\delta g(r')\,\mathrm{d}\sigma + O_P(\delta)\|g\|_1$. For the last statement, apply Theorem 11.13: along an extremizing sequence $\int_0^\delta f_k(1/2+\sigma)\,\mathrm{d}\sigma \to C$. □

The factor $c_P = \|K_{r_0}\|_2^{-2}$ is the price of removing the small primes, and nothing else is. What remains for Conjecture 1 is thus whether ℓ attains its norm, and whether the local constant $\lim_P \liminf_{\delta\to 0} C^{(\delta)}_{>P}/c_P$, in which every coordinate is small, is decided by the Gaussian limit; for that, kernels at many scales would have to be shown to be asymptotically optimal, which needs a central limit theorem for $\log|f|$ under $|f|\,\mathrm{d}m$ along concentrating sequences.

Open

(1) Prove $C > \pi$ deterministically, by bounding the Gaussian integral of Theorem 11.3 for one explicit quadratic.
(2) Prove Conjecture 1. By Theorem 11.13 and Proposition 11.16 this splits in two: does ℓ attain its norm on $\mathscr{H}^1_0$, and is the local constant at $\sigma = 0$, with every coordinate small, decided by kernels at many scales?
(3) Is the window scaling limit of J at $\sigma \asymp 1/\log P$ governed by critical multiplicative chaos, and does it give a per-unit-scale formula for C?

APPENDIX A

Computations

Several results rest on a computation. The scripts that check them are published at `https://github.com/Oisov/math-preprints`, folder `numerics/book`, and `python3 run_all.py` there reruns all of them and compares their output with the values printed here; it takes about two minutes. Interval computations use ball arithmetic (Arb, through `python-flint`); the others use exact rational or directed dyadic arithmetic in the Python standard library.

Result	What is computed	Arithmetic
Theorem 2.8	the analytic margin at $\sigma = 1/100$ and Schur rows on 268 cells of $[1/100, 2]$	Arb
Proposition 3.7	a 192-term Legendre witness for $C_0 > 2\pi + 2.8 \cdot 10^{-7}$, and the rank-two bound $C_0 < 2\pi + 10^{-3}$	exact dyadic
Remark 5.6	entries of 128×128 Bloch matrices at 129 nodes, their inertia by exact LDL^* factorization with residual bounds, and the operator error budget	Arb, exact dyadic and rational
Theorems 8.6 and 9.2	the displayed polynomial identities and the signs of their coefficients	exact (symbolic)
Proposition 9.14	Schur tests with verified weights, Hilbert–Schmidt tails, the integral behind 10^{216}, and the L^1 norm of the witness for C_2	Arb

The probabilistic certificate of remark 11.10 is published in the same folder, with its scripts, the fixed quadratic as exact binary64 data, and the log of the certified run. It is not a proof. Its failure bound $2 \cdot 10^{-30}$ assumes that the pseudo-random generator behaves like independent draws and that floating-point arithmetic obeys its rounding model, and an exact rerun needs the numpy version of the original run.

APPENDIX B

Problems and their status

The book states eleven numbered problems: eight from the literature and three that arise here. The table records what is known about each, as of October 2026; several of the answers rest on recent preprints. Of the eleven, two are answered in this book, one of them computer-assisted (Problems 4 and 10), three have been answered elsewhere, one of them also by a second proof here and with a new Bergman analogue (Problems 1, 3 and 7), four are answered in part, and two are open.

	Source	Status
Problem 1	Saksman–Seip [49, Problem 2.1]	Answered, modulo a preprint: yes exactly for $p \geq 2$. No for $0 < p < 2$ by Harper [25]; yes for $p > 2$ announced by Chen et al. [19]. Here: the rate $N_p(\varepsilon) \lesssim \varepsilon^{-1/2}$ for every p (Theorem 3.4).
Problem 2	Perfekt–Pushnitski [46]	Partial. Yes for every partially averaged curve measure, in three geometries (Theorems 4.2 and 6.14), and for energy-separated families (Proposition 2.13).
Problem 3	Saksman–Seip [49, Problem 3.2]	Answered by Perfekt [45]; second proof in Corollary 4.19; the Bergman analogue is answered in Theorem 6.15.
Problem 4	This book	Answered: yes under regular variation (Theorem 5.4; under C^2 regularity, with the finer spectral type, Ma–Lu–Zu [38]); no in general, by a computer-assisted gap for one prime set (Proposition 5.5 and remark 5.6).
Problem 5	Brevig–Perfekt [14, Question 3]	Partial. Such forms exist for every $p > 4$ (Theorem 7.6; also Bampouras [4]); open for $2 < p \leq 4$, where the natural families are sharp at 4 (chapter 8). In two variables none exist, for any $p < \infty$ [8].
Problem 6	This book	Open. Every family treated in chapter 8 satisfies the inequality.
Problem 7	Brevig–Perfekt [15]	Answered, in preprints: for balls by Bampouras–Perfekt [7], for convex sets free of lines by Bampouras [5]. The constant exceeds 1 and grows exactly exponentially on polydiscs (Proposition 8.15 and Corollary 8.16). For S_p, $2 < p \leq 4$, open on the disc; no threshold for polygons.
Problem 8	This book	Open. $C < \infty$ by Perfekt [45]; $\pi \leq C \leq 5.515$ (Proposition 9.14); a probabilistic certificate, not a proof, indicates $C \geq 3.145$ (remark 11.10).
Problem 9	Perfekt [45]	Partial. True on every piece of Theorem 10.2, for products across prime sets (Proposition 10.4), and with constant $3.1407 < \pi$ on primes up to 10^{216} (Proposition 9.14).
Problem 10	Bondarenko–Brevig–Saksman–Seip [11]	Answered: yes, and for all p, β and prime densities (Corollary 10.10 and Theorem 10.9).
Problem 11	Saksman–Seip [49, Problem 4.5]	Partial. On $\mathscr{H}^1$, $\log\log N \lesssim \|S_N\|$ was known, and $\|S_N\| \lesssim \log N / \log\log N$ [11]; on $\mathscr{H}^\infty$ the order is $\log N$ [20]. An output-sensitive form of the BBSS bound here (Proposition 10.12).

Further open questions

(1) *Two-point stability.* Is the factor 2 in Corollary 1.8 necessary (chapter 1)?
(2) *What Hellinger–Bessel loses.* Is there a geometric condition, between ℓ^2-bounded affinities and separation, under which the Gram condition is necessary (Proposition 2.13)?
(3) *Vertical lines.* The rate of $N_p(\varepsilon)$ for $p < 2$, and the exact value of C_0 in Proposition 3.6 (chapter 3).
(4) *Spectra.* Which measures have $s_n \asymp a_n^*$ without an index change? Certify the spectral gaps of Proposition 5.5. The exponent of the bounded-rank Nehari constant, between 0.174 and 1/2 (chapter 5).
(5) *Bergman spaces.* The sharp two-point constants: in the Bergman geometry beyond the first zero of $K_x + K_{-x}$ (up to it the square bound is sharp), and in the Gauss geometry, where the square bound always fails, the constant itself (Proposition 6.8); and whether Bergman Dirichlet kernels cancel for $\beta > 1$ (chapter 6).
(6) *Near $p = 2$.* Is there a dimension- and degree-free interval above 2 with no strict Gaussian blocks (chapter 8)?
(7) *Fixed dimension.* The S_p Nehari constant on $\mathbb{T}^d$ is finite for every p, by Nehari's theorem on $\mathbb{D}^d$; does it grow with d for some $p \le 4$ (chapter 7)? Does the S_2 constant for the ball B^n grow with n, as it does for polydiscs (chapter 8)?
(8) *Spectral type.* For prime sets of density θ, is the critical spectrum absolutely continuous, and when are there eigenvalues above it? For all primes in the Bergman geometry this is known (chapter 5).
(9) *The sharp constant.* A deterministic proof of $C_{\mathrm{Gauss}} > \pi$; does ℓ attain its norm on $\mathscr{H}_0^1$ (Corollary 11.14); is $C = C_{\mathrm{Gauss}}$ (Conjecture 1)?

Bibliography

1. A. Aleman, J.-F. Olsen, and E. Saksman, *Fourier multipliers for Hardy spaces of Dirichlet series*, *Int. Math. Res. Not. IMRN* (2014), no. 16, 4368–4378, DOI: 10.1093/imrn/rnt080.
2. M. Bailleul and P. Lefèvre, *Some Banach spaces of Dirichlet series*, *Studia Math.* **226** (2015), 17–55, arXiv: 1311.3845.
3. R. Balasubramanian, B. Calado, and H. Queffélec, *The Bohr inequality for ordinary Dirichlet series*, *Studia Math.* **175** (2006), no. 3, 285–304.
4. K. Bampouras, *Failure of Nehari's theorem for multiplicative Hankel forms in Schatten classes II*, 2026, arXiv: 2610.06728.
5. ———, *Helson inequality, Hankel operators, and weak factorization on Paley–Wiener spaces of convex domains*, 2026, arXiv: 2609.10730.
6. ———, *Nehari's theorem and Hardy's inequality for Paley–Wiener spaces*, *J. Funct. Anal.* **290** (2026), 111408.
7. K. Bampouras and K.-M. Perfekt, *Boundary-weighted Fourier inequalities for convex domains*, 2026, arXiv: 2608.19806.
8. ———, *Recovering product BMO from Schatten Hankel operators*, 2026, arXiv: 2604.26722.
9. F. Bayart and O. F. Brevig, *Composition operators and embedding theorems for some function spaces of Dirichlet series*, *Math. Z.* **293** (2019), no. 3–4, 989–1014, DOI: 10.1007/s00209-018-2215-x.
10. F. Bayart et al., *Contractive inequalities for Bergman spaces and multiplicative Hankel forms*, *Trans. Amer. Math. Soc.* **371** (2019), no. 1, 681–707, DOI: 10.1090/tran/7290.
11. A. Bondarenko et al., *Linear space properties of H^p spaces of Dirichlet series*, *Trans. Amer. Math. Soc.* **372** (2019), no. 9, 6677–6702, DOI: 10.1090/tran/7898.
12. O. F. Brevig, *Linear functions and duality on the infinite polytorus*, *Collectanea Mathematica* **70** (2019), no. 3, 493–500, DOI: 10.1007/s13348-019-00243-8.
13. ———, *Minimal norm Hankel operators*, *Proc. Amer. Math. Soc.* **150** (2022), no. 12, 4383–4392, DOI: 10.1090/proc/15969.
14. O. F. Brevig and K.-M. Perfekt, *Failure of Nehari's theorem for multiplicative Hankel forms in Schatten classes*, *Studia Math.* **228** (2015), no. 2, 101–108, DOI: 10.4064/sm228-2-1.
15. ———, *The Nehari problem for the Paley–Wiener space of a disc*, Preprint, 2022.

16. O. F. Brevig et al., *The multiplicative Hilbert matrix*, *Adv. Math.* **302** (2016), 410–432.
17. H. Brezis and E. Lieb, *A relation between pointwise convergence of functions and convergence of functionals*, *Proc. Amer. Math. Soc.* **88** (1983), no. 3, 486–490.
18. S. Chavan, C. K. Sahu, and K. B. Sinha, *Helson matrices induced by measures*, arXiv:2407.04124, 2024.
19. B. Chen et al., *The local embedding problem for Hardy spaces of Dirichlet series*, Version 2, 13 September 2026, (2026), arXiv: 2609.11560v2.
20. Y. Chen and X. Fu, *A note on the partial sum of bounded Dirichlet series*, 2026, arXiv: 2608.17989.
21. B. J. Cole and T. W. Gamelin, *Representing measures and Hardy spaces for the infinite polydisk algebra*, *Proc. London Math. Soc. (3)* **53** (1986), no. 1, 112–142.
22. A. Defant and I. Schoolmann, $\mathcal{H}_p$*-theory of general Dirichlet series*, *J. Fourier Anal. Appl.* **25** (2019), no. 6, 3220–3258.
23. S. H. Ferguson and M. T. Lacey, *A characterization of product BMO by commutators*, *Acta Math.* **189** (2002), no. 2, 143–160.
24. M. Gerspach, *Low pseudomoments of the Riemann zeta function and its powers*, *Int. Math. Res. Not. IMRN* (2022), no. 1, 625–664, DOI: 10.1093/imrn/rnaa159, arXiv: 1909.10224.
25. A. J. Harper, *Moments of random multiplicative functions, I: Low moments, better than square-root cancellation, and critical multiplicative chaos*, *Forum Math. Pi* **8** (2020), e1, DOI: 10.1017/fmp.2019.7, arXiv: 1703.06654.
26. P. Hartman, *On completely continuous Hankel matrices*, *Proc. Amer. Math. Soc.* **9** (1958), 862–866, DOI: 10.1090/S0002-9939-1958-0108684-8.
27. H. Helson, *Hankel forms and sums of random variables*, *Studia Math.* **176** (2006), no. 1, 85–92.
28. H. Helson and D. Lowdenslager, *Prediction theory and Fourier series in several variables*, *Acta Math.* **99** (1958), 165–202.
29. W. Hoeffding, *Probability inequalities for sums of bounded random variables*, *J. Amer. Statist. Assoc.* **58** (1963), 13–30, DOI: 10.1080/01621459.1963.10500830.
30. S. Janson, J. Peetre, and R. Rochberg, *Hankel forms and the Fock space*, *Rev. Mat. Iberoamericana* **3** (1987), no. 1, 61–138, DOI: 10.4171/RMI/46.
31. S. Janson, H. Upmeier, and R. Wallstén, *Schatten-norm identities for Hankel operators*, *J. Funct. Anal.* **119** (1994), no. 1, 210–216.
32. M. I. Kadets and M. G. Snobar, *Certain functionals on the Minkowski compactum*, *Mat. Zametki* **10** (1971), 453–457.
33. S. Kakutani, *On equivalence of infinite product measures*, *Ann. of Math. (2)* **49** (1948), 214–224.
34. J. Komlós, *A generalization of a problem of Steinhaus*, *Acta Math. Acad. Sci. Hungar.* **18** (1967), 217–229.
35. A. Kulikov, *Functionals with extrema at reproducing kernels*, *Geom. Funct. Anal.* **32** (2022), 938–949.

36. M. Lacey and E. Terwilleger, *Hankel operators in several complex variables and product BMO*, *Houston J. Math.* **35** (2009), no. 1, 159–183.
37. D. H. Luecking, *Trace ideal criteria for Toeplitz operators*, *J. Funct. Anal.* **73** (1987), no. 2, 345–368.
38. J. Ma, J. Lu, and X. Zu, *Helson forms and integral operators on Bergman spaces of Dirichlet series*, Version 2, 26 August 2026, 2026, arXiv: 2608.22882.
39. N. Miheisi and A. Pushnitski, *A Helson matrix with explicit eigenvalue asymptotics*, *J. Funct. Anal.* **275** (2018), no. 4, arXiv:1709.06326, 967–987.
40. A. Nikeghbali and M. Yor, *The Barnes G function and its relations with sums and products of generalized gamma convolution variables*, *Electron. Commun. Probab.* **14** (2009), 396–411.
41. A. Nishry and A. Wennman, *The forbidden region for random zeros: appearance of quadrature domains*, *Comm. Pure Appl. Math.* (), DOI: 10.1002/cpa.22142.
42. J.-F. Olsen and E. Saksman, *On the boundary behaviour of the Hardy spaces of Dirichlet series and a frame bound estimate*, *J. Reine Angew. Math.* **663** (2012), 33–66.
43. J.-F. Olsen, *Local properties of Hilbert spaces of Dirichlet series*, *J. Funct. Anal.* **261** (2011), no. 9, 2669–2696.
44. J. Ortega-Cerdà and K. Seip, *A lower bound in Nehari's theorem on the polydisc*, *J. Anal. Math.* **118** (2012), no. 1, 339–342, DOI: 10.1007/s11854-012-0038-y.
45. K.-M. Perfekt, *A Fejér–Riesz inequality for Dirichlet series*, Version 2, 14 September 2026, (2026), arXiv: 2609.03855v2.
46. K.-M. Perfekt and A. Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, *Proc. Lond. Math. Soc. (3)* **116** (2018), no. 1, 101–134, DOI: 10.1112/plms.12068.
47. ———, *On the spectrum of the multiplicative Hilbert matrix*, *Ark. Mat.* **56** (2018), no. 1, 163–183, DOI: 10.4310/ARKIV.2018.v56.n1.a10.
48. J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, *Illinois J. Math.* **6** (1962), 64–94.
49. E. Saksman and K. Seip, *Some open questions in analysis for Dirichlet series*, in: *Recent progress on operator theory and approximation in spaces of analytic functions*, vol. 679, Contemp. Math. Amer. Math. Soc., Providence, RI, 2016, 179–191.
50. D. Sarason, *Algebras of functions on the unit circle*, *Bull. Amer. Math. Soc.* **79** (1973), 286–299, DOI: 10.1090/S0002-9904-1973-13144-1.
51. Ø. Søvik, *Hankel forms and Nehari's theorem*, Master's thesis, Norwegian University of Science and Technology, 2017.
52. A. Stan, *A Hölder inequality for holomorphic functions*, *J. Inequal. Pure Appl. Math.* **4** (2003), no. 4, Article 75.
53. J.-K. Wang, *Note on a theorem of Nehari on Hankel forms*, *Proc. Amer. Math. Soc.* **24** (1970), 103–105.
54. H. Widom, *Hankel matrices*, *Trans. Amer. Math. Soc.* **121** (1966), 1–35, DOI: 10.2307/1994330.